

8		7		6		5		4		3		2		1																																																																																																																																																																																																																																																																																																																																																																																																																																																																													
1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%. 2. ALL CAPACITANCE VALUES ARE IN MICROFARADS. 3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.												REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE																																																																																																																																																																																																																																																																																																																																																																																																																																																																												
												7	0011175308	ENGINEERING RELEASED	2018-02-05																																																																																																																																																																																																																																																																																																																																																																																																																																																																												
D32/D33 Top MLB: EVT (D32 Build)																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
LAST_MODIFICATION=Wed Jan 31 16:37:18 2018																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
<table><tr><th>PAGE</th><th>CSA</th><th>CONTENTS</th><th>SYNC</th><th>DATE</th><th>PAGE</th><th>CSA</th><th>CONTENTS</th><th>SYNC</th><th>DATE</th></tr><tr><td>1</td><td>1</td><td>TABLE OF CONTENTS</td><td></td><td></td><td>46</td><td>60</td><td>I/O: LDCM</td><td>test_mlb</td><td>06/06/2017</td></tr><tr><td>2</td><td>2</td><td>SYSTEM:BOM Tables</td><td>test_mlb</td><td>10/13/2016</td><td>47</td><td>61</td><td>I/O: Gecko</td><td>test_mlb</td><td>10/17/2016</td></tr><tr><td>3</td><td>3</td><td>SYSTEM:BOM Tables FF Specific</td><td></td><td>08/09/2017</td><td>48</td><td>62</td><td>I/O: USB PD</td><td>test_mlb</td><td>10/13/2016</td></tr><tr><td>4</td><td>4</td><td>SYSTEM: Mechanical Components</td><td></td><td></td><td>49</td><td>63</td><td>I/O: Hydra</td><td>test_mlb</td><td>10/13/2016</td></tr><tr><td>5</td><td>5</td><td>SYSTEM: Testpoints (Top)</td><td>test_mlb</td><td>10/13/2016</td><td>50</td><td>64</td><td>I/O: B2B Dock</td><td>test_mlb</td><td>10/13/2016</td></tr><tr><td>6</td><td>6</td><td>BOOTSTRAPPING</td><td>test_mlb</td><td>10/13/2016</td><td>51</td><td>65</td><td>B2B: Interposer Bot</td><td></td><td>08/30/2017</td></tr><tr><td>7</td><td>10</td><td>SOC: JTAG,USB,XTAL</td><td>test_mlb</td><td>10/17/2016</td><td>52</td><td>66</td><td>SYSTEM: AP I2C</td><td></td><td></td></tr><tr><td>8</td><td>11</td><td>SOC: PCIE</td><td></td><td>04/07/2017</td><td>53</td><td>67</td><td>SYSTEM: ISP I2C</td><td></td><td></td></tr><tr><td>9</td><td>12</td><td>SOC: MIPI</td><td></td><td></td><td>54</td><td>68</td><td>SYSTEM: AOP/SMC I2C</td><td></td><td></td></tr><tr><td>10</td><td>13</td><td>SOC: LPDP</td><td>test_mlb</td><td>10/13/2016</td><td>55</td><td>70</td><td>SYSTEM: SOC/PMU GPIOs</td><td></td><td>05/09/2017</td></tr><tr><td>11</td><td>14</td><td>SOC: SERIAL</td><td>test_mlb</td><td>04/05/2017</td><td>56</td><td>71</td><td>SYSTEM: AOP GPIOs</td><td></td><td>05/09/2017</td></tr><tr><td>12</td><td>15</td><td>SOC: GPIO & UART</td><td>test_mlb</td><td>04/05/2017</td><td>57</td><td>81</td><td>Interposer: Pins 1-144</td><td></td><td>08/29/2017</td></tr><tr><td>13</td><td>16</td><td>SOC: AOP</td><td></td><td></td><td>58</td><td>82</td><td>Interposer: Pins 145-285</td><td></td><td>08/30/2017</td></tr><tr><td>14</td><td>17</td><td>SOC: POWER (1/3)</td><td></td><td></td><td>59</td><td>83</td><td>Interposer: Top Aliases</td><td></td><td>08/17/2017</td></tr><tr><td>15</td><td>18</td><td>SOC: POWER (2/3)</td><td></td><td></td><td>60</td><td>85</td><td>Interposer: Pins 286-359</td><td></td><td>08/30/2017</td></tr><tr><td>16</td><td>19</td><td>SOC: POWER (3/3)</td><td>test_mlb</td><td>10/17/2016</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>17</td><td>20</td><td>SOC: DEV BOARD ALIASES</td><td></td><td>04/17/2017</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>18</td><td>21</td><td>SOC: LPDP ALIASES</td><td></td><td>08/17/2017</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>19</td><td>26</td><td>NAND</td><td>test_mlb</td><td>03/22/2017</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>20</td><td>27</td><td>SYSTEM POWER: PMU Bucks (1/4)</td><td>test_mlb</td><td>03/10/2017</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>21</td><td>28</td><td>SYSTEM POWER: PMU Bucks (2/4)</td><td>test_mlb</td><td>06/01/2017</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>22</td><td>29</td><td>SYSTEM POWER: PMU LDOs (3/4)</td><td>test_mlb</td><td>03/10/2017</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>23</td><td>30</td><td>SYSTEM POWER: PMU (4/4)</td><td>test_mlb</td><td>03/10/2017</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>24</td><td>31</td><td>SYSTEM POWER: Boost</td><td>test_mlb</td><td>10/13/2016</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>25</td><td>32</td><td>SYSTEM POWER: B2B Battery</td><td>test_mlb</td><td>10/13/2016</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>26</td><td>33</td><td>SYSTEM POWER: Charger</td><td>test_mlb</td><td>10/13/2016</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>27</td><td>35</td><td>SYSTEM POWER: B2B Cyclone + Button</td><td>test_mlb</td><td>10/13/2016</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>28</td><td>36</td><td>SENSORS</td><td>test_mlb</td><td>10/13/2016</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>29</td><td>37</td><td>CAMERA: PMU (1/2)</td><td>test_mlb</td><td>10/13/2016</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>30</td><td>38</td><td>CAMERA: PMU (2/2)</td><td>test_mlb</td><td>03/22/2017</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>31</td><td>39</td><td>CAMERA: B2B Wide (TX)</td><td>test_mlb</td><td>10/13/2016</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>32</td><td>40</td><td>CAMERA: B2B Tele [MT]</td><td>test_mlb</td><td>10/13/2016</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>33</td><td>41</td><td>CAMERA: Strobe Drivers</td><td>test_mlb</td><td>03/22/2017</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>34</td><td>42</td><td>CAMERA: B2B Fcam</td><td>test_mlb</td><td>10/13/2016</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>35</td><td>43</td><td>CAMERA: B2B Strobe + Hold Button</td><td>test_mlb</td><td>03/22/2017</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>36</td><td>44</td><td>PEARL: Power</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>37</td><td>45</td><td>PEARL: B2B Romeo + Juliet</td><td>test_mlb</td><td>10/13/2016</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>38</td><td>46</td><td>PEARL: B2B Rosaline + Sensor</td><td>test_mlb</td><td>10/13/2016</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>39</td><td>47</td><td>AUDIO: CODEC (1/2)</td><td>test_mlb</td><td>10/13/2016</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>40</td><td>48</td><td>AUDIO: CODEC (2/2)</td><td>test_mlb</td><td>10/13/2016</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>41</td><td>49</td><td>AUDIO: SOUTH SPKAMP</td><td></td><td>04/05/2017</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>42</td><td>50</td><td>AUDIO: NORTH SPKAMP</td><td></td><td>04/05/2017</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>43</td><td>51</td><td>ARC: AMP</td><td></td><td>04/05/2017</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>44</td><td>57</td><td>CG: B2B Display</td><td>test_mlb</td><td>10/13/2016</td><td></td><td></td><td></td><td></td><td></td></tr><tr><td>45</td><td>59</td><td>I/O: Overvoltage Cut-Off Circuit</td><td></td><td></td><td></td><td></td><td></td><td></td><td></td></tr></table>																PAGE	CSA	CONTENTS	SYNC	DATE	PAGE	CSA	CONTENTS	SYNC	DATE	1	1	TABLE OF CONTENTS			46	60	I/O: LDCM	test_mlb	06/06/2017	2	2	SYSTEM:BOM Tables	test_mlb	10/13/2016	47	61	I/O: Gecko	test_mlb	10/17/2016	3	3	SYSTEM:BOM Tables FF Specific		08/09/2017	48	62	I/O: USB PD	test_mlb	10/13/2016	4	4	SYSTEM: Mechanical Components			49	63	I/O: Hydra	test_mlb	10/13/2016	5	5	SYSTEM: Testpoints (Top)	test_mlb	10/13/2016	50	64	I/O: B2B Dock	test_mlb	10/13/2016	6	6	BOOTSTRAPPING	test_mlb	10/13/2016	51	65	B2B: Interposer Bot		08/30/2017	7	10	SOC: JTAG,USB,XTAL	test_mlb	10/17/2016	52	66	SYSTEM: AP I2C			8	11	SOC: PCIE		04/07/2017	53	67	SYSTEM: ISP I2C			9	12	SOC: MIPI			54	68	SYSTEM: AOP/SMC I2C			10	13	SOC: LPDP	test_mlb	10/13/2016	55	70	SYSTEM: SOC/PMU GPIOs		05/09/2017	11	14	SOC: SERIAL	test_mlb	04/05/2017	56	71	SYSTEM: AOP GPIOs		05/09/2017	12	15	SOC: GPIO & UART	test_mlb	04/05/2017	57	81	Interposer: Pins 1-144		08/29/2017	13	16	SOC: AOP			58	82	Interposer: Pins 145-285		08/30/2017	14	17	SOC: POWER (1/3)			59	83	Interposer: Top Aliases		08/17/2017	15	18	SOC: POWER (2/3)			60	85	Interposer: Pins 286-359		08/30/2017	16	19	SOC: POWER (3/3)	test_mlb	10/17/2016						17	20	SOC: DEV BOARD ALIASES		04/17/2017						18	21	SOC: LPDP ALIASES		08/17/2017						19	26	NAND	test_mlb	03/22/2017						20	27	SYSTEM POWER: PMU Bucks (1/4)	test_mlb	03/10/2017						21	28	SYSTEM POWER: PMU Bucks (2/4)	test_mlb	06/01/2017						22	29	SYSTEM POWER: PMU LDOs (3/4)	test_mlb	03/10/2017						23	30	SYSTEM POWER: PMU (4/4)	test_mlb	03/10/2017						24	31	SYSTEM POWER: Boost	test_mlb	10/13/2016						25	32	SYSTEM POWER: B2B Battery	test_mlb	10/13/2016						26	33	SYSTEM POWER: Charger	test_mlb	10/13/2016						27	35	SYSTEM POWER: B2B Cyclone + Button	test_mlb	10/13/2016						28	36	SENSORS	test_mlb	10/13/2016						29	37	CAMERA: PMU (1/2)	test_mlb	10/13/2016						30	38	CAMERA: PMU (2/2)	test_mlb	03/22/2017						31	39	CAMERA: B2B Wide (TX)	test_mlb	10/13/2016						32	40	CAMERA: B2B Tele [MT]	test_mlb	10/13/2016						33	41	CAMERA: Strobe Drivers	test_mlb	03/22/2017						34	42	CAMERA: B2B Fcam	test_mlb	10/13/2016						35	43	CAMERA: B2B Strobe + Hold Button	test_mlb	03/22/2017						36	44	PEARL: Power								37	45	PEARL: B2B Romeo + Juliet	test_mlb	10/13/2016						38	46	PEARL: B2B Rosaline + Sensor	test_mlb	10/13/2016						39	47	AUDIO: CODEC (1/2)	test_mlb	10/13/2016						40	48	AUDIO: CODEC (2/2)	test_mlb	10/13/2016						41	49	AUDIO: SOUTH SPKAMP		04/05/2017						42	50	AUDIO: NORTH SPKAMP		04/05/2017						43	51	ARC: AMP		04/05/2017						44	57	CG: B2B Display	test_mlb	10/13/2016						45	59	I/O: Overvoltage Cut-Off Circuit							
PAGE	CSA	CONTENTS	SYNC	DATE	PAGE	CSA	CONTENTS	SYNC	DATE																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
1	1	TABLE OF CONTENTS			46	60	I/O: LDCM	test_mlb	06/06/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
2	2	SYSTEM:BOM Tables	test_mlb	10/13/2016	47	61	I/O: Gecko	test_mlb	10/17/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
3	3	SYSTEM:BOM Tables FF Specific		08/09/2017	48	62	I/O: USB PD	test_mlb	10/13/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
4	4	SYSTEM: Mechanical Components			49	63	I/O: Hydra	test_mlb	10/13/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
5	5	SYSTEM: Testpoints (Top)	test_mlb	10/13/2016	50	64	I/O: B2B Dock	test_mlb	10/13/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
6	6	BOOTSTRAPPING	test_mlb	10/13/2016	51	65	B2B: Interposer Bot		08/30/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
7	10	SOC: JTAG,USB,XTAL	test_mlb	10/17/2016	52	66	SYSTEM: AP I2C																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
8	11	SOC: PCIE		04/07/2017	53	67	SYSTEM: ISP I2C																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
9	12	SOC: MIPI			54	68	SYSTEM: AOP/SMC I2C																																																																																																																																																																																																																																																																																																																																																																																																																																																																																				
10	13	SOC: LPDP	test_mlb	10/13/2016	55	70	SYSTEM: SOC/PMU GPIOs		05/09/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
11	14	SOC: SERIAL	test_mlb	04/05/2017	56	71	SYSTEM: AOP GPIOs		05/09/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
12	15	SOC: GPIO & UART	test_mlb	04/05/2017	57	81	Interposer: Pins 1-144		08/29/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
13	16	SOC: AOP			58	82	Interposer: Pins 145-285		08/30/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
14	17	SOC: POWER (1/3)			59	83	Interposer: Top Aliases		08/17/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
15	18	SOC: POWER (2/3)			60	85	Interposer: Pins 286-359		08/30/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																		
16	19	SOC: POWER (3/3)	test_mlb	10/17/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
17	20	SOC: DEV BOARD ALIASES		04/17/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
18	21	SOC: LPDP ALIASES		08/17/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
19	26	NAND	test_mlb	03/22/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
20	27	SYSTEM POWER: PMU Bucks (1/4)	test_mlb	03/10/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
21	28	SYSTEM POWER: PMU Bucks (2/4)	test_mlb	06/01/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
22	29	SYSTEM POWER: PMU LDOs (3/4)	test_mlb	03/10/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
23	30	SYSTEM POWER: PMU (4/4)	test_mlb	03/10/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
24	31	SYSTEM POWER: Boost	test_mlb	10/13/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
25	32	SYSTEM POWER: B2B Battery	test_mlb	10/13/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
26	33	SYSTEM POWER: Charger	test_mlb	10/13/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
27	35	SYSTEM POWER: B2B Cyclone + Button	test_mlb	10/13/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
28	36	SENSORS	test_mlb	10/13/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
29	37	CAMERA: PMU (1/2)	test_mlb	10/13/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
30	38	CAMERA: PMU (2/2)	test_mlb	03/22/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
31	39	CAMERA: B2B Wide (TX)	test_mlb	10/13/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
32	40	CAMERA: B2B Tele [MT]	test_mlb	10/13/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
33	41	CAMERA: Strobe Drivers	test_mlb	03/22/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
34	42	CAMERA: B2B Fcam	test_mlb	10/13/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
35	43	CAMERA: B2B Strobe + Hold Button	test_mlb	03/22/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
36	44	PEARL: Power																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
37	45	PEARL: B2B Romeo + Juliet	test_mlb	10/13/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
38	46	PEARL: B2B Rosaline + Sensor	test_mlb	10/13/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
39	47	AUDIO: CODEC (1/2)	test_mlb	10/13/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
40	48	AUDIO: CODEC (2/2)	test_mlb	10/13/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
41	49	AUDIO: SOUTH SPKAMP		04/05/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
42	50	AUDIO: NORTH SPKAMP		04/05/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
43	51	ARC: AMP		04/05/2017																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
44	57	CG: B2B Display	test_mlb	10/13/2016																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							
45	59	I/O: Overvoltage Cut-Off Circuit																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
BOM:639-03991 (Ultimate) BOM:639-03992 (Extreme) BOM:639-03990 (Max) MCO:056-05750																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
<table><tr><th>PART#</th><th>QTY</th><th>DESCRIPTION</th><th>REFERENCE DESIGNATOR(S)</th><th>CRITICAL</th><th>BOM OPTION</th></tr><tr><td>051-02545</td><td>1</td><td>SCH,MLB_TOP,D32</td><td>SCH</td><td>CRITICAL</td><td>?</td></tr><tr><td>820-00997</td><td>1</td><td>PCB,MLB_TOP,D32</td><td>PCB</td><td>CRITICAL</td><td>?</td></tr></table>																PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION	051-02545	1	SCH,MLB_TOP,D32	SCH	CRITICAL	?	820-00997	1	PCB,MLB_TOP,D32	PCB	CRITICAL	?																																																																																																																																																																																																																																																																																																																																																																																																																																																										
PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
051-02545	1	SCH,MLB_TOP,D32	SCH	CRITICAL	?																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
820-00997	1	PCB,MLB_TOP,D32	PCB	CRITICAL	?																																																																																																																																																																																																																																																																																																																																																																																																																																																																																						
TABLE OF CONTENTS																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
<table><tr><td colspan="2">DRAWING TITLE</td><td colspan="2">SCH,MLB,TOP,D32</td></tr><tr><td colspan="2">DRAWING NUMBER</td><td>051-02545</td><td>SIZE D</td></tr><tr><td colspan="2">REVISION</td><td colspan="2">7.0.0</td></tr><tr><td colspan="2">BRANCH</td><td colspan="2"></td></tr><tr><td colspan="2">PAGE</td><td colspan="2">1 OF 85</td></tr><tr><td colspan="2">SHEET</td><td colspan="2">1 OF 60</td></tr></table>																DRAWING TITLE		SCH,MLB,TOP,D32		DRAWING NUMBER		051-02545	SIZE D	REVISION		7.0.0		BRANCH				PAGE		1 OF 85		SHEET		1 OF 60																																																																																																																																																																																																																																																																																																																																																																																																																																																					
DRAWING TITLE		SCH,MLB,TOP,D32																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
DRAWING NUMBER		051-02545	SIZE D																																																																																																																																																																																																																																																																																																																																																																																																																																																																																								
REVISION		7.0.0																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
BRANCH																																																																																																																																																																																																																																																																																																																																																																																																																																																																																											
PAGE		1 OF 85																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
SHEET		1 OF 60																																																																																																																																																																																																																																																																																																																																																																																																																																																																																									
8		7		6		5		4		3		2		1																																																																																																																																																																																																																																																																																																																																																																																																																																																																													

Display CMC's

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
155S00415	155S00391	ALT_PARTS	ALL	CMC, 350SM, 7HGz, MUR

NAND Ultimate

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00340	1	HYNIX, 3DV4, ULTIMATE	U2600	CRITICAL	ULTIMATE

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00359	335S00340	ALT_PARTS	U2600	TOSHIBA, BICS3, ULT
335S00286	335S00340	ALT_PARTS	U2600	SANDISK, BICS3, ULT
335S00288	335S00340	ALT_PARTS	U2600	SAMSUNG, 3DV4, ULT

Global R/C Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0648	138S0652	ALT_PARTS	ALL	CAP, XSR, 4, 7UF, 5, 3V, 0, 60PM, 0402, TAIYO
138S0739	138S0706	ALT_PARTS	ALL	CAP, CER, XSR, 0, 220F, 20V, 0, 3V, 20M
138S00849	138S0831	ALT_PARTS	ALL	CAP, CER, XSR, 0, 220F, 20V, 0, 3V, 500M

Yangtze Inductors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00872	152S00918	ALT_PARTS	ALL	IND, MLD, 0, 47UH, TDK
152S00847	152S00918	ALT_PARTS	ALL	IND, MLD, 0, 47UH, CYN

CRITICAL PART#	COMMENT
152S00918	IND, MLD, 0, 22UH, 20M, 5, 0A, 40MOHM, H=-.65, 1608

Extreme

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00342	1	HYNIX, 3DV4, Extreme	U2600	CRITICAL	EXTREME

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00247	335S00342	ALT_PARTS	U2600	SANDISK, BICS3, SUPREME
335S00276	335S00342	ALT_PARTS	U2600	SAMSUNG, 3DV4, SUPREME
335S00358	335S00342	ALT_PARTS	U2600	TOSHIBA, 3DV4, SUPREME

Max

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
335S00343	1	HYNIX, 3DV4, MAX	U2600	CRITICAL	MAX

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00339	335S00343	ALT_PARTS	U2600	SAMSUNG, 3DV4, MAX

Denali Inductors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00878	152S00831	ALT_PARTS	ALL	IND, MLD, 0, 22UH, 20M, 5, 0A, 30MOHM, H=-.65, 1608
152S00818	152S00831	ALT_PARTS	ALL	IND, MLD, 0, 22UH, 20M, 5, 0A, 30MOHM, H=-.65, 1608, CYN
152S00835	152S00822	ALT_PARTS	ALL	IND, MLD, 0, 47UH, 20M, 4, 5A, 40MOHM, H=-.65, 2012
152S00827	152S00822	ALT_PARTS	ALL	IND, MLD, 0, 47UH, 20M, 4, 5A, 30MOHM, H=-.65, 2012
152S00877	152S00817	ALT_PARTS	ALL	IND, MLD, 0, 1UH, 20M, 0A, 20MOHM, H=-.6, 2012
152S00829	152S00817	ALT_PARTS	ALL	IND, MLD, 0, 1UH, 20M, 0, 0A, 22MOHM, H=-.6, 2012
152S00825	152S00823	ALT_PARTS	ALL	IND, MLD, 1UH, 20M, 0A, 60MO, H=-.65, 2012
152S00833	152S00819	ALT_PARTS	ALL	IND, MLD, 1UH, 20M, 2, 1A, 60MO, H=-.65, 2012
152S00824	152S00819	ALT_PARTS	ALL	IND, MLD, 1UH, 20M, 2A, 60MO, H=-.65, 2012
152S00834	152S00820	ALT_PARTS	ALL	IND, MLD, 0, 47UH, 20M, 3, 0A, 270MO, H=-.65, 2012
152S00828	152S00820	ALT_PARTS	ALL	IND, MLD, 0, 47UH, 20M, 3, 2A, 60MO, H=-.65, 2012
152S00826	152S00821	ALT_PARTS	ALL	IND, MLD, 1UH, 20M, 2, 1A, 60MO, H=-.65, 2012
152S00866	152S00821	ALT_PARTS	ALL	IND, MLD, 1UH, 20M, 2, 1A, 60MO, H=-.65, 2012

CRITICAL PART#	COMMENT
152S00831	IND, MLD, 0, 22UH, 20M, 5, 0A, 40MOHM, H=-.65, 1608
152S00822	IND, MLD, 0, 47UH, 20M, 4, 5A, 47MOHM, H=-.8, 2012
152S00817	IND, MLD, 0, 1UH, 20M, 9, 4A, 22MOHM, H=-.65, 1608
152S00823	IND, MLD, 1UH, 20M, 3, 0A, 60MO, H=-.65, 2016
152S00819	IND, MLD, 1UH, 20M, 1, 7A, 60MO, H=-.65, 2012
152S00820	IND, MLD, 0, 47UH, 20M, 3, 2A, 42MO, H=-.60, 2012
152S00821	IND, MLD, 1UH, 20M, 2, 2A, 60MO, H=-.60, 2012

XTAL Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
197S0612	197S00118	ALT_PARTS	Y1000	XTAL, 24M, 1512
197S00120	197S00118	ALT_PARTS	Y1000	XTAL, 24M, 1512

NEON Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00721	152S00876	ALT_PARTS	L4100, L4120	TY, 1M0

ANSEL Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00716	152S00875	ALT_PARTS	L3700	TY, 1M0

PAGE TITLE			
SYSTEM:BOM Tables			
0	DRAWING NUMBER		SIZE
	051-02545		D
	REVISION		
	7.0.0		
	BRANCH		
	PAGE		
	2 OF 85		
	SHEET		
	2 OF 60		



8

7

6

5

4

3

2

1

EEEE Codes

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7691	1	EEEE FOR (MLB_TOP,639-83991,ULTIMATE)	EEEE_HMV1	CRITICAL	ULTIMATE
825-7691	1	EEEE FOR (MLB_TOP,639-83992,EXTREME)	EEEE_HMV2	CRITICAL	EXTREME
825-7691	1	EEEE FOR (MLB_TOP,639-83998,MAX)	EEEE_HMV8	CRITICAL	MAX

Cyprus OMIT

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
339S80518	1	CYPRUS 4GB Micron	U1000	CRITICAL	SOC

Cyprus ALTs

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
339S80511	339S80518	ALT_PARTS	U1000	CYPRUS 4GB Hynix
339S80512	339S80518	ALT_PARTS	U1000	CYPRUS 4GB Samsung

Combo Stiffener

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
604-19651	1	Combo Stiffener	ST0401	CRITICAL	ALL

SYNC_MASTER=			SYNC_DATE=06/09/2017		
PAGE TITLE					
SYSTEM:BOM Tables FF Specific					
			DRAWING NUMBER		SIZE
			051-02545		D
			REVISION		
			7.0.0		
			BRANCH		
			PAGE		
			SHEET		
			3 OF 85		
			3 OF 60		

8

7

6

5

4

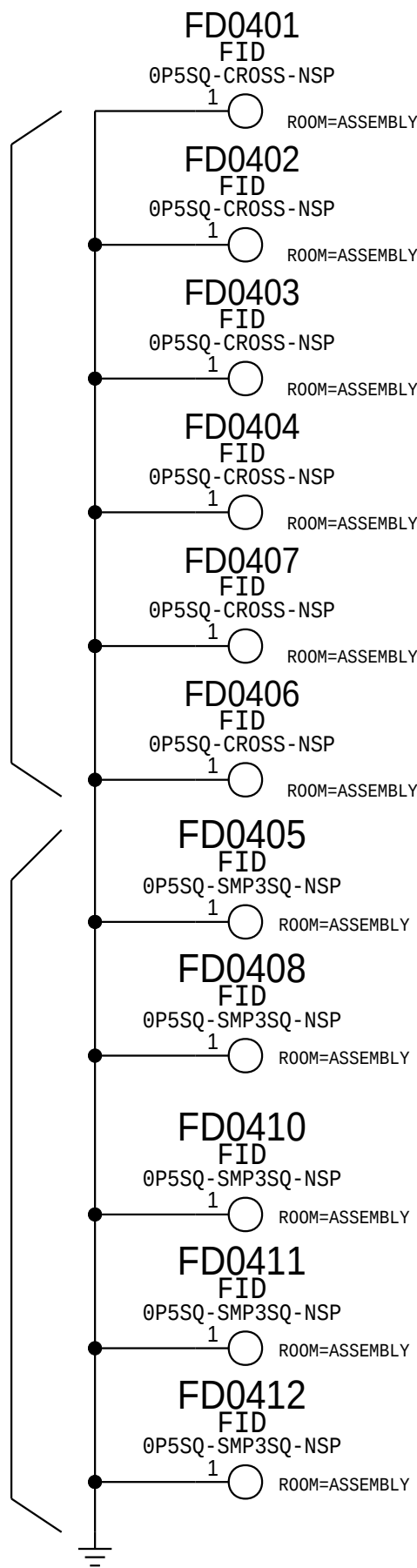
3

2

1

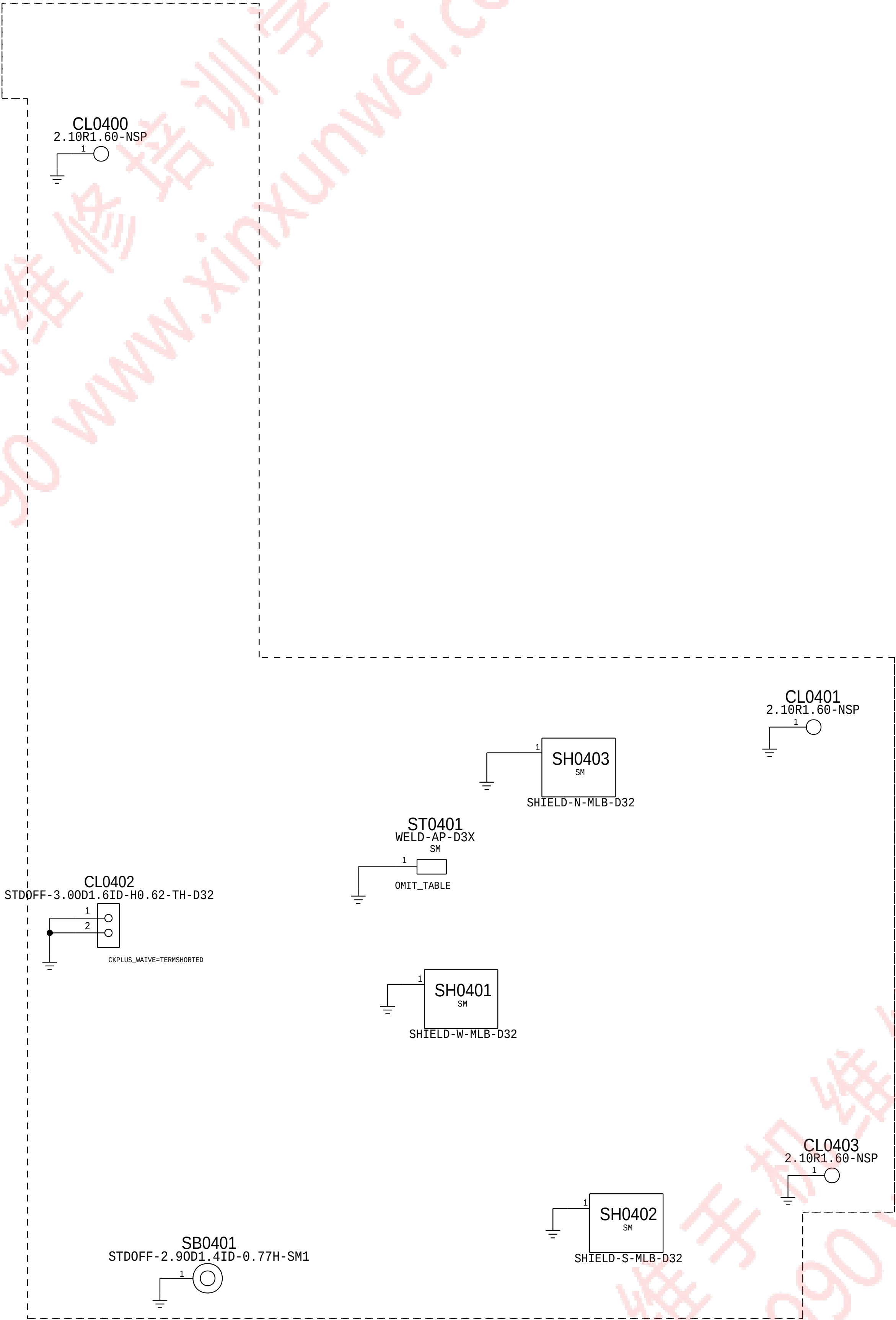


FIDUCIALS



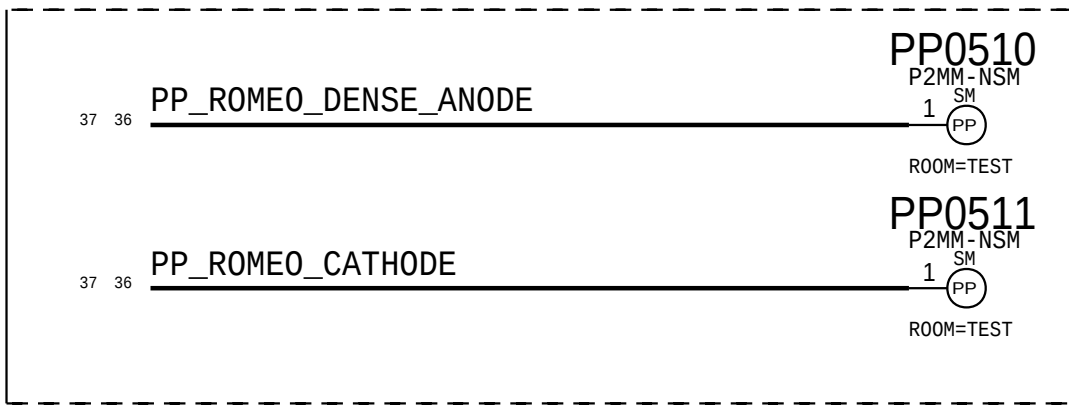
Crosses

Squares

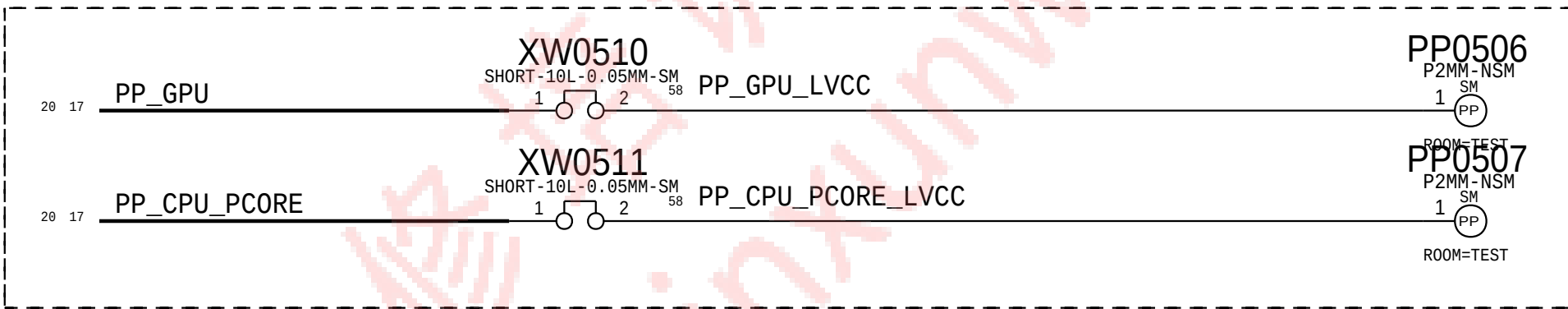


PAGE TITLE		
SYSTEM: Mechanical Components		
	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	4 OF 85
IV ALL RIGHTS RESERVED	SHEET	4 OF 60

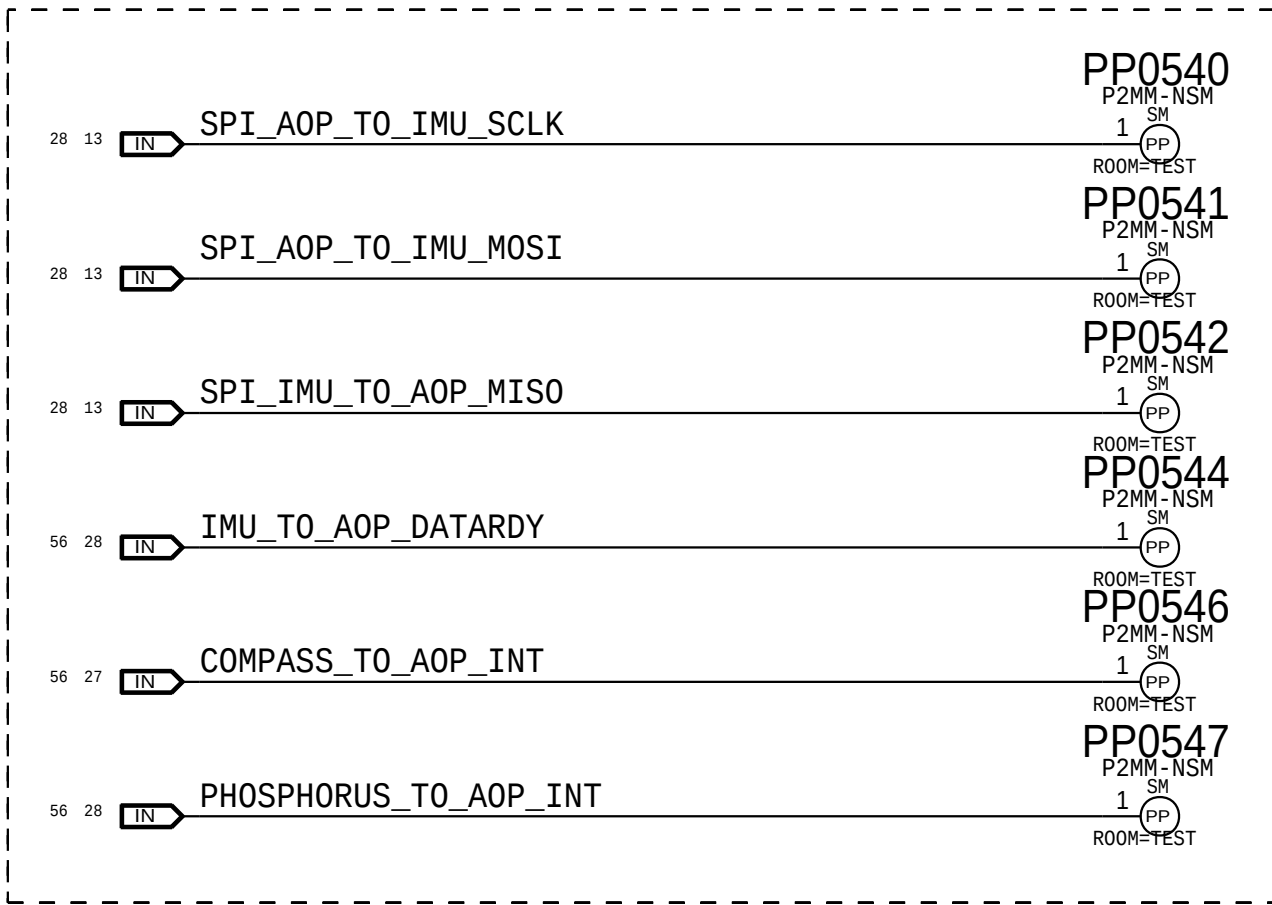
PEARL



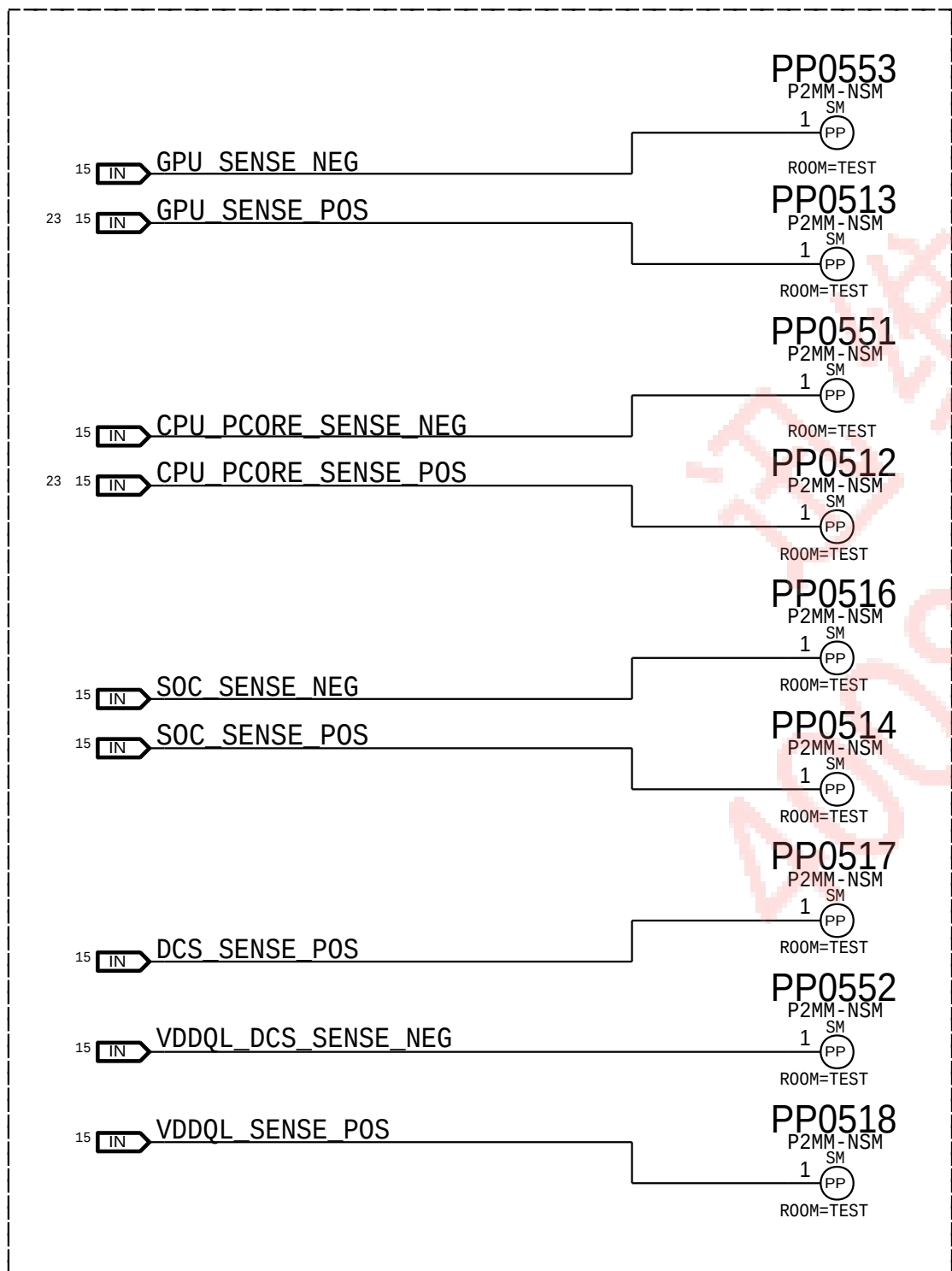
LVCC



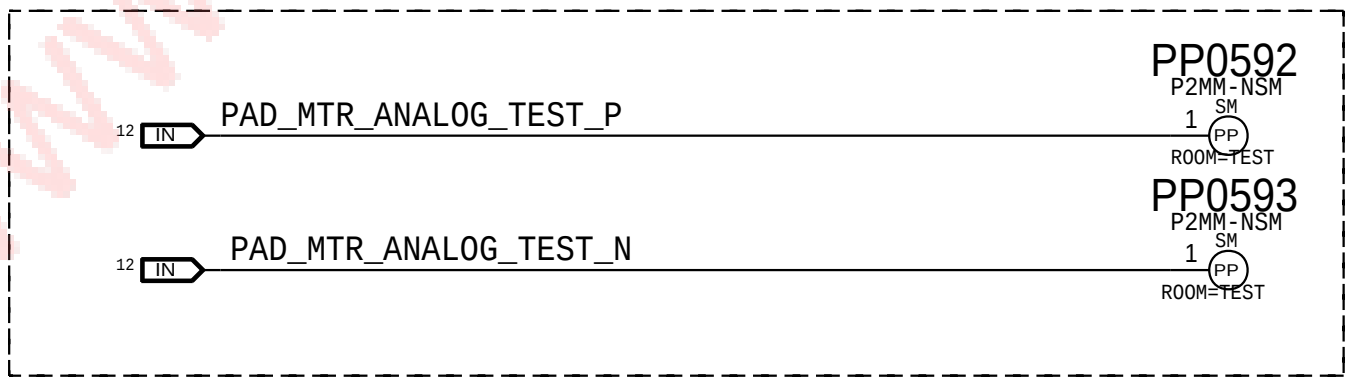
Sensors



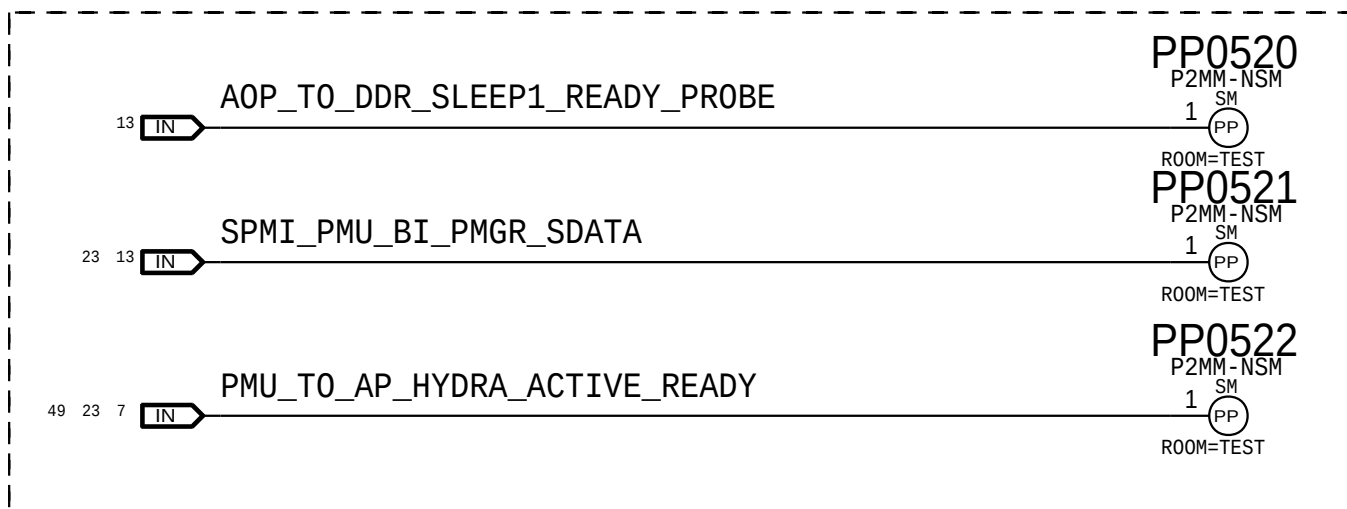
BUMP SENSE



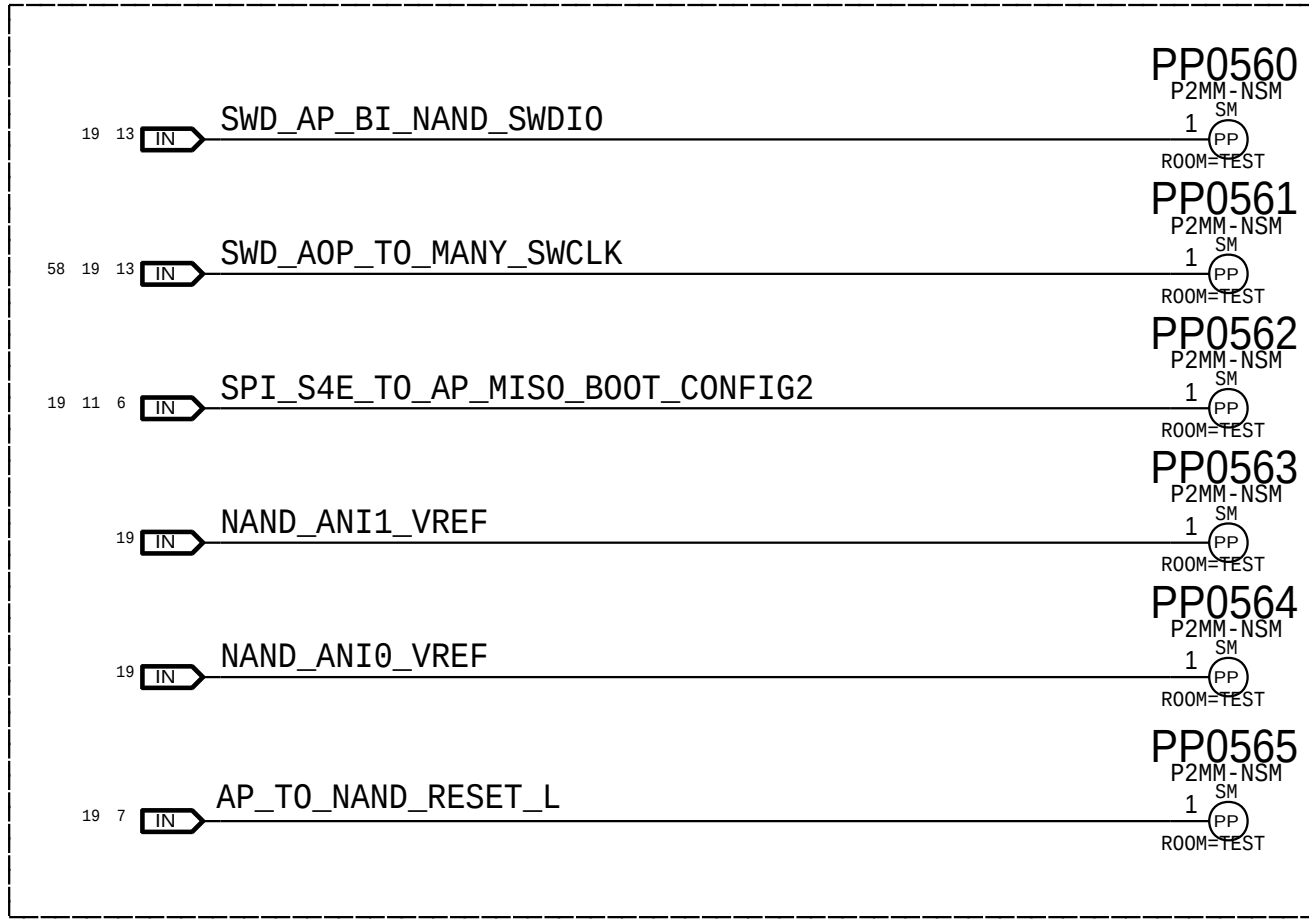
METROLOGY



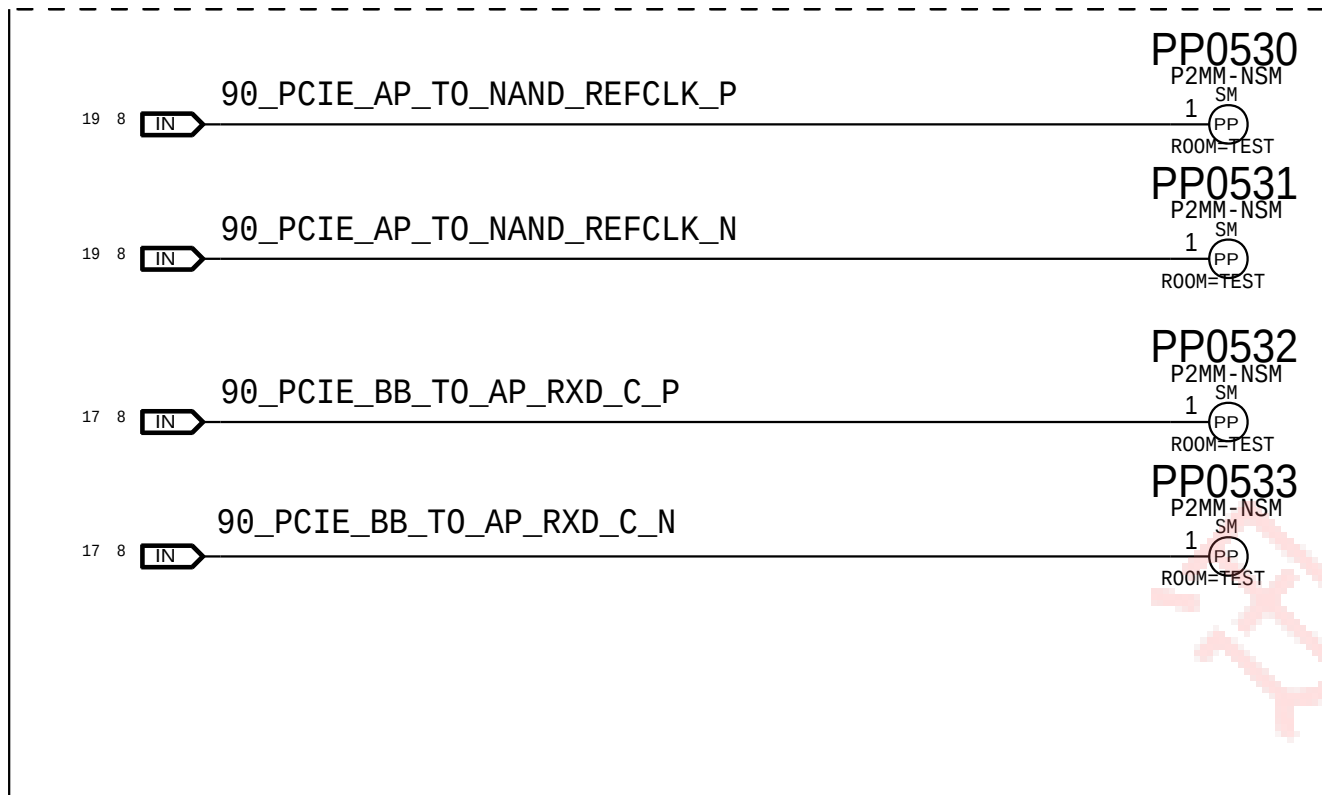
PMU



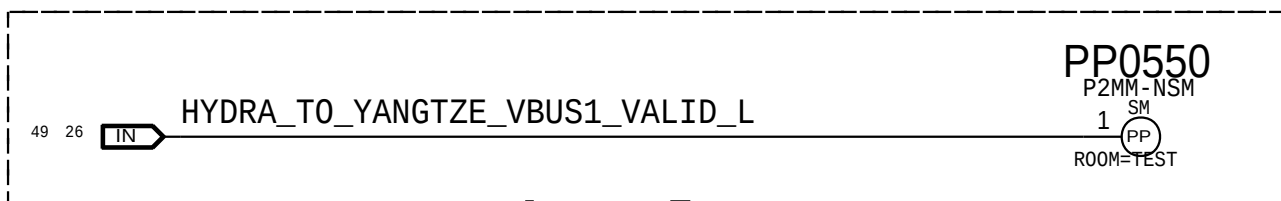
NAND



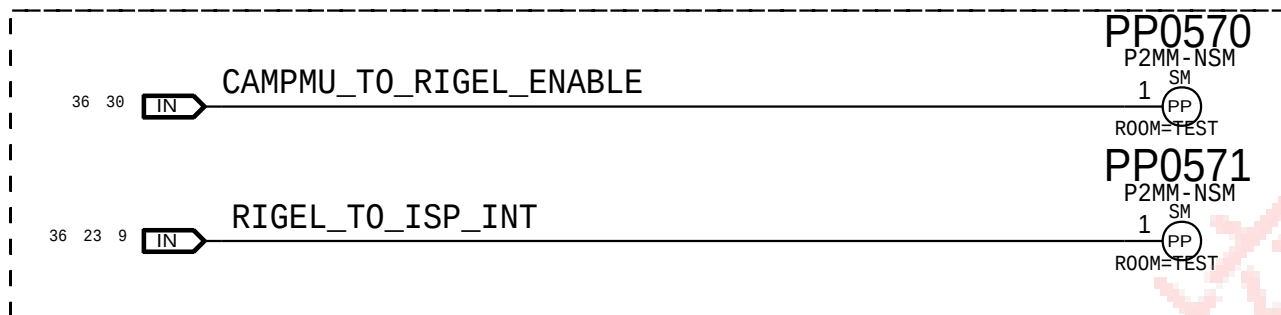
PCIE Refclk



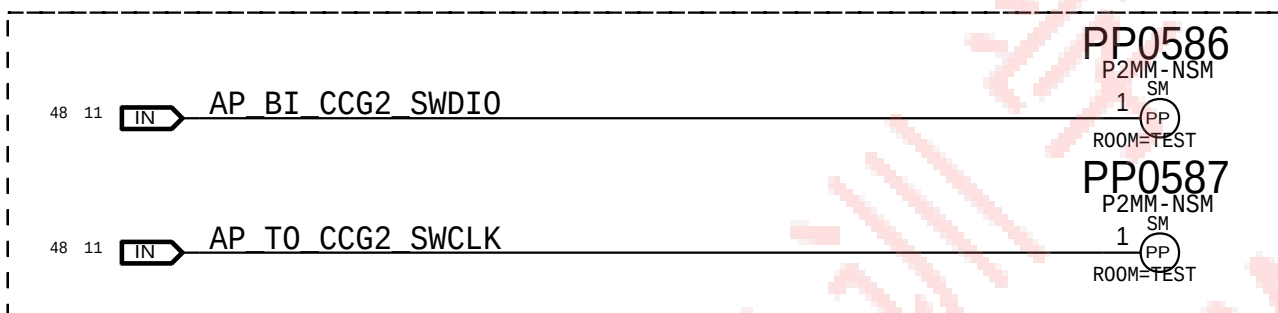
Hydra VBUS



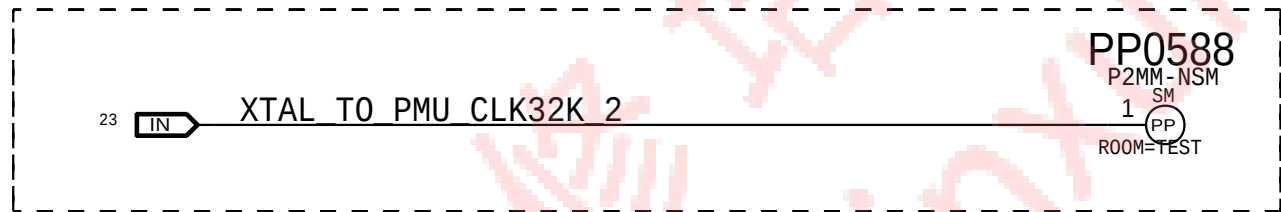
Rigel



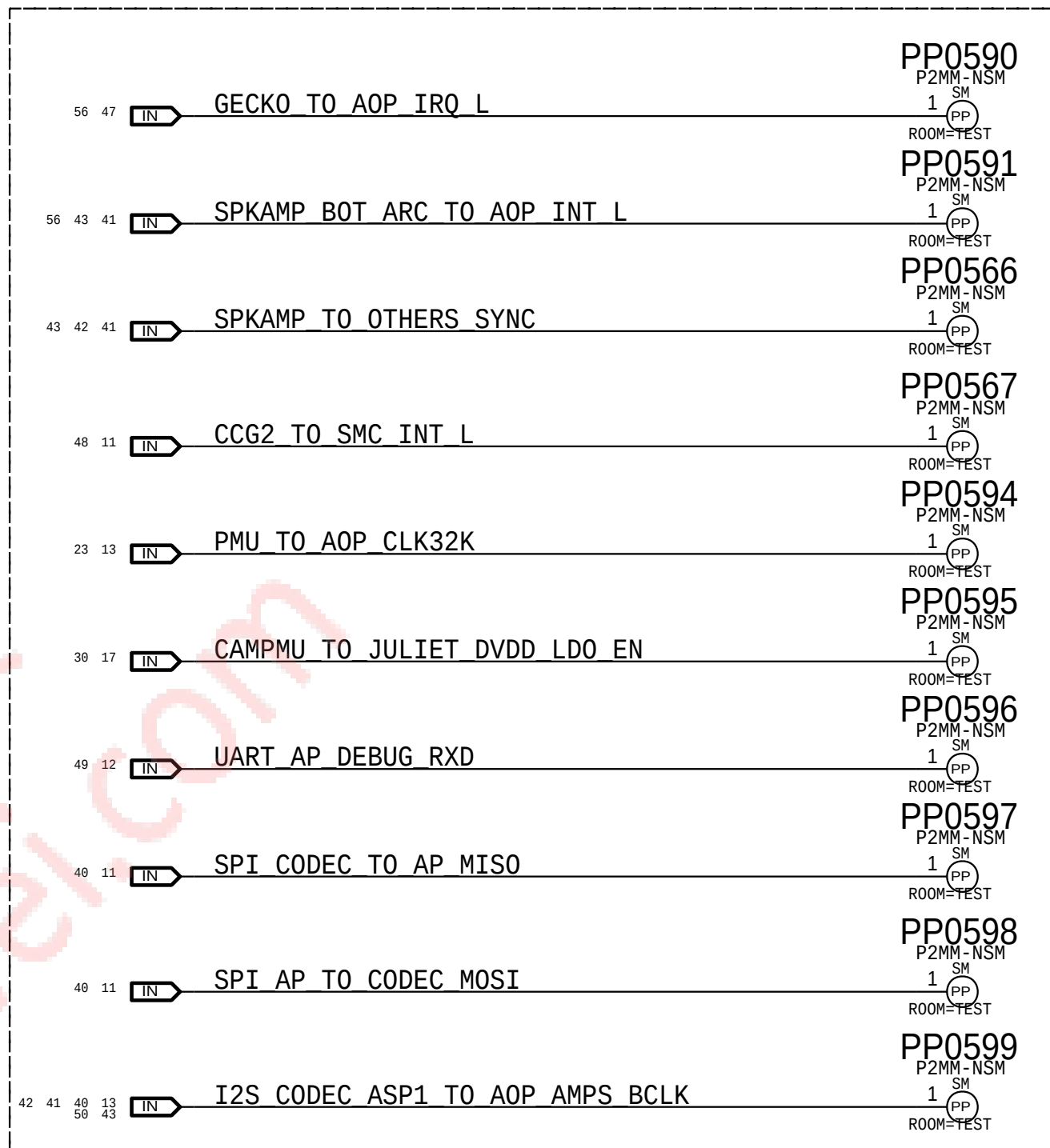
CCG SWD



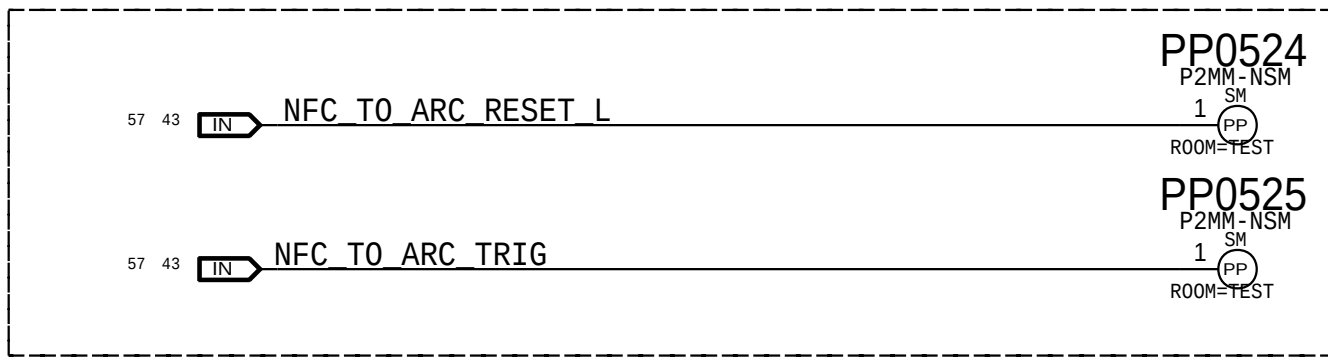
PMU XTAL



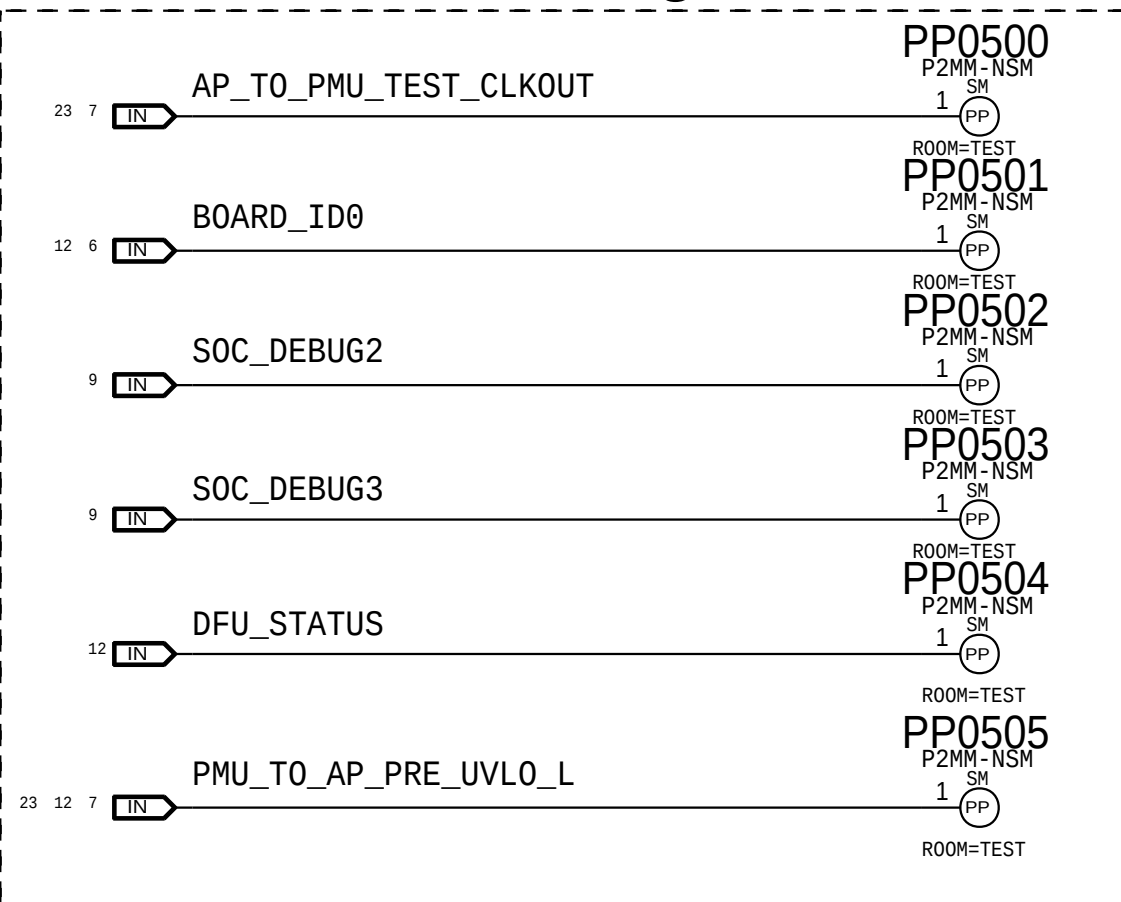
VALIDATION PP's



WALLET MODE



SOC Debug



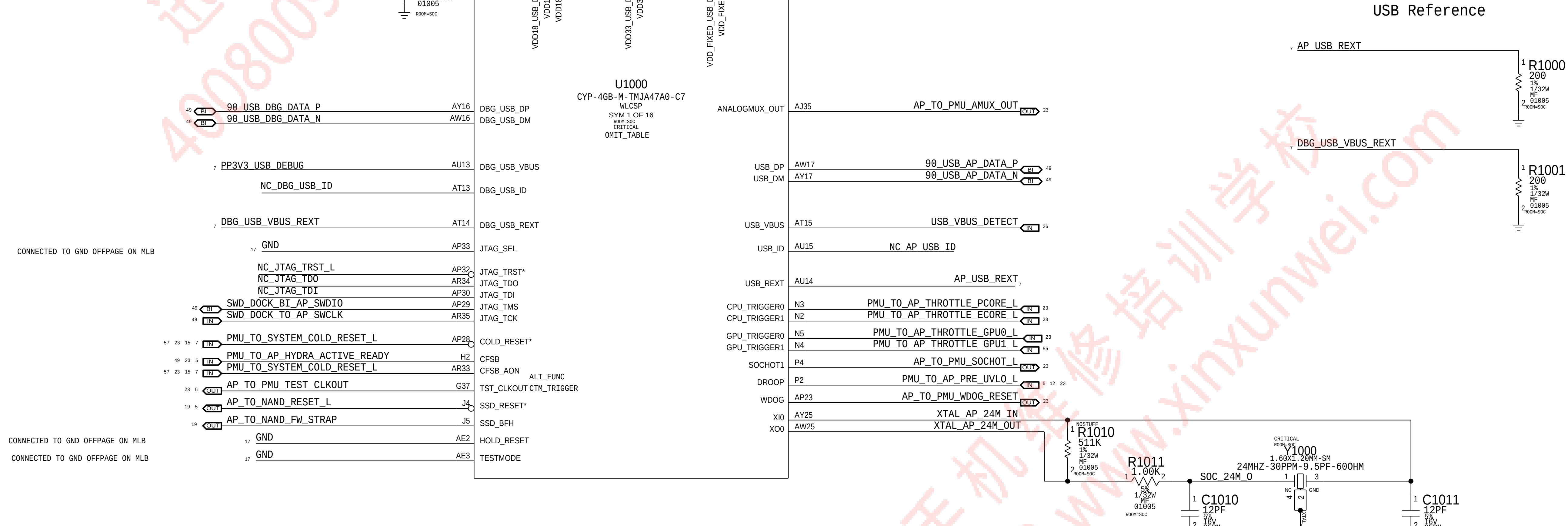
PAGE TITLE		
SYSTEM: Testpoints (Top)		
	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	5 OF 85
	SHEET	5 OF 60

BOOTSTRAPPING:BOARD REV
BOARD ID
BOOT CONFIG



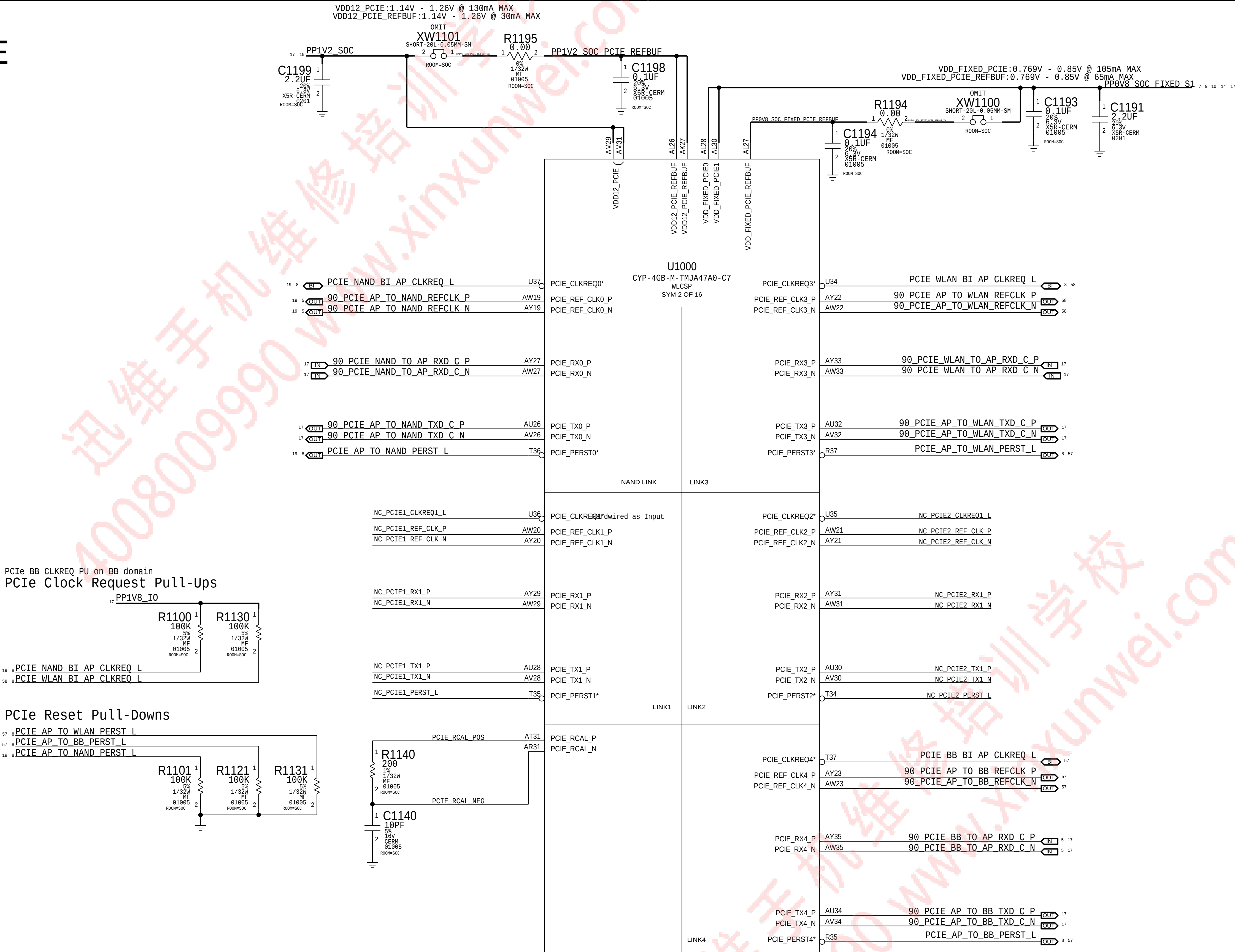
Boot Config [2:0]			
Float = Low PU = High			
	2	1	0
SPI NOR on SPI0 12 MHz	0	0	0
SPI NOR on SPI0 12 MHz Test	0	0	1
SPI NAND on SPI0 12 MHz	0	1	0
SPI NAND on SPI0 12 MHz Test	0	1	1
SPI NOR on SPI0 40 MHz	1	0	0
SPI NOR on SPI0 40 MHz Test	1	0	1
SPI NOR on SPI0 6 MHz	1	1	0
SPI NOR on SPI0 6 MHz Test	1	1	1

PAGE TITLE		051-02545		051-02545	
BOOTSTRAPPING		051-02545		051-02545	
		REVISION		7.0.0	
		BRANCH			
		PAGE		6 OF 85	
		SHEET		6 OF 60	



PAGE TITLE		S0C: JTAG, USB, XTAL	
	DRAWING NUMBER		SIZE
	051-02545		D
	REVISION		
	7.0.0		
	BRANCH		
	PAGE		
	10 OF 85		
	SHEET		
	7 OF 85		

SOC - PCIE



PAGE TITLE			
SOC: PCIE			
	DRAWING NUMBER	051-02545	SIZE
	REVISION	7.0.0	D
	BRANCH		
	PAGE		
	11 OF 85		
	SHEET		
	8 OF 60		

SOC - MIPI

NEED MIPI LANE AND POLAIRTY SWAPPING MAP

(Analog)
VDD_FIXED_MIPI0 0.769V - 0.85V @ TBDmA MAX
VDD_FIXED_MIPI1 0.769V - 0.85V @ TBDmA MAX
VDD_FIXED_MIPI2_PLL 0.769V - 0.85V @ TBDmA MAX

VDD18_MIPI*:1.62V - 1.98V @ TBDmA MAX

PP0V8_SOC_FIXED_S1

PP1V8_I0

C1291
2.2UF
X5R-CERM
0204

C1290
0.1UF
X5R-CERM
0204

C1296
0.1UF
X5R-CERM
0204

C1295
2.2UF
X5R-CERM
0204

1

2

2

1

2

2

1

2

2

1

2

2

E17

AM9

AF9

G18

AL10

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

VDD_FIXED_MIPI0

VDD_FIXED_MIPI1

VDD_FIXED_MIPI2_PLL

VDD18_MIPI0

VDD18_MIPI1

MIPI lanes can all flip polarity for routing purposes

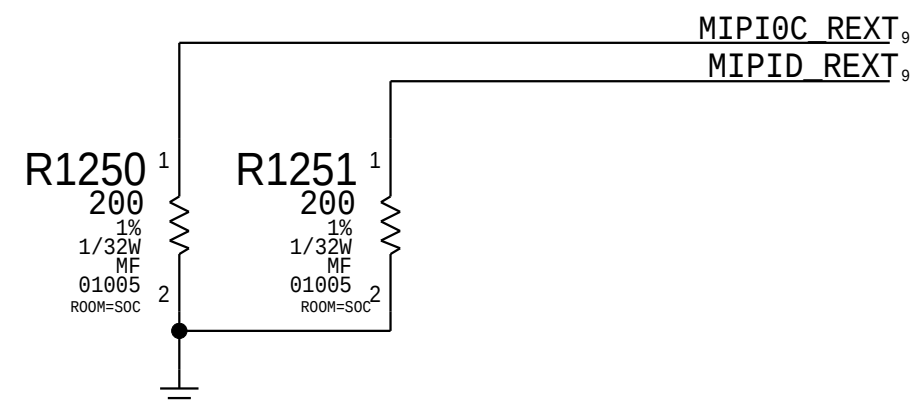
GNDED offpage on MLB

Display MIPI

Juliet MIPI

GNDED offpage on MLB

MIPI Reference



PAGE TITLE		
SOC: MIPI		
	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	12 OF 85
	SHEET	9 OF 60



SOC - LPDP

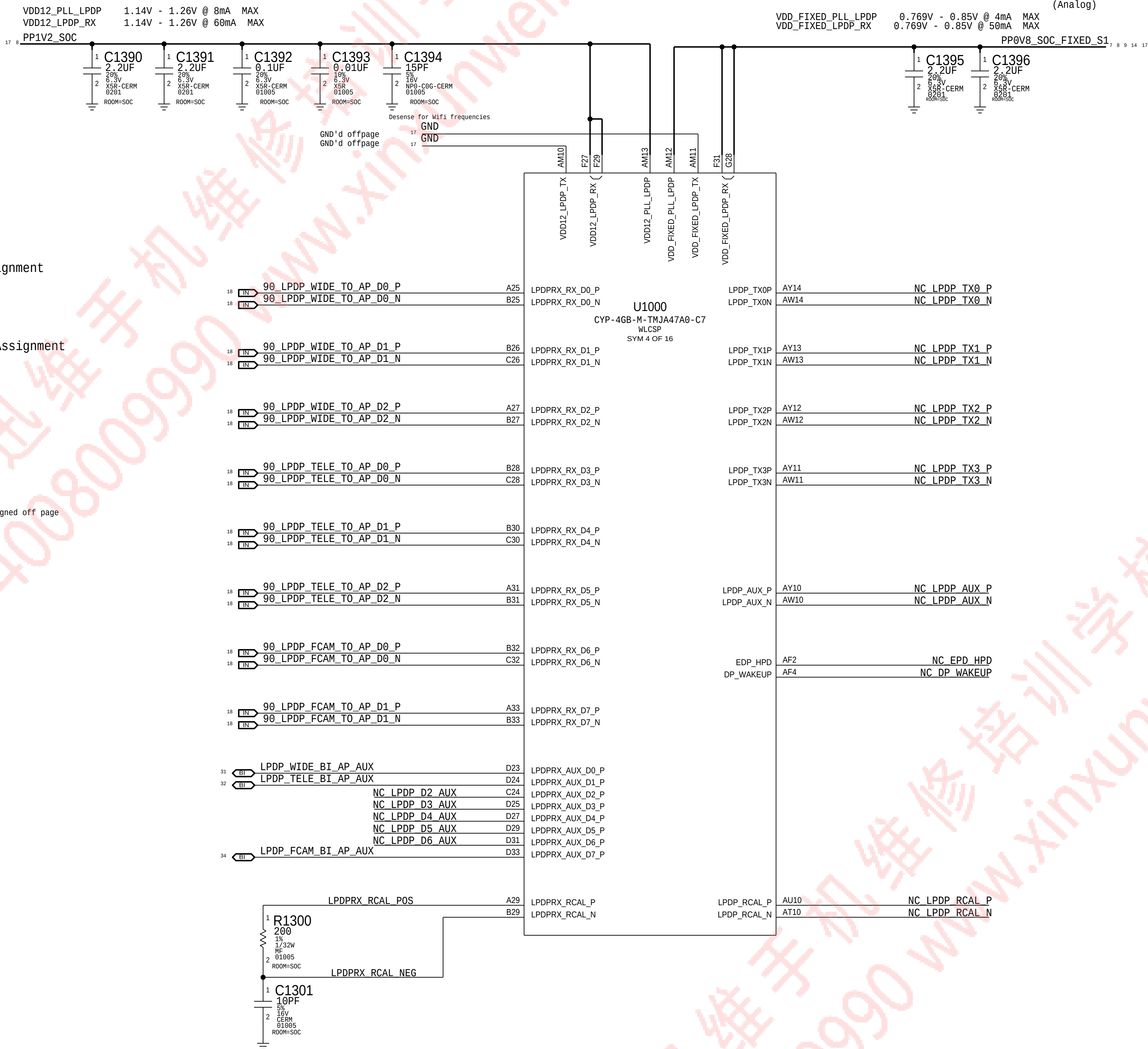
Dan LPDP Lane Assignment

Wide: 0-2
Tele: 3-5
Fcam: 6-7

Justin LPDP Lane Assignment

Wide: 2-4
Tele: 5-7
Fcam: 0-1

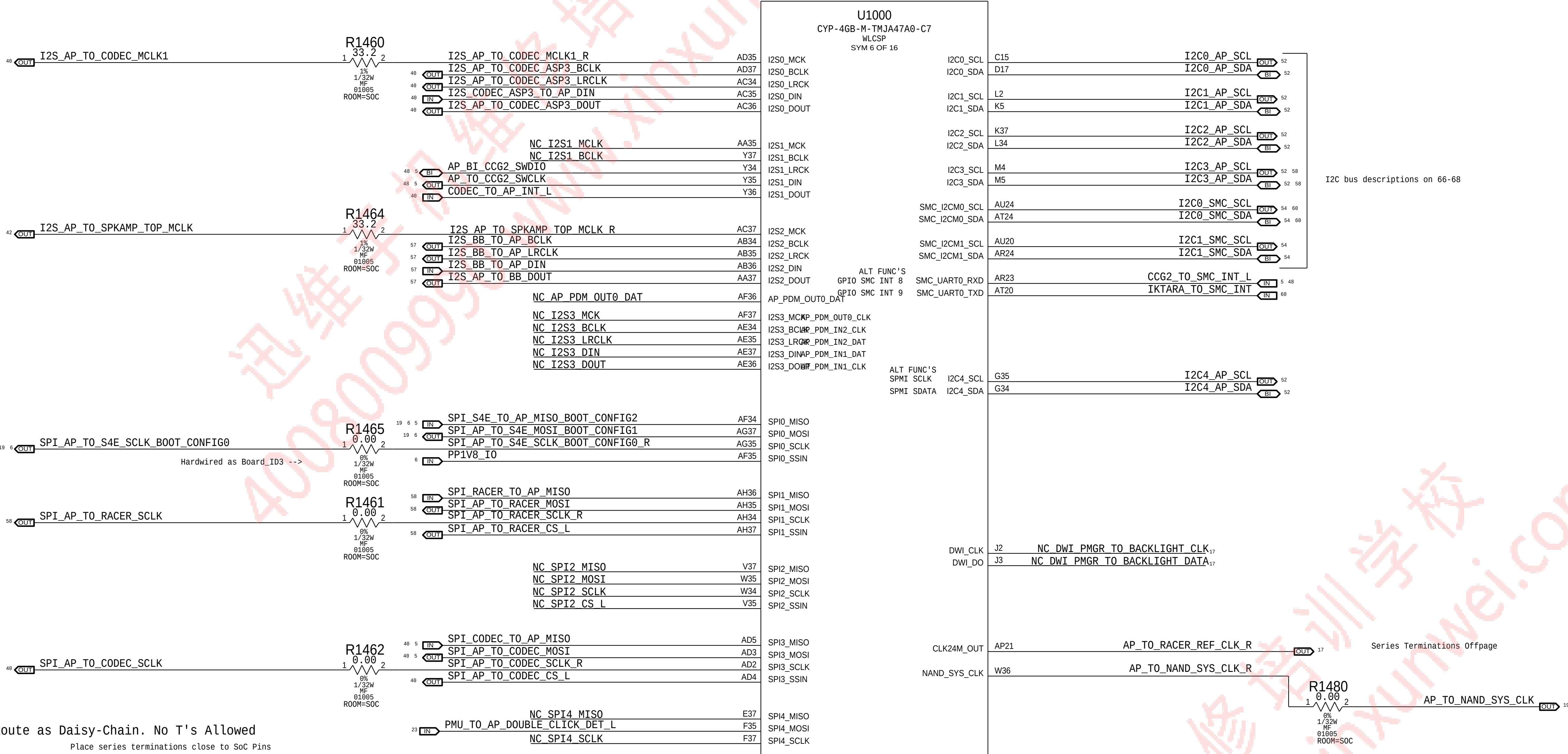
LPD Assigned off page



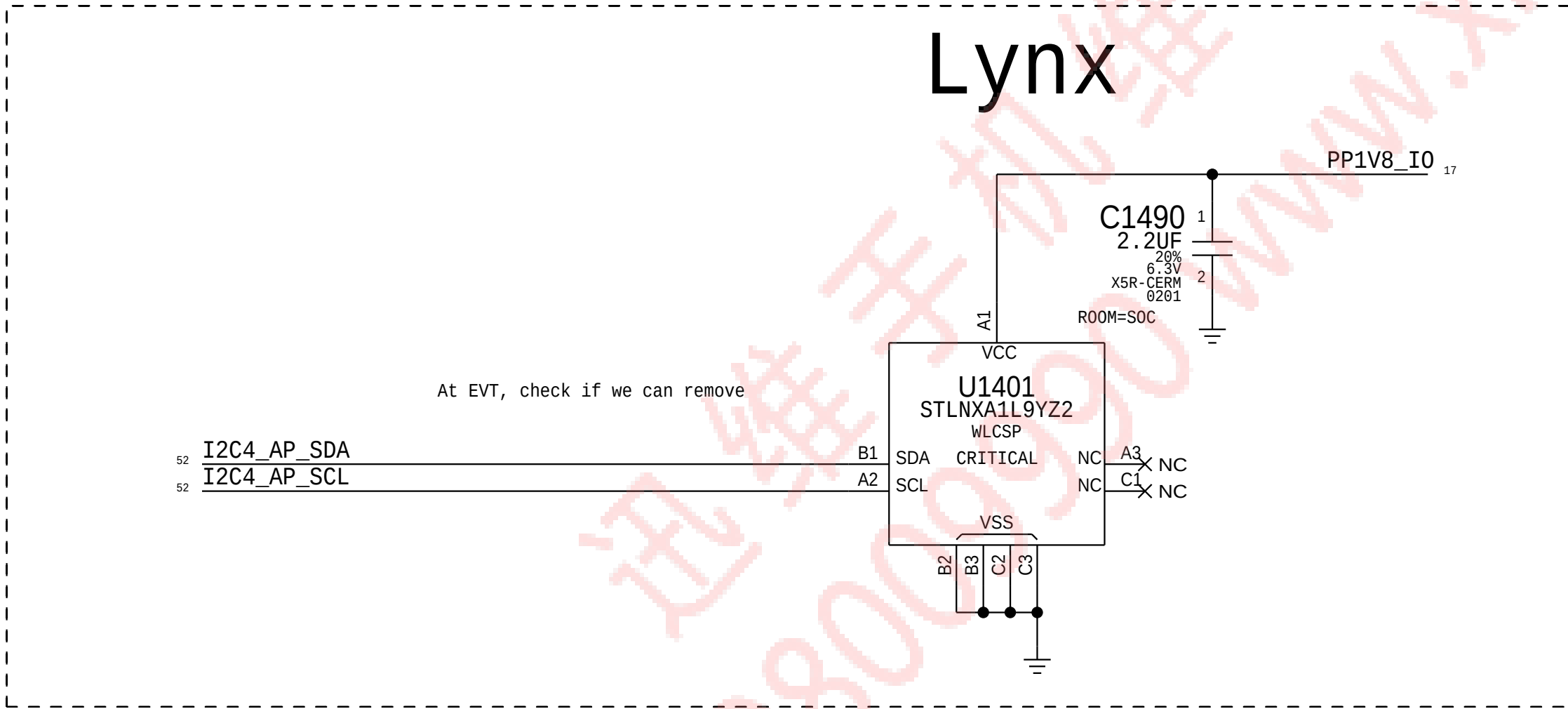
SOC: LPDP

DRAWING NUMBER	051-02545	SIZE	D
REVISION	7.0.0	BRANCH	
PAGE	13 OF 85	SHEET	10 OF 60

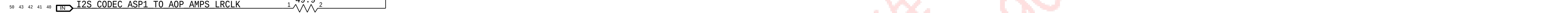
SOC - SERIAL INTERFACES



SPI: Route as Daisy-Chain. No T's Allowed
Place series terminations close to SoC Pins



PAGE TITLE		
SOC: SERIAL		
DRAWING NUMBER	051-02545	SIZE
REVISION	7.0.0	D
BRANCH		
PAGE	14 OF 85	
SHEET	11 OF 60	



PAGE TITLE			SOC: AOP	
0	DRAWING NUMBER		051-02545	
	REVISION		7.0.0	
	BRANCH			
	PAGE		16 OF 85	
	SHEET		13 OF 60	

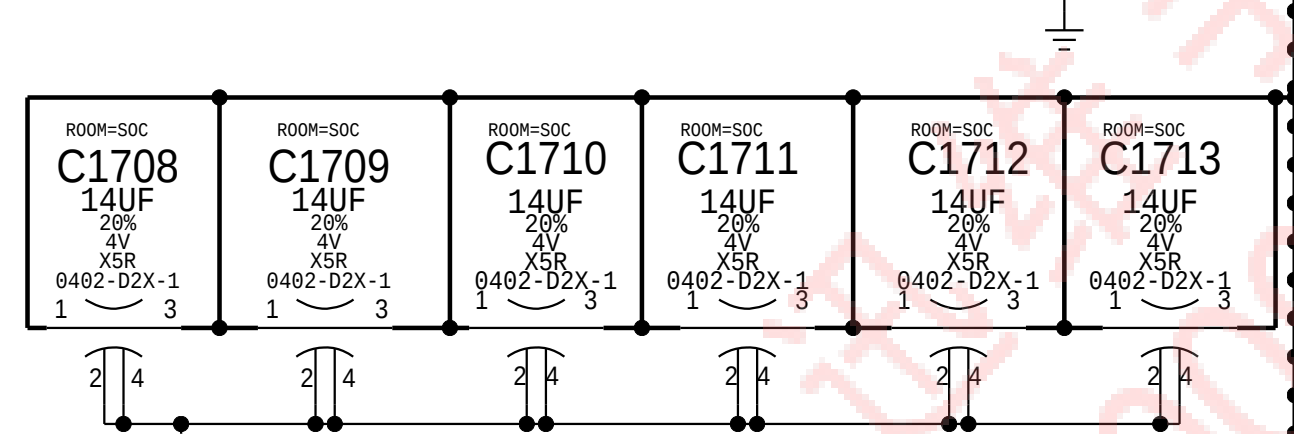
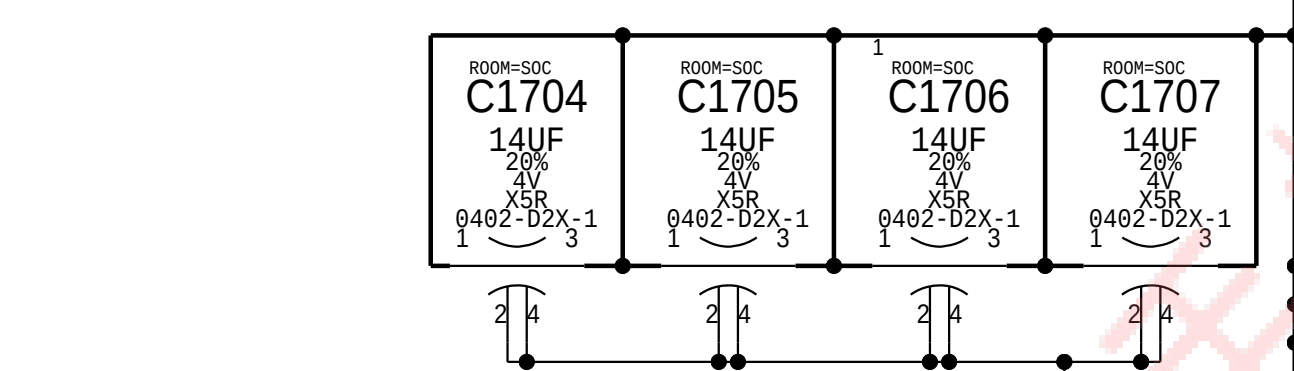
SOC - CPU, GPU & SOC RAILS

1.06V @ 13.8A MAX
0.905V @ 12.9A MAX
0.527V @ 2.4A MAX
PP_CPU_PCORE

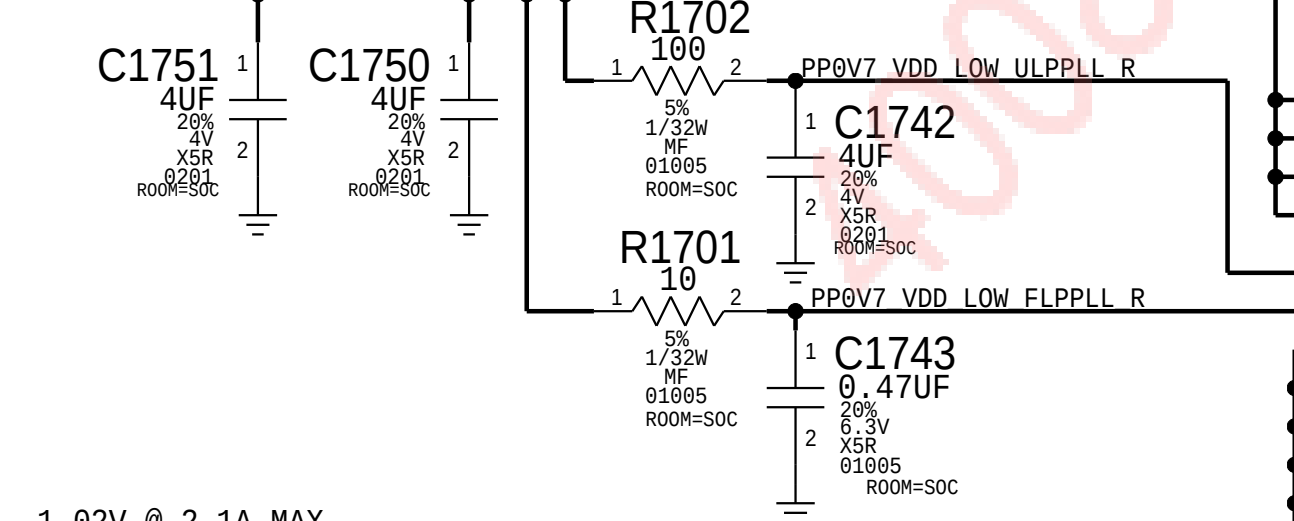


Remote sense XW's for Buck0 Buck1 and Buck11 live off page for dev board compapability

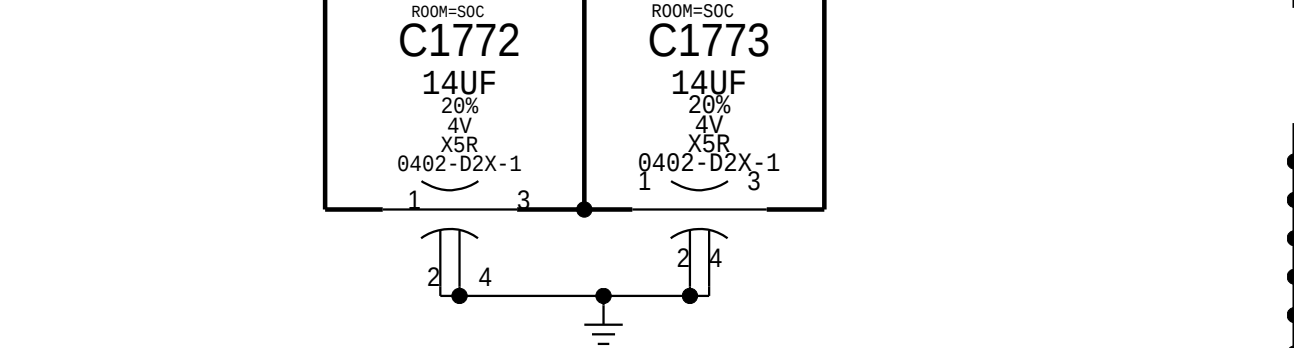
1.06V @ 14.5A MAX
0.725V @ 6.3A MAX
0.570V @ 3.1A MAX
PP_GPU



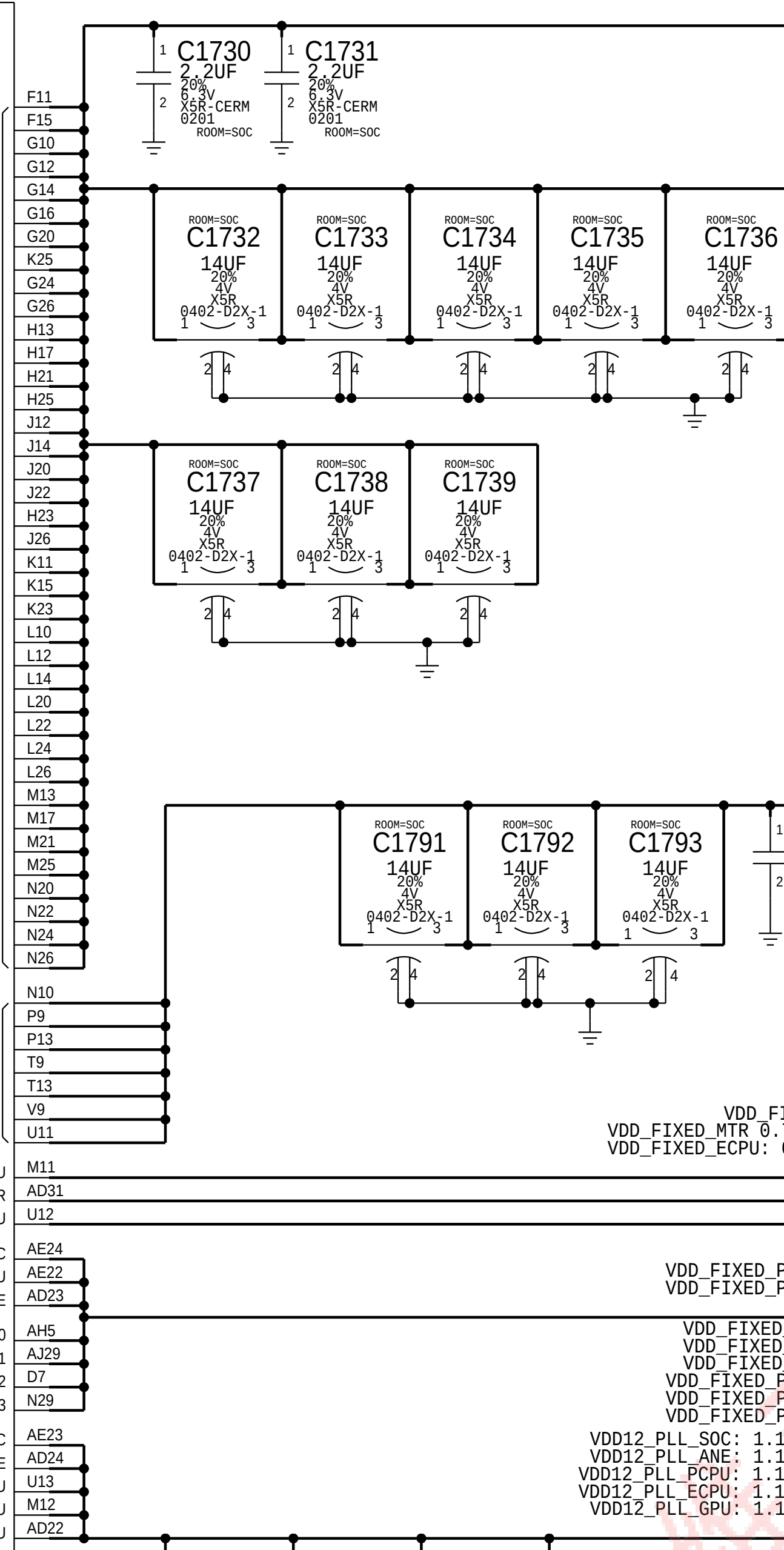
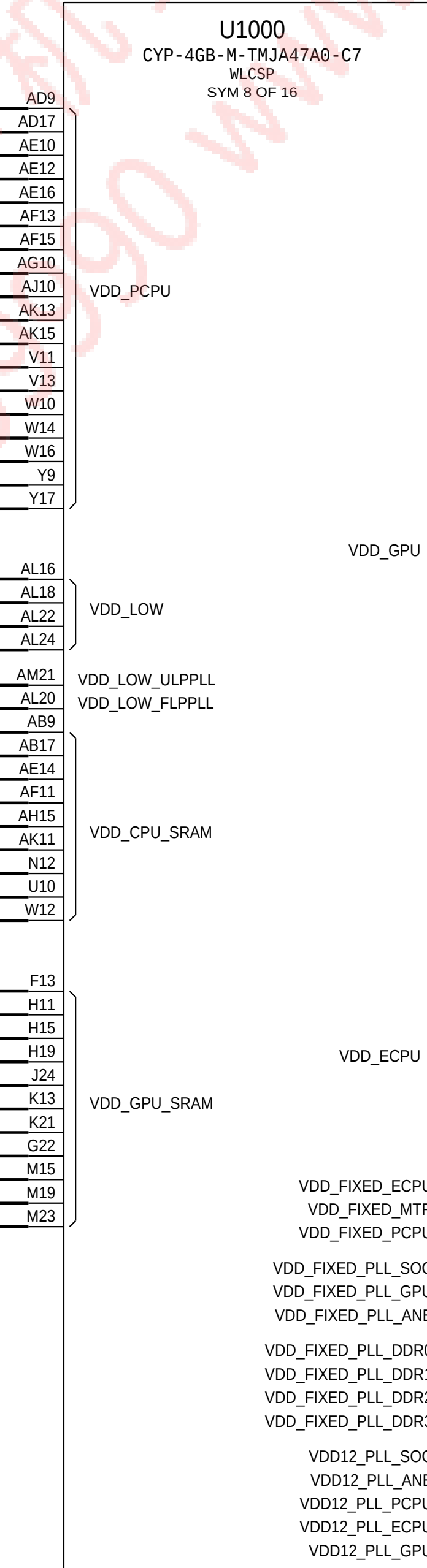
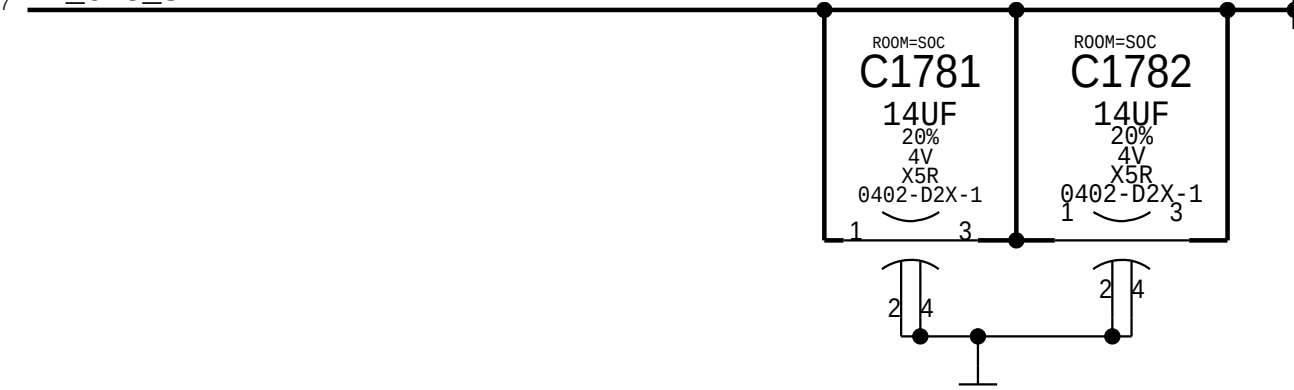
VDD_LOW: 0.691V - 0.756V @ 75mA MAX
VDD_LOW_ULPPLL: 0.691V - 0.756V @ 0.3mA MAX
PP0V7_VDD_LOW_S2



1.02V @ 2.1A MAX
0.975V @ 1.4A MAX
0.765V @ 0.33A MAX
PP_CPU_SRAM



1.06V @ 0.6A MAX
0.725V @ 0.41A MAX
0.685V @ 0.39A MAX
PP_GPU_SRAM

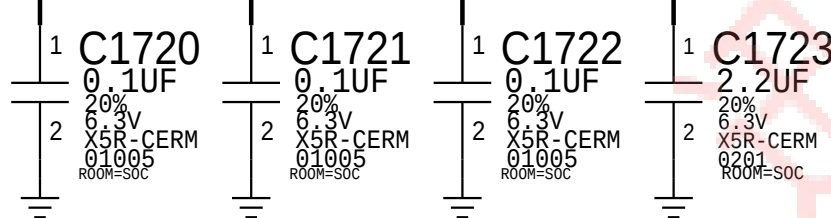


0.945V @ 2.9A MAX
0.626V @ 1.2A MAX
0.517V @ 0.62A MAX
PP_CPU_ECORE

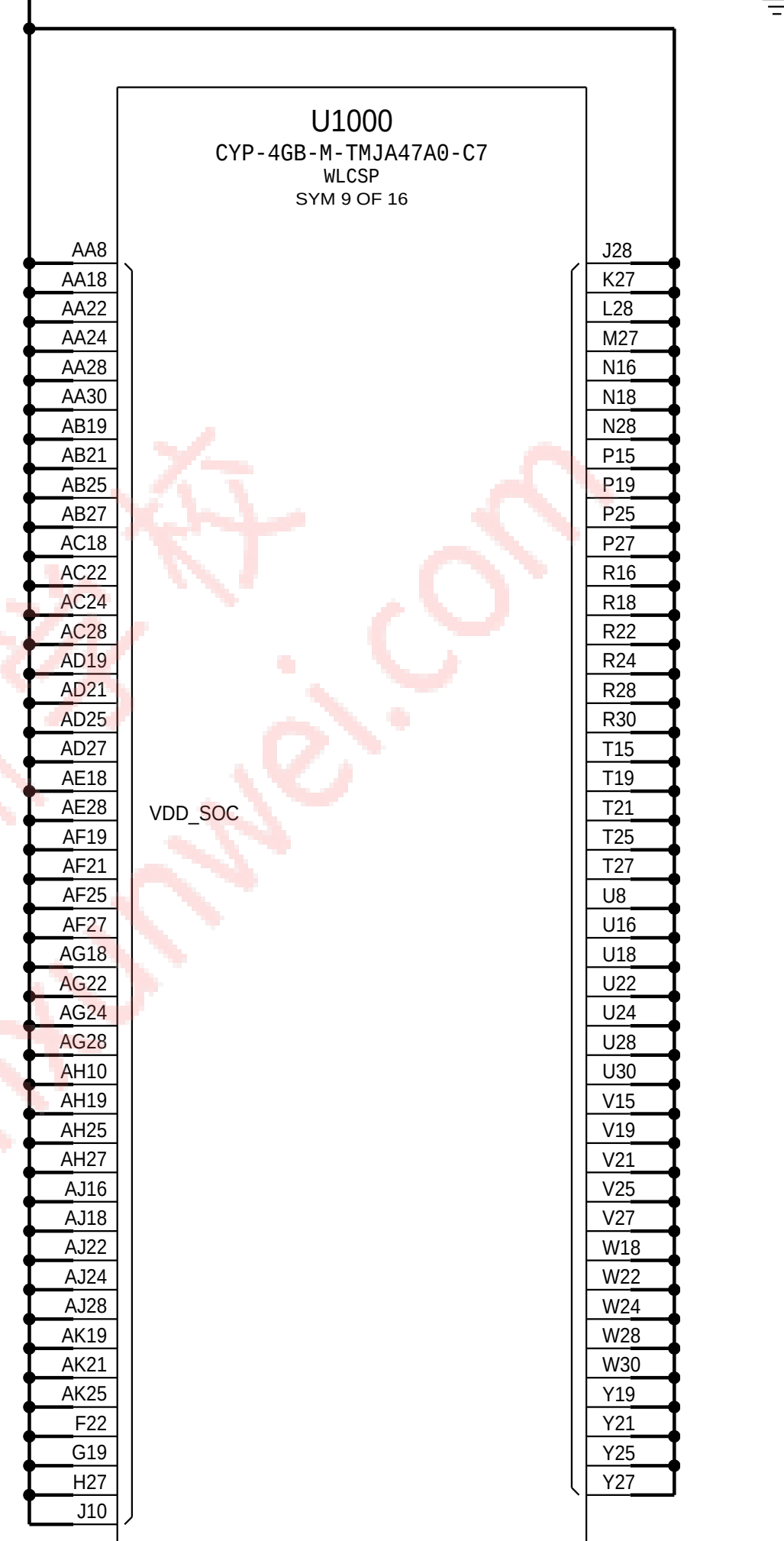
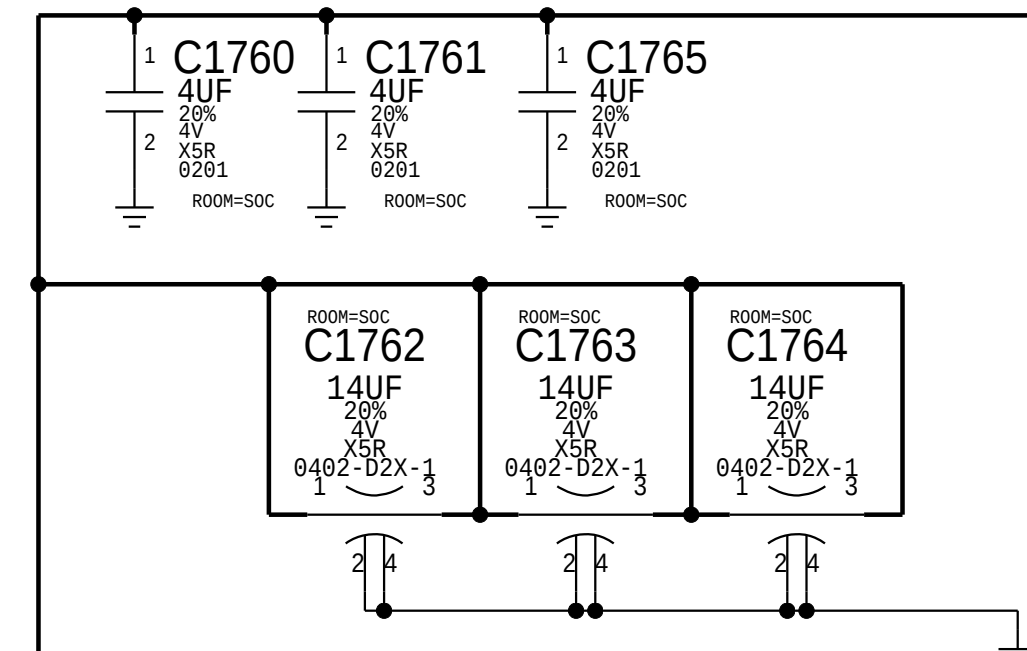
VDD_FIXED_PCPU: 0.81V @ 5mA
VDD_FIXED_MTR: 0.769V - 0.85V @ TBDmA
VDD_FIXED_ECPU: 0.769V - 0.85V @ 5mA
PP0V8_SOC_FIXED_S1
PP0V8_SOC_FIXED_S1
PP0V8_SOC_FIXED_S1

VDD_FIXED_PLL_DDR3: 0.81V @ 8mA
VDD_FIXED_PLL_DDR2: 0.81V @ 8mA
PP0V8_SOC_FIXED_S1
VDD_FIXED_PLL_SOC: 0.81V @ 9mA
VDD_FIXED_PLL_GPU: 0.81V @ 5mA
VDD_FIXED_PLL_ANE: 0.81V @ 5mA
VDD_FIXED_PLL_DDR0: 0.81V @ 5mA
VDD_FIXED_PLL_DDR1: 0.81V @ 8mA
VDD_FIXED_PLL_LPDP: 0.81V @ 2mA

VDD12_PLL_SOC: 1.14 - 1.26V @ 8mA MAX
VDD12_PLL_ANE: 1.14 - 1.26V @ 7mA MAX
VDD12_PLL_PCPU: 1.14 - 1.26V @ 7mA MAX
VDD12_PLL_ECPU: 1.14 - 1.26V @ 7mA MAX
VDD12_PLL_GPU: 1.14 - 1.26V @ 7mA MAX
PP1V2_SOC



0.783V @ 4.2A MAX
0.661V @ 2.6A MAX
0.595V @ 2.1A MAX
PP_SOC_S1



SOC: POWER (1/3)

DRAWING NUMBER	SIZE
051-02545	D
REVISION	7.0.0
BRANCH	
PAGE	17 OF 85
SHEET	14 OF 60

PAGE TITLE		SOC: POWER (2/3)	
		DRAWING NUMBER	SIZE
		051-02545	D
		REVISION	
		7.0.0	
		BRANCH	
		PAGE	18 OF 85
		SHEET	15 OF 60



SOC - POWER SUPPLIES

D

C

B

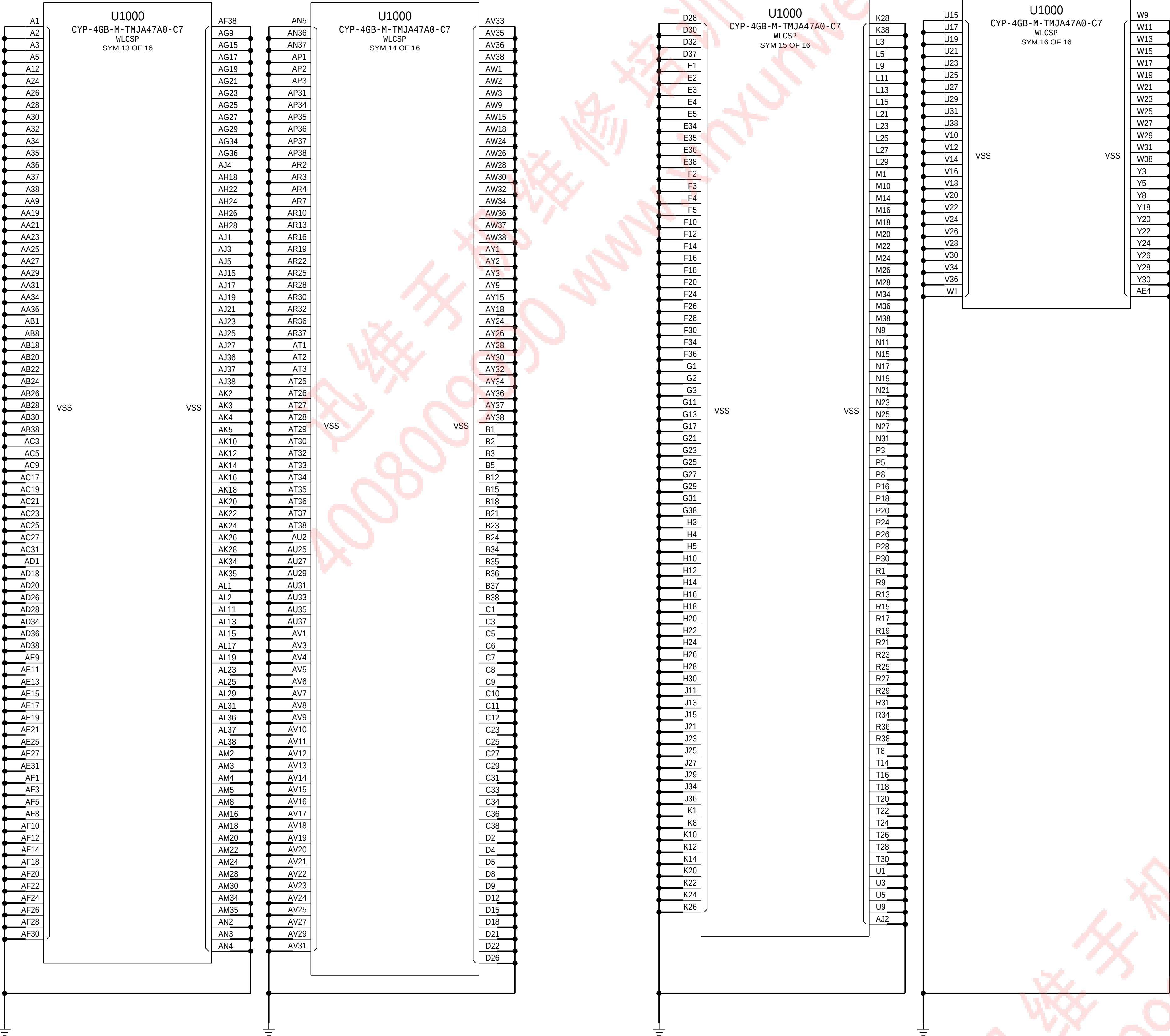
A

D

C

B

A



PAGE TITLE		
SOC: POWER (3/3)		
0	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	
	19 OF 85	
	SHEET	
		16 OF 60

Medusa Compatibility

PCIE Series Caps

FF Touch Compatibility

FF Display Compatibility

Dev Board Power Compatibility

Dev Board Compability GNDs

UF Dam Caps

Dev Board Compatibility FB Xw's

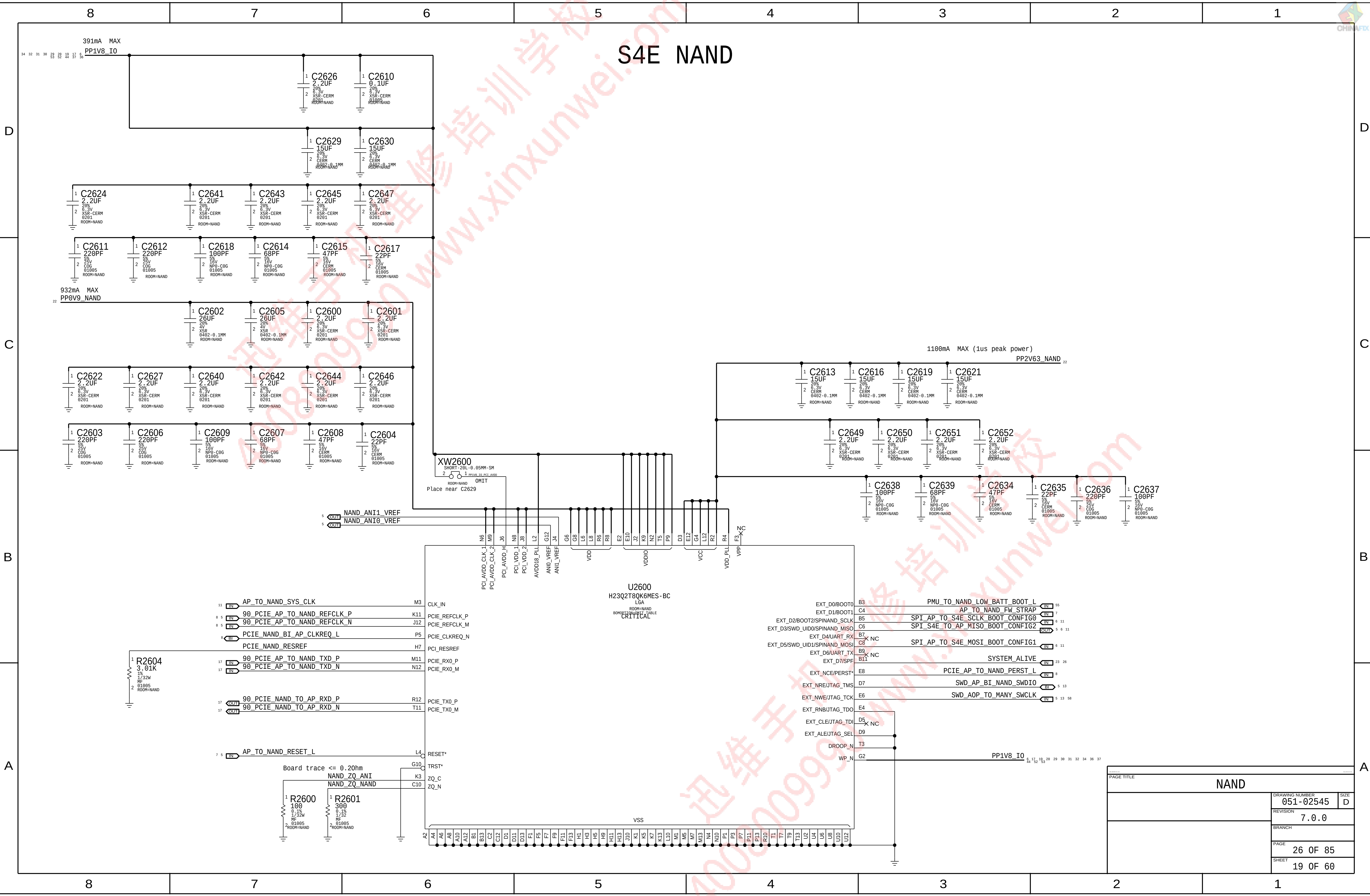
Place near SOC Balls

SOC: DEV BOARD ALIASES

DRAWING NUMBER	051-02545	SIZE	D
REVISION	7.0.0	BRANCH	
PAGE	20 OF 85	SHEET	17 OF 60

PAGE TITLE		SYMC_DATE=06/17/2017	
SOC: LPDP ALIASES			
	DRAWING NUMBER 051-02545		SIZE D
	REVISION 7.0.0		
	BRANCH		
	PAGE 21 OF 85		
	SHEET 18 OF 60		

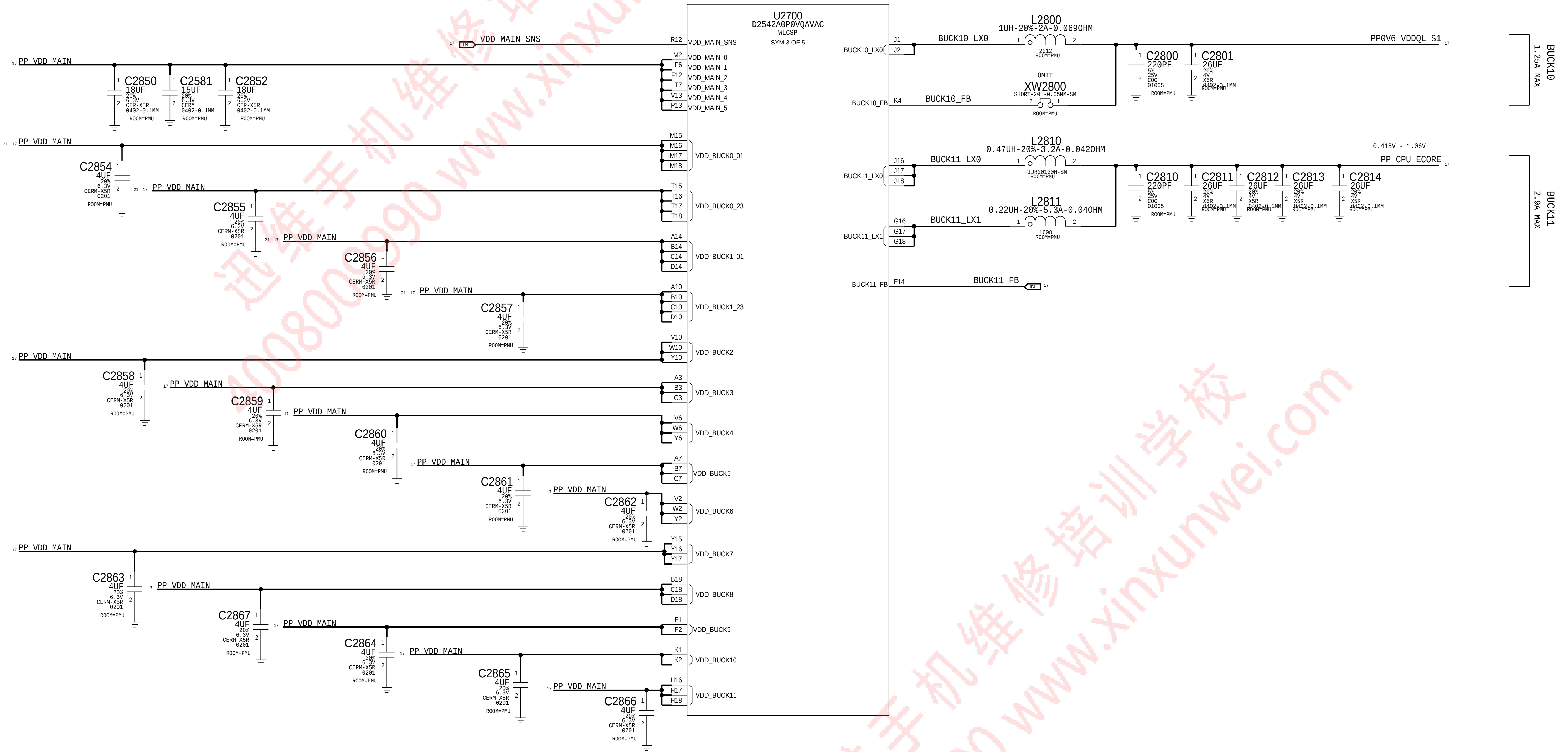
S4E NAND



PAGE TITLE		
NAND		
DRAWING NUMBER	051-02545	SIZE
REVISION	7.0.0	D
BRANCH		
PAGE	26 OF 85	
SHEET	19 OF 60	



PMU - BUCKS



SYSTEM POWER: PMU Bucks (2/4)		
DRAWING NUMBER	051-02545	SIZE D
REVISION	7.0.0	
BRANCH		
PAGE	28 OF 85	
SHEET	21 OF 60	



PMU - LD0s

D

D

C

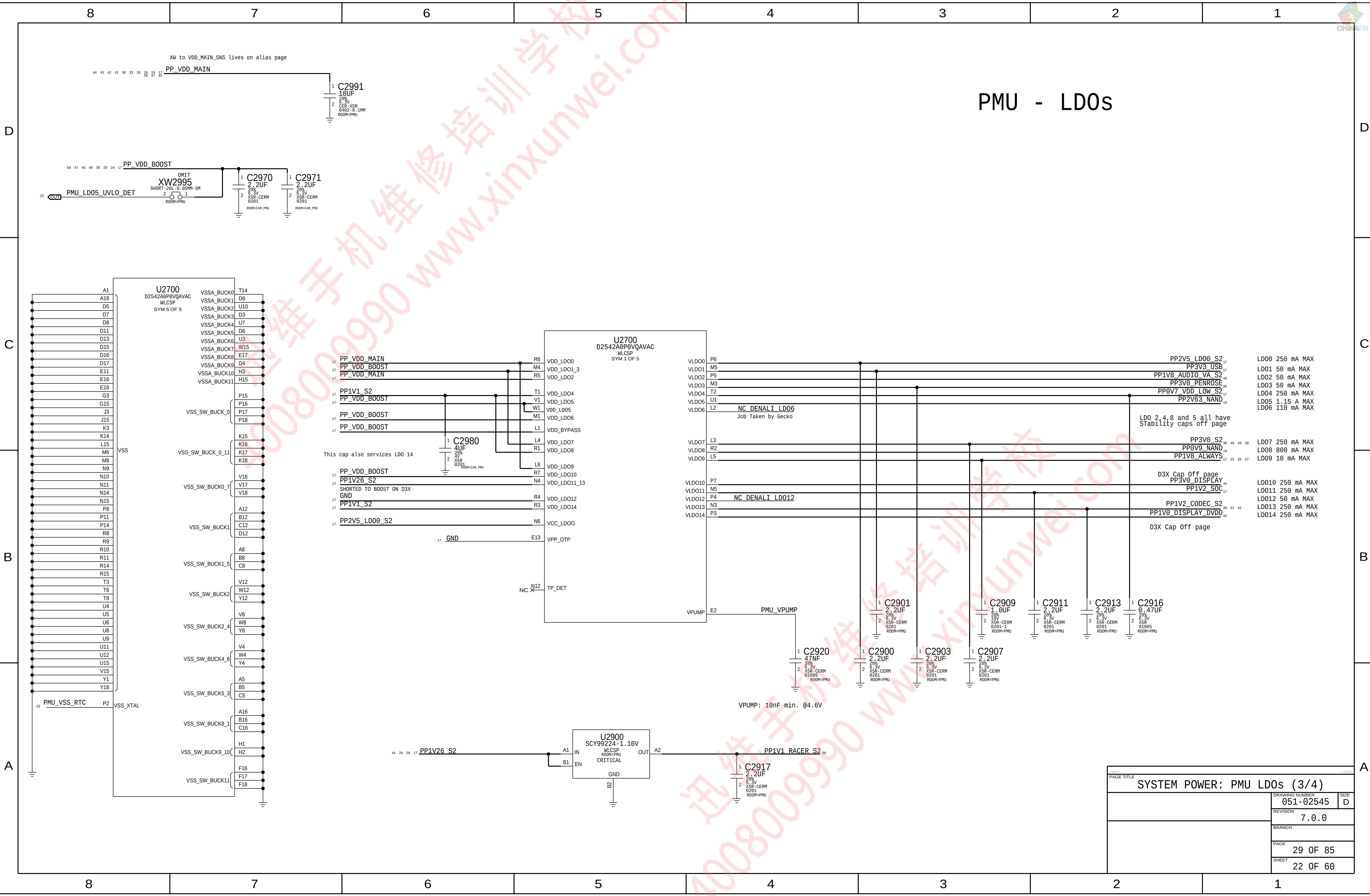
C

B

B

A

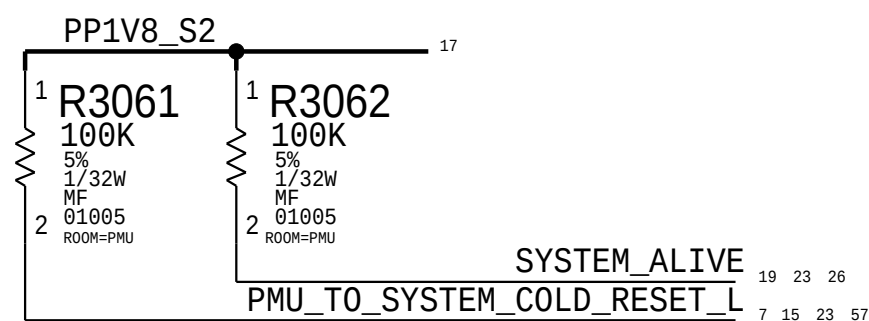
A



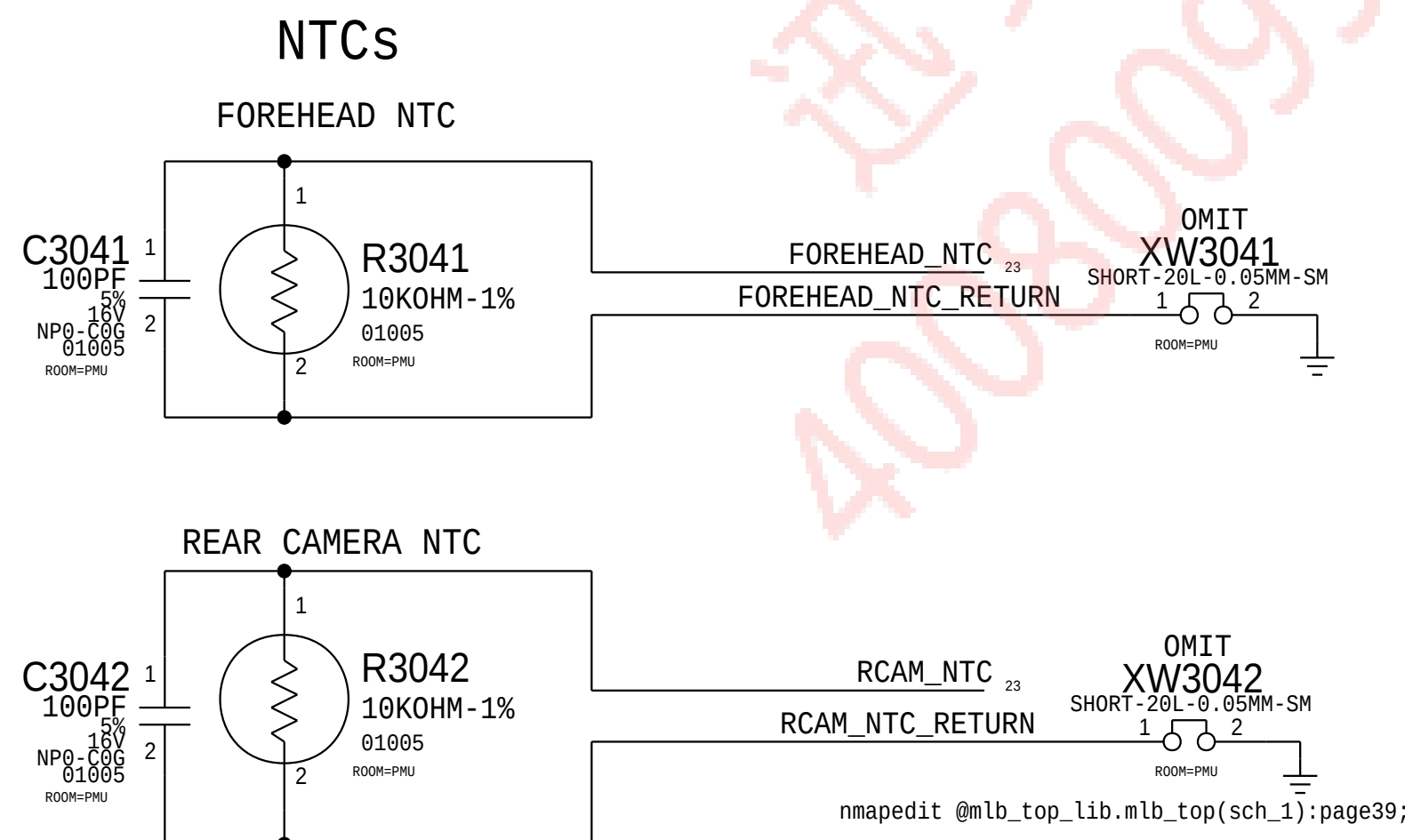
PAGE TITLE		
SYSTEM POWER: PMU LD0s (3/4)		
	DRAWING NUMBER	051-02545
	SIZE	D
	REVISION	7.0.0
	BRANCH	
	PAGE	29 OF 85
	SHEET	22 OF 60

NTCs

NOTE (1):INPUT PULL-DOWN 100-300k
NOTE (2):INPUT PULL-DOWN 1M
NOTE (3):INPUT PULL-UP OR DOWN 100k-300k
NOTE (4):OUTPUT OPEN-DRAIN, REQUIRES PULL-UP

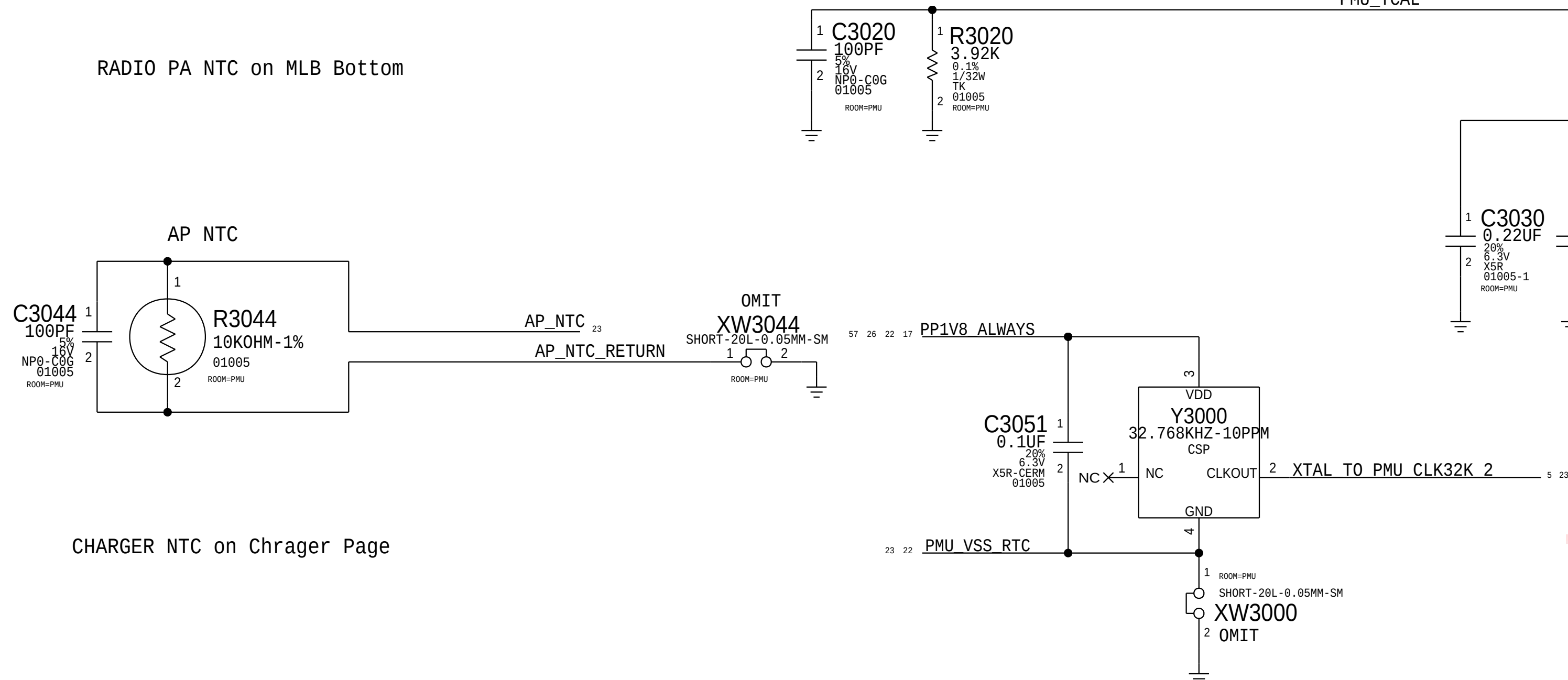


Only has DS control when powered by VBUCK3
Only has DS control when powered by VBUCK3

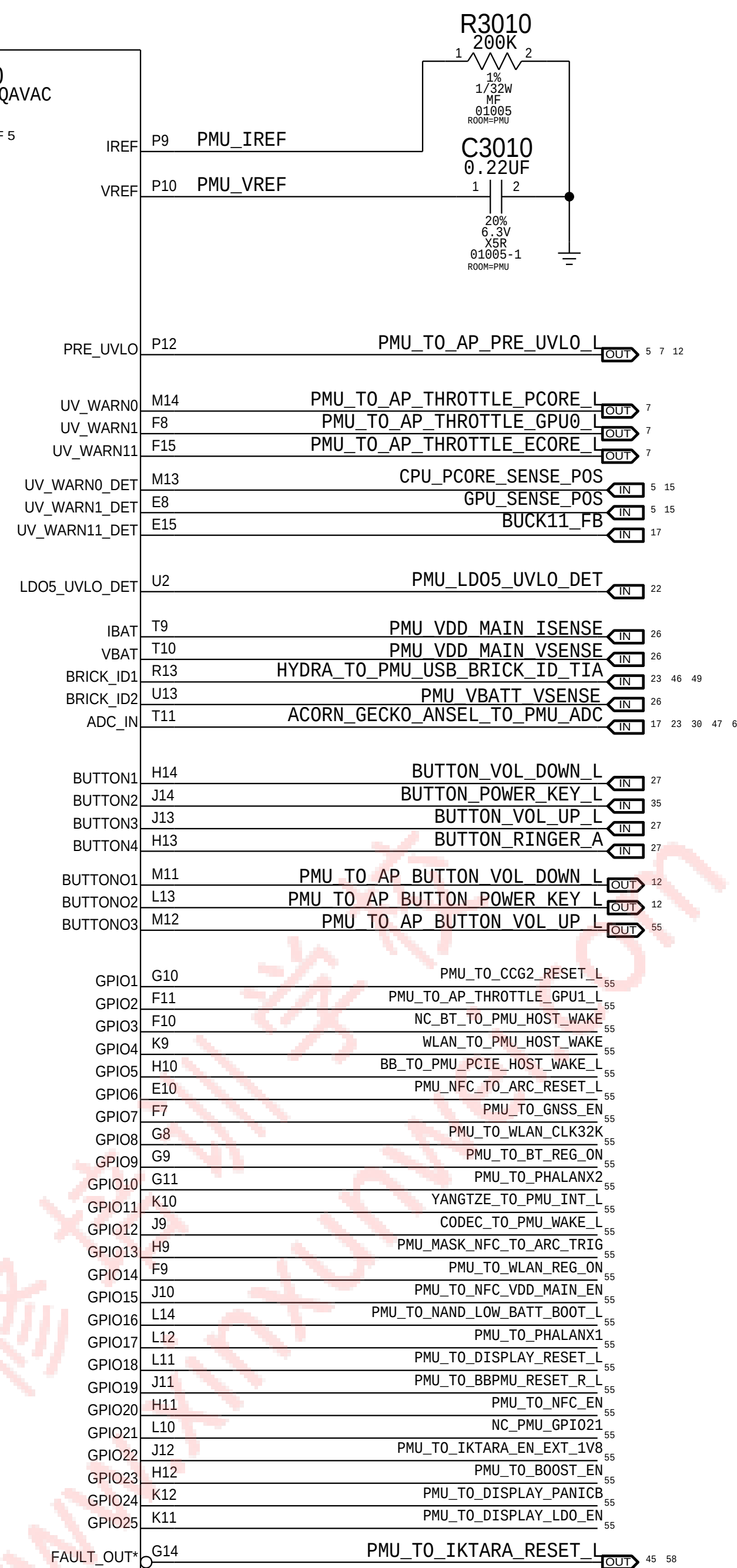
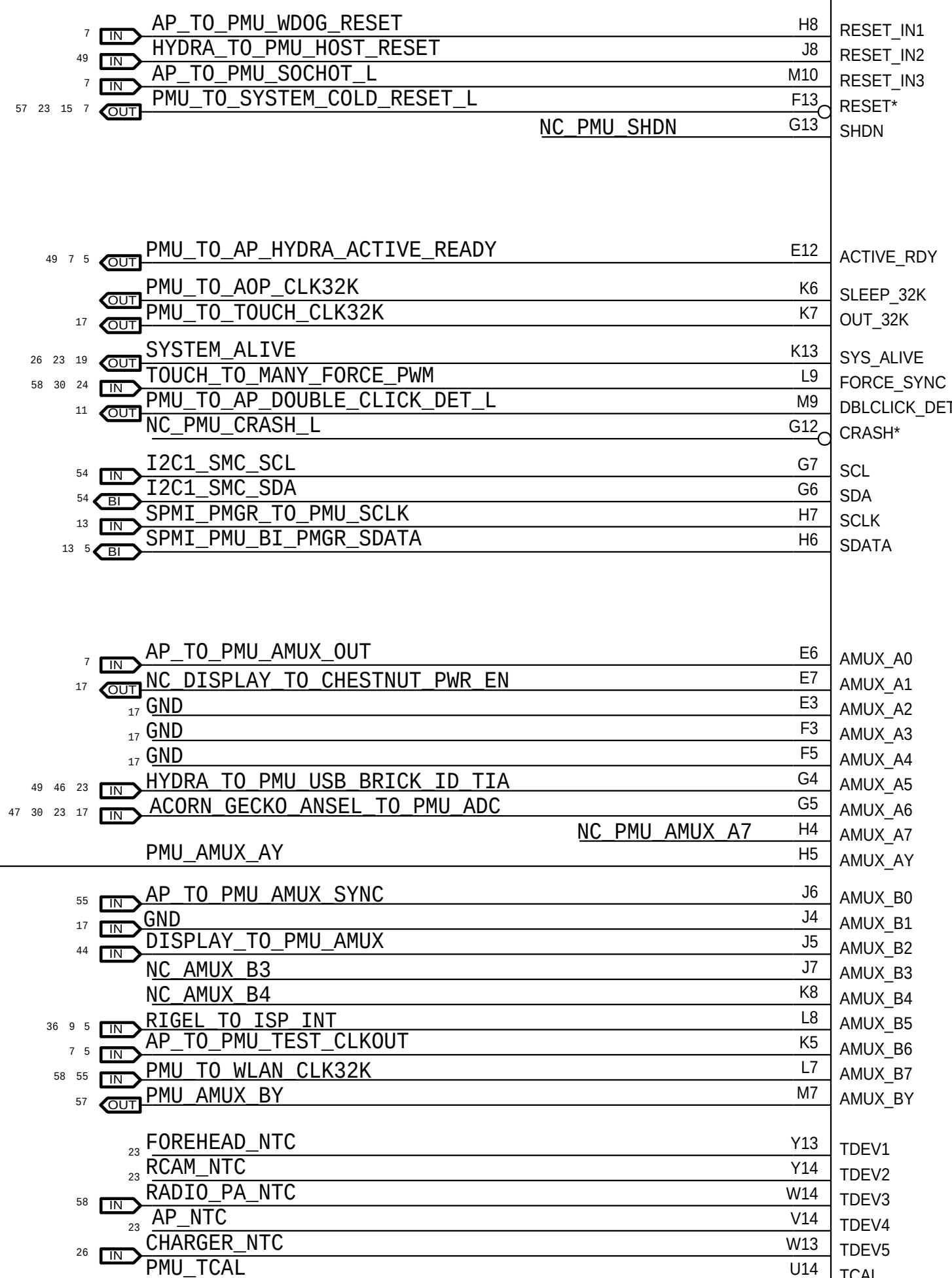


```
nmapedit @mlb_top_lib.mlb_top(sch_1):page39;
```

RADIO PA NTC on MLB Bottom

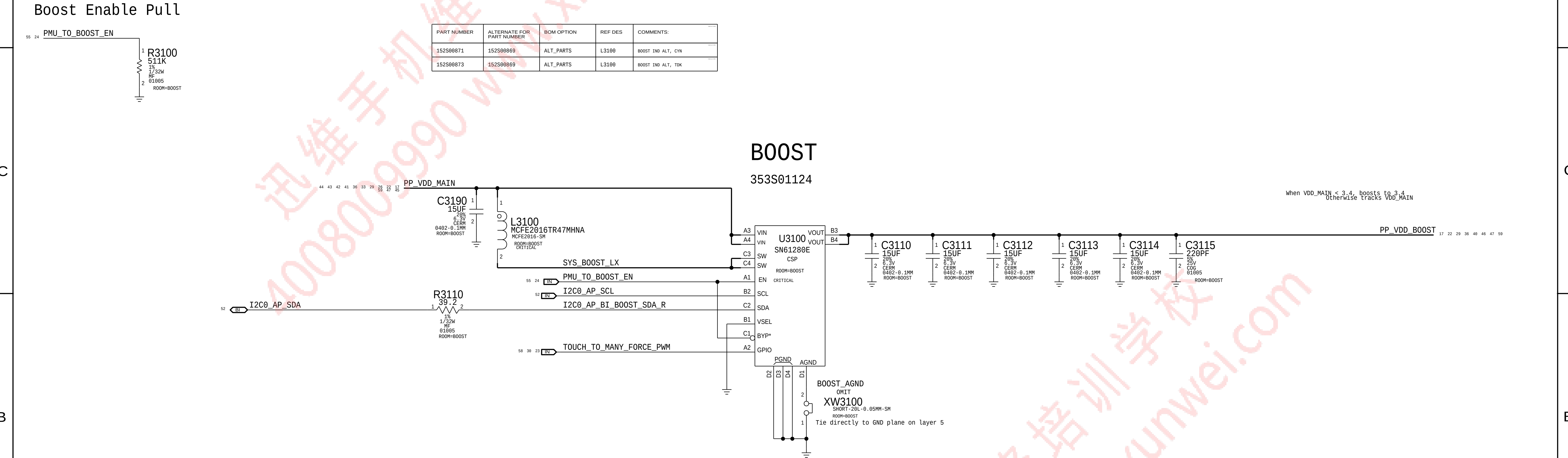


NOTE: 100PF CAPS ARE THE SAMPLING CAPS FOR PMU ADC



DRAWING NO.		SHEET NO.	
PAGE TITLE SYSTEM POWER: PMU (4/4)			
	DRAWING NUMBER	051-02545	SIZE D
	REVISION	7.0.0	
	BRANCH		
	PAGE	30 OF 85	
	SHEET	23 OF 60	

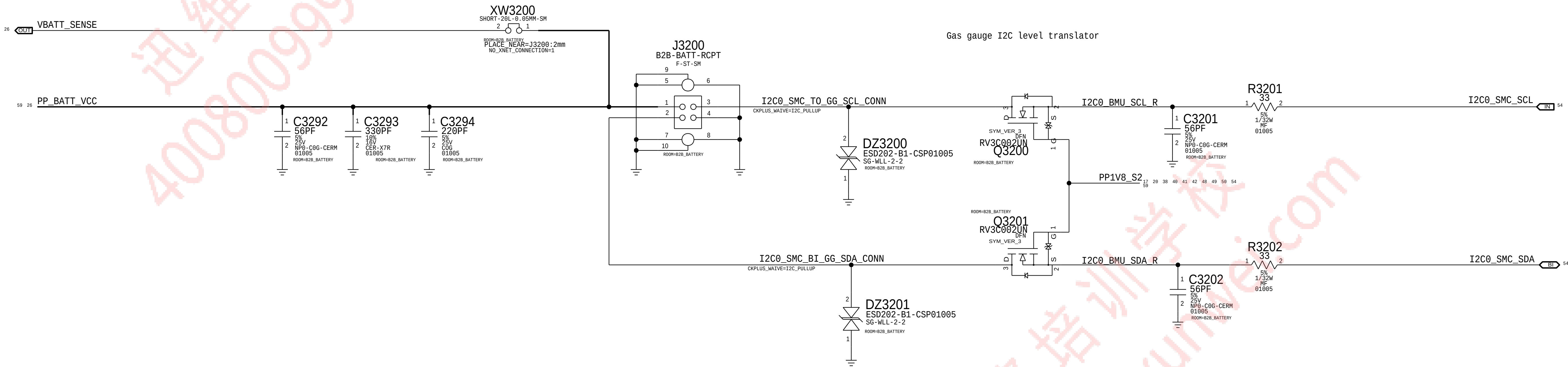
PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00871	152S00869	ALT_PARTS	L3100	BOOST IND ALT, CYN
152S00873	152S00869	ALT_PARTS	L3100	BOOST IND ALT, YDK



SYSTEM POWER: Boost		
	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	31 OF 85
	SHEET	24 OF 60

BATTERY CONNECTOR

Rcpt: 516S00232
Plug: 516S00233



PAGE TITLE		
SYSTEM POWER: B2B Battery		
DRAWING NUMBER	051-02545	SIZE
		D
REVISION		7.0.0
BRANCH		
PAGE		32 OF 85
SHEET		25 OF 60

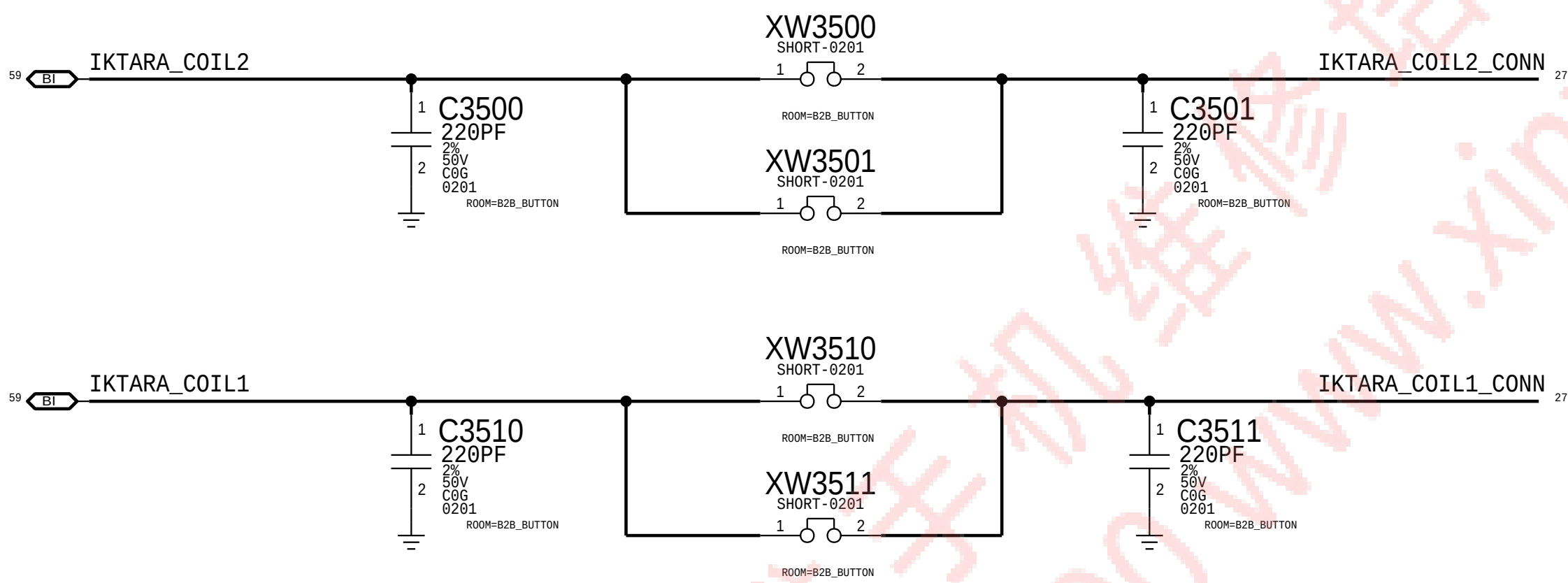
NON-ZRB ALTS

The left diagram is a circuit for a 'ROOM-CHARGER'. It features a 100PF capacitor (C3370) connected to a 10KOHM-1% resistor (R3370). The resistor is connected to a 20L-0.05MM-SM inductor (XW3370). The inductor is connected to a 'BATTERY_NTC' and 'BATTERY_NTC_RETURN' line. The return line is connected to a 'ROOM-CHARGER' terminal. The right diagram is a circuit for a 'ROOM-PRU'. It features a 100PF capacitor (C3045) connected to a 10KOHM-1% resistor (R3045). The resistor is connected to a 20L-0.05MM-SM inductor (XW3045). The inductor is connected to a 'CHARGER_NTC' and 'CHARGER_NTC_RETURN' line. The return line is connected to a 'ROOM-PRU' terminal.

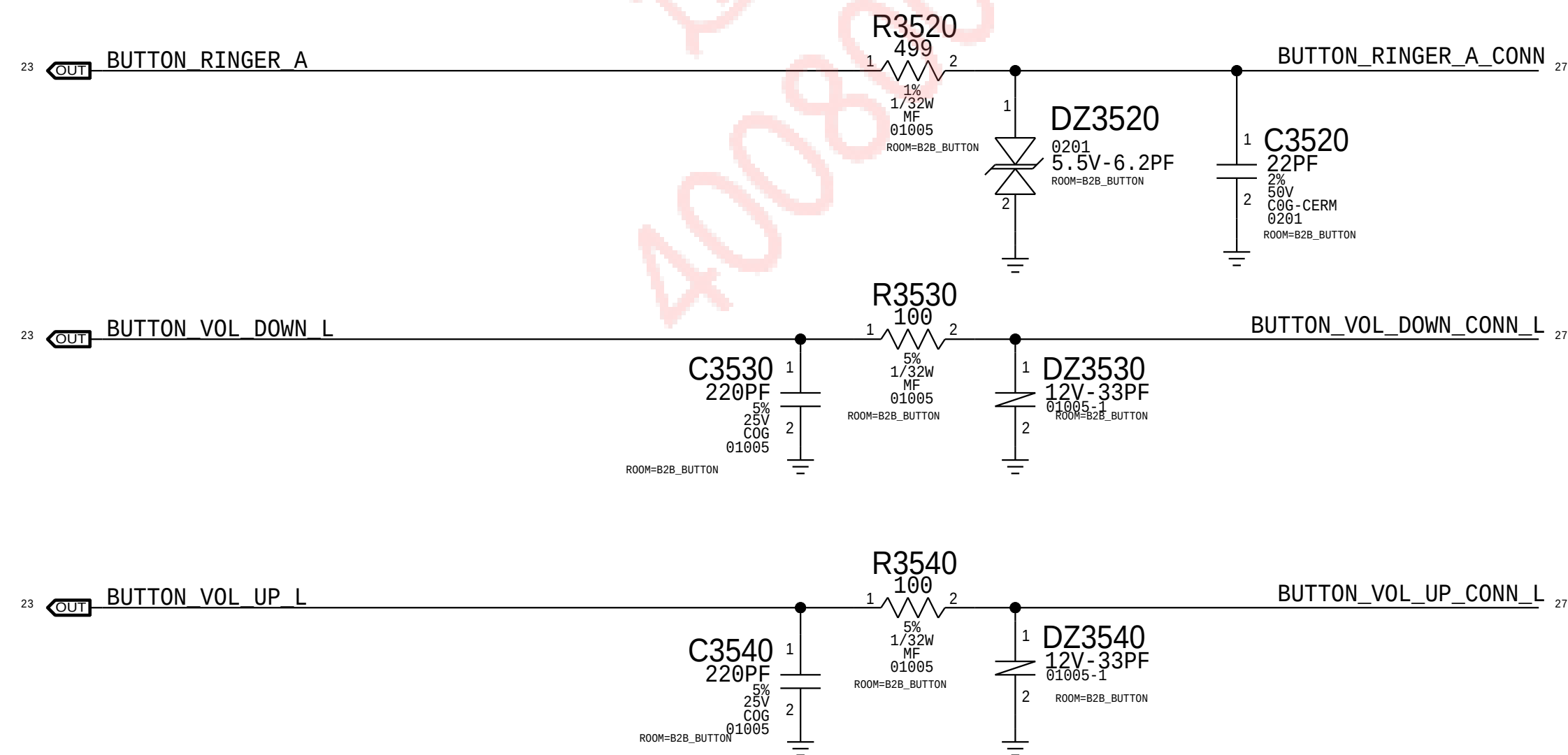
Cyclone + Button Connector

Rcpt: 516S00289 <-- This one on MLB
Plug: 516S00290

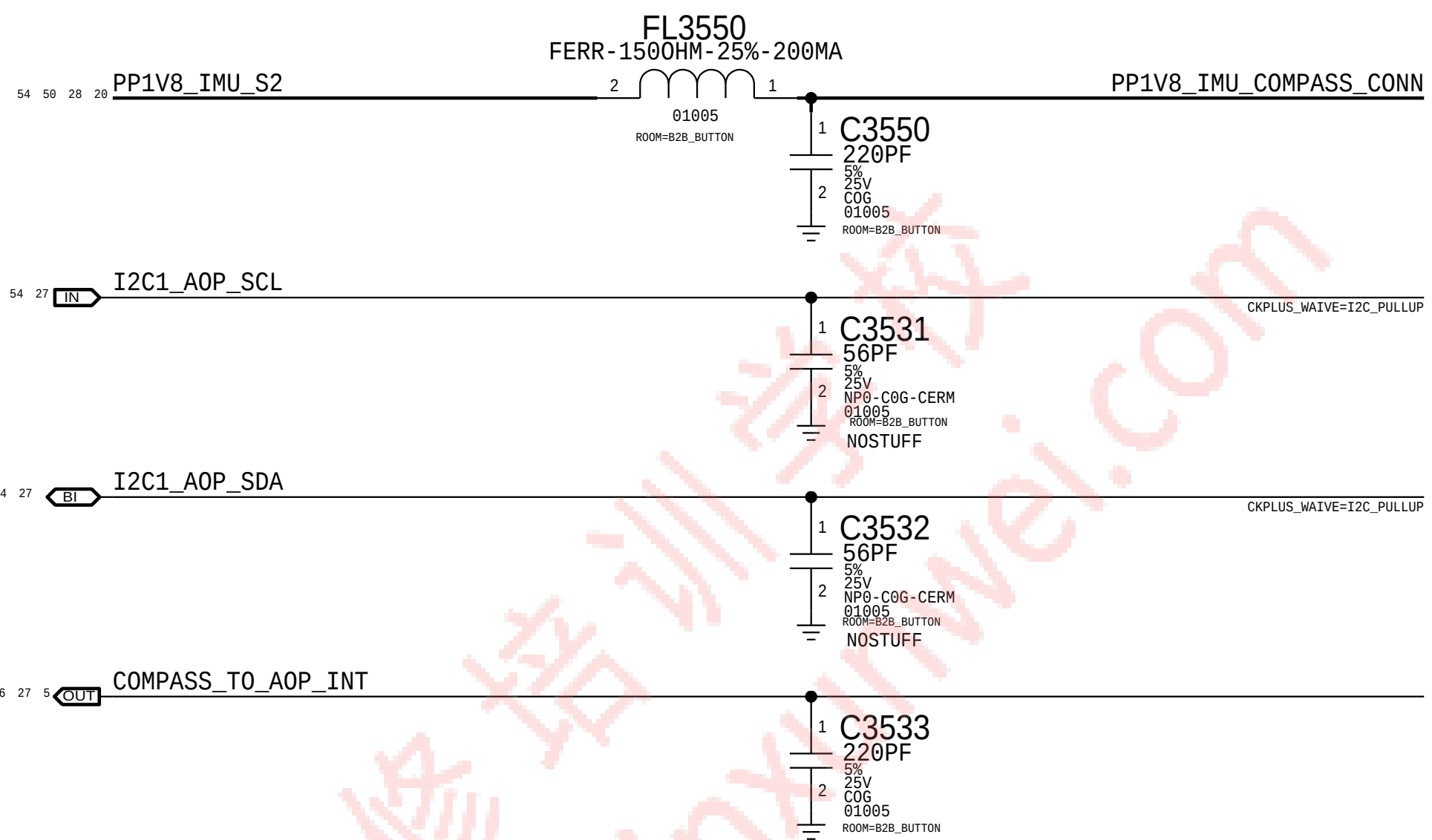
Cyclone Filtering



BUTTONS



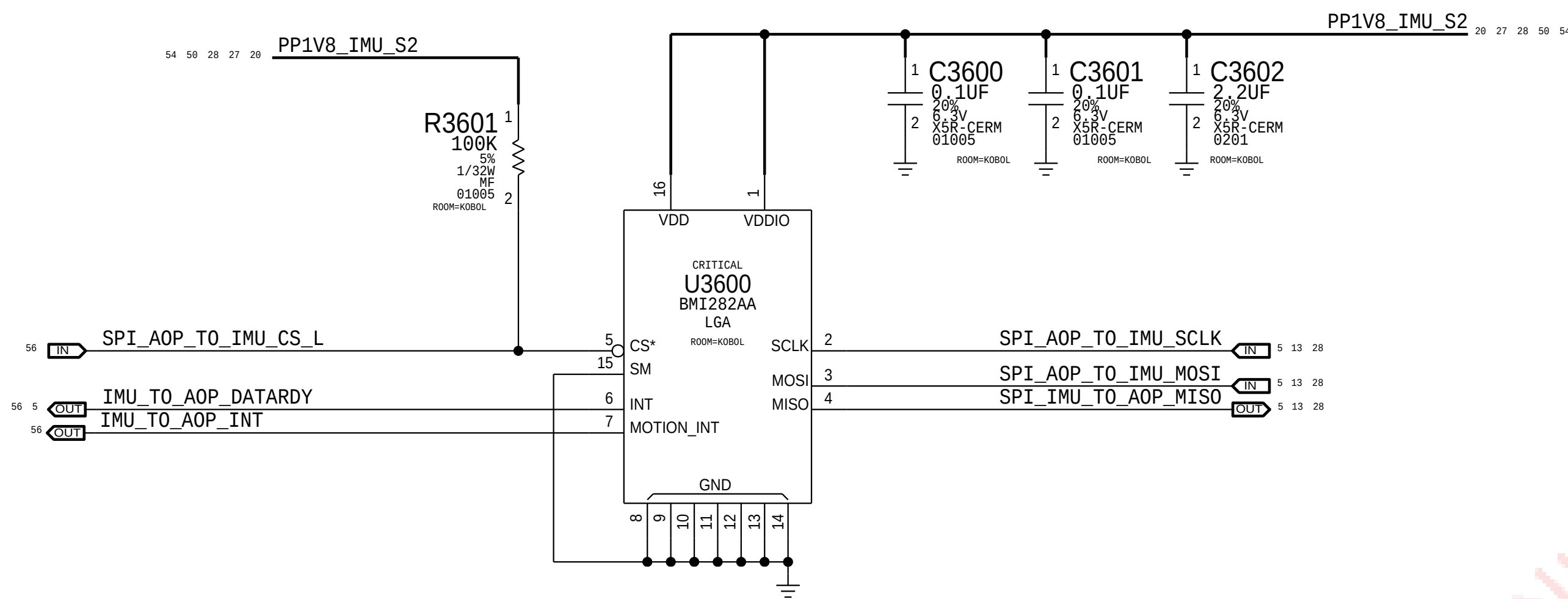
Compass



PAGE TITLE		
SYSTEM POWER: B2B Cyclone + Button		
DRAWING NUMBER	051-02545	
	SIZE D	
REVISION	7.0.0	
	BRANCH	
PAGE	35 OF 85	
	SHEET 27 OF 60	

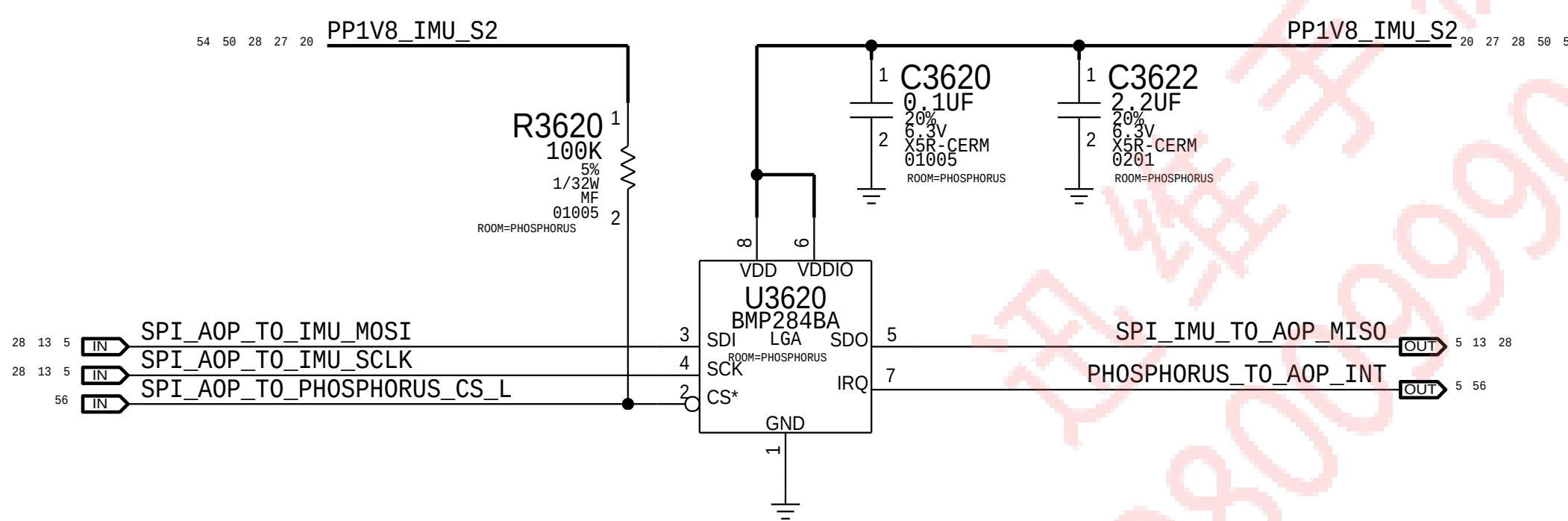
Kobol - Accel & Gyro

APN: 338S00367



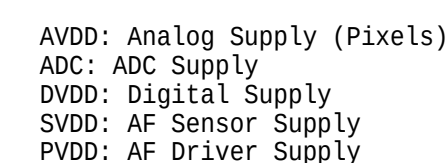
Phosphorus

BOSCH (APN:338S00334)



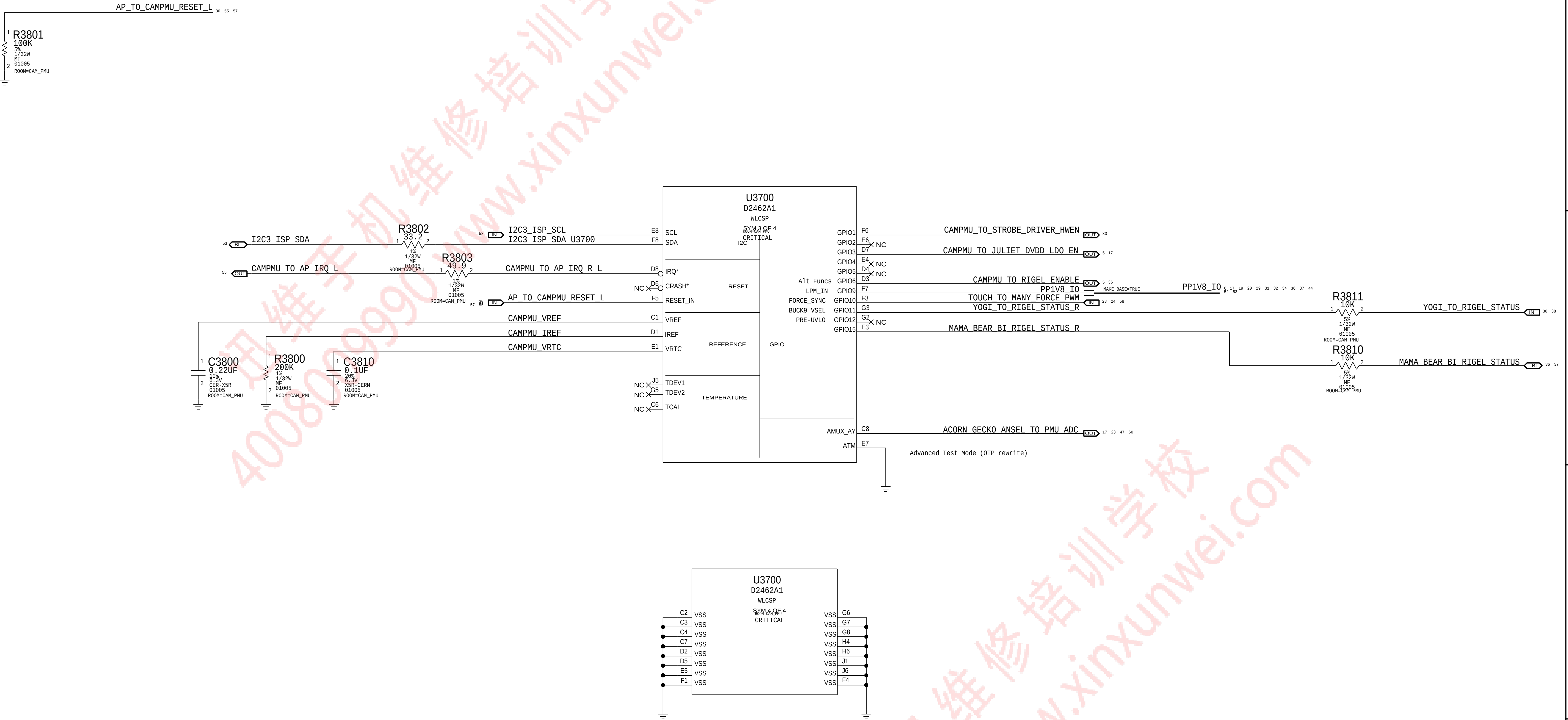
PAGE TITLE		
SENSORS		
	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	36 OF 85
	SHEET	28 OF 60
	SIZE	D

5	
---	--



PAGE TITLE		CAMERA: PMU (1/2)	
		DRAWING NUMBER	SIZE
		051-02545	D
		REVISION	
		7.0.0	
		BRANCH	
		PAGE	
		37 OF 85	
		SHEET	
		29 OF 60	

Pull Downs

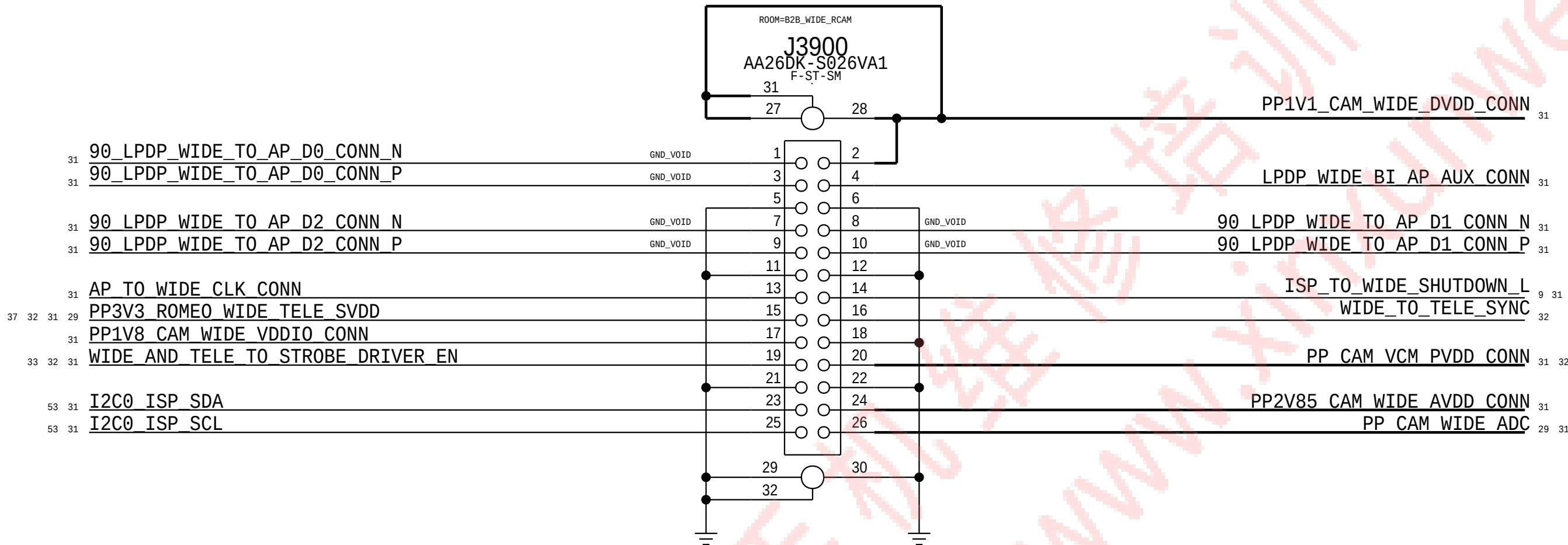


PAGE TITLE		
CAMERA: PMU (2/2)		
	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	38 OF 85
	SHEET	30 OF 60
	SIZE	D

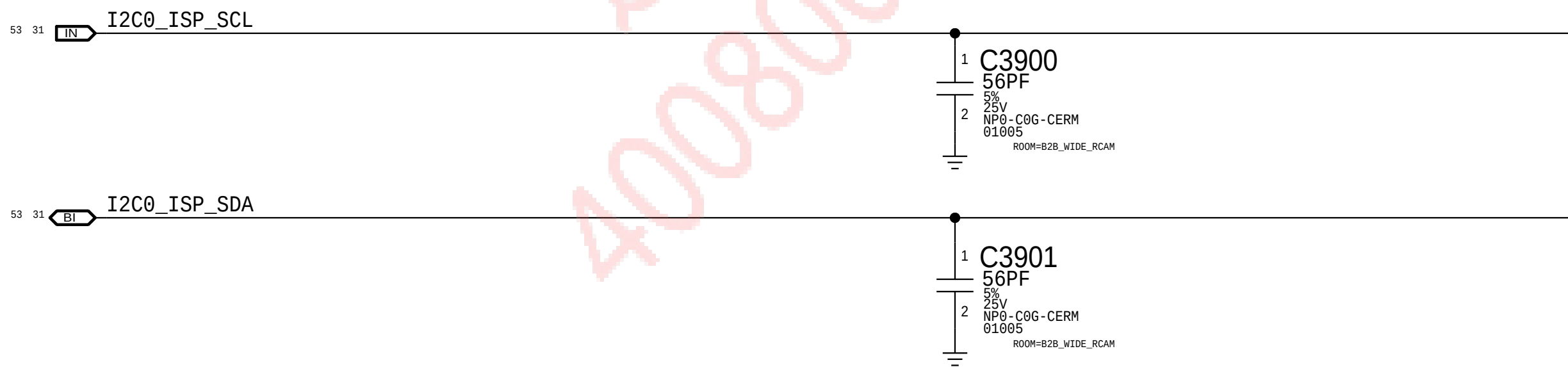


Wide Camera Connector

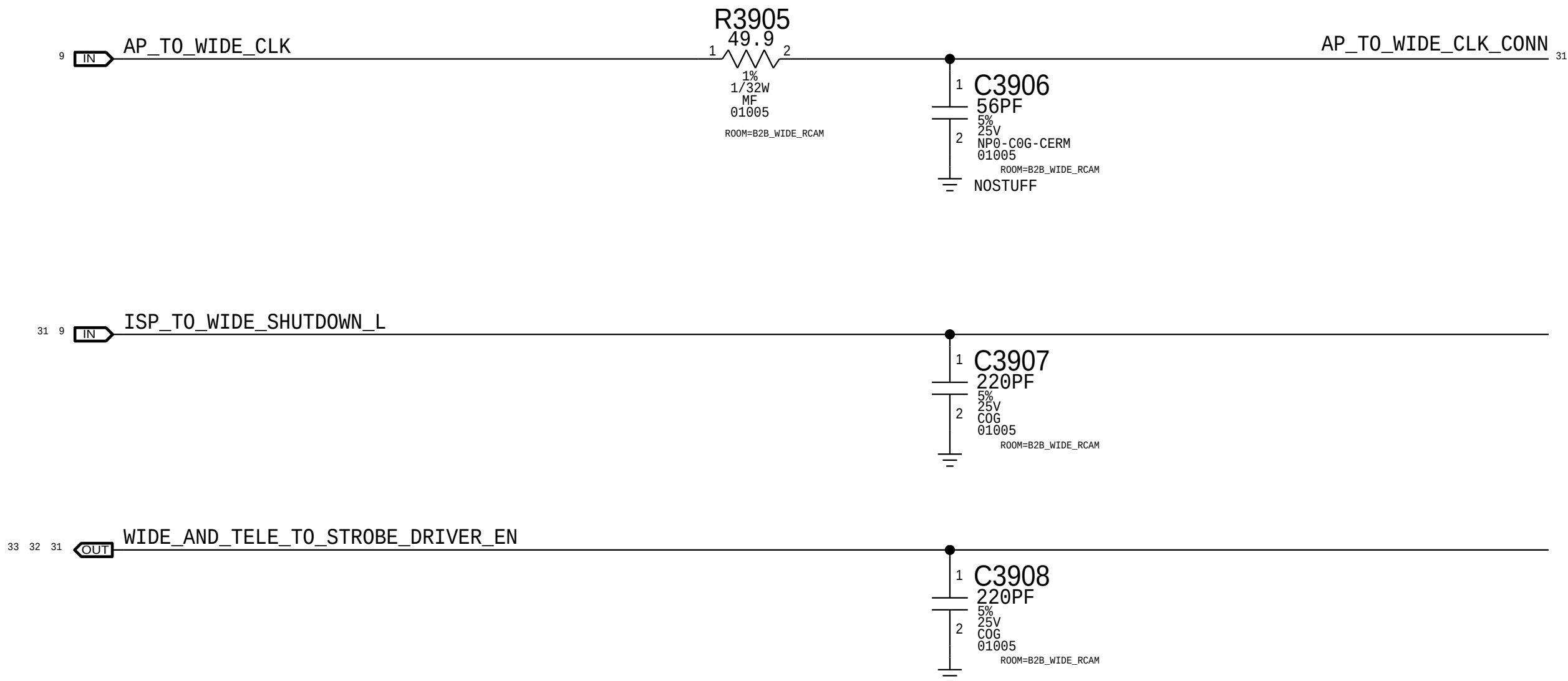
Rcpt: 516S00313 <-- This one on MLB
Plug: 516S00314



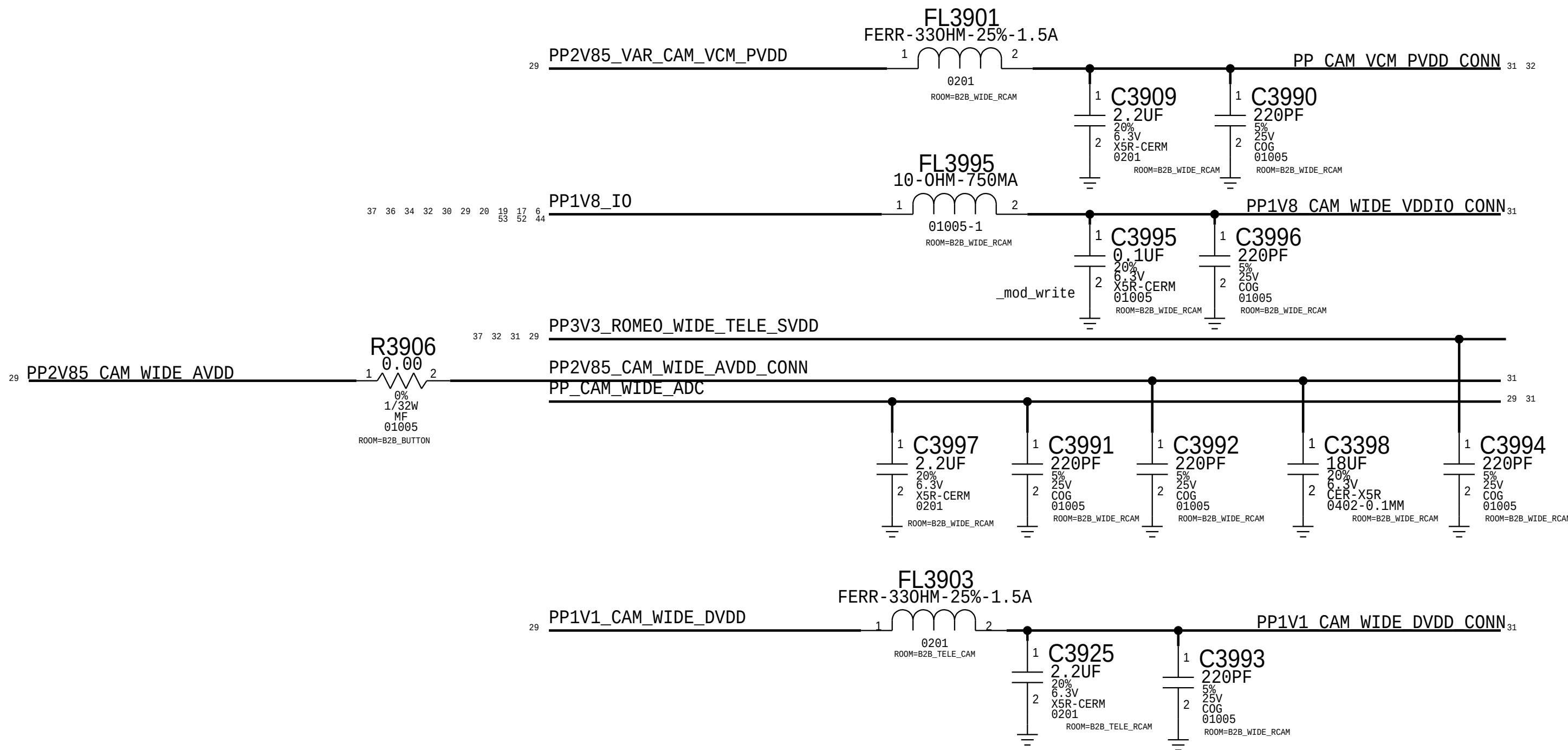
ISP I2C



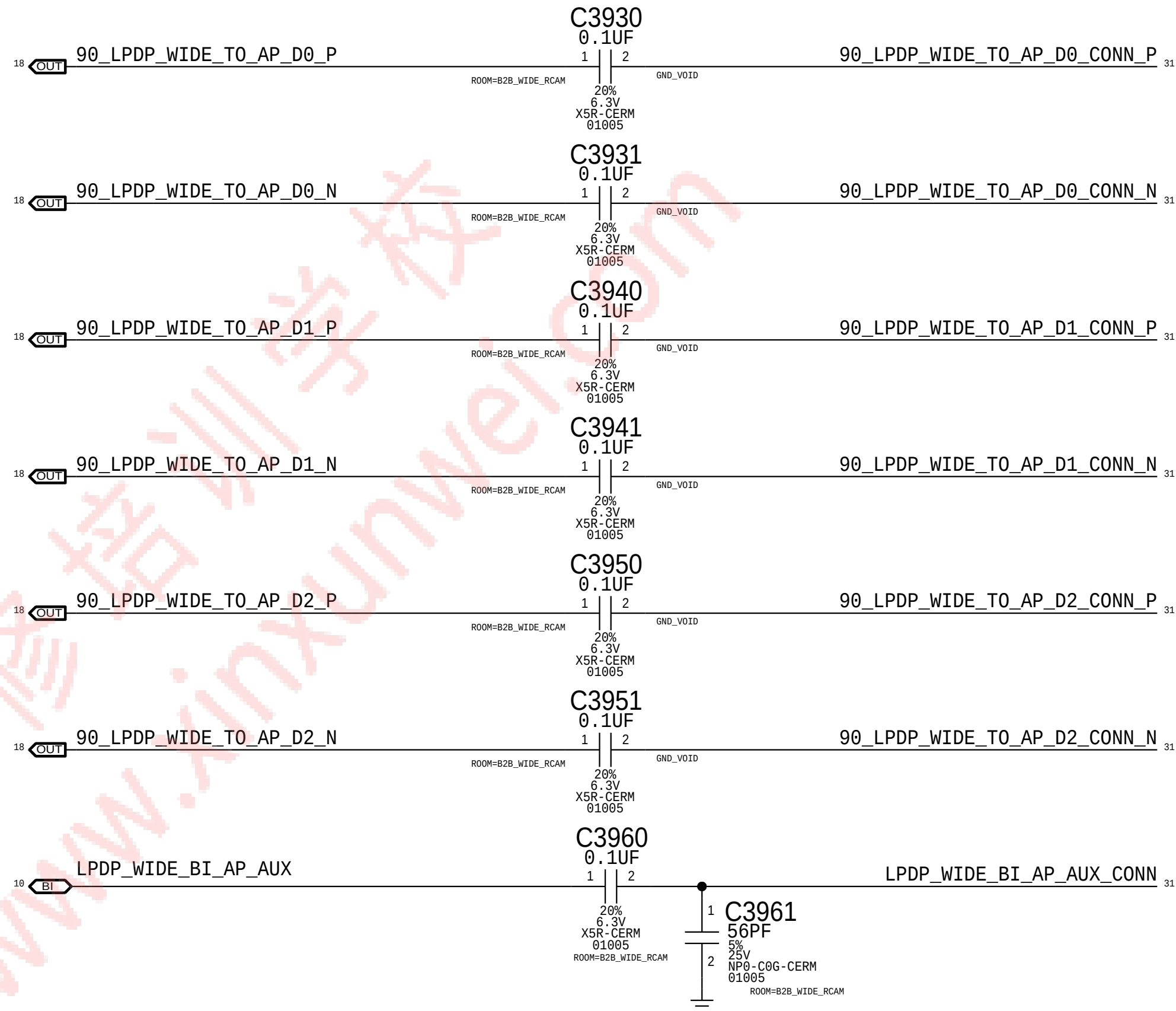
I/O Filters



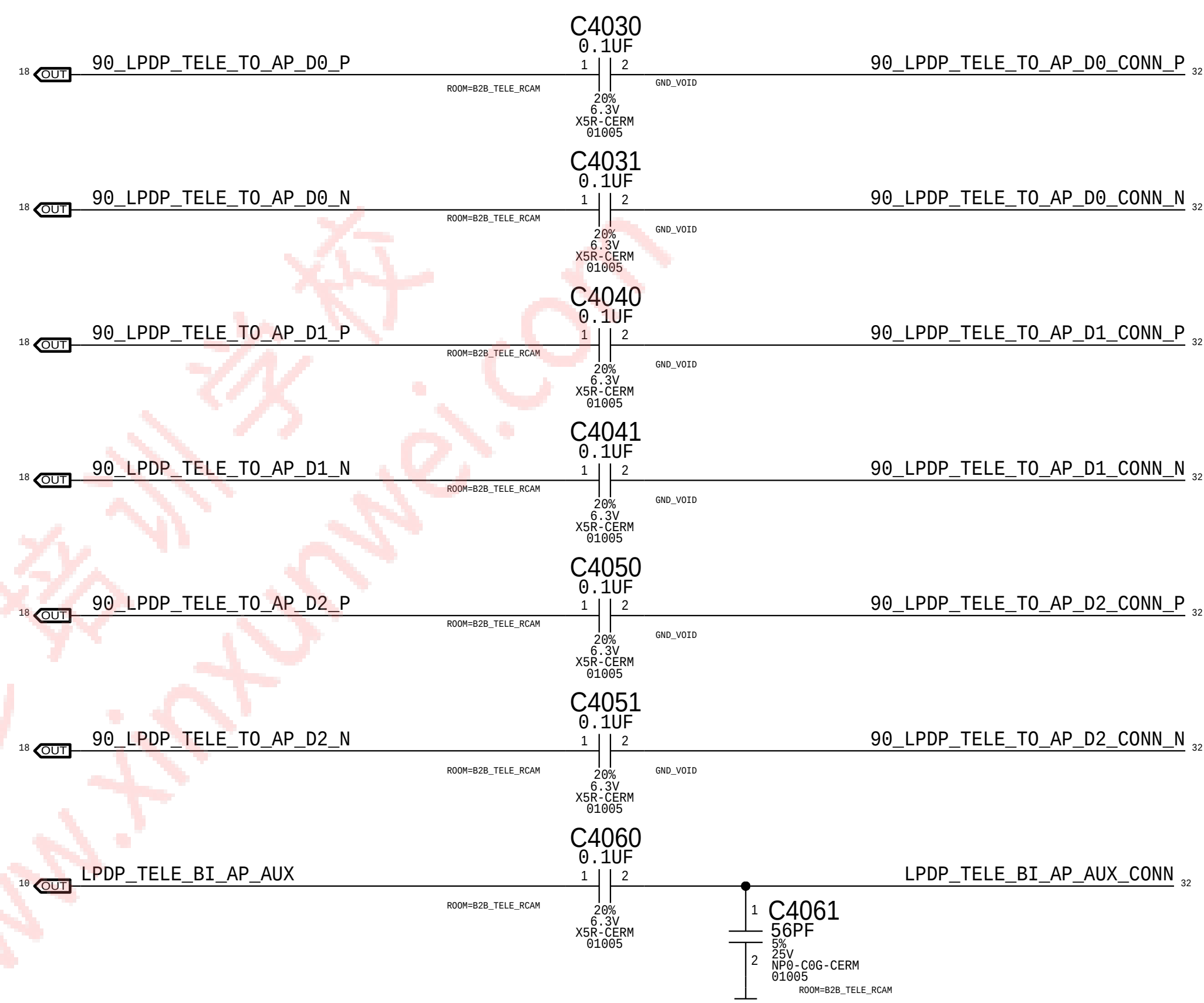
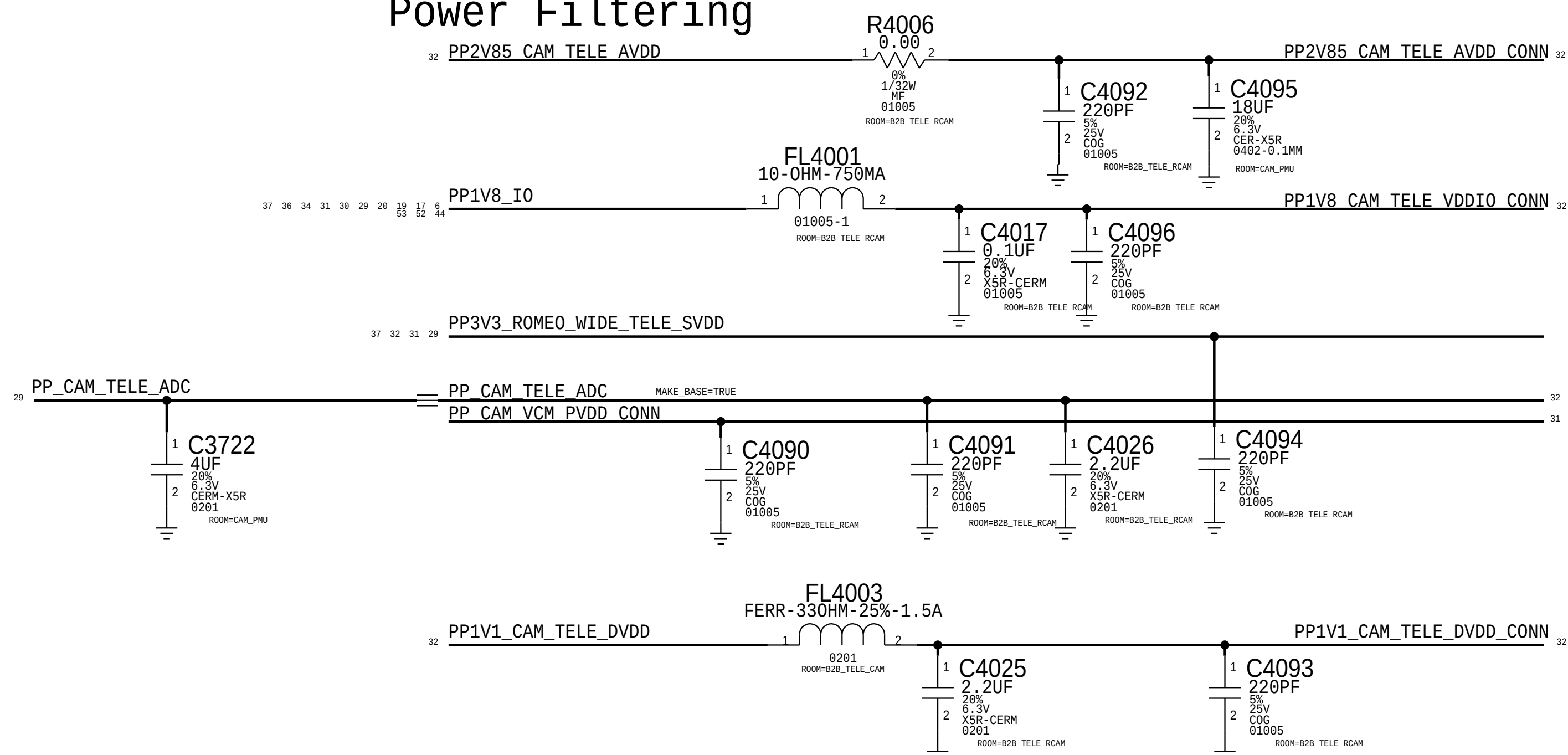
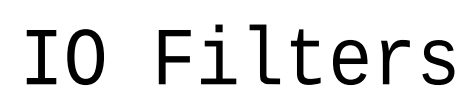
Power Filtering



LPDP Filters

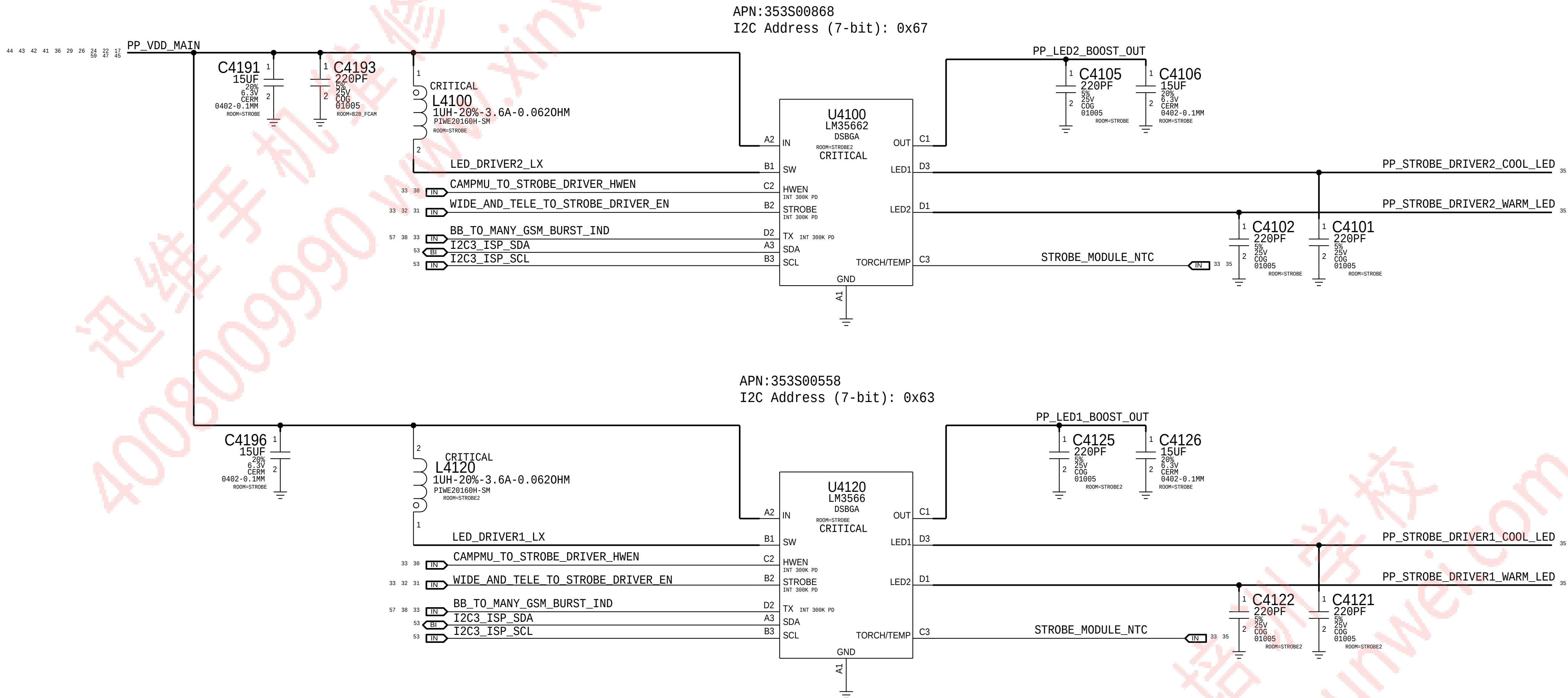


PAGE TITLE		
CAMERA: B2B Wide (TX)		
DRAWING NUMBER	051-02545	SIZE
REVISION	7.0.0	D
BRANCH		
PAGE	39 OF 85	
SHEET	31 OF 60	



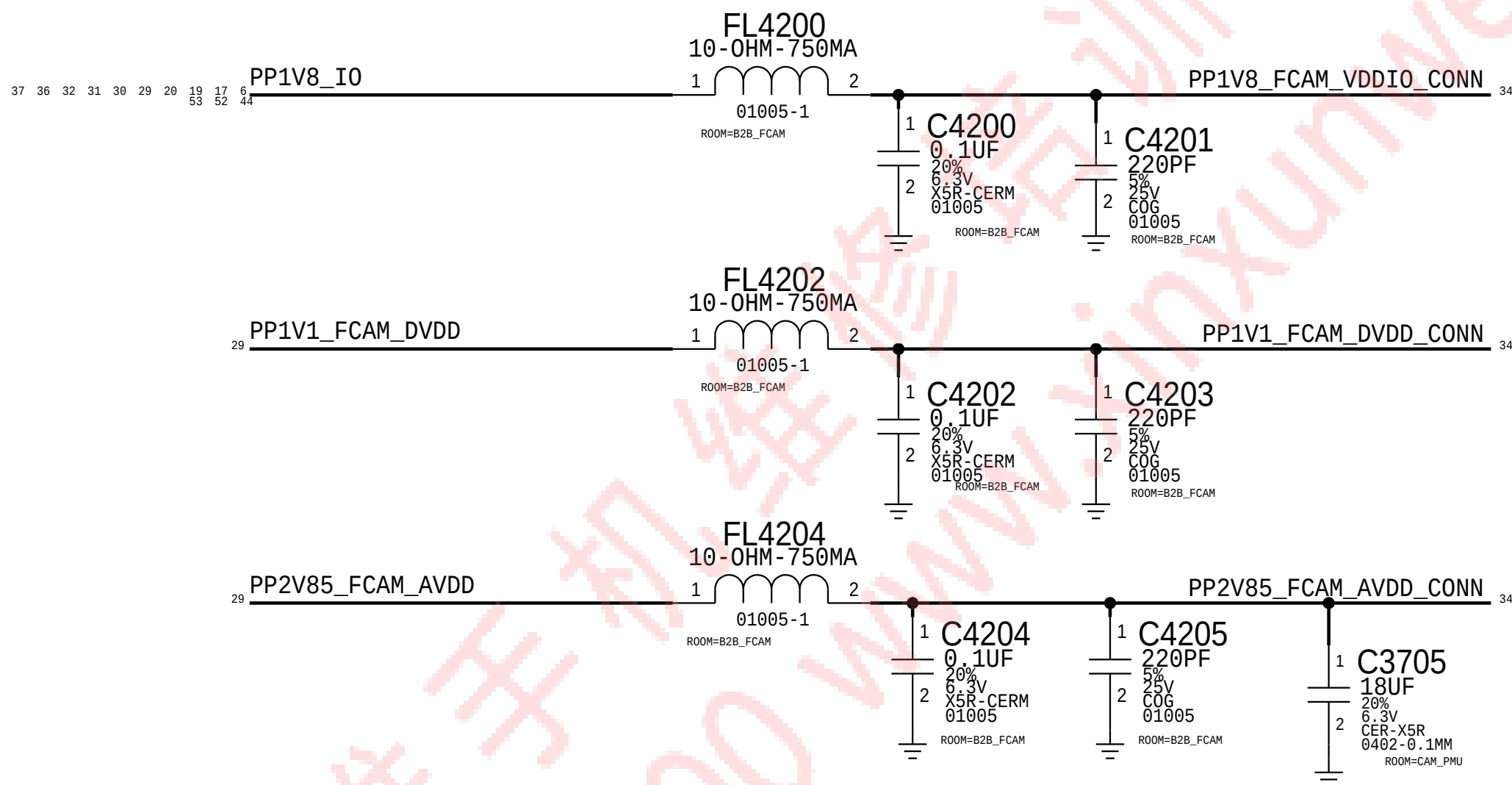
PAGE TITLE		CAMERA: B2B Tele [MT]	
	DRAWING NUMBER	051-02545	SIZE D
	REVISION	7.0.0	
	BRANCH		
	PAGE	40 OF 85	
	SHEET	32 OF 60	

LED STROBE DRIVERS (NEON)



PAGE TITLE		
CAMERA: Strobe Drivers		
DRAWING NUMBER	051-02545	SIZE
		D
REVISION		7.0.0
BRANCH		
PAGE		41 OF 85
SHEET		33 OF 60

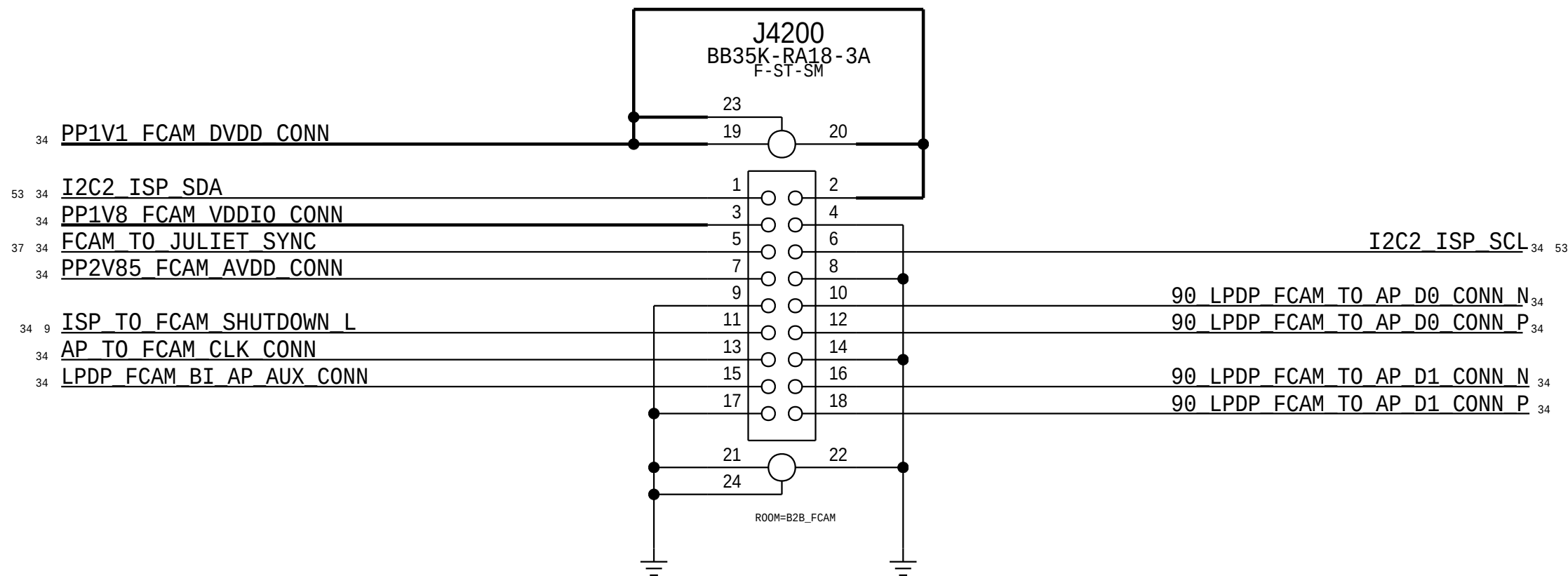
LONG ISLAND POWER



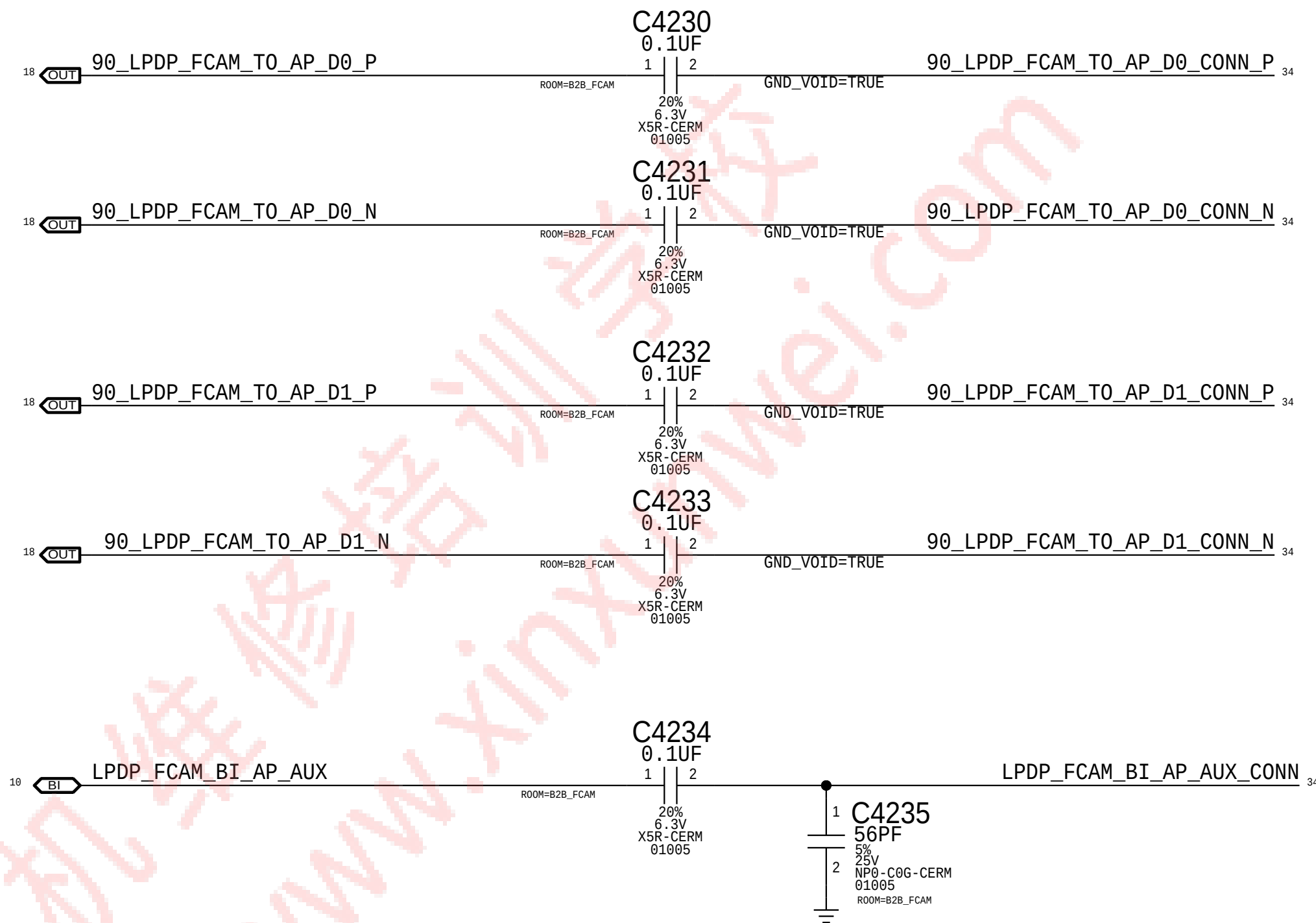
FCAM Connector

Rcpt: 516S00244
Plug: 516S00245

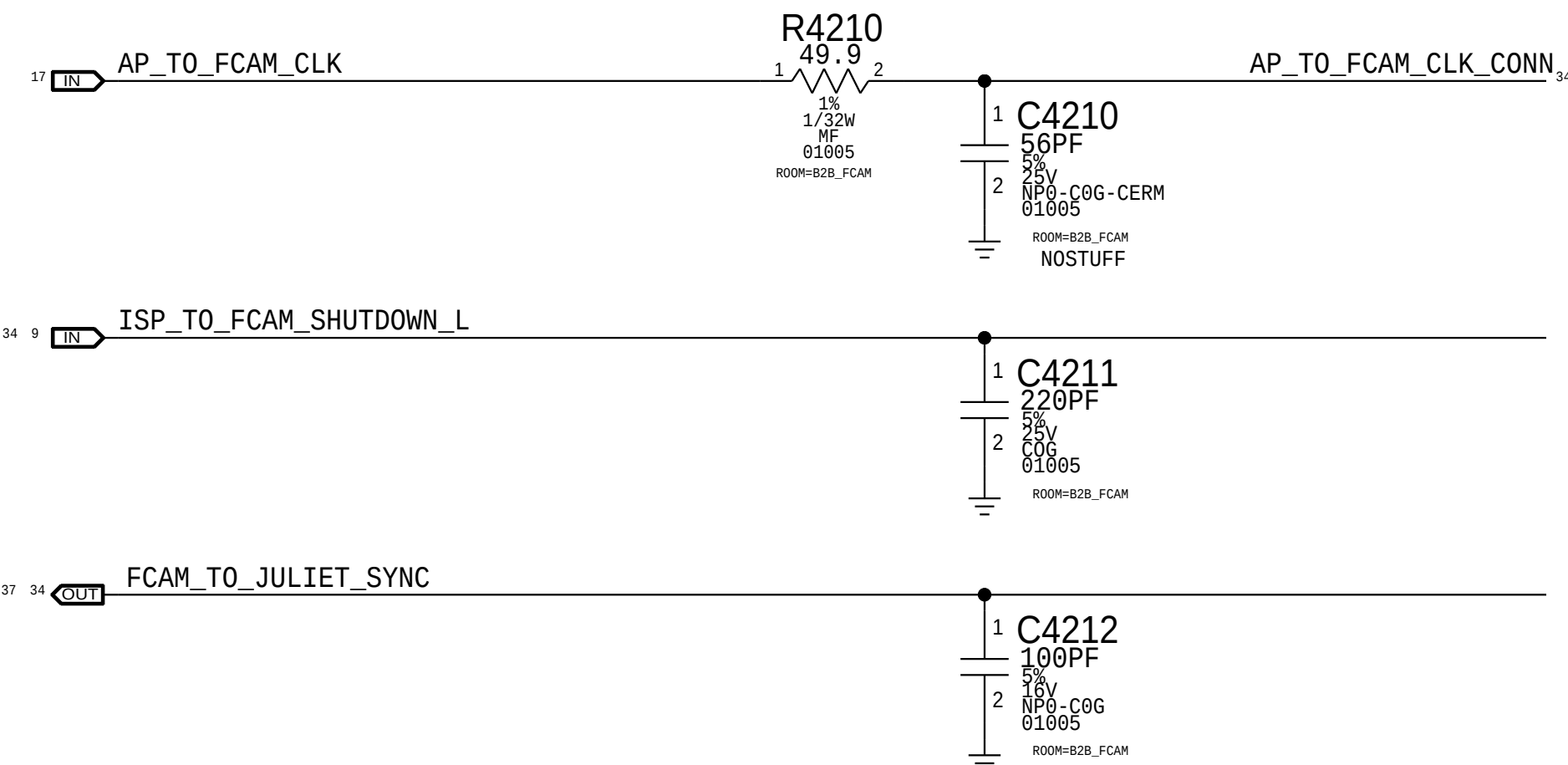
<-- This one on MLB



LPDP FILTERS

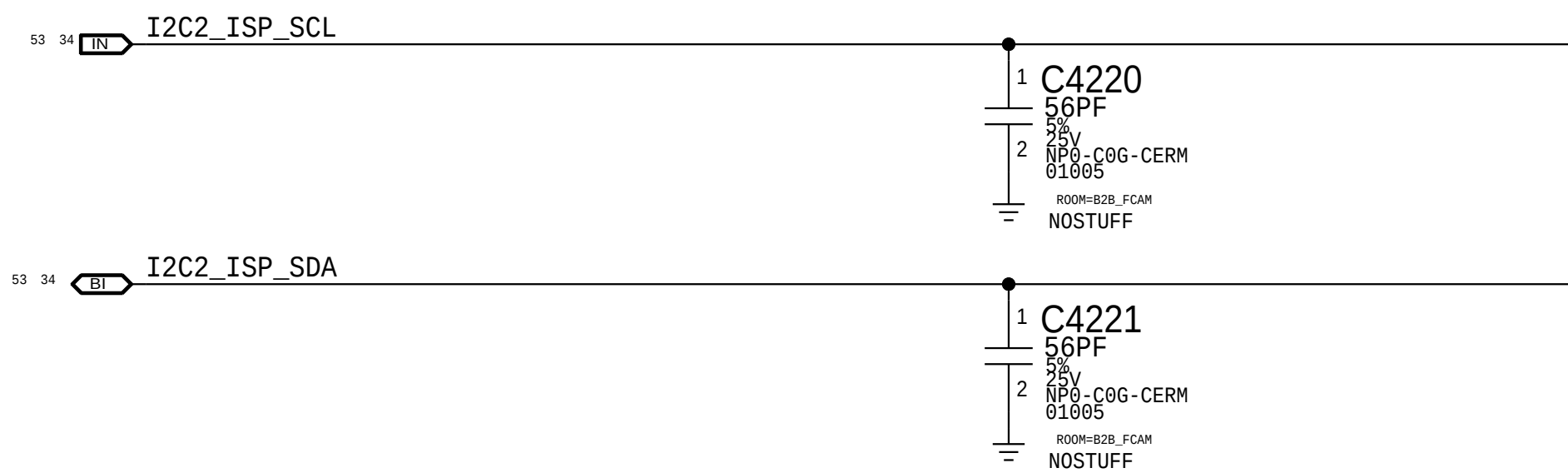


FCAM I/O



ISP I2C2

NOTE: SAME I2C as FCAM



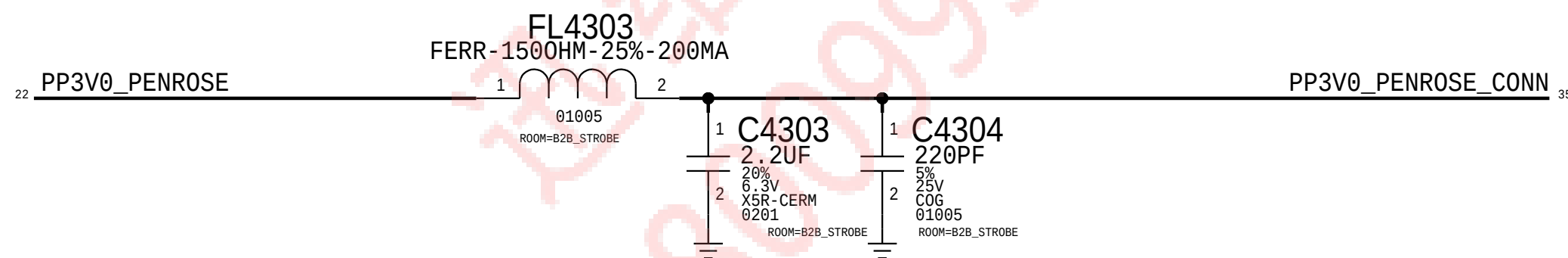
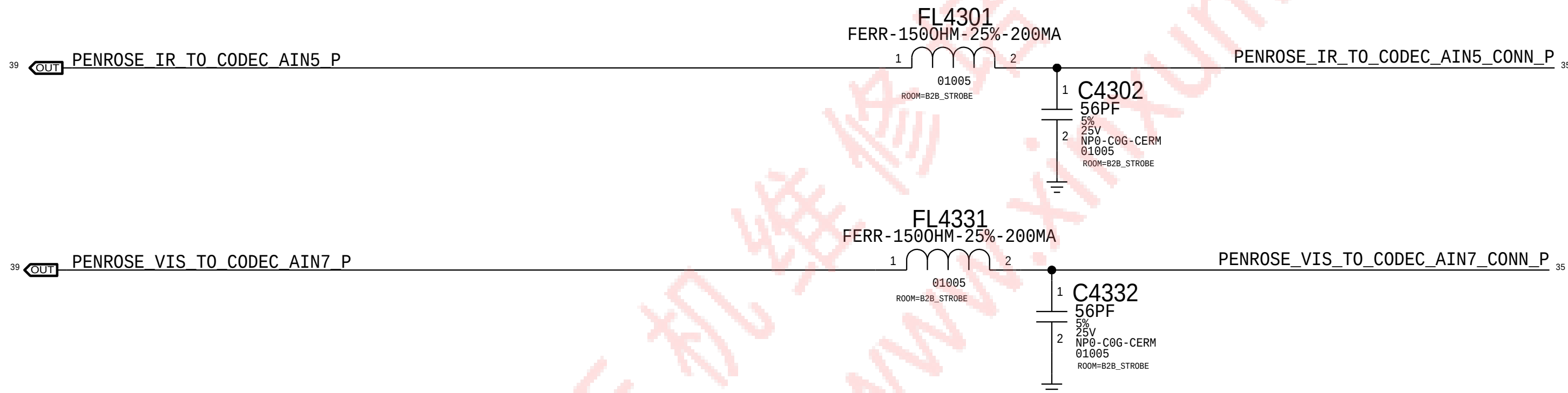
CAMERA: B2B Fcam		
DRAWING NUMBER	051-02545	SIZE
REVISION	7.0.0	D
BRANCH		
PAGE	42 OF 85	
SHEET	34 OF 60	



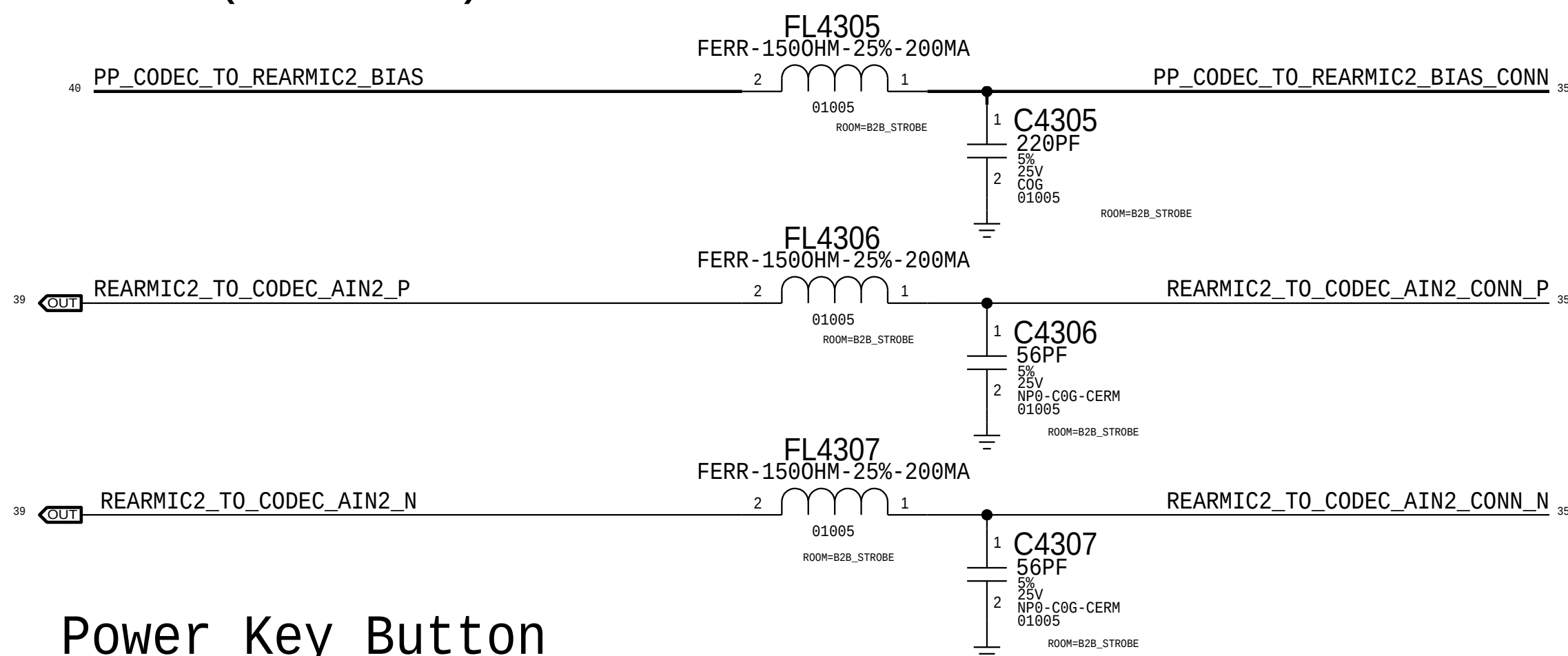
Strobe Connector

Rcpt: 516S00381 <-- This one on MLB
Plug: 516S00382

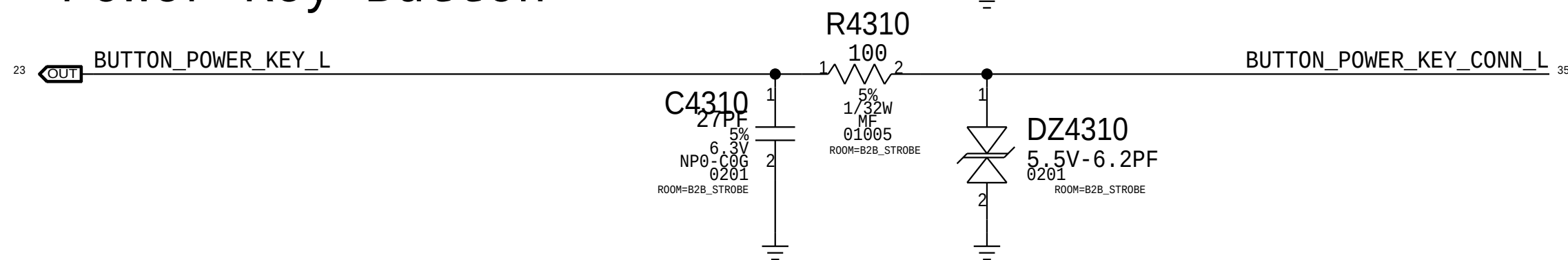
PENROSE



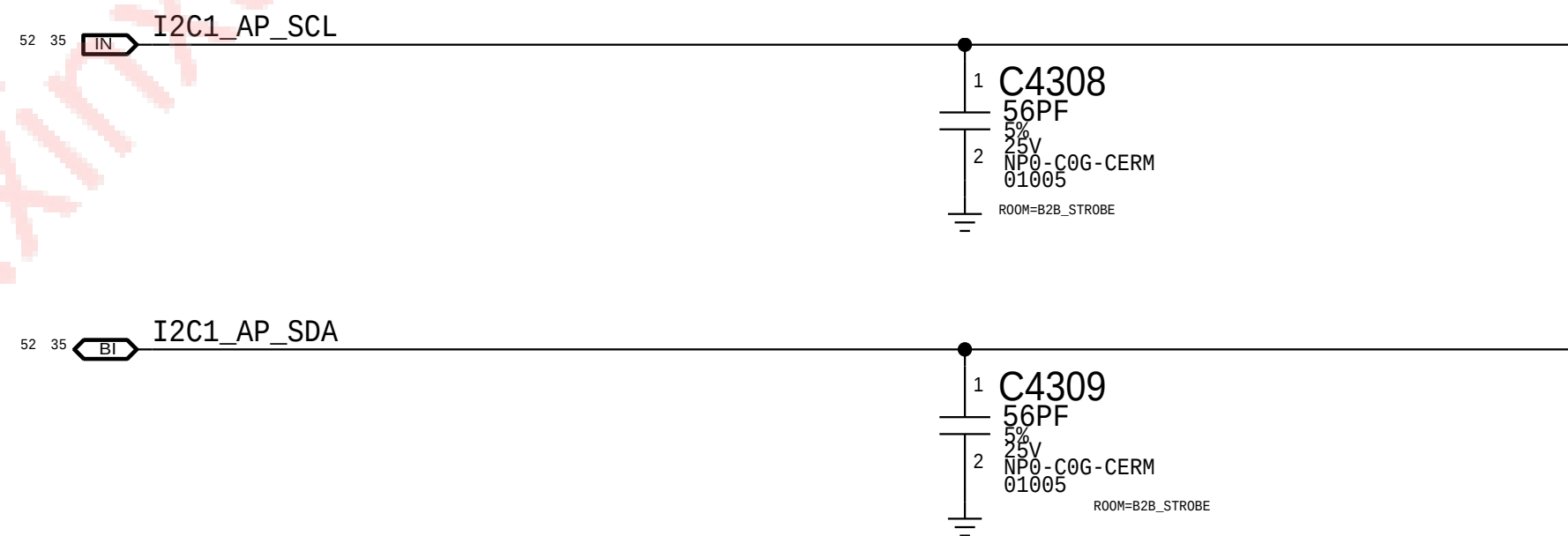
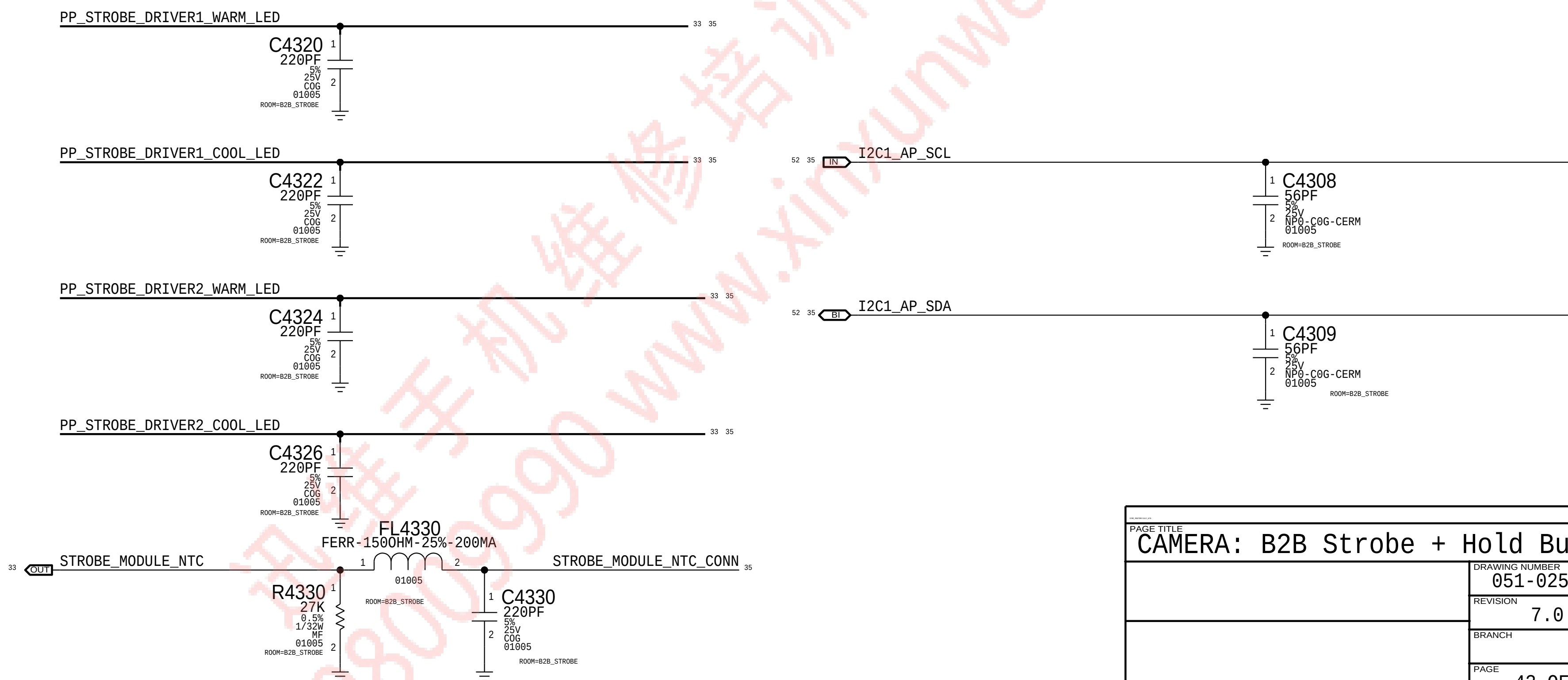
MIC2 (ANC REF)



Power Key Button



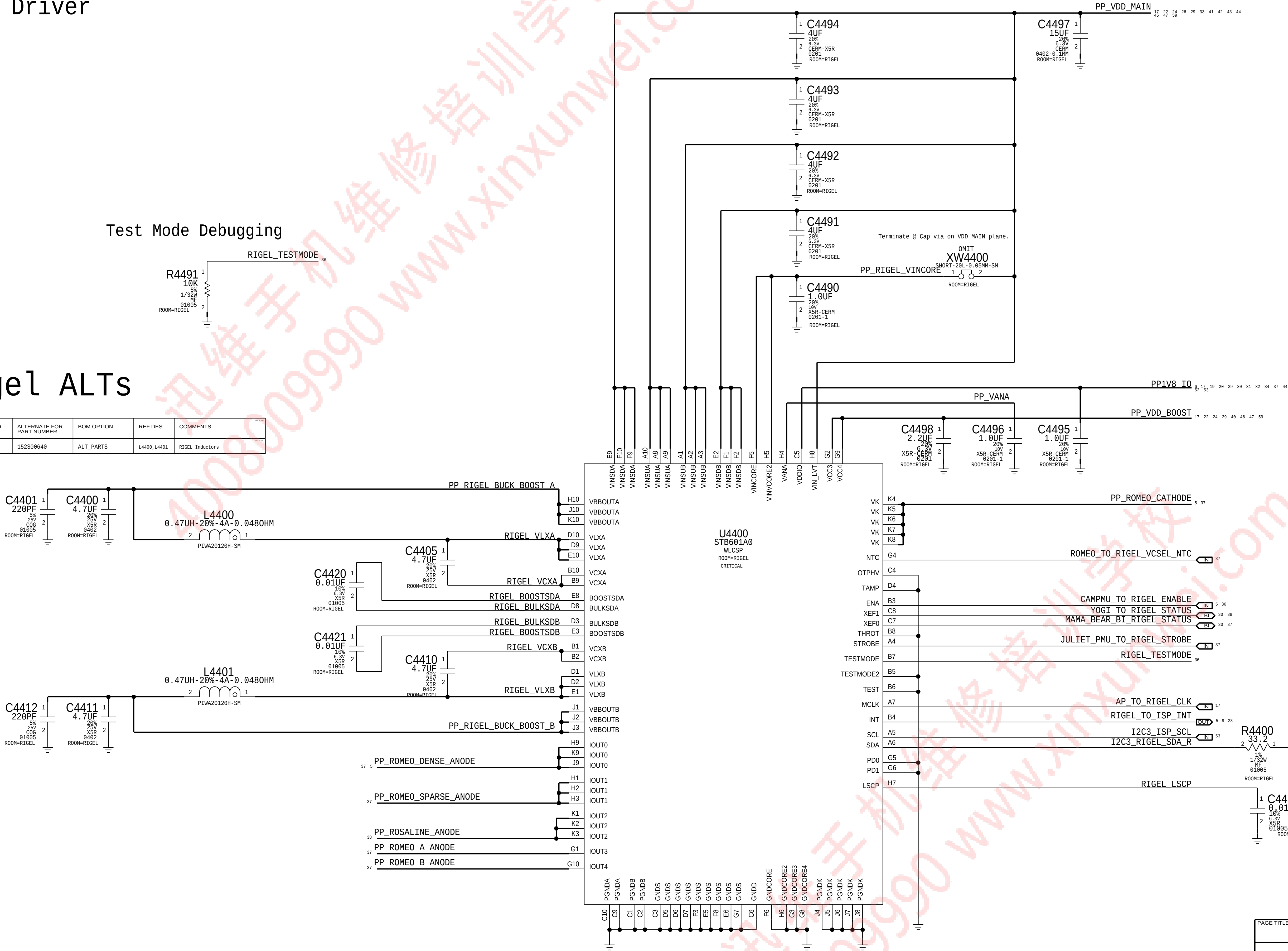
Strobe Filtering



PAGE TITLE		
CAMERA: B2B Strobe + Hold Button		
DRAWING NUMBER	051-02545	SIZE
REVISION	7.0.0	D
BRANCH		
PAGE		
43 OF 85		
SHEET		
35 OF 60		

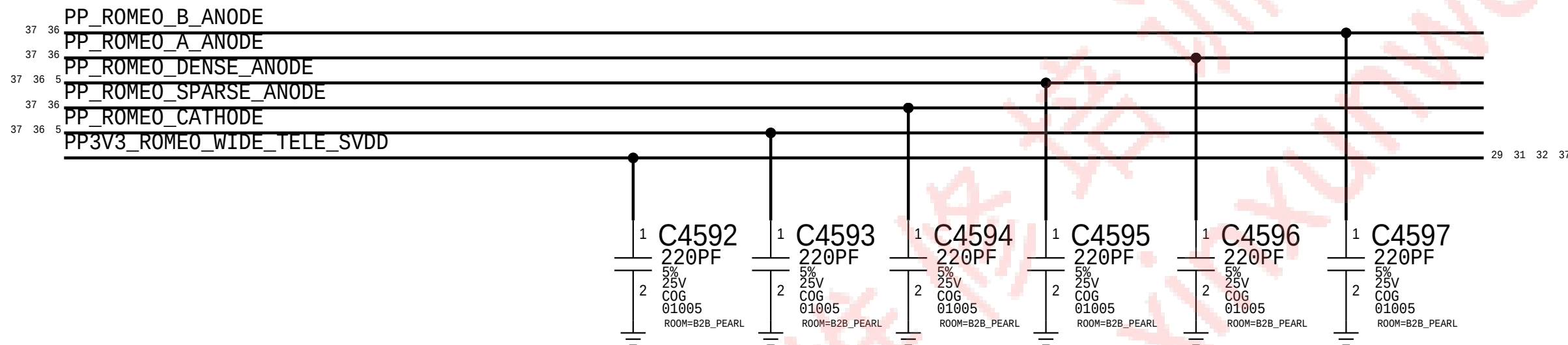
Rigel ALTs

PAGE TITLE		
PEARL: Power		
	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	44 OF 85
	SHEET	36 OF 60
SIZE		
D		

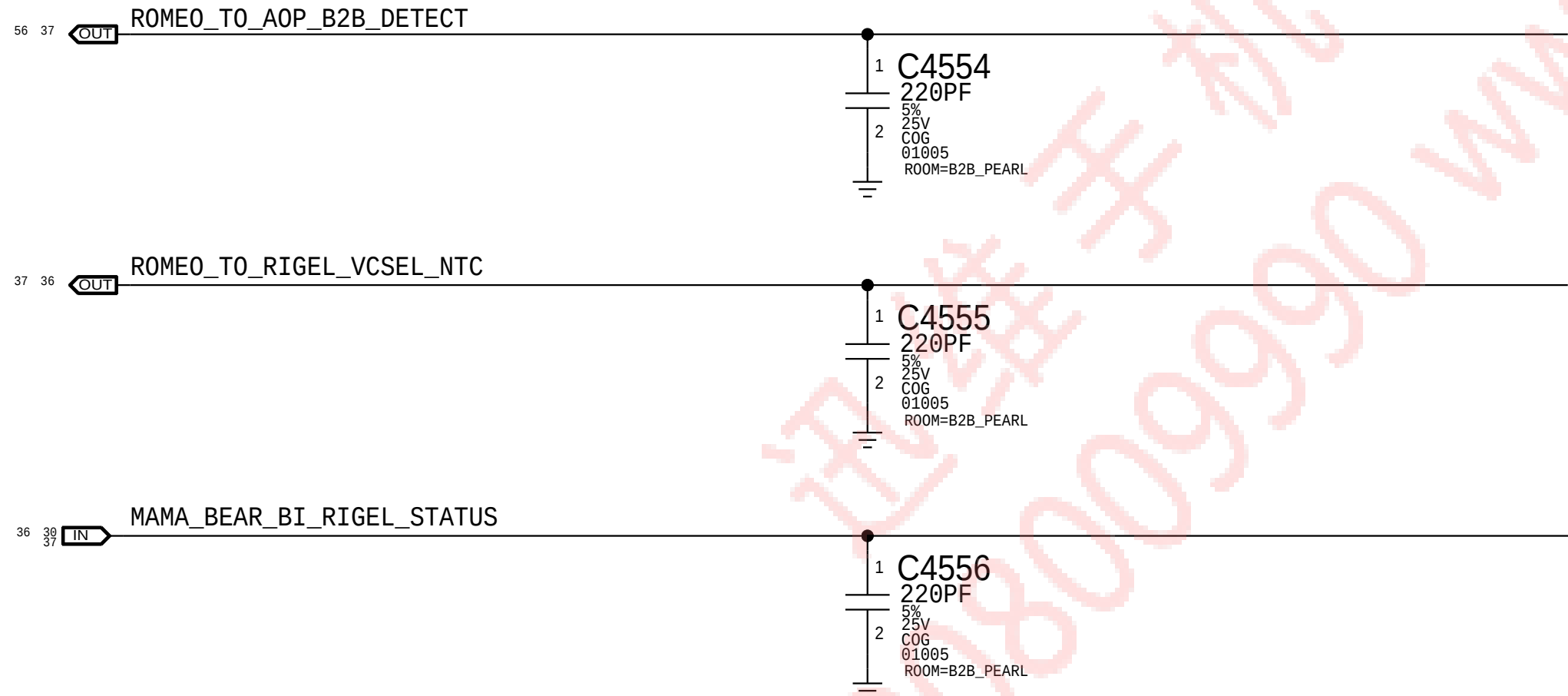




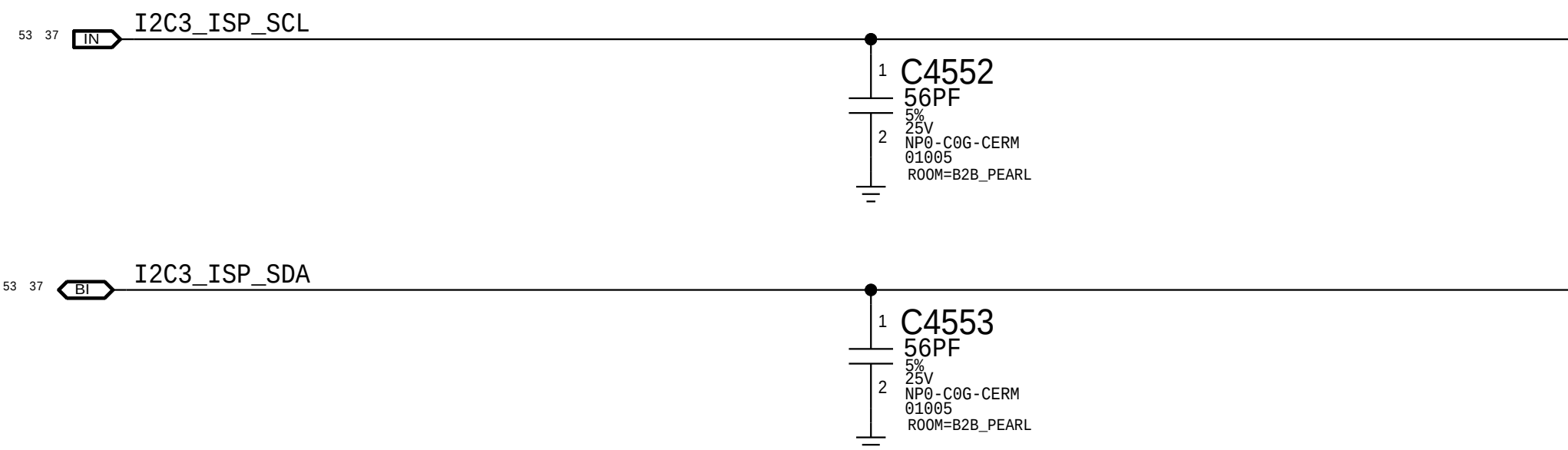
Romeo Power Filtering



Romeo I/O

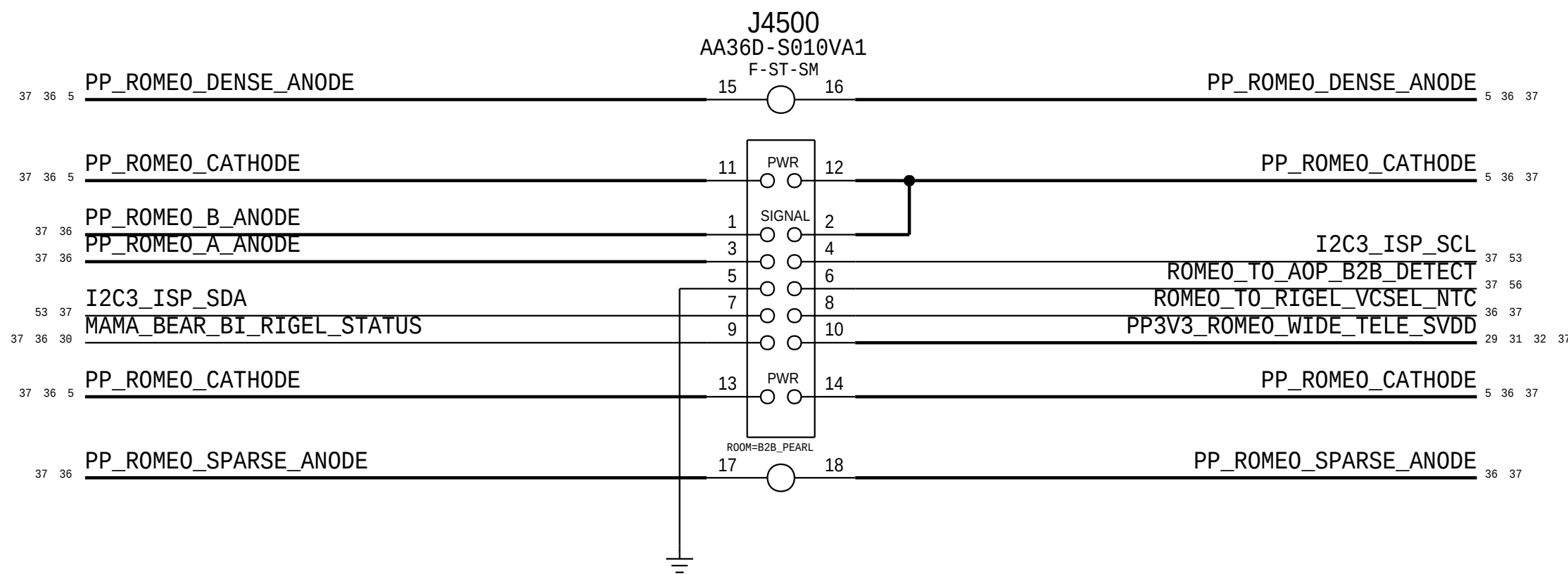


ISP I2C3



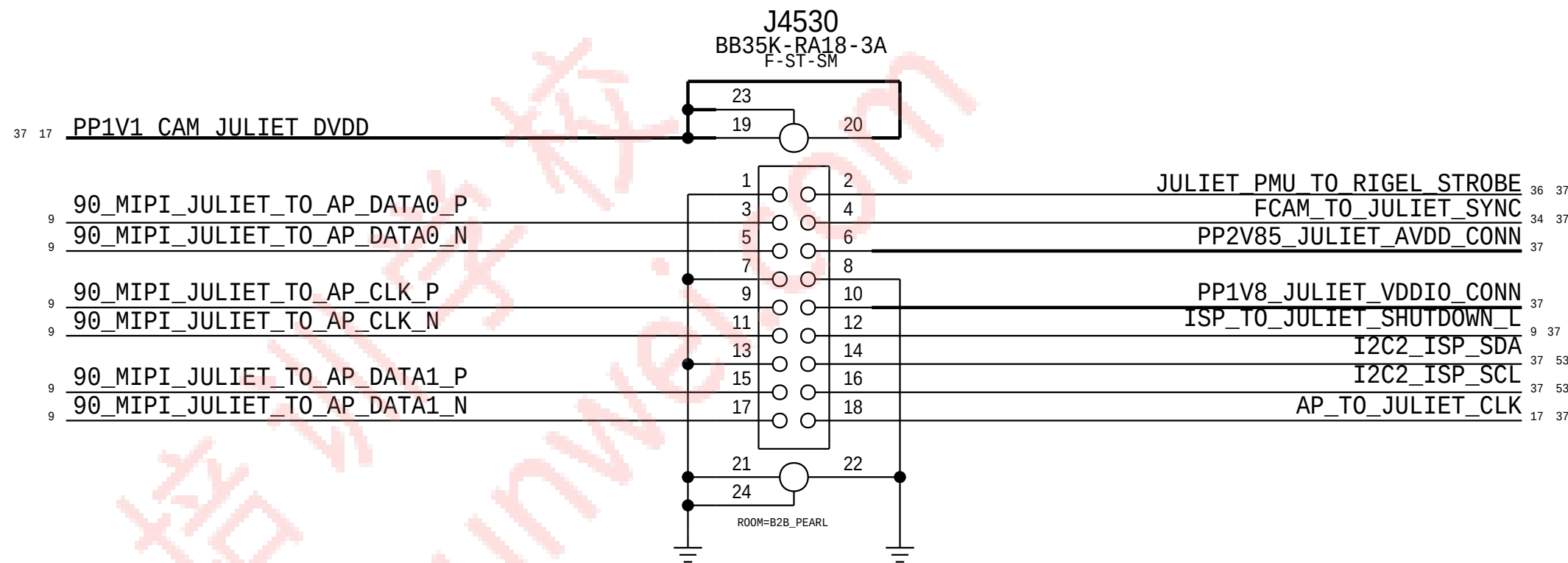
Romeo Connector

Rcpt: 516S00267 <-- This one on MLB
Plug: 516S00268

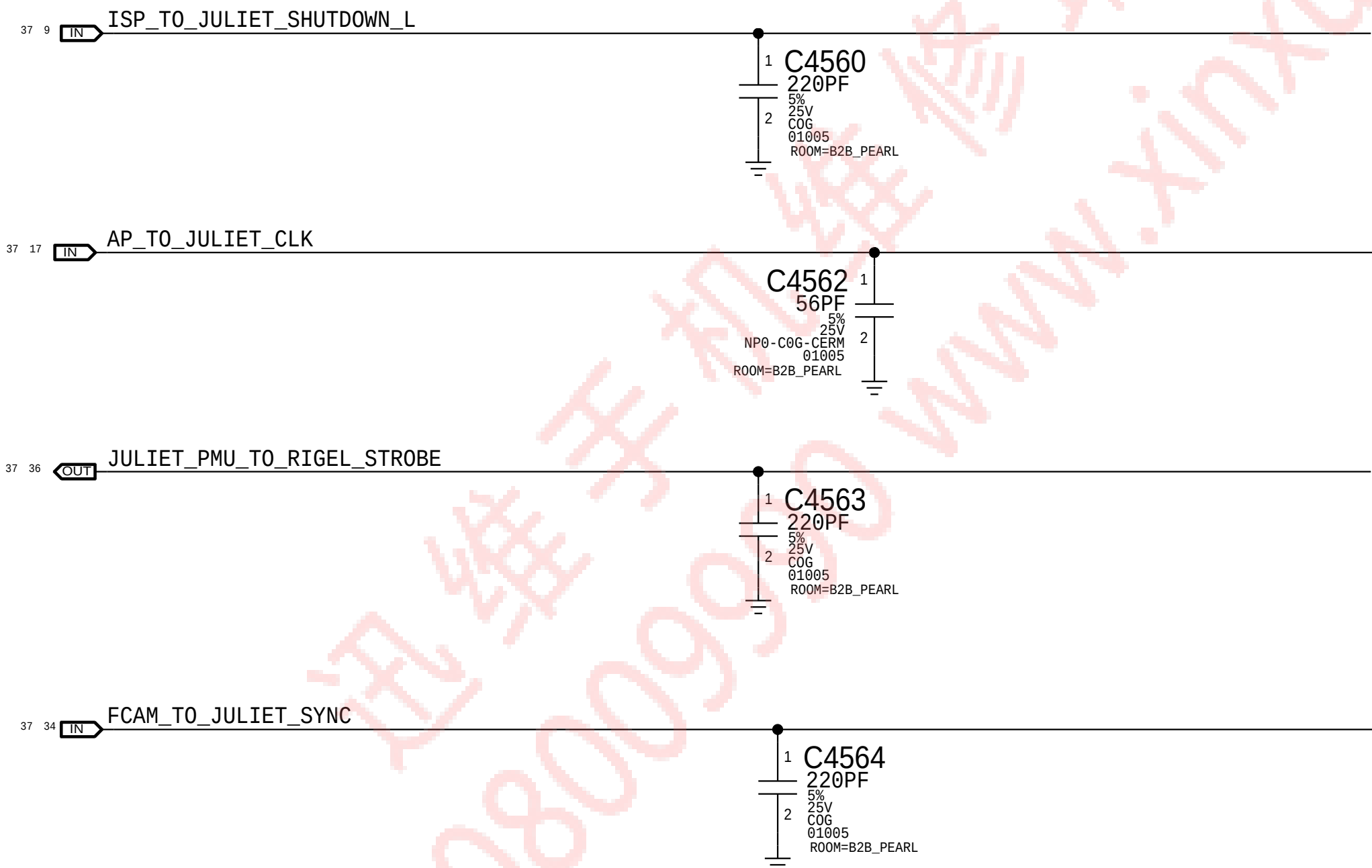
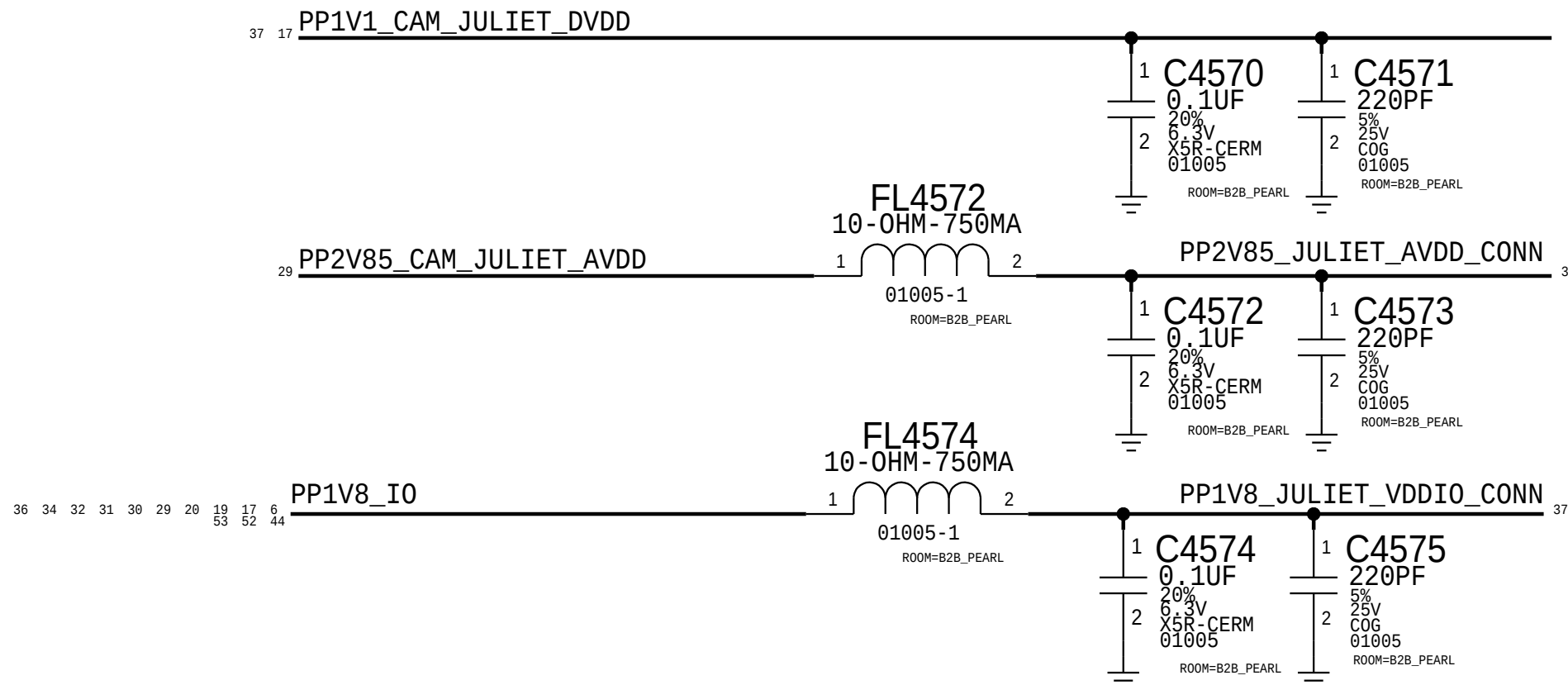


Juliet Connector

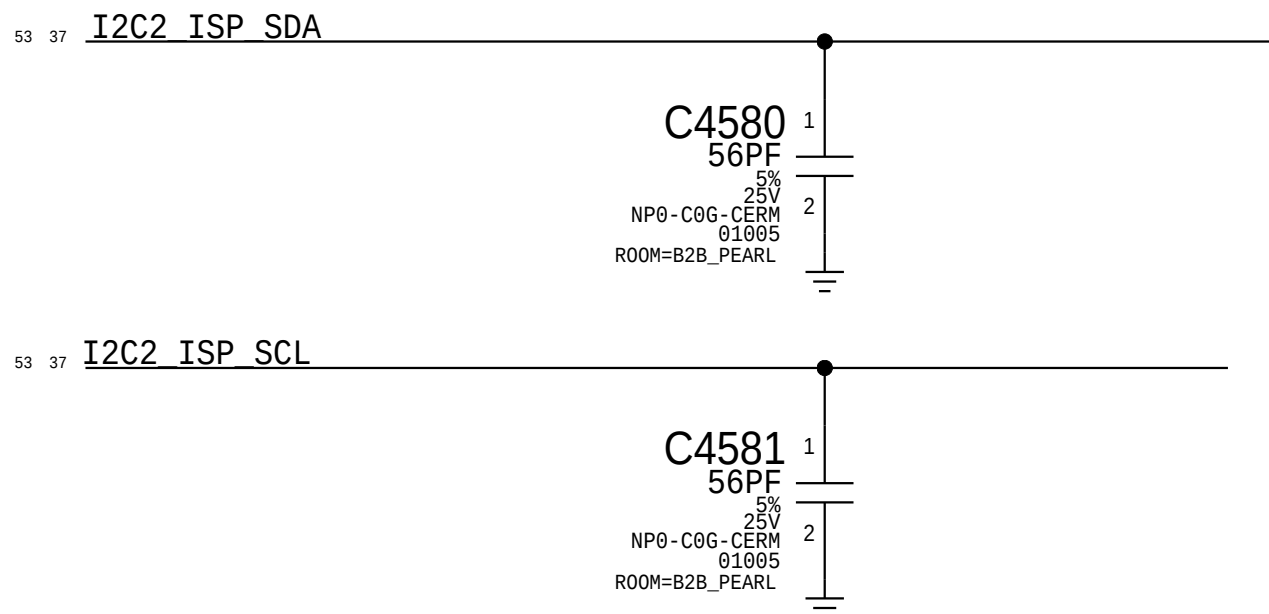
Rcpt: 516S00244 <-- This one on MLB
Plug: 516S00245



Juliet Power and I/O



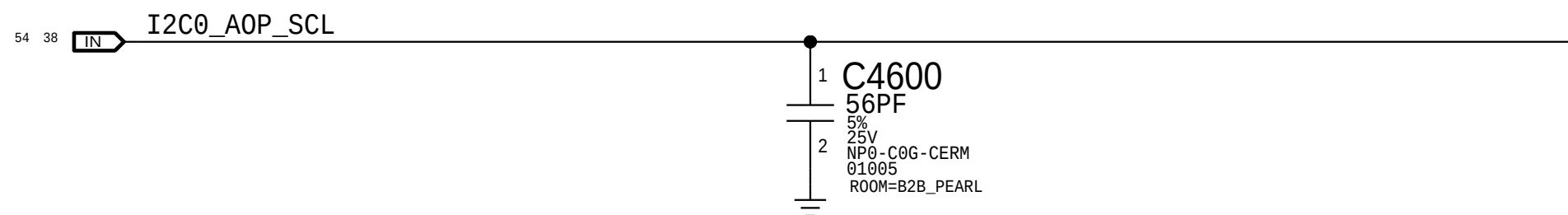
NOTE: SAME I2C as FCAM



PAGE TITLE		
PEARL: B2B Romeo + Juliet		
	DRAWING NUMBER	SIZE
	051-02545	D
	REVISION	7.0.0
	BRANCH	
	PAGE	
	45 OF 85	
	SHEET	
	37 OF 60	



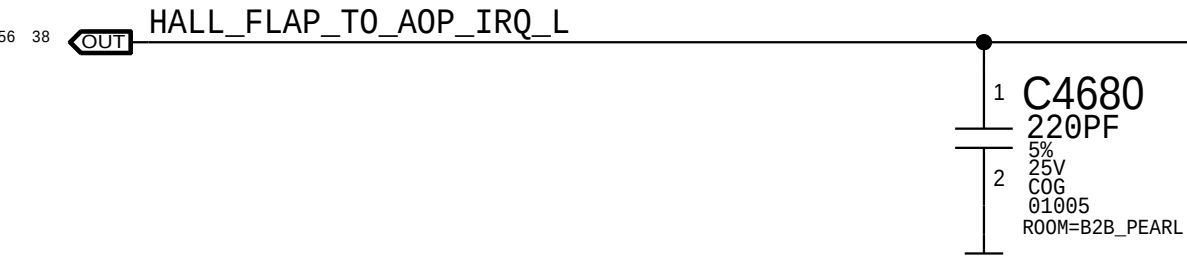
AOP I2C



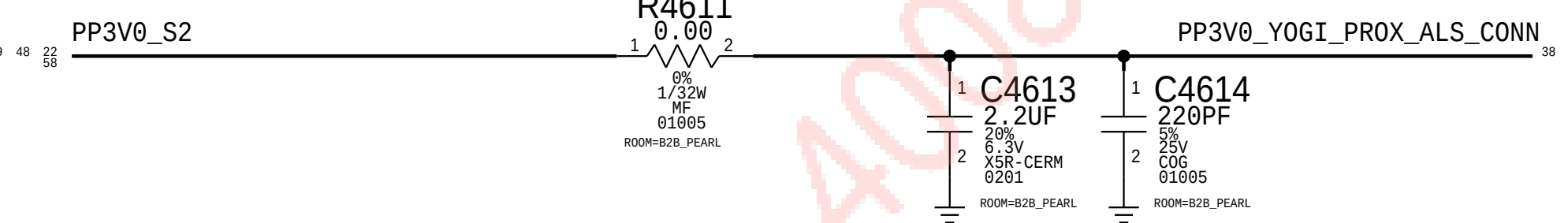
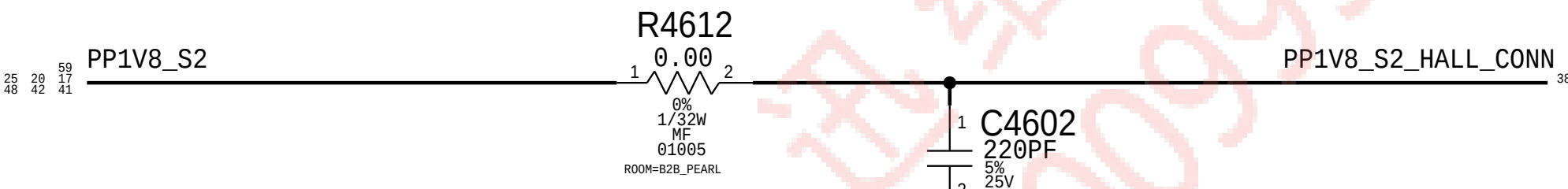
I2C0_AOP_SDA



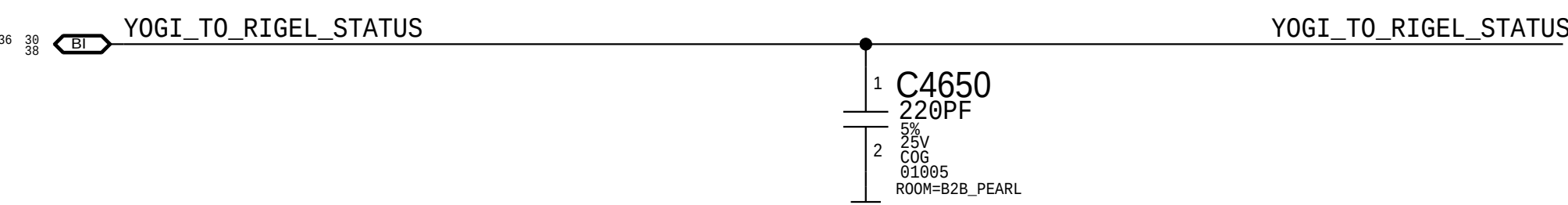
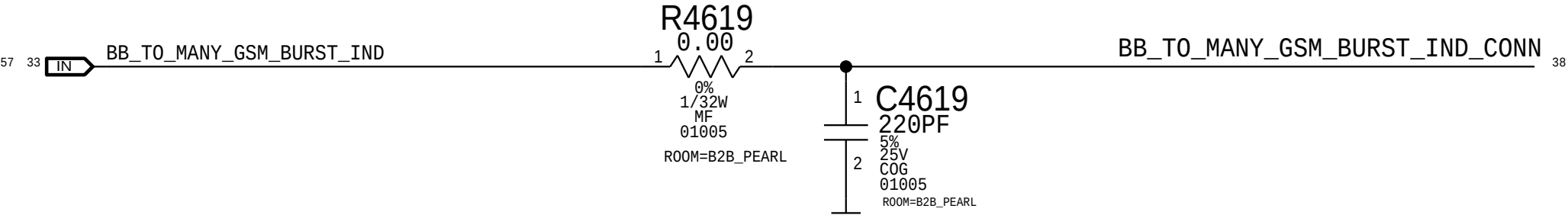
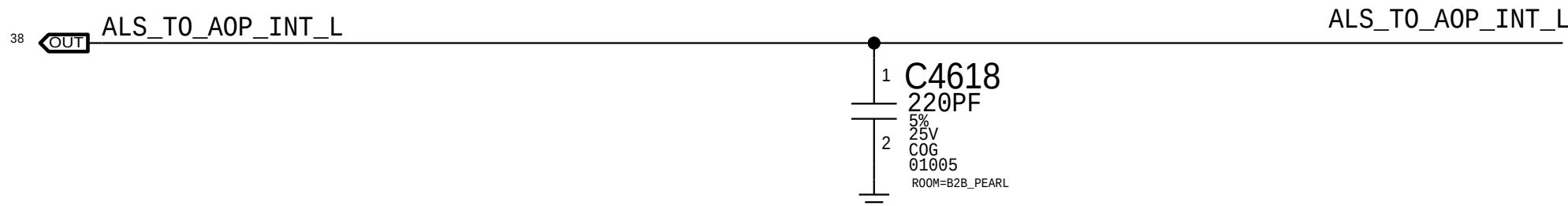
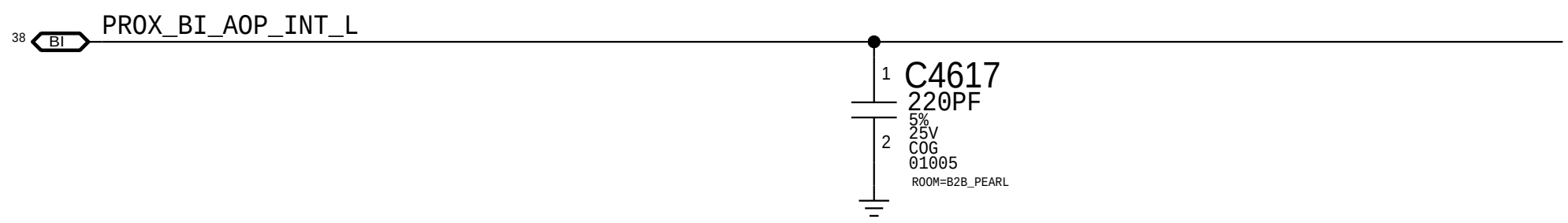
HALL I/Os



PROX & HALL POWER

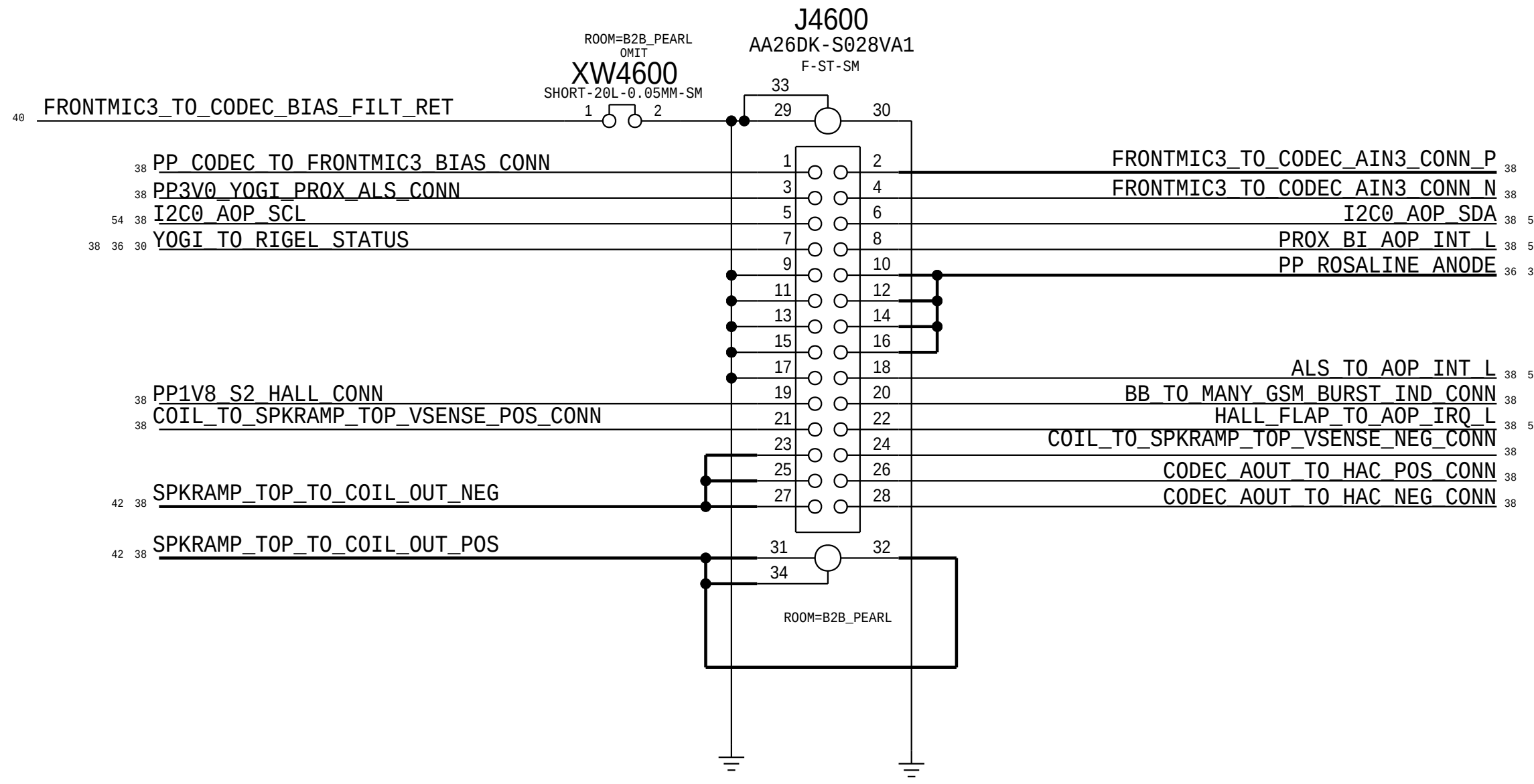


PROX/ALS/YOGI I/O

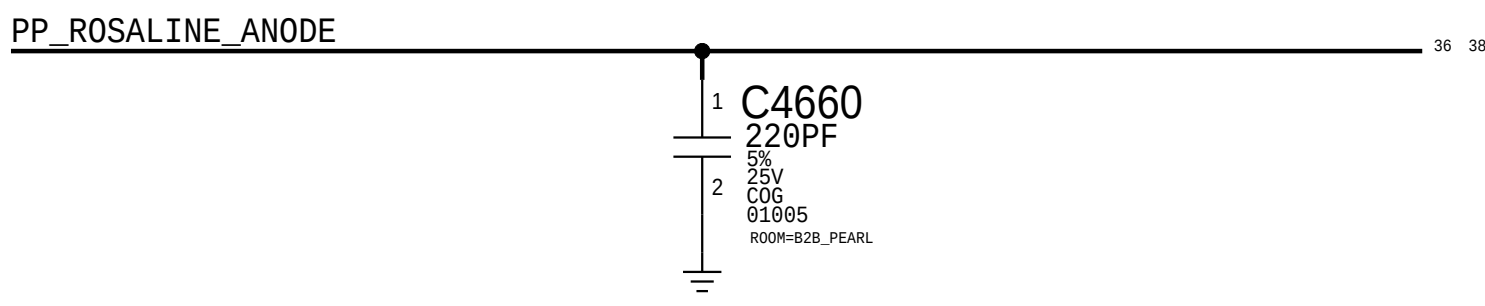


Rosaline + Sensor Connector

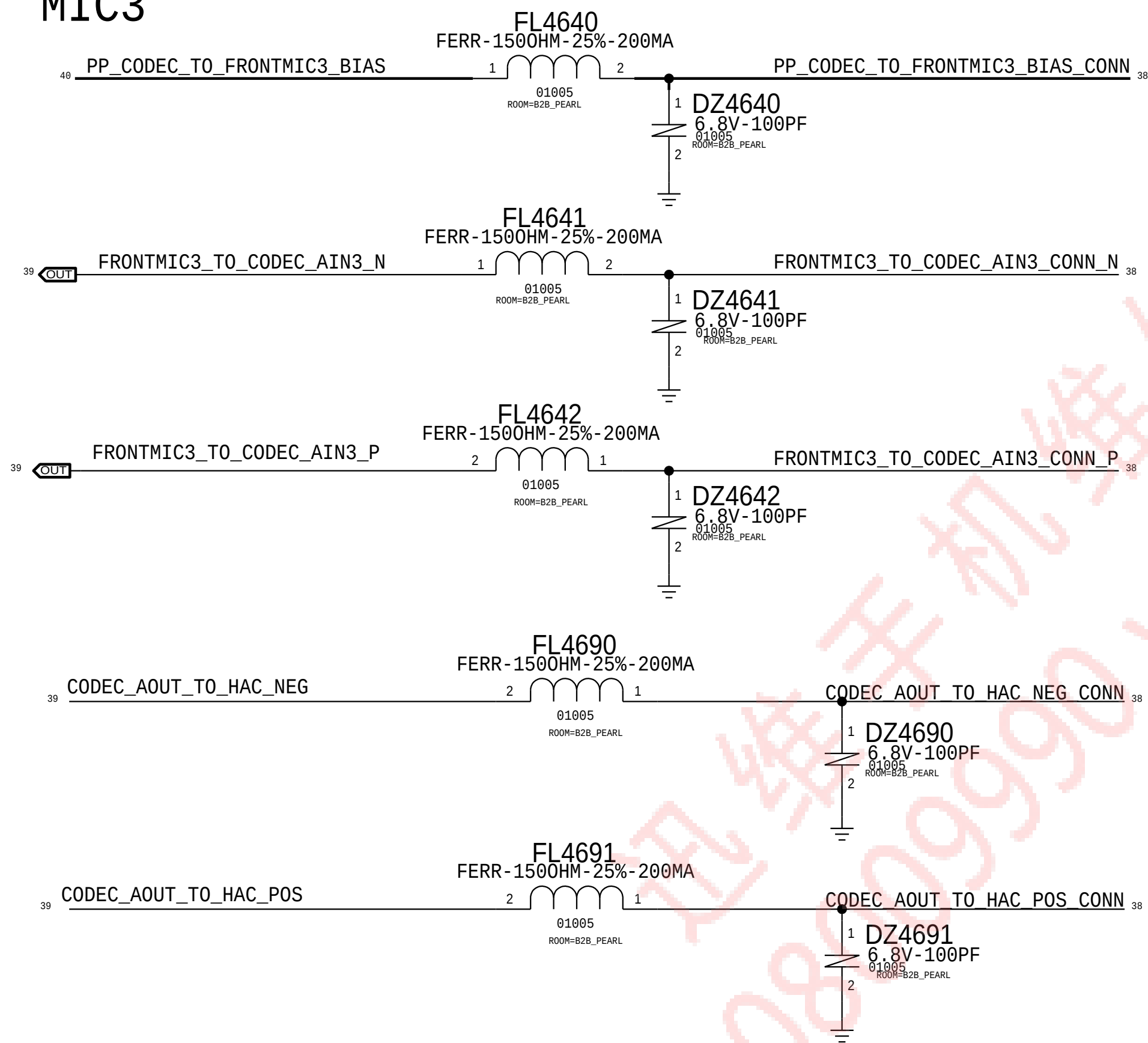
Rcpt: 516S00325 <-- This one on MLB
Plug: 516S00326



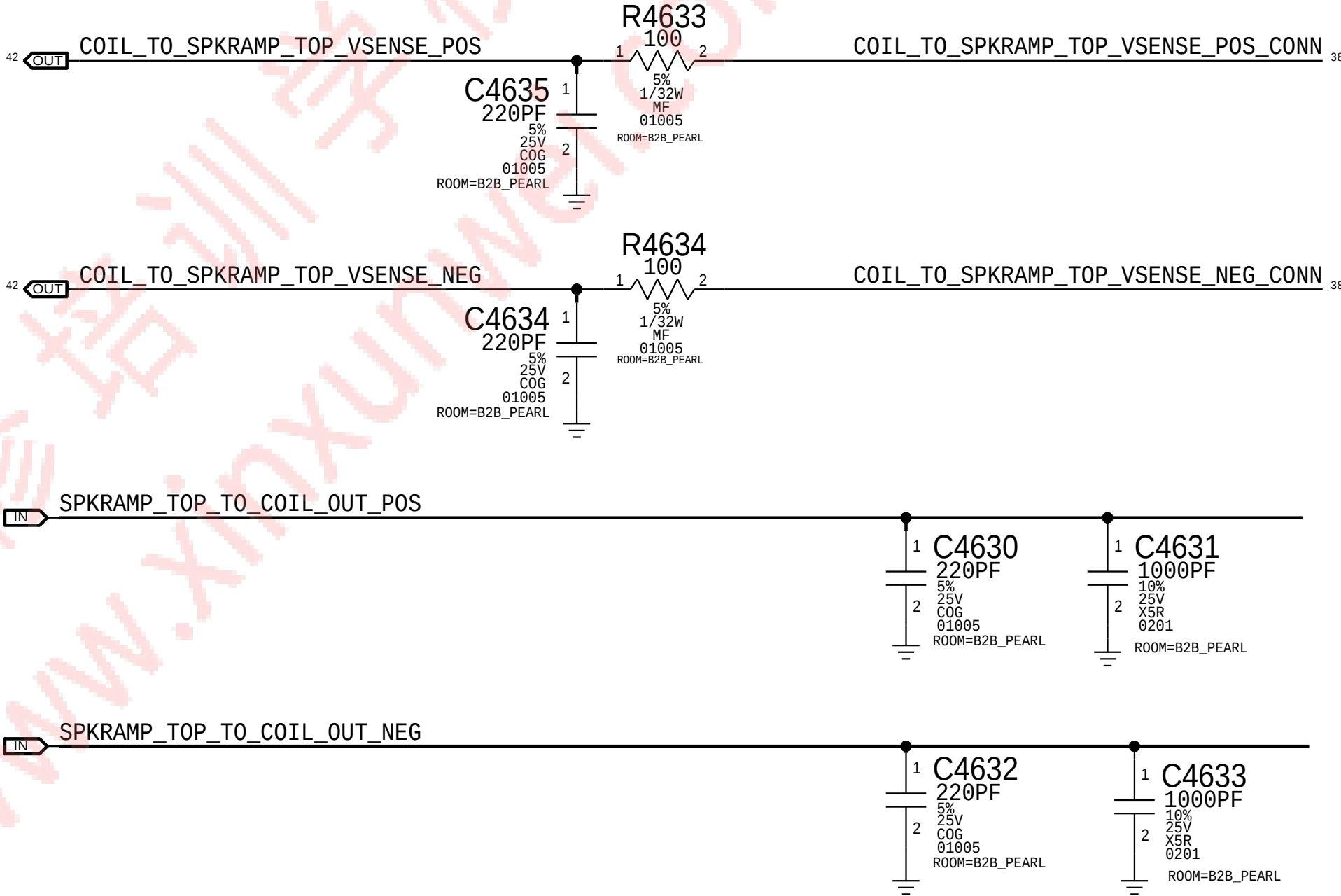
Yogi Signals



MIC3

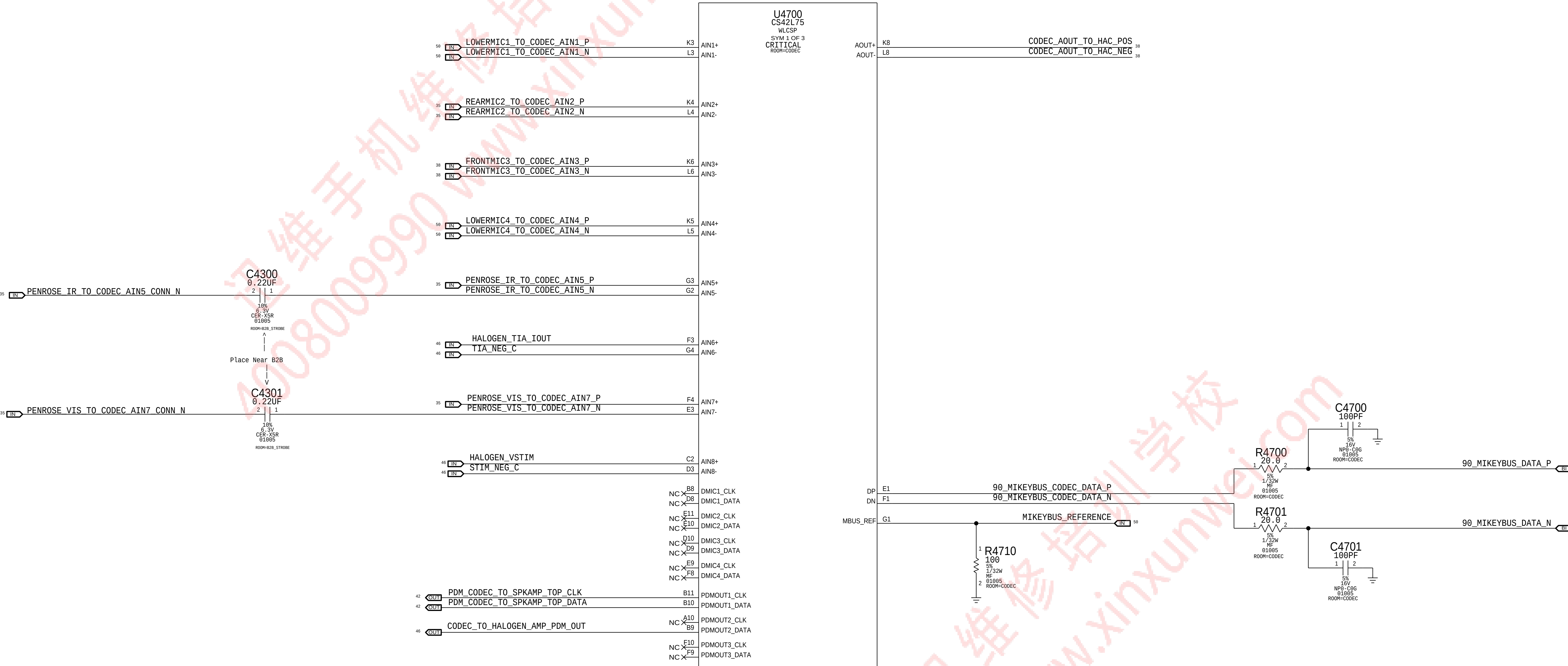


SPEAKER2



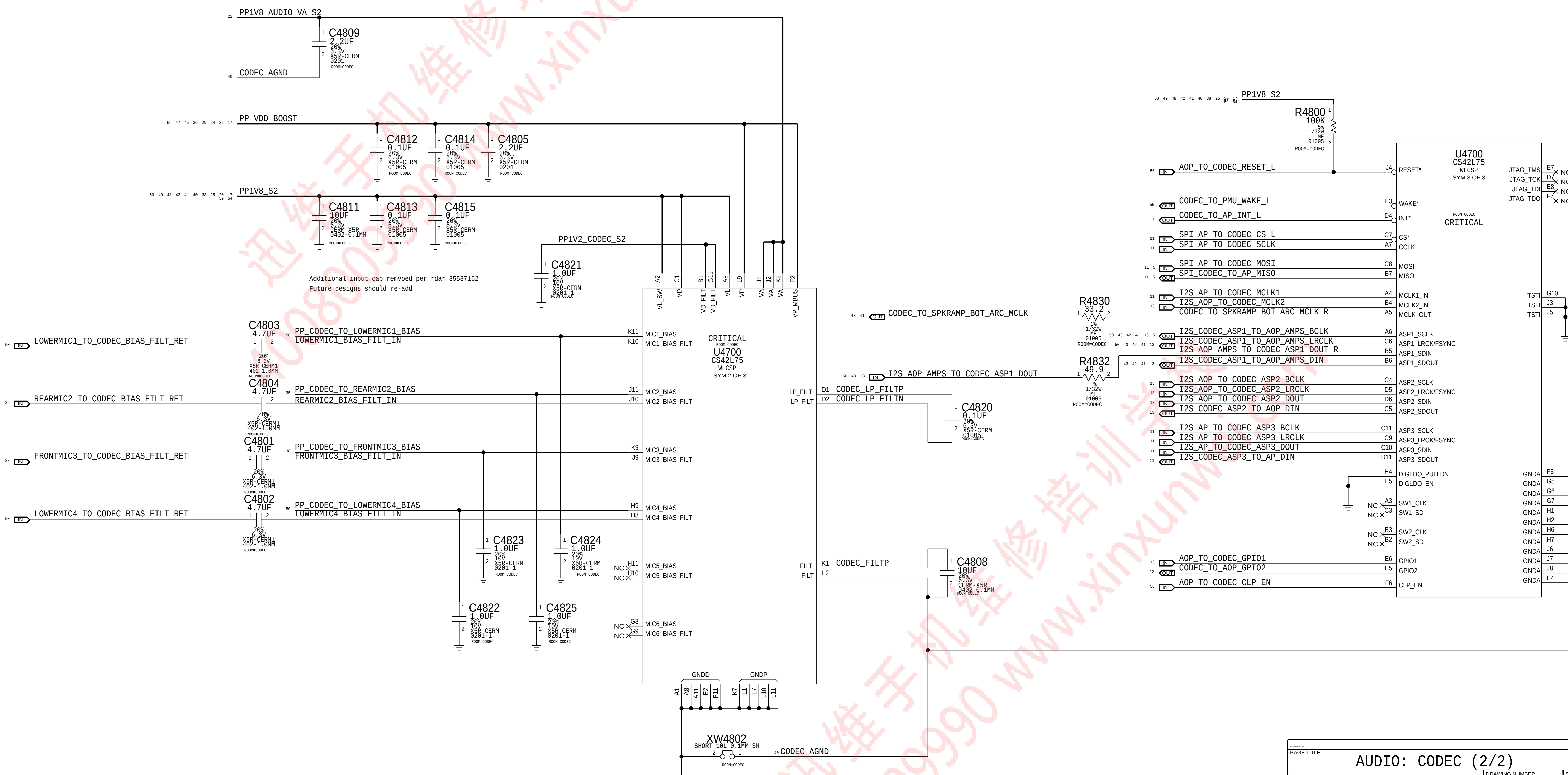
PAGE TITLE		
PEARL: B2B Rosaline + Sensor		
DRAWING NUMBER	051-02545	SIZE
REVISION	7.0.0	D
BRANCH		
PAGE	46	OF 85
SHEET	38	OF 60

CALLAN AUDIO CODEC (ANALOG INPUTS & OUTPUTS)

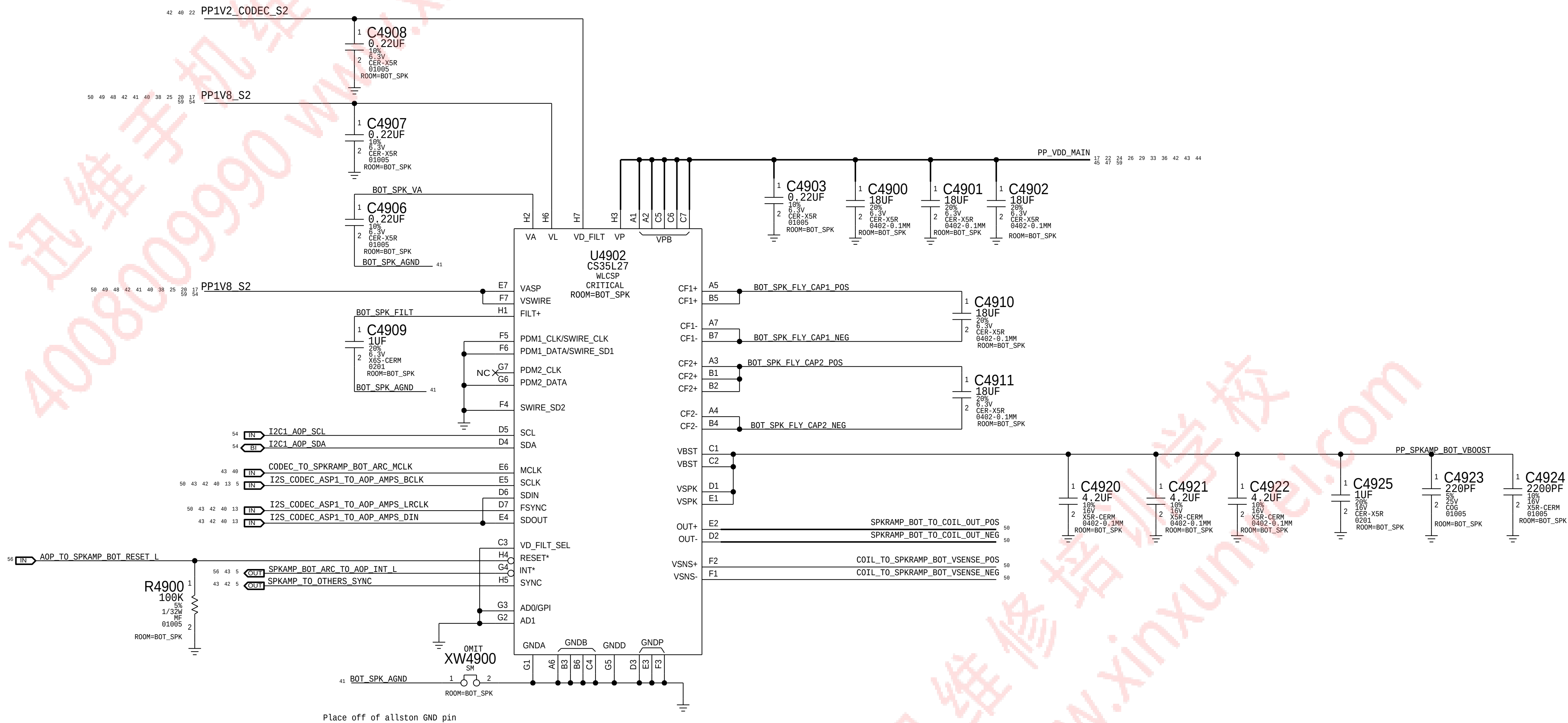
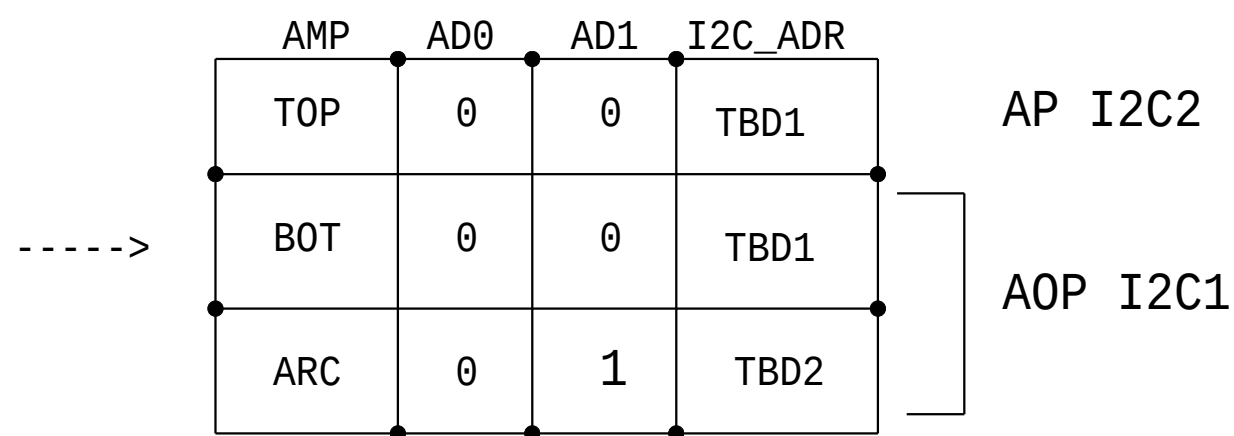


PAGE TITLE		
AUDIO: CODEC (1/2)		
0	DRAWING NUMBER	051-02545
	SIZE	D
	REVISION	7.0.0
	BRANCH	
	PAGE	47 OF 85
	SHEET	39 OF 60

CALLAN AUDIO CODEC (POWER & I/O)



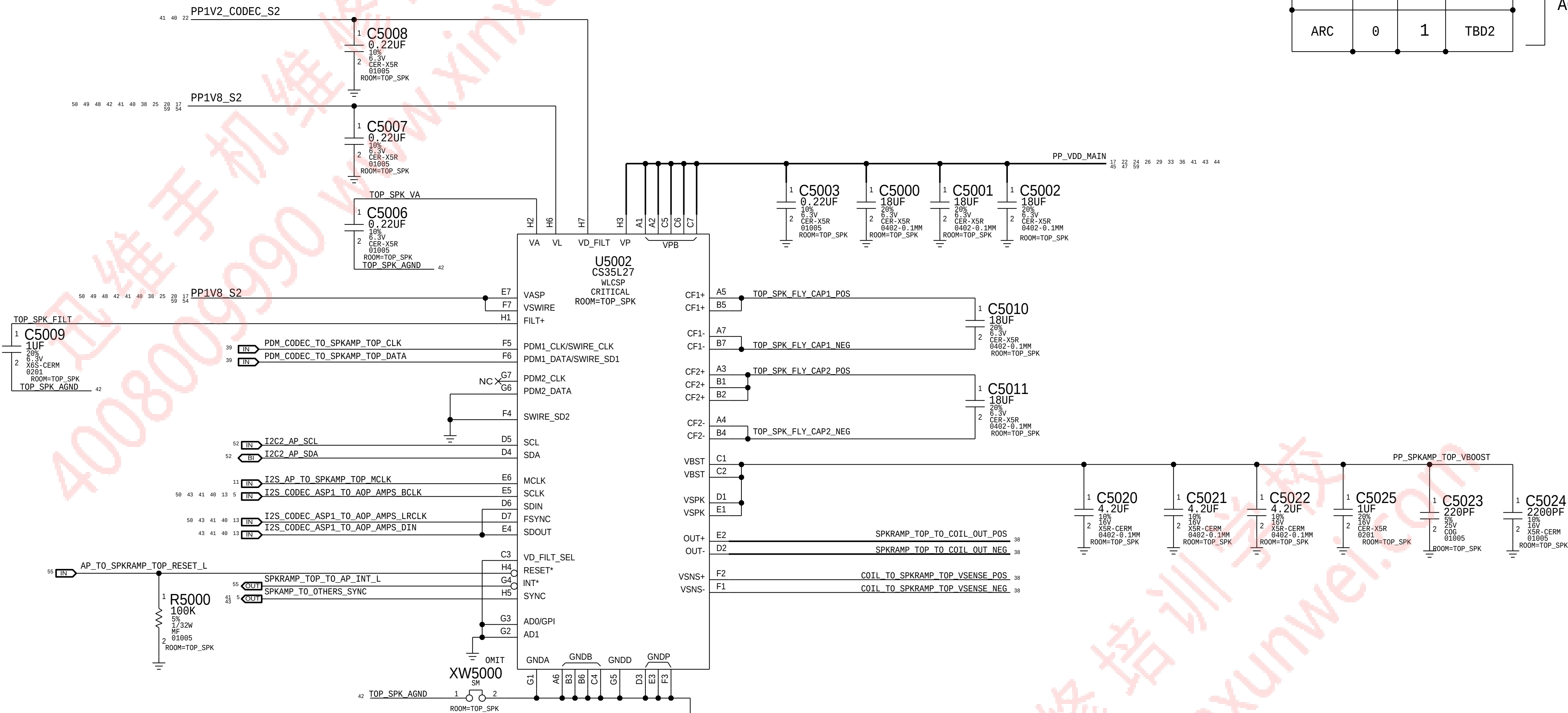
PAGE TITLE		
AUDIO: CODEC (2/2)		
DRAWING NUMBER	051-02545	SIZE
REVISION	7.0.0	D
BRANCH		
PAGE	48 OF 85	
SHEET	40 OF 60	



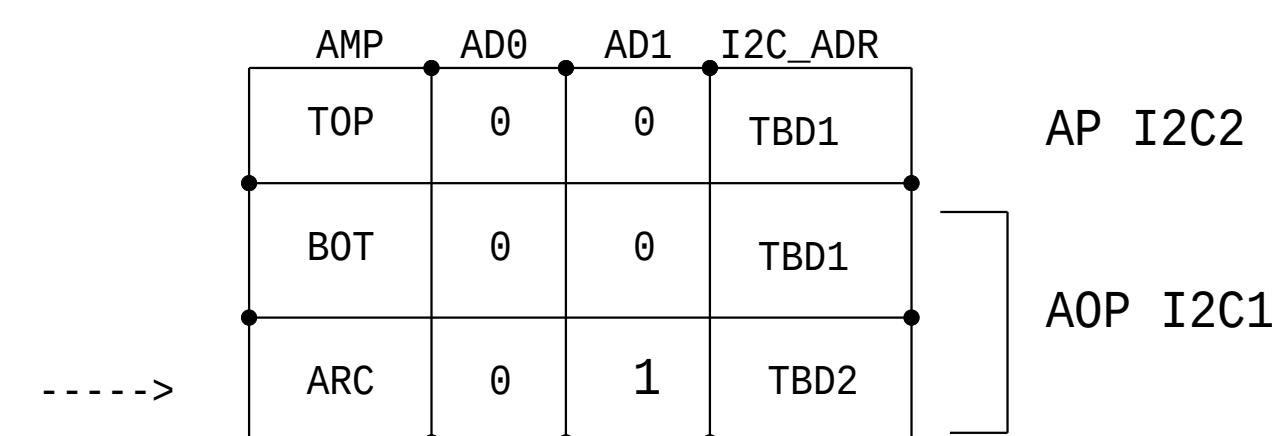
PAGE TITLE			SYNC_DATE=04/05/2017	
AUDIO: SOUTH SPKAMP			DRAWING NUMBER	SIZE
			051-02545	D
			REVISION	7.0.0
			BRANCH	
			PAGE	49 OF 85
			SHEET	41 OF 60



	AMP	AD0	AD1	I2C_ADR	
----->	TOP	0	0	TBD1	AP I2C2
	BOT	0	0	TBD1	AOP I2C1
	ARC	0	1	TBD2	



PAGE TITLE			SYNC_DATE=04/05/2017		
AUDIO: NORTH SPKAMP			DRAWING NUMBER	051-02545	SIZE
			REVISION	7.0.0	D
			BRANCH		
			PAGE	50 OF 85	
			SHEET	42 OF 60	



PAGE TITLE		DRAWING NUMBER		SIZE	
ARC : AMP		051-02545		D	
		REVISION		7.0.0	
		BRANCH			
		PAGE		51 OF 85	
		SHEET		43 OF 60	



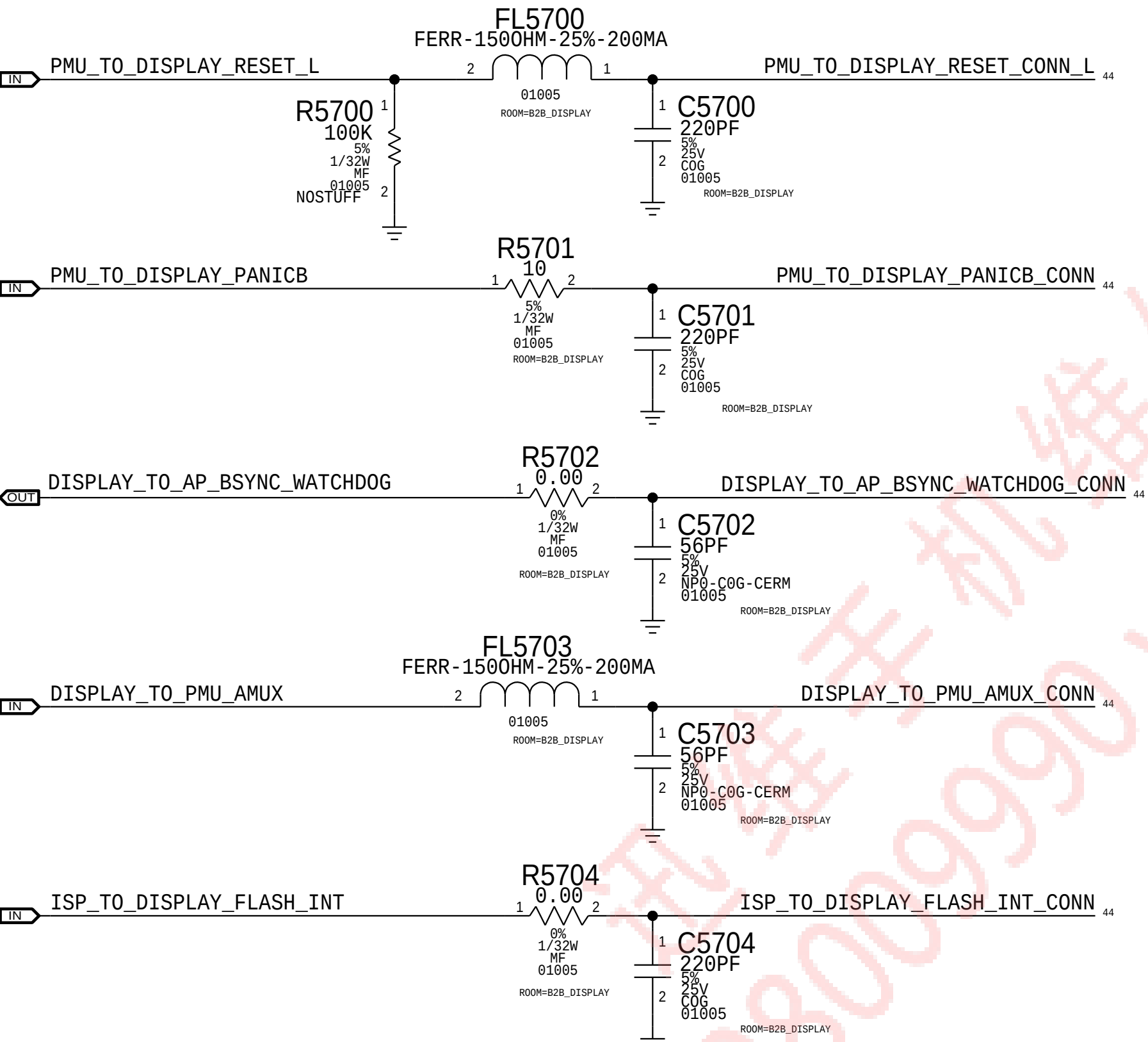
Display Control Signals

Display 1V0 LD0 for D33 second display vendor

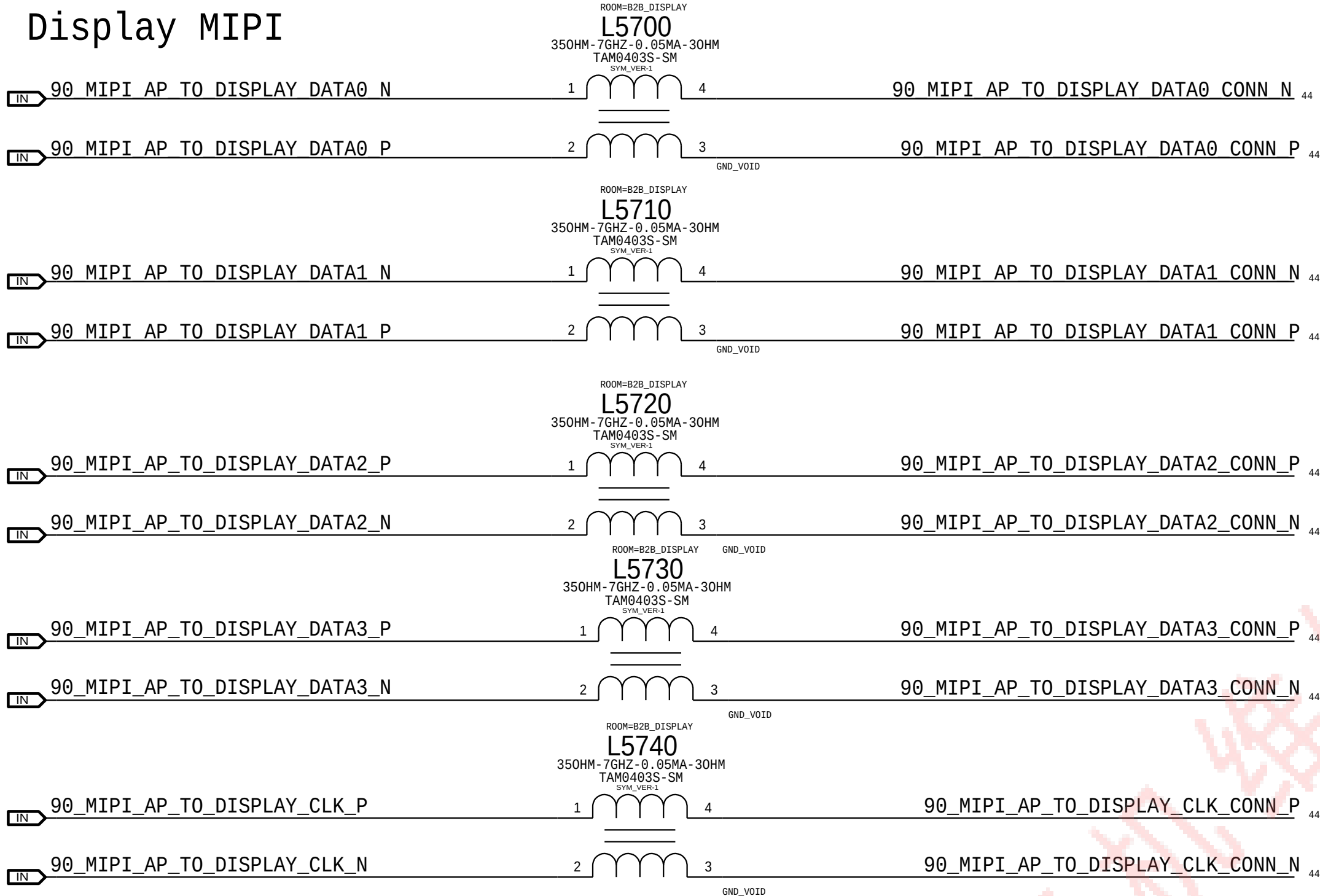
rdar: #29872369

Display Flex Connector

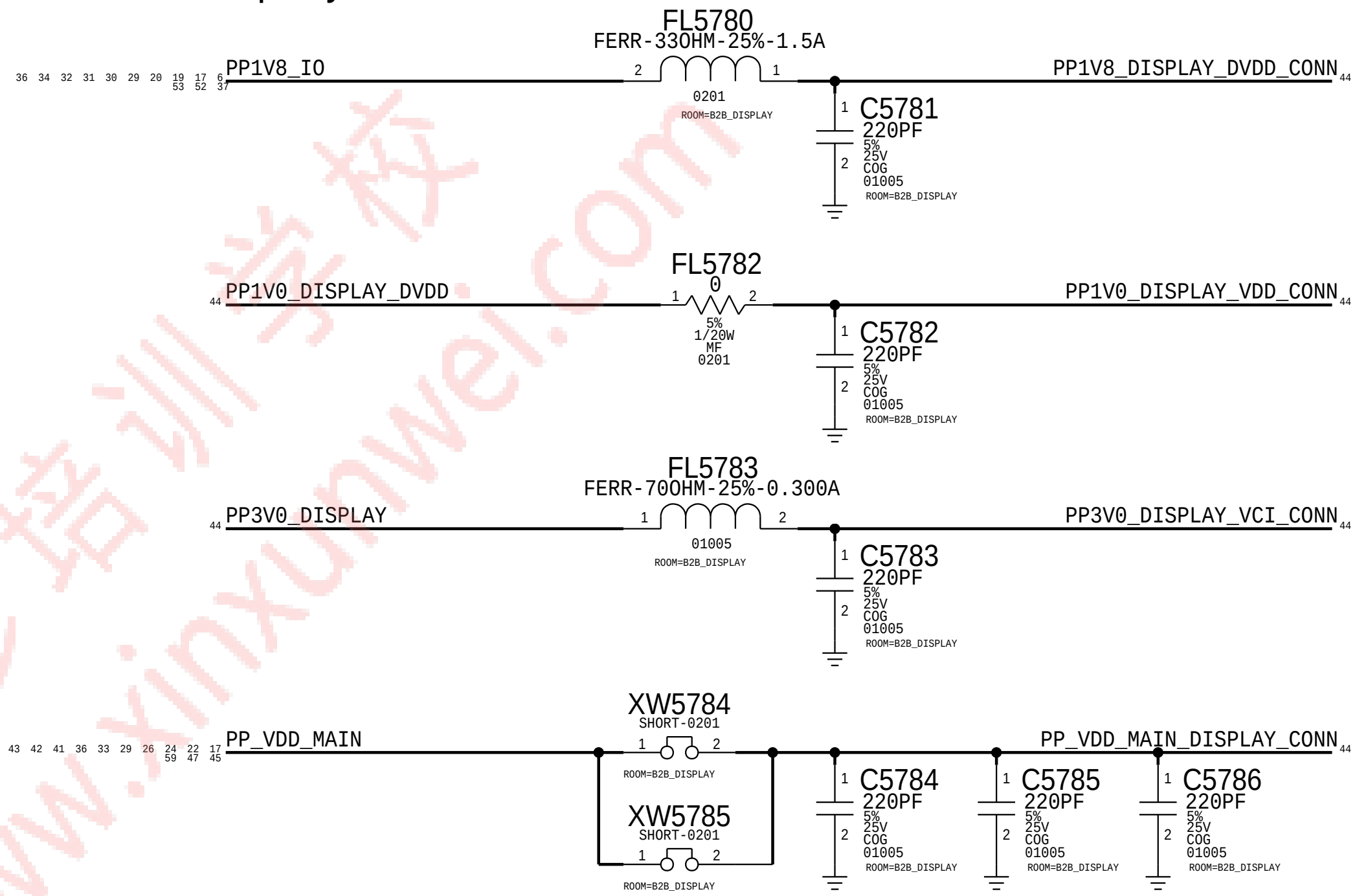
Rcpt: 516S00210 <-- This one on MLB
Plug: 516S00211



Display MIPI

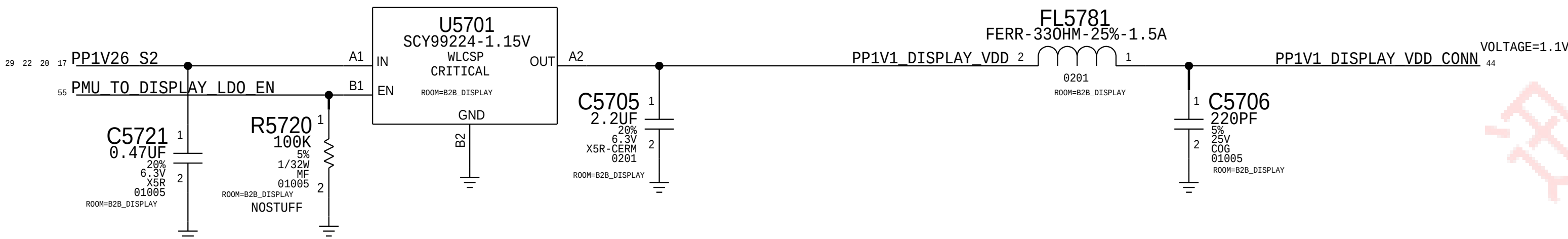


Display Power



1.2V LD0 is for LGC test chip

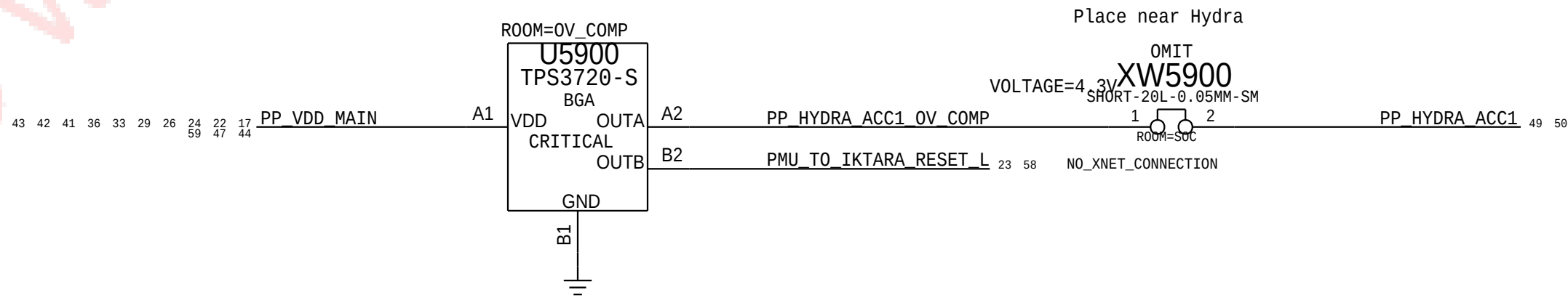
Once normal panel is available switch to 1.1V



PAGE TITLE		
CG: B2B Display		
DRAWING NUMBER	051-02545	SIZE
REVISION	7.0.0	D
BRANCH		
PAGE	57 OF 85	
SHEET	44 OF 60	



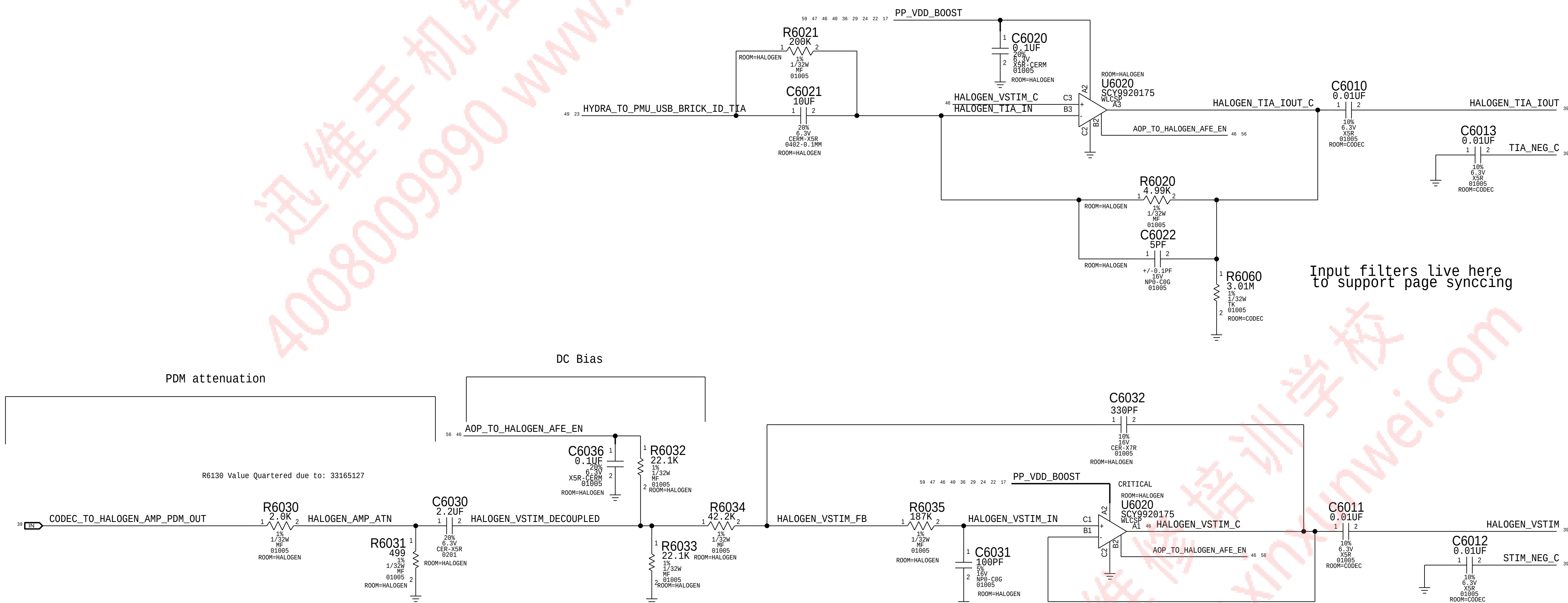
VDD_MAIN 0V CUT-OFF CIRCUIT



PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S01375	353S01300	ALT_PARTS	U5900	ON SEMI

PAGE TITLE		
I/O: Overvoltage Cut-Off Circuit		
	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	59 OF 85
	SHEET	45 OF 60

LDCM



Input filters live here
to support page syncning

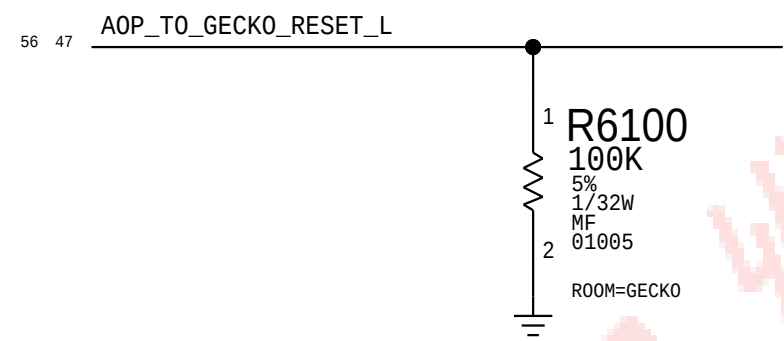
AIN6

Codec

AIN8

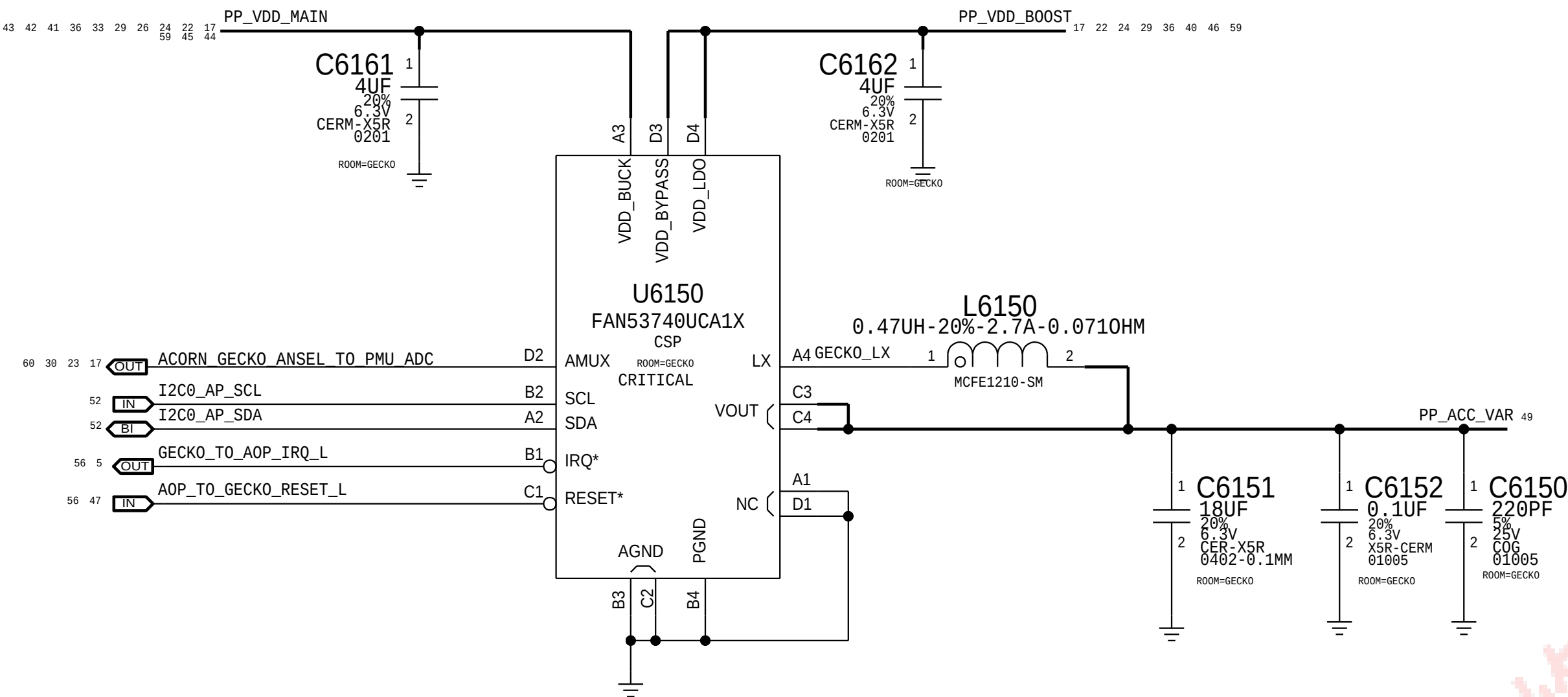
PAGE TITLE		I/O: LDCM	
	DRAWING NUMBER		SIZE
	051-02545		D
	REVISION		
	7.0.0		
	BRANCH		
	PAGE		
	60 OF 85		
	SHEET		
	46 OF 60		

Gecko Reset Pull Down



Gecko

I2C ADDRESS: 0X52

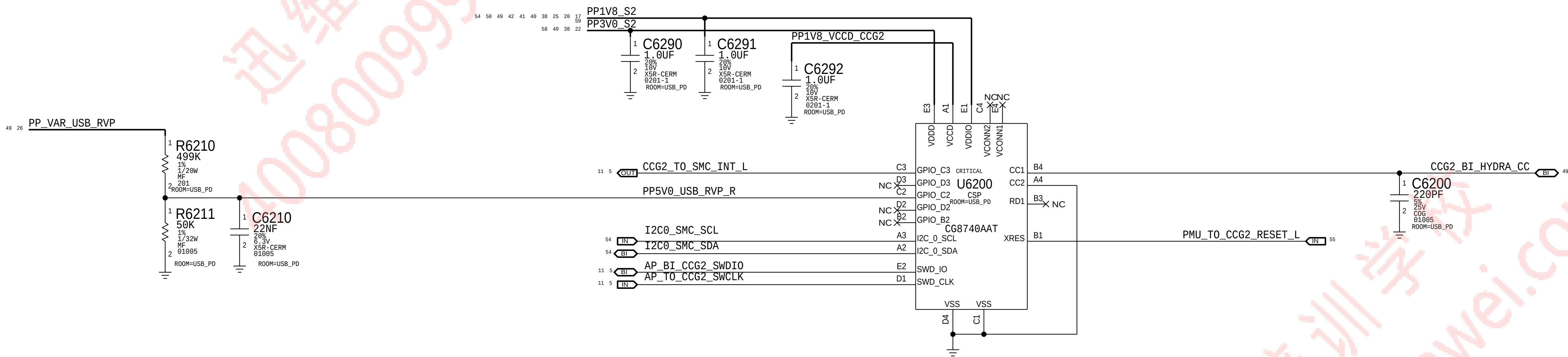


IND Alternate

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00854	152S00853	ALT_PARTS	L6150	IND, PWR, 0.47uH, 20%, 2.7A, CY
152S00855	152S00853	ALT_PARTS	L6150	IND, PWR, 0.47uH, 20%, 2.7A, Murata

PAGE TITLE		
I/O: Gecko		
	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	61 OF 85
	SHEET	47 OF 60

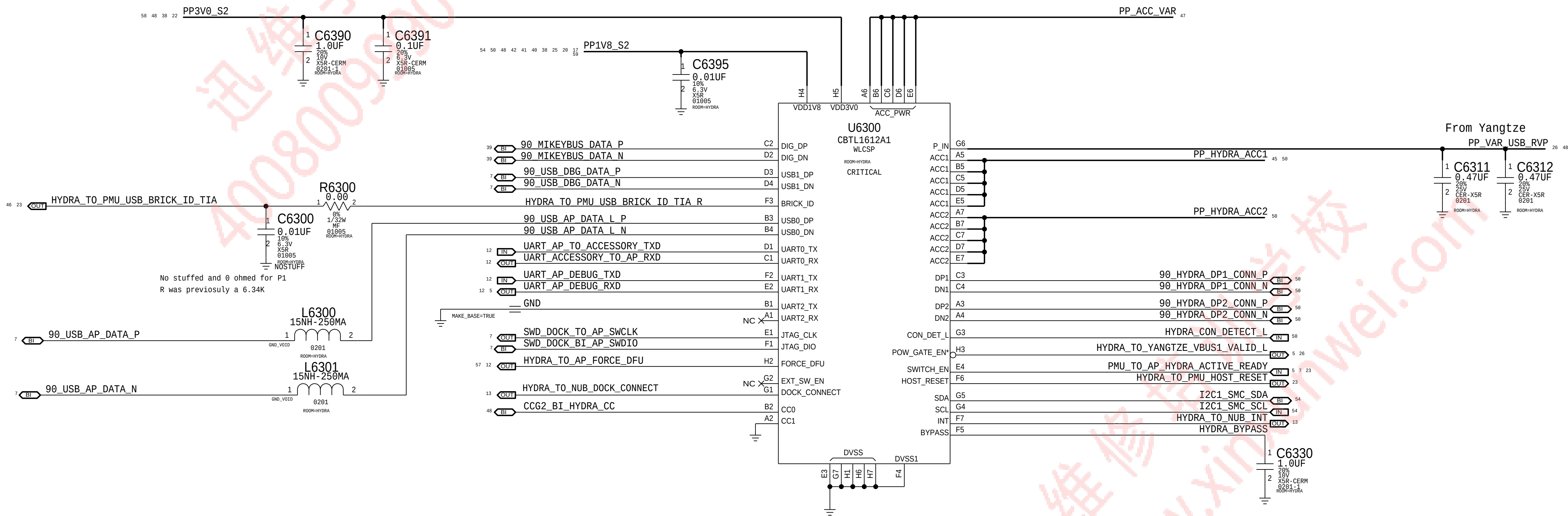
USB-PD



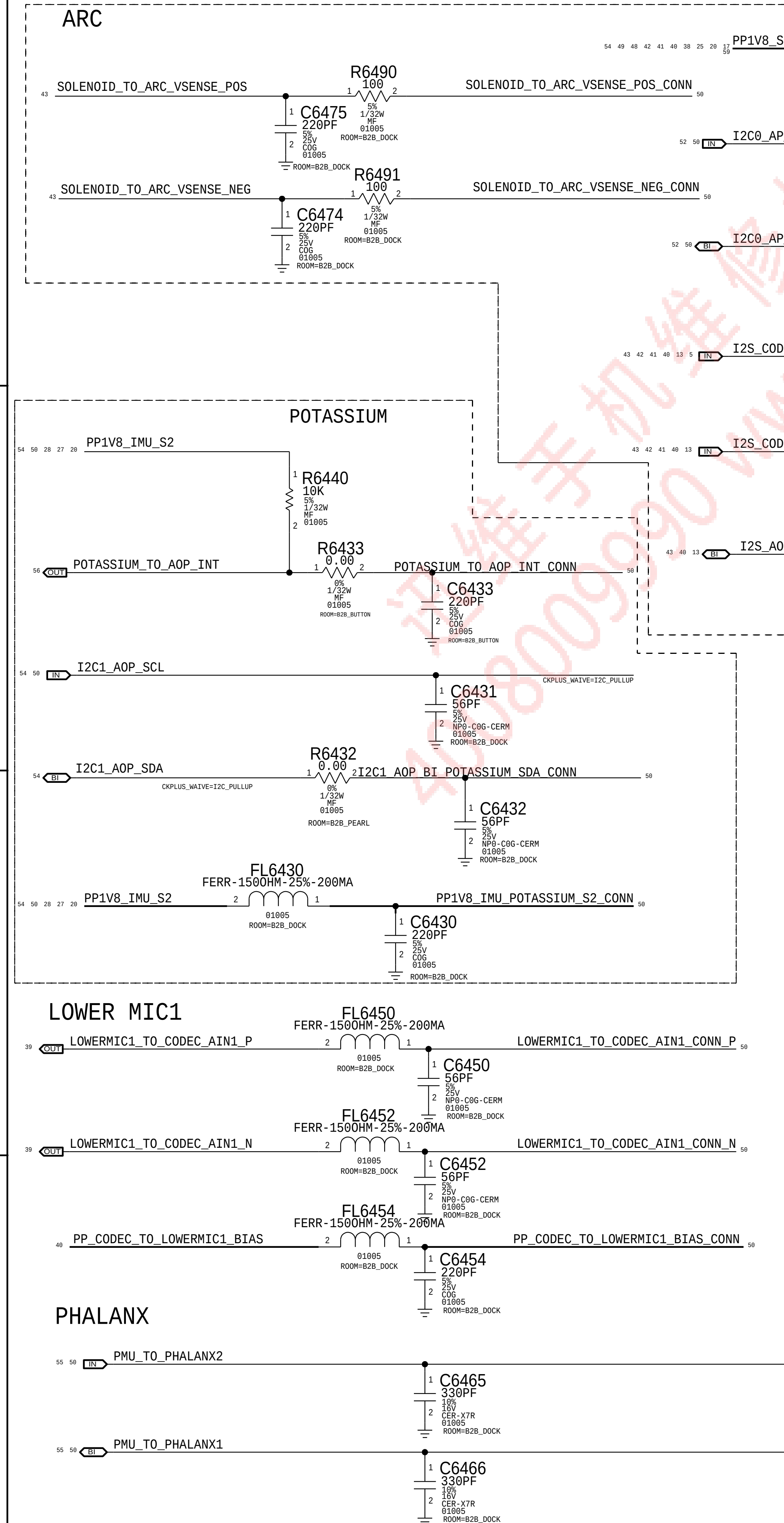
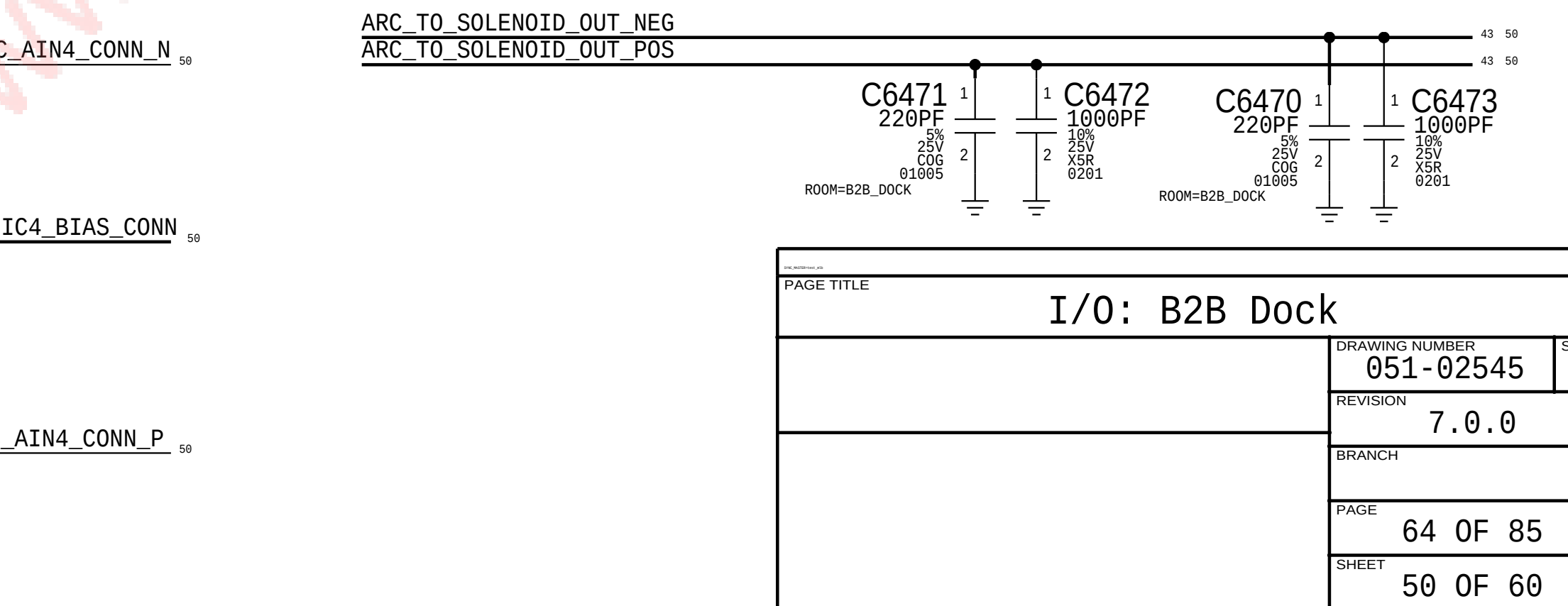
PAGE TITLE		
I/O: USB PD		
	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	62 OF 85
	SHEET	48 OF 60

Hydra

I2C Address: 0011010X



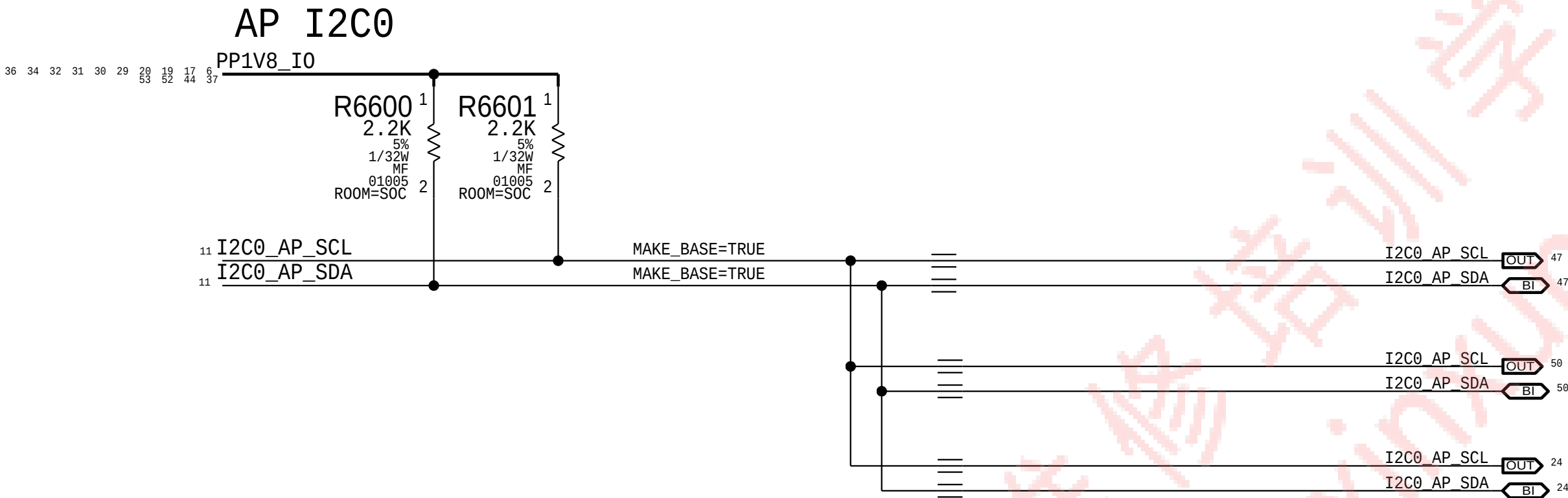
PAGE TITLE		
I/O: Hydra		
DRAWING NUMBER	051-02545	SIZE
REVISION	7.0.0	D
BRANCH		
PAGE	63 OF 85	
SHEET	49 OF 60	



AP I2C

D

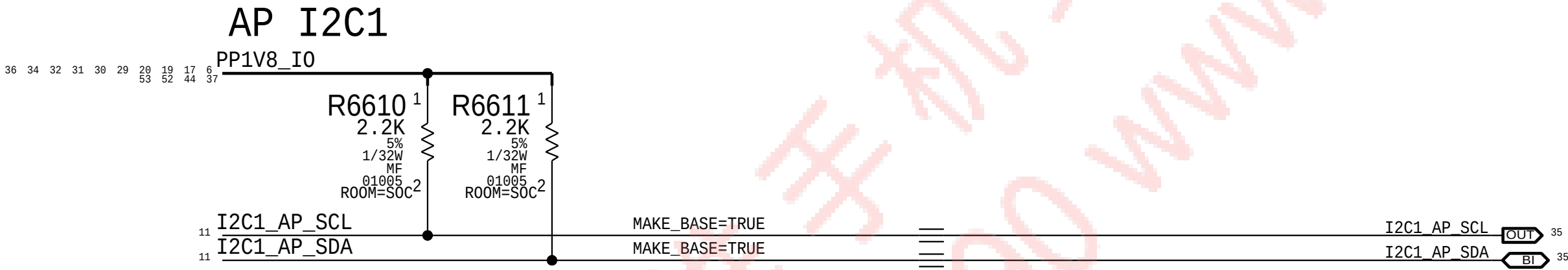
D



Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AP I2C0	PP1V8_I0	400 kHz	GECK0	0x52	1010 010X	0xA4, 0xA5	-	1 MHz	TOP MLB
			SAKONNET	0x08	0001 000X	0x10, 0x11	-	1 MHz	Dock Flex
			BOOST	0x75	1110 101X	0xEA, 0xEB	-	400 KHz	TOP MLB
			ARC EEPROM	0x50	1010 000X	0xA0, 0xA1	-	400 KHz	Dock Flex

C

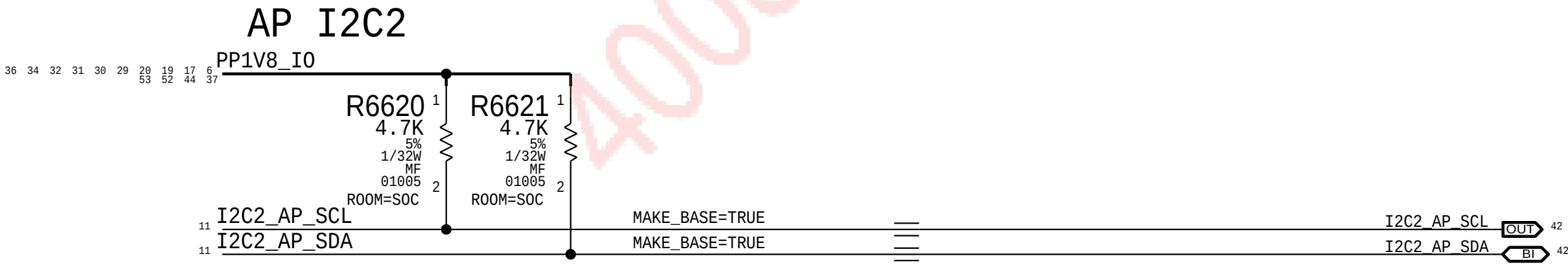
C



Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AP I2C1	PP1V8_I0	100 kHz	MIC2	0x56	1010 100X	0xA8, 0xA9	-	1 MHz	Strobe Flex

B

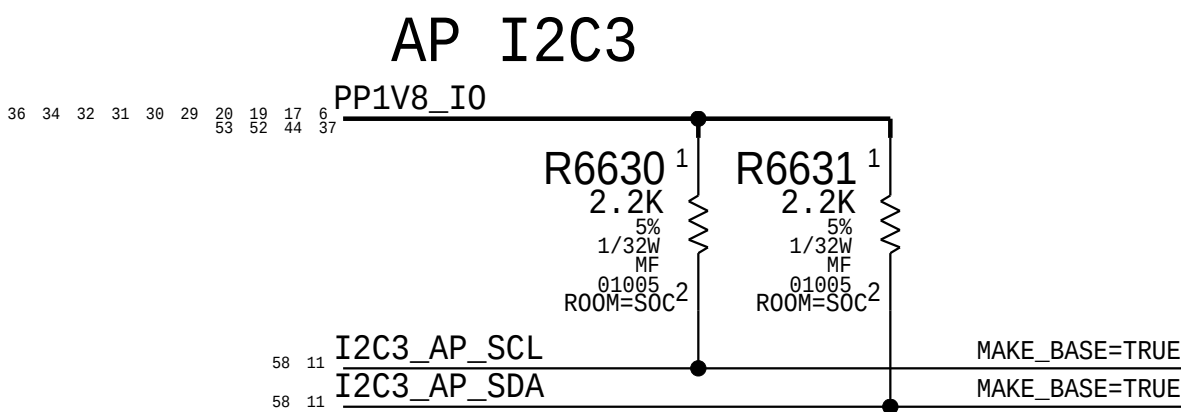
B



Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AP I2C2	PP1V8_I0	1 MHz	Top Speaker Amp	0x40	1000 000X	0x80, 0x81	-	1 MHz	Top MLB

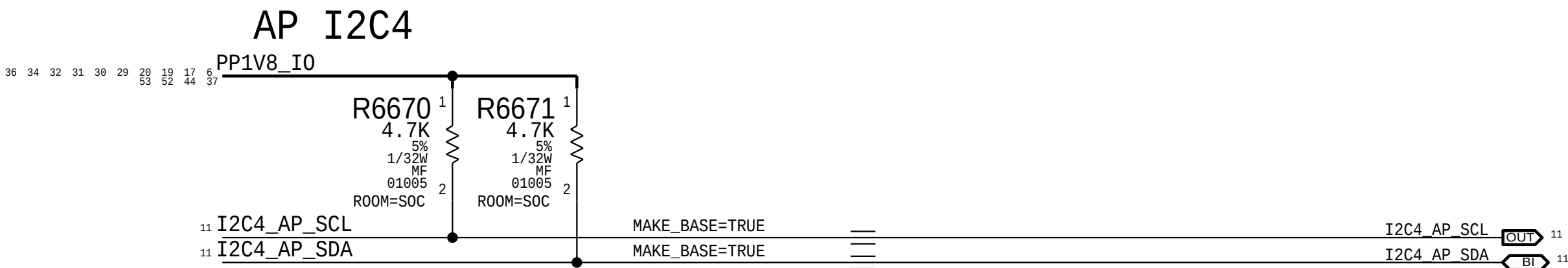
A

A



Acorn and Touch EEPROM Live on Bottom Board

Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AP I2C3	PP1V8_I0	400 kHz	ACORN	0x2A	0101 010X	0x54, 0x55	-	1 MHz	Bot MLB
			TOUCH EEPROM	0x51	1010 001X	0xA2, 0xA3	-	1 MHz	Touch Flex



Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Location
AP I2C4	PP1V8_I0	400 kHz	LYNX	0x71	Top MLB

PAGE TITLE		
SYSTEM: AP I2C		
	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	66 OF 85
	SHEET	52 OF 60

8

7

6

5

4

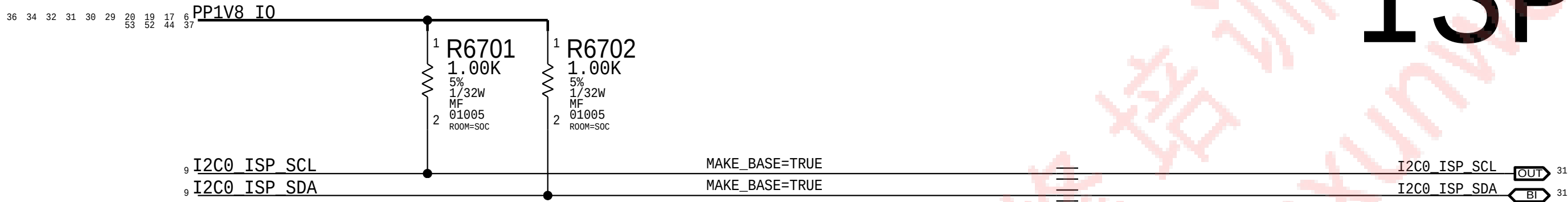
3

2

1

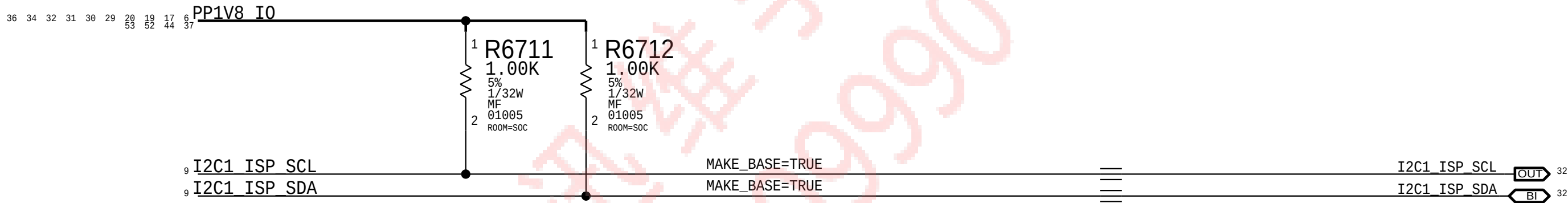
ISP I2C

ISP I2C0



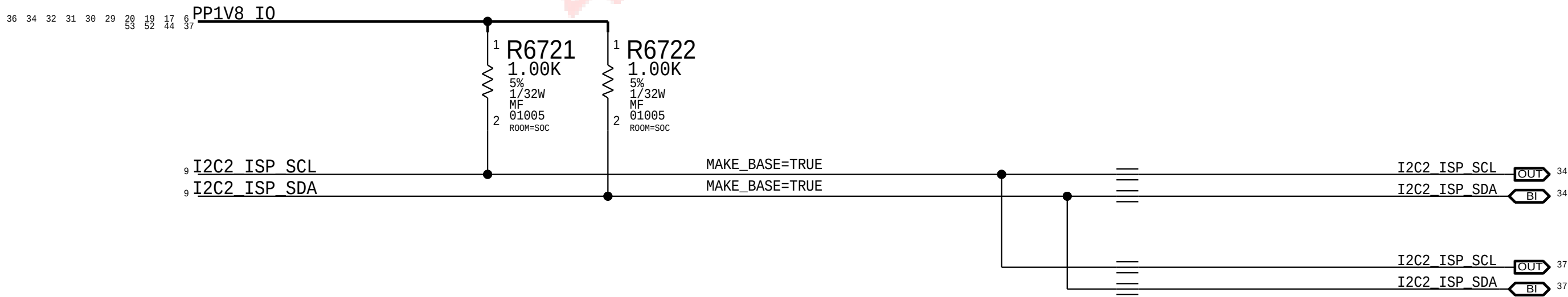
Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
ISP I2C0	PP1V8_I0	1 MHz	Austin	0X10	0010 000X	0x20, 0x21	-	1 MHz	Wide Cam
			Raman	0X3C	0111 100X	0x78, 0x79	-	1 MHz	Wide Cam

ISP I2C1



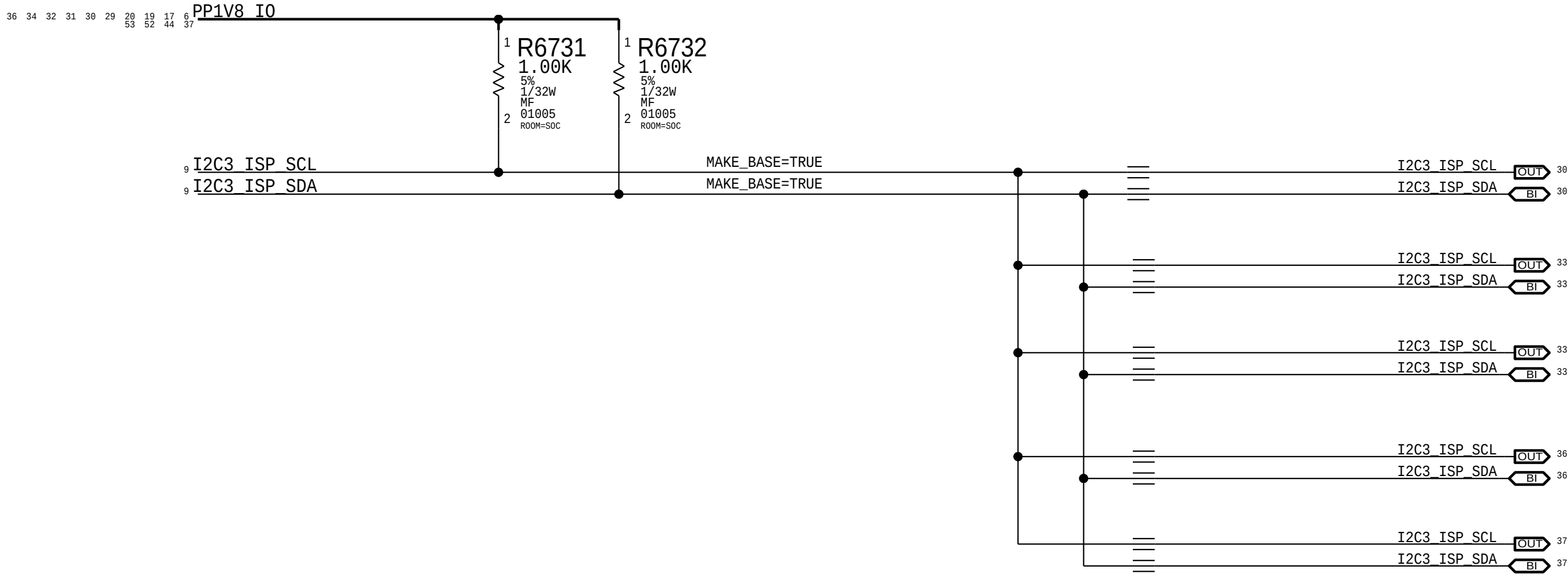
Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
ISP I2C1	PP1V8_I0	1 MHz	Billings	0x20	0100 000X	0x40, 0x41	-	1 MHz	Tele Cam
			Grunberg+	0x1C	0011 100X	0x38, 0x39	-	1 MHz	Tele Cam

ISP I2C2



Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
ISP I2C2	PP1V8_I0	1 MHz	Yonkers	0x10	0010 000X	0x20, 0x21	-	1 MHz	Fcam
			Flatiron	0x70	1110 000X	0xE0, 0xE1	-	1 MHz	Fcam
			Savage	0x18	0011 000X	0x30, 0x31	-	1 MHz	Juliet Flex

ISP I2C3

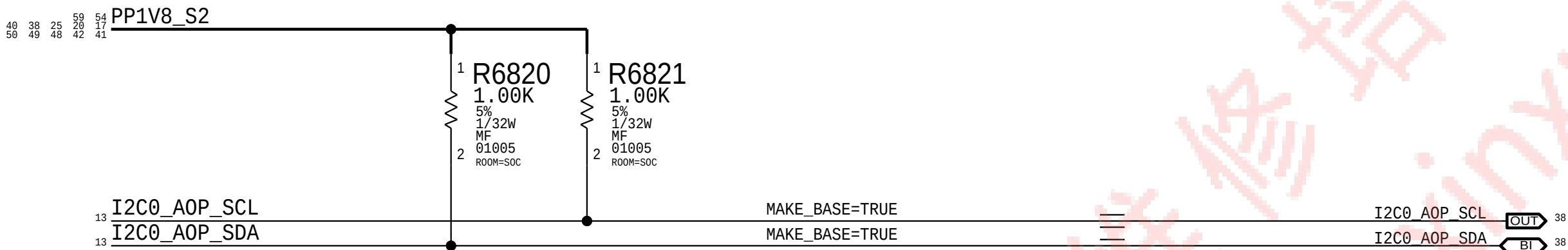


Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
ISP I2C3	PP1V8_I0	1 MHz	Ansel	0x40	1000 000X	0x80, 0x81	-	1 MHz	Top Board
			Neon	0x63	1100 011X	0xC6, 0xC7	-	1 MHz	Top Board
			Neon	0x67	1100 111X	0xCE, 0xCF	-	1 MHz	Top Board
			Rigel	0x55	1100 011X	0xAA, 0xAB	-	1 MHz	Top Board
			Mama Bear	0x50	1010 000X	0xA0, 0xA1	-	1 MHz	Romeo Flex

PAGE TITLE		
SYSTEM: ISP I2C		
0	DRAWING NUMBER	051-02545
	REVISION	7.0.0
	BRANCH	
	PAGE	
	67 OF 85	
	SHEET	
	53 OF 60	

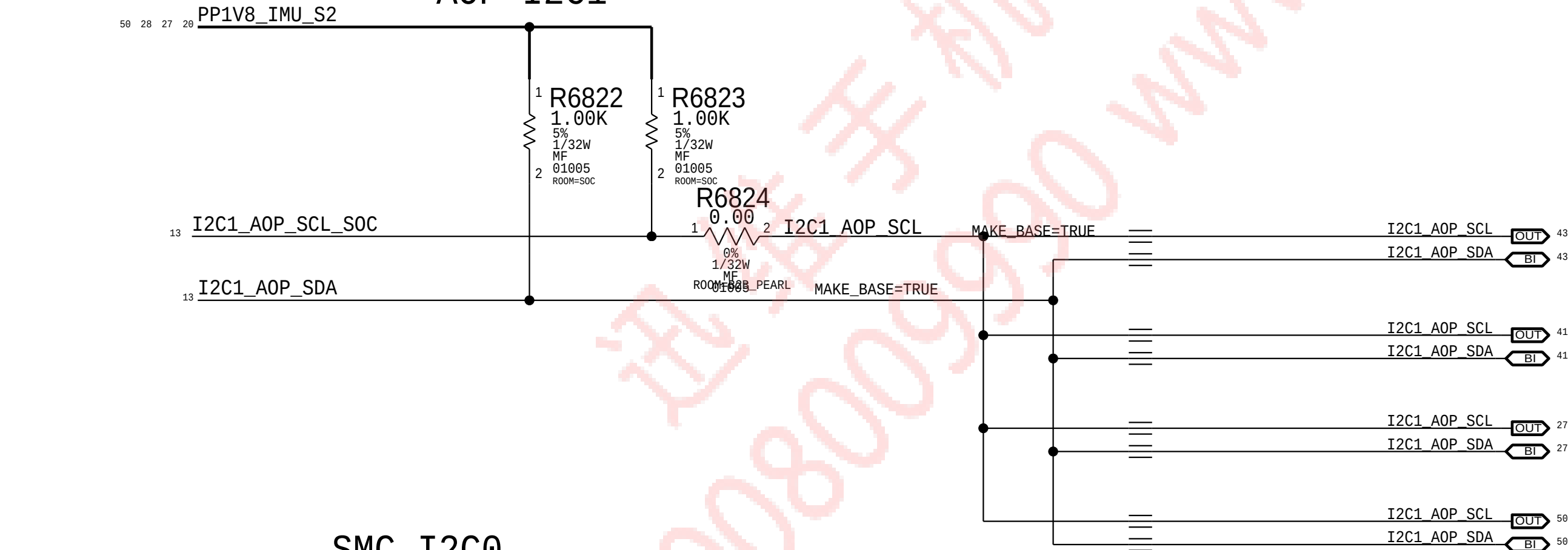
AOP/SMC I2C

AOP I2C0



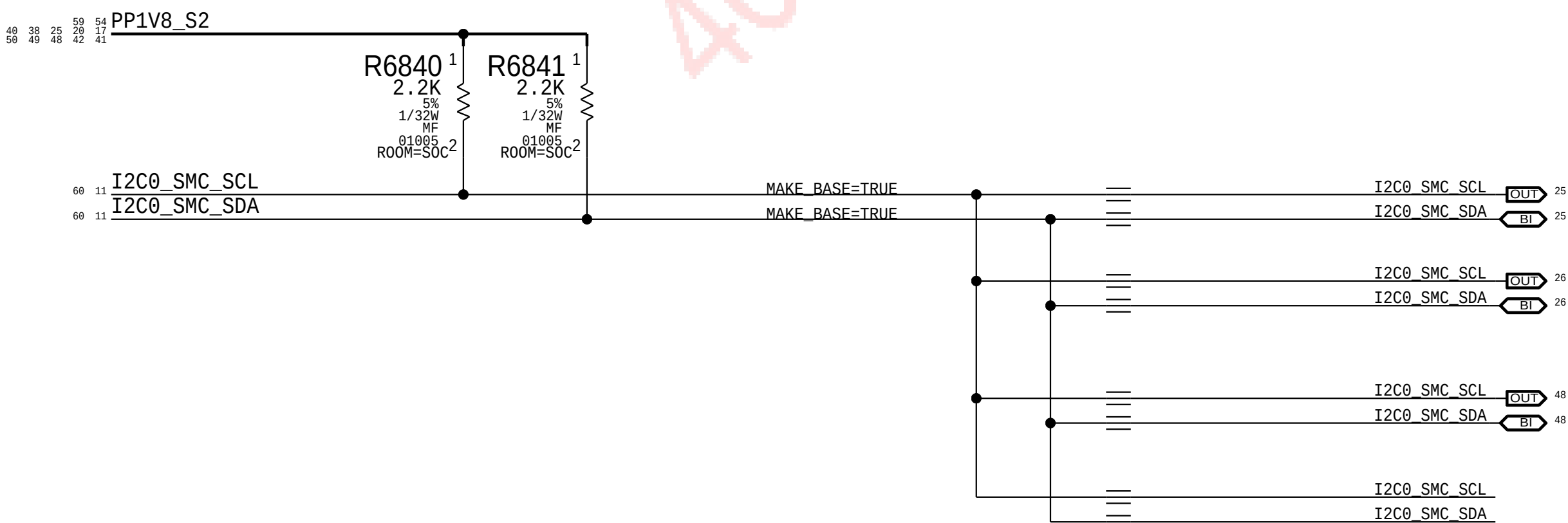
Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AOP I2C0	PP1V8_S2	750 kHz	Doppler	0x58	1011 000X	0xB0, 0xB1	-	1 MHz	Sensor Flex
			Blackbird	0x29	0101 001X	0x52, 0x53	-	1 MHz	Sensor Flex
			Yogi	0x33	0110 011X	0x66, 0x67	-	1 MHz	Sensor Flex

AOP I2C1



Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
AOP I2C1	PP1V8_IMU_S2	400 kHz	Arc	0x42	1000 001X	0x82, 0x83	-	1 MHz	Top Board
			Bottom Speaker	0x40	1000 000X	0x80, 0x81	-	1 MHz	Top Board
			Moly	0x0E	0001 110X	0x1C, 0x1D	-	1 MHz	Button Cyclone
			Potassium	0x76	1110 110X	0xEC, 0xED	-	1 MHz	Dock Flex

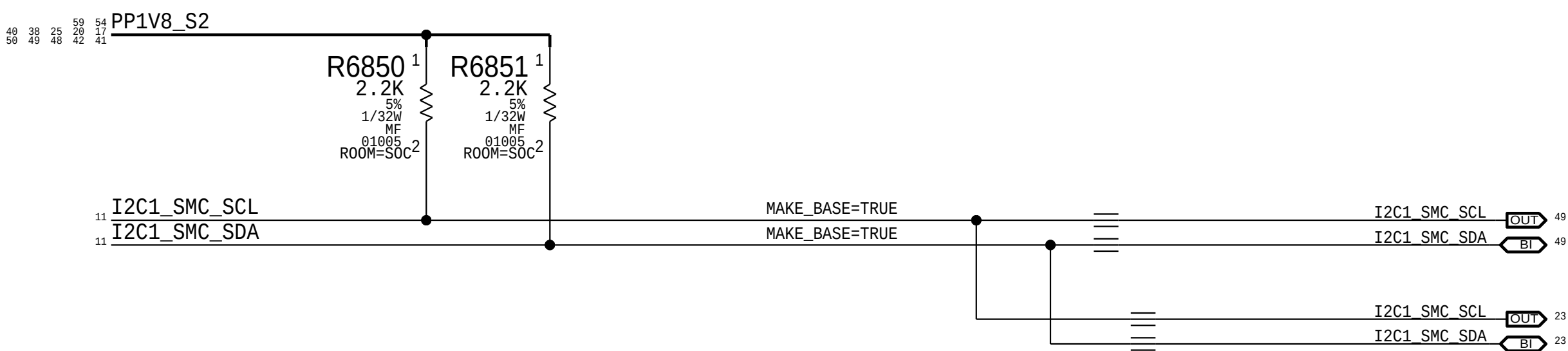
SMC I2C0



Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
SMC I2C0	PP1V8_S2	400 kHz	Yangtze	0x71	1110 001X	0xE2, 0xE3	-	400 KHz	Top Board
			Iktara	0x39	0111 001X	0x72, 0x73	-	400 KHz	Bot Board
			CCG2	0x12	0010 010X	0x24, 0x25	-	1 MHz	Top Board
			Gas Guage	0x55	0010 010X	0xAA, 0xAB	-	1 MHz	BMU Flex
			Roswell	0x10	0100 000X	0x20, 0x21	-	400 KHz	BMU Flex

Lives on bottom board

SMC I2C1



Bus Name	Bus Voltage	Bus Speed	Device	7-Bit Addr.	Binary	8-Bit Addr.	Min Speed	Max Speed	Location
SMC I2C1	PP1V8_S2	400 kHz	Hydra	0x1A	0011 010X	0x34, 0x35	-	400 KHz	Top Board
			Denali	0x74	1110 100X	0xE8, 0xE9	-	400 KHz	Top Board

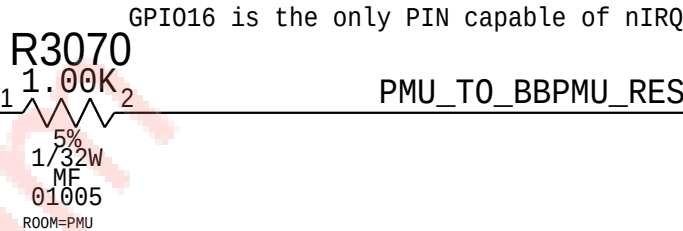
AP/PMU GPIOs

SOC

GPIO_0	12	AP_TO_BT_DEVICE_WAKE	AP_TO_BT_DEVICE_WAKE	58
GPIO_1	12	BOARD_REV0	BOARD_REV0	6
GPIO_2	12	BOARD_REV1	BOARD_REV1	6
GPIO_3	12	BOARD_REV2	BOARD_REV2	6
GPIO_4	12	AP_TO_PMU_AMUX_SYNC	AP_TO_PMU_AMUX_SYNC	23
GPIO_5	12	BOARD_REV3	BOARD_REV3	6
GPIO_6	12	AP_CANARY1	AP_CANARY1	66
GPIO_7	12	AP_BUTTON_VOL_UP_L	PMU_TO_AP_BUTTON_VOL_UP_L	23
GPIO_8	12	NC_AP_GPIO8	NC_AP_GPIO8	MAKE_BASE=TRUE
GPIO_9	12	AP_TO_BBPMU_RADIO_ON_L	AP_TO_BBPMU_RADIO_ON_L	57
GPIO_10	12	AP_TO_SPKRAMP_TOP_RESET_L	AP_TO_SPKRAMP_TOP_RESET_L	42
GPIO_11	12	AP_TO_NFC_FW_DWLD_REQ	AP_TO_NFC_FW_DWLD_REQ	57
GPIO_12	15	AP_TO_BB_PEAK_POWER_INDICATOR	AP_TO_BB_PEAK_POWER_INDICATOR	57
GPIO_13	12	AP_TO_NFC_DEV_WAKE	AP_TO_NFC_DEV_WAKE	57
GPIO_14	12	CAMPMU_TO_AP_IRQ_L	CAMPMU_TO_AP_IRQ_L	38
GPIO_15	12	AP_TO_GNSS_TIME_MARK	AP_TO_GNSS_TIME_MARK	57
GPIO_16	15	SPKRAMP_TOP_TO_AP_INT_L	SPKRAMP_TOP_TO_AP_INT_L	42
GPIO_17	12	BB_TO_AP_COEX	BB_TO_AP_COEX	57
GPIO_18	12	BT_TO_AP_TIME_SYNC	BT_TO_AP_TIME_SYNC	58
GPIO_19	12	AP_TO_BB_RESET_L	AP_TO_BB_RESET_L	57
GPIO_20	12	BB_TO_AP_PEAK_POWER_INDICATOR	BB_TO_AP_PEAK_POWER_INDICATOR	57
GPIO_21	12	BB_TO_AP_RESET_DETECT_L	BB_TO_AP_RESET_DETECT_L	57
GPIO_22	12	AP_TO_BB_COREDUMP_TRIG	AP_TO_BB_COREDUMP_TRIG	57
GPIO_23	12	AP_TO_CAMPMU_RESET_L	AP_TO_CAMPMU_RESET_L	38
GPIO_24	12	AP_TO_BB_COEX	AP_TO_BB_COEX	57
GPIO_25	12	DISPLAY_TO_AP_PANEL_ID	DISPLAY_TO_AP_PANEL_ID	44
GPIO_26	12	AP_CANARY2	AP_CANARY2	58
GPIO_27	12	NC_AP_GPIO27	NC_AP_GPIO27	MAKE_BASE=TRUE
GPIO_28	12	NC_AP_GPIO28	NC_AP_GPIO28	MAKE_BASE=TRUE
GPIO_29	12	AP_TO_RACER_RESET_L	AP_TO_RACER_RESET_L	57
GPIO_30	15	GNSS_TO_AP_LOW_PWR_IND	GNSS_TO_AP_LOW_PWR_IND	57

PMU

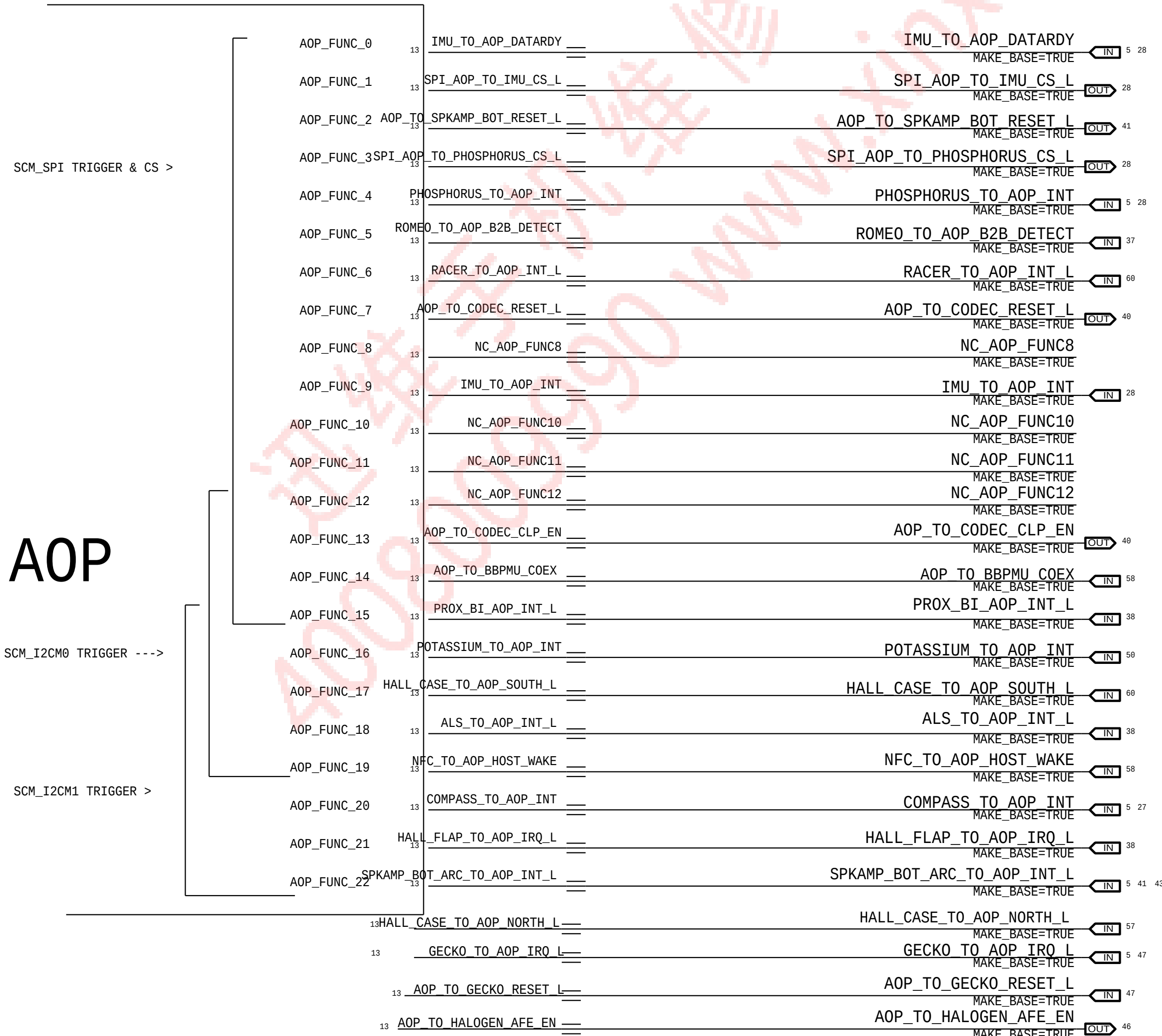
GPIO_1	23	PMU_TO_CCG2_RESET_L	PMU_TO_CCG2_RESET_L	48
GPIO_2	23	PMU_TO_AP_THROTTLE_GPU1_L	PMU_TO_AP_THROTTLE_GPU1_L	7
GPIO_3	23	NC_BT_TO_PMU_HOST_WAKE	NC_BT_TO_PMU_HOST_WAKE	MAKE_BASE=TRUE
GPIO_4	23	WLAN_TO_PMU_HOST_WAKE	WLAN_TO_PMU_HOST_WAKE	58
GPIO_5	23	BB_TO_PMU_PCIE_HOST_WAKE_L	BB_TO_PMU_PCIE_HOST_WAKE_L	58
GPIO_6	23	PMU_NFC_TO_ARC_RESET_L	PMU_NFC_TO_ARC_RESET_L	43
GPIO_7	23	PMU_TO_GNSS_EN	PMU_TO_GNSS_EN	58
GPIO_8	23	PMU_TO_WLAN_CLK32K	PMU_TO_WLAN_CLK32K	23
GPIO_9	23	PMU_TO_BT_REG_ON	PMU_TO_BT_REG_ON	58
GPIO_10	23	PMU_TO_PHALANX2	PMU_TO_PHALANX2	50
GPIO_11	23	YANGTZE_TO_PMU_INT_L	YANGTZE_TO_PMU_INT_L	26
GPIO_12	23	CODEC_TO_PMU_WAKE_L	CODEC_TO_PMU_WAKE_L	46
GPIO_13	23	PMU_MASK_NFC_TO_ARC_TRIG	PMU_MASK_NFC_TO_ARC_TRIG	43
GPIO_14	23	PMU_TO_WLAN_REG_ON	PMU_TO_WLAN_REG_ON	58
GPIO_15	23	PMU_TO_NFC_VDD_MAIN_EN	PMU_TO_NFC_VDD_MAIN_EN	57
GPIO_16	23	PMU_TO_NAND_LOW_BATT_BOOT_L	PMU_TO_NAND_LOW_BATT_BOOT_L	19
GPIO_17	23	PMU_TO_PHALANX1	PMU_TO_PHALANX1	50
GPIO_18	23	PMU_TO_DISPLAY_RESET_L	PMU_TO_DISPLAY_RESET_L	44
GPIO_19	23	PMU_TO_BBPMU_RESET_R_L	PMU_TO_BBPMU_RESET_R_L	58
GPIO_20	23	PMU_TO_NFC_EN	PMU_TO_NFC_EN	58
GPIO_21	23	NC_PMU_GPIO21	NC_PMU_GPIO21	MAKE_BASE=TRUE
GPIO_22	23	PMU_TO_IKTARA_EN_EXT_1V8	PMU_TO_IKTARA_EN_EXT_1V8	66
GPIO_23	23	PMU_TO_BOOST_EN	PMU_TO_BOOST_EN	24
GPIO_24	23	PMU_TO_DISPLAY_PANICB	PMU_TO_DISPLAY_PANICB	44
GPIO_25	23	PMU_TO_DISPLAY_LDO_EN	PMU_TO_DISPLAY_LDO_EN	44



PAGE TITLE		SYSTEM: SOC/PMU GPIOs	
	DRAWING NUMBER	051-02545	SIZE
	REVISION	7.0.0	
	BRANCH		
	PAGE	70	OF 85
	SHEET	55	OF 60



AOP GPIOs



PAGE TITLE			SYSTEM: AOP GPIOs		
		DRAWING NUMBER		SIZE	
		051-02545		D	
		REVISION		7.0.0	
		BRANCH			
		PAGE		71 OF 85	
		SHEET		56 OF 60	

4

THIS SIDE HAS ATTRIBUTE
MAKE **BASE=TRUE**

51	PCIE AP TO WLAN PERST_L	WAKE_BASO/TRUE		PCIE AP TO WLAN PERST_L	8
51	GND	WAKE_BASO/TRUE	GND		57 59
51	AP TO RACER RESET_L	WAKE_BASO/TRUE		AP TO RACER RESET_L	55
51	GND	WAKE_BASO/TRUE	GND		57 59
51	AP TO WLAN TIME SYNC	WAKE_BASO/TRUE		AP TO WLAN TIME SYNC	12
51	GND	WAKE_BASO/TRUE	GND		57 59
51	GNSS TO AP LOW PWR_IND	WAKE_BASO/TRUE		GNSS TO AP LOW PWR_IND	55
51	GND	WAKE_BASO/TRUE	GND		57 59
51	HYDRA TO AP_FORCE_DFU	WAKE_BASO/TRUE		HYDRA TO AP_FORCE_DFU	12 40
51	GND	WAKE_BASO/TRUE	GND		57 59
51	PP1V8_S2	WAKE_BASO/TRUE		PP1V8_S2	59
51	PP1V8_S2	WAKE_BASO/TRUE		PP1V8_S2	59
51	GND	WAKE_BASO/TRUE	GND		57 59
51	GND	WAKE_BASO/TRUE	GND		59
51	GND	WAKE_BASO/TRUE	GND		59
51	GND	WAKE_BASO/TRUE	GND		59
51	GND	WAKE_BASO/TRUE	GND		57 58 59
51	INTERPOSER_PIN_90	WAKE_BASO/TRUE	INTERPOSER_PIN_90		
51	GND	WAKE_BASO/TRUE	GND		59
51	AP TO BB COEX	WAKE_BASO/TRUE		AP TO BB COEX	55
51	BB TO AP COEX	WAKE_BASO/TRUE		BB TO AP COEX	55
51	GND	WAKE_BASO/TRUE	GND		59
51	AP TO NFC_DEV_WAKE	WAKE_BASO/TRUE		AP TO NFC_DEV_WAKE	55
51	AP TO NFC_FW_DWLD_REQ	WAKE_BASO/TRUE		AP TO NFC_FW_DWLD_REQ	55
51	GND	WAKE_BASO/TRUE	GND		59
51	PMU TO NFC_VDD_MAIN_EN	WAKE_BASO/TRUE		PMU TO NFC_VDD_MAIN_EN	55
51	UART_AOP_TO_BB_TXD	WAKE_BASO/TRUE		UART_AOP_TO_BB_TXD	13
51	UART_AP_TO_GNSS_TXD	WAKE_BASO/TRUE		UART_AP_TO_GNSS_TXD	12
51	GND	WAKE_BASO/TRUE	GND		59
51	AP TO BB_COREDUMP_TRIG	WAKE_BASO/TRUE		AP TO BB_COREDUMP_TRIG	55
51	UART_AP_TO_NFC_TXD	WAKE_BASO/TRUE		UART_AP_TO_NFC_TXD	12
51	UART_NFC_TO_AP_RXD	WAKE_BASO/TRUE		UART_NFC_TO_AP_RXD	12
51	BB TO AP_RESET_DETECT_L	WAKE_BASO/TRUE		BB TO AP_RESET_DETECT_L	55
51	GND	WAKE_BASO/TRUE	GND		59
51	BOARD_ID2	WAKE_BASO/TRUE		BOARD_ID2	6 12
51	AP TO_GNSS_TIME_MARK	WAKE_BASO/TRUE		AP TO_GNSS_TIME_MARK	55
51	NC_INTERPOSER_109	WAKE_BASO/TRUE		NC_INTERPOSER_109	55
51	AP_BB_PEAKE_POWER_INDICATOR	WAKE_BASO/TRUE		AP_BB_PEAKE_POWER_INDICATOR	55
51	AP TO_BBPMU_RADIO_ON_L	WAKE_BASO/TRUE		AP TO_BBPMU_RADIO_ON_L	55
51	PP_VDD_MAIN	WAKE_BASO/TRUE		PP_VDD_MAIN	59
51	PP_VDD_MAIN	WAKE_BASO/TRUE		PP_VDD_MAIN	59
51	PP_VDD_MAIN	WAKE_BASO/TRUE		PP_VDD_MAIN	59
51	GND	WAKE_BASO/TRUE	GND		57 58 59
51	90_PCIE_BB_TO_AP_RXD_N	WAKE_BASO/TRUE		90_PCIE_BB_TO_AP_RXD_N	17
51	90_PCIE_BB_TO_AP_RXD_P	WAKE_BASO/TRUE		90_PCIE_BB_TO_AP_RXD_P	17
51	GND	WAKE_BASO/TRUE	GND		57 59
51	90_PCIE_AP_TO_BB_TXD_N	WAKE_BASO/TRUE		90_PCIE_AP_TO_BB_TXD_N	17
51	90_PCIE_AP_TO_BB_TXD_P	WAKE_BASO/TRUE		90_PCIE_AP_TO_BB_TXD_P	17
51	GND	WAKE_BASO/TRUE	GND		57 59
51	90_PCIE_AP_TO_BB_REFCLK_P	WAKE_BASO/TRUE		90_PCIE_AP_TO_BB_REFCLK_P	8
51	90_PCIE_AP_TO_BB_REFCLK_N	WAKE_BASO/TRUE		90_PCIE_AP_TO_BB_REFCLK_N	8
51	GND	WAKE_BASO/TRUE	GND		57 59
51	UART_BB_TO_AOP_RXD	WAKE_BASO/TRUE		UART_BB_TO_AOP_RXD	13
51	UART_GNSS_TO_AP_RXD	WAKE_BASO/TRUE		UART_GNSS_TO_AP_RXD	12
51	PCIE_AP_TO_BB_PERST_L	WAKE_BASO/TRUE		PCIE_AP_TO_BB_PERST_L	8
51	UART_NFC_TO_AP_CTS_L	WAKE_BASO/TRUE		UART_NFC_TO_AP_CTS_L	12
51	GND	WAKE_BASO/TRUE	GND		57 59
51	UART_AP_TO_NFC_RTS_L	WAKE_BASO/TRUE		UART_AP_TO_NFC_RTS_L	12
51	PMU_AMUX_BY	WAKE_BASO/TRUE		PMU_AMUX_BY	23
51	PMU_AMUX_AY	WAKE_BASO/TRUE		PMU_AMUX_AY	23
51	GND	WAKE_BASO/TRUE	GND		57 59
51	PCIE_BB_BI_AP_CLKREQ_L	WAKE_BASO/TRUE		PCIE_BB_BI_AP_CLKREQ_L	8
51	NC_INT_135	WAKE_BASO/TRUE		NC_INT_135	55
51	BB_AP_PEAKE_POWER_INDICATOR	WAKE_BASO/TRUE		BB_AP_PEAKE_POWER_INDICATOR	55
51	GND	WAKE_BASO/TRUE	GND		57 59
51	PP_VDD_MAIN	WAKE_BASO/TRUE		PP_VDD_MAIN	59
51	PP_VDD_MAIN	WAKE_BASO/TRUE		PP_VDD_MAIN	59
51	PP_VDD_MAIN	WAKE_BASO/TRUE		PP_VDD_MAIN	59
51	GND	WAKE_BASO/TRUE	GND		57 58 59
51	GND	WAKE_BASO/TRUE	GND		57 58 59
51	GND	WAKE_BASO/TRUE	GND		57 58 59
51	GND	WAKE_BASO/TRUE	GND		57 58 59

PAGE TITLE		DRAWING NUMBER		SIZE	
Interposer: Pins 1-144		051-02545		D	
		REVISION			
		7.0.0			
		BRANCH			
		PAGE		81 OF 85	
		SHEET		57 OF 60	

[illegible]

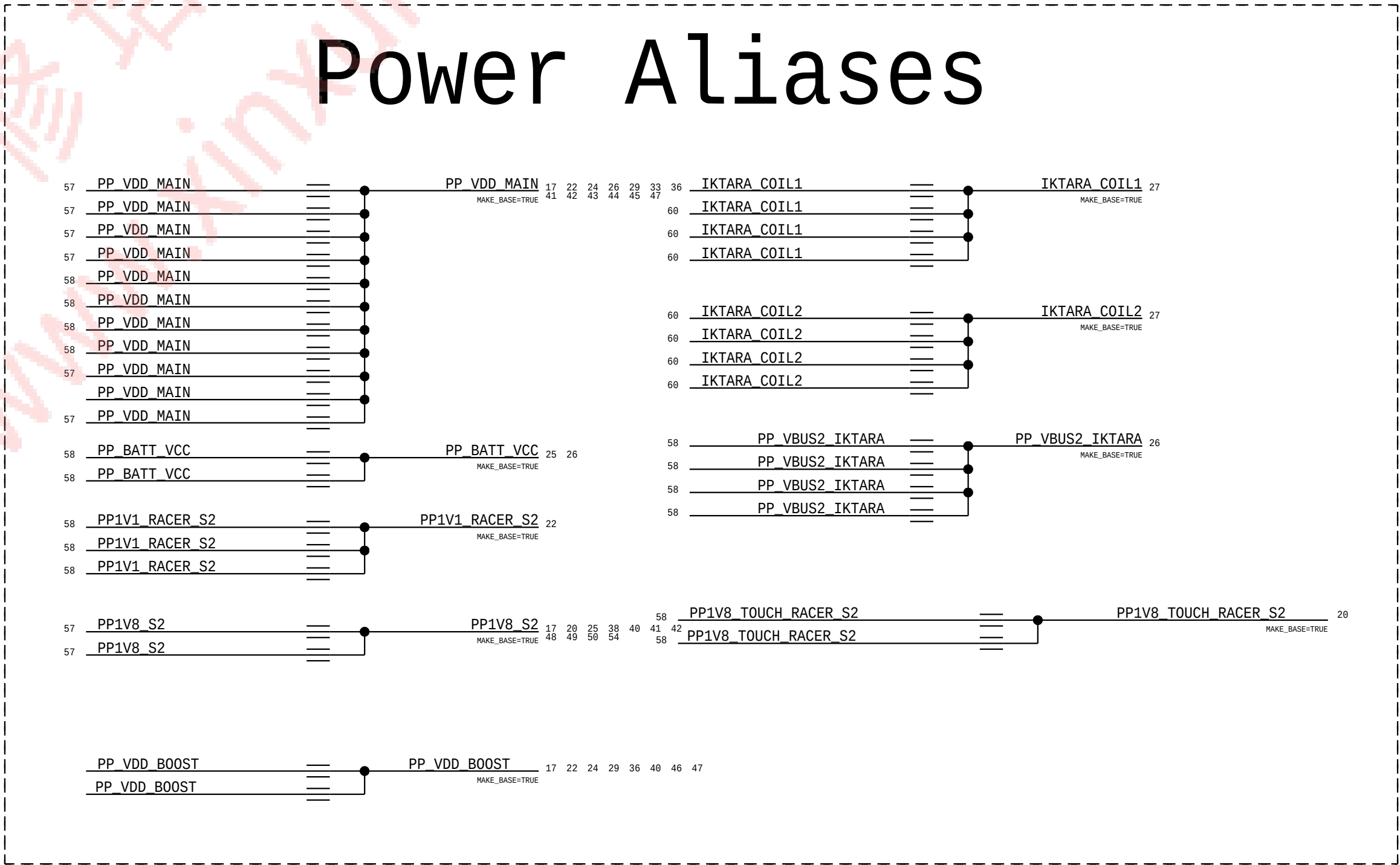
THIS SIDE HAS ATTRIBUTE
MAKE BASE=TRUE

[illegible]

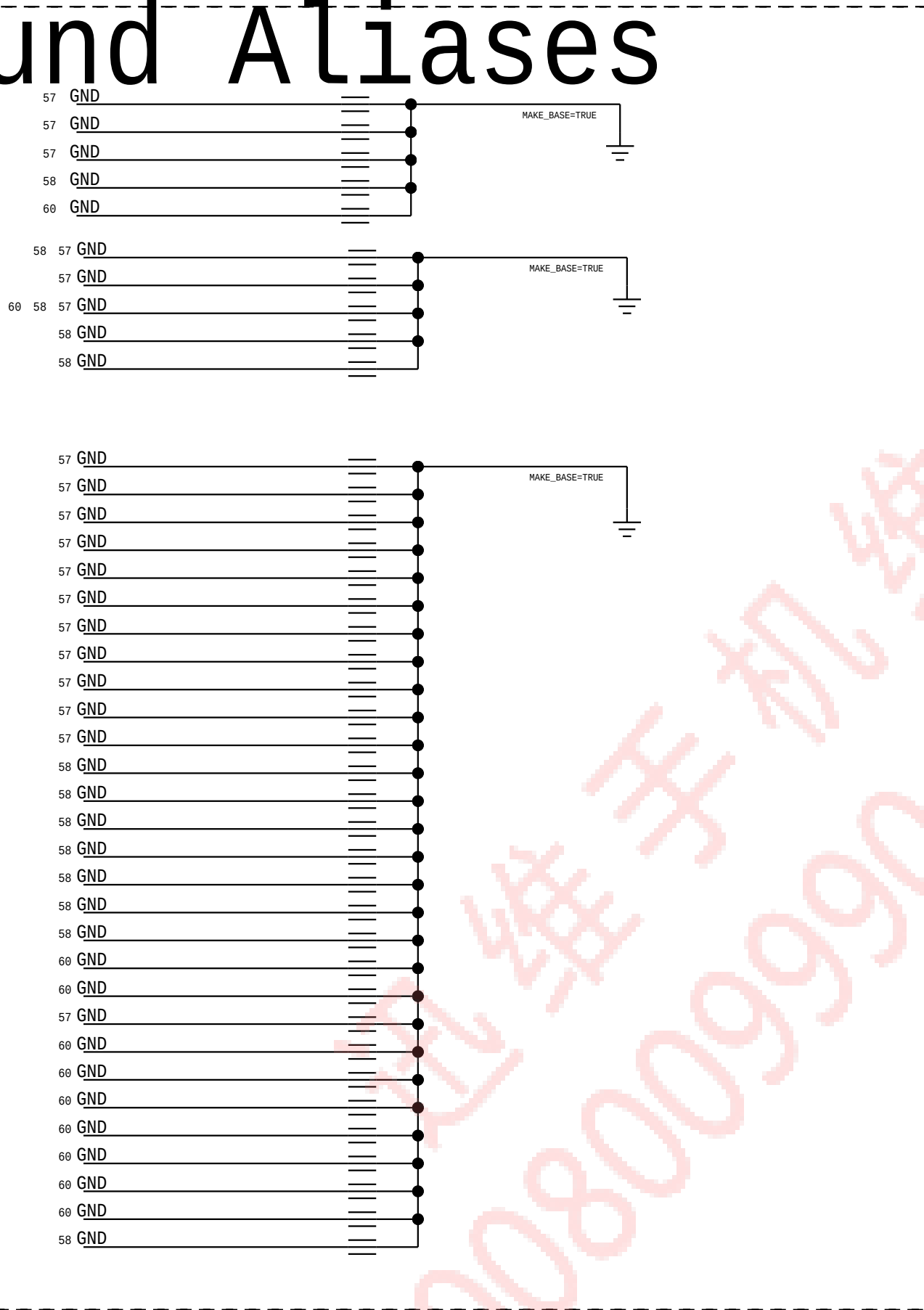
PAGE TITLE		SYNCH DATE=88/38/2817	
Interposer: Pins 145-285		DRAWING NUMBER	SIZE
		051-02545	D
		REVISION	
		7.0.0	
		BRANCH	
		PAGE	
		82 OF 85	
		SHEET	
		58 OF 60	

Interposer Top Level Aliases

Power Aliases



Ground Aliases



PAGE TITLE		
Interposer: Top Aliases		
	DRAWING NUMBER	051-02545
	REVISION	7.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH	
	PAGE	83 OF 85
	SHEET	59 OF 60

THIS SIDE HAS ATTRIBUTE
MAKE_BASE=TRUE

[illegible]



EEEE Codes

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
825-7691	1	EEEE (MLB_BOT_ICE, JP, 639-84872)	EEEE_X8XR	CRITICAL	JP
825-7691	1	EEEE (MLB_BOT_ICE, ROW, 639-84346)	EEEE_HYFC	CRITICAL	ROW

Global Capacitors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00148	138S00149	BOM_TABLE_ALTS	ALL	0402-3T, 10.5uF@1V, Kyocera
138S00150	138S00149	BOM_TABLE_ALTS	ALL	0402-3T, 10.5uF@1V, SEMCO
138S00151	138S00149	BOM_TABLE_ALTS	ALL	0402-3T, 10.5uF@1V, TY

CRITICAL PART#	COMMENT
138S00149	0402-3T, 10.5uF@1V

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00143	138S00144	BOM_TABLE_ALTS	ALL	0402, 16uF@1V, Kyocera
138S00163	138S00144	BOM_TABLE_ALTS	ALL	0402, 16uF@1V, Taiyo

CRITICAL PART#	COMMENT
138S00144	0402, 16uF@1V

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00138	138S00139	BOM_TABLE_ALTS	ALL	0201, 3uF@1V, Kyocera
138S00164	138S00139	BOM_TABLE_ALTS	ALL	0201, 3uF@1V, Taiyo

CRITICAL PART#	COMMENT
138S00139	0201, 3uF@1V

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00221	138S00146	BOM_TABLE_ALTS	ALL	0402, 5.1uF@3V, Kyocera

CRITICAL PART#	COMMENT
138S00146	0402, 5.1uF@3V

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00140	138S00141	BOM_TABLE_ALTS	ALL	0201, 1.1uF@3V, Kyocera
138S00142	138S00141	BOM_TABLE_ALTS	ALL	0201, 1.1uF@3V, SEMCO
138S00166	138S00141	BOM_TABLE_ALTS	ALL	0201, 1.1uF@3V, Taiyo

CRITICAL PART#	COMMENT
138S00141	0201, 1.1uF@3V

Global Ferrites

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00558	152S00557	BOM_TABLE_ALTS	ALL	IND, ML2, 0.47UH, 20%, 2.5A, 80MD, 1600
155S00194	155S00610	BOM_TABLE_ALTS	ALL	FERR BD, 150 OHM, 25%, 200MA, 0.7 DCR, 01005
155S00200	155S00610	BOM_TABLE_ALTS	ALL	FERR BD, 150 OHM, 25%, 200MA, 0.7 DCR, 01005
155S00338	155S00661	BOM_TABLE_ALTS	ALL	FERR BD, 33 OHM, 25%, 1.5A, 1000000 DCR, 0201

CRITICAL PART#	COMMENT
152S00557	IND, ML2, 0.47UH, 20%, 2.5A, 80MD, 1600
155S00610	FERR BD, 150 OHM, 25%, 200MA, 0.7 DCR, 01005

CRITICAL PART#	COMMENT
155S00661	FERR BD, 33 OHM, 25%, 1.5A, 550000H DCR, 0201

Global R/C Alternates

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00648	138S00652	BOM_TABLE_ALTS	ALL	CAP, XSR, 4.7UF, 6.3V, 0.65MM, 0402, TAIYO
138S00024	138S00986	BOM_TABLE_ALTS	ALL	CAP, CER, 3-TERM, 1.0UF, 50V, 0.65MM, 0402, TAIYO/TDK
138S0706	138S0739	BOM_TABLE_ALTS	ALL	CAP, CER, 1UF, 20%, 10V, XSR, 0201, MURATA
138S0945	138S0739	BOM_TABLE_ALTS	ALL	CAP, CER, 1UF, 20%, 10V, XSR, 0201, KYOCERA
138S0739	138S0706	BOM_TABLE_ALTS	ALL	CAP, CER, XSR, 0.22UF, 20%, 6.3V, 20%
132S0436	132S0400	BOM_TABLE_ALTS	ALL	CAP, CER, XSR, 0.22UF, 20%, 6.3V, 01005
138S00049	138S00831	BOM_TABLE_ALTS	ALL	CAP, CER, XSR, 2.2UF, 20%, 6.3V, 0201
138S00133	138S00128	BOM_TABLE_ALTS	ALL	CAP, XSR, 0.47UF, 20%, 6.3V, KYO, 01005
138S00116	138S00071	BOM_TABLE_ALTS	ALL	CAP, XSR, 4UF, 20%, 6.3V, 0201, T=0.55MM
138S00117	138S00071	BOM_TABLE_ALTS	ALL	CAP, XSR, 4UF, 20%, 6.3V, 0201, T=0.55MM
138S00048	138S00003	BOM_TABLE_ALTS	ALL	CAP, XSR, 100PF, 20%, 6.3V, 0.65MM, RECT, 0402
131S00172	131S00164	BOM_TABLE_ALTS	ALL	CAP, CER, C0G, 220PF, 5%, 25V, 01005
131S00173	131S00164	BOM_TABLE_ALTS	ALL	CAP, CER, C0G, 220PF, 5%, 25V, 01005
155S00338	155S00402	BOM_TABLE_ALTS	ALL	FERR BD, 33 OHM, 25%, 1.5A, TY
155S00437	155S00402	BOM_TABLE_ALTS	ALL	FERR BD, 33 OHM, 25%, 1.5A, TYK

CRITICAL PART#	COMMENT
138S00652	CAP, XSR, 4.7UF, 6.3V, 0.65MM, 0402, TAIYO
138S00986	CAP, CER, 3-TERM, 1.0UF, 50V, 0.65MM, 0402, TAIYO/TDK
138S0739	CAP, CER, 1UF, 20%, 10V, XSR, 0201, MURATA

CRITICAL PART#	COMMENT
138S0706	CAP, CER, XSR, 0.22UF, 20%, 6.3V, 20%
132S0400	CAP, CER, XSR, 0.22UF, 20%, 6.3V, 01005
138S0031	CAP, CER, XSR, 2.2UF, 20%, 6.3V, 0201
138S00128	CAP, XSR, 0.47UF, 20%, 6.3V, KYO, 01005
138S00071	CAP, XSR, 4UF, 20%, 6.3V, 0201, T=0.55MM

CRITICAL PART#	COMMENT
138S00003	CAP, XSR, 15UF, 20%, 6.3V, 0.65MM, HRZTL, 0402
131S00164	CAP, CER, C0G, 220PF, 5%, 16V, 01005

Global Inductors

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
152S00653	152S00651	BOM_TABLE_ALTS	ALL	IND, 1.2UH, 3A, 2016, 0.65Z
152S00654	152S00652	BOM_TABLE_ALTS	ALL	IND, 1.2UH, 3A, 2016, 0.8Z

CRITICAL PART#	COMMENT
152S00651	IND, 1.2UH, 3A, 2016, 0.65Z
152S00652	IND, 1.2UH, 3A, 2016, 0.8Z

Iktara

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
116S00002	6	RES, High Current 0 Ohm	C3401, C3402, C3403, C3404, C3405, C3406	CRITICAL	UNBALANCED CAPS
132S00186	1	CAP, 0.33uF, 10%, 100V, X6S	C3451	CRITICAL	UNBALANCED CAPS
132S00187	3	CAP, 0.047uF, 10%, 100V, X6S	C3452, C3453, C3456	CRITICAL	UNBALANCED CAPS

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
685-00183	685-00182	BOM_TABLE_ALTS	SUBBOM_DS	DI000A, MLB_BOT, DI0005, DI0001, 1001

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
685-00182	1	SUBBOM, MLB_BOT, DI0005, DI0005, X891	SUBBOM_DS	CRITICAL	COMMON
371S00133	4	DI0005, SHOTTKY DIODE, 30V, 2A, 0603	D3400, D3401, D3402, D3403	CRITICAL	DIODES_DS
371S00132	4	ONSEMI, SHOTTKY DIODE, 30V, 2A, 0603	D3400, D3401, D3402, D3403	CRITICAL	ONSEMI_DS

Hall Effect Alts

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
353S01327	353S3697	BOM_TABLE_ALTS	ALL	ROWH, RES

Multi-Vendor Criticals

CRITICAL PART#	COMMENT	CRITICAL PART#	COMMENT
138S00979	CAP, CER, XSR, 18UF, 20%, 10V, 0402, H=0.65MM	132S00008	CAP, CER, 0.1UF, 10%, 50V, X7R, 0402
138S00683	CAP, CER, XSR, 1UF, 10%, 25V, 0402	131S00004	CAP, CER, 27PF, 5%, C0G, 25V, 0201
132S00663	CAP, CER, XSR, 1UF, 10%, 25V, 0402	131S00307	CAP, CER, NP0/C0G, 100PF, 5%, 16V, 01005
132S0288	CAP, CER, XSR, 0.1UF, 10%, 10V, 0201	131S00053	CAP, CER, C0G, 220PF, 5%, 10V, 01005
132S0275	CAP, CER, XSR, 470PF, 10%, 10V, 01005	117S00055	RES, MF, 1/20W, 2W OHM, 5, 0201, SMD
132S0245	CAP, CER, XSR, 0.01UF, 10%, 6.3V, 01005	107S0257	THERMISTOR, NTC, 10K OHM, 1%, B=3435, 01005
138S00692	CAP, CER, XSR, 1UF, 20%, 6.3V, 0201	131S00170	CAP, CER, C0G, 220PF, 5%, 25V, 01005
		131S00643	CAP, CER, NP0/C0G, 50PF, 5%, 25V, 01005
		131S0316	CAP, CER, XSR, 0.1UF, 20%, 6.3V, 01005

PAGE TITLE			SYNC_DATE=12/13/2018
SYSTEM:BOM Tables			
		DRAWING NUMBER	051-02613
		REVISION	4.0.0
		BRANCH	
		PAGE	2 OF 85
		SHEET	2 OF 49
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			



NOTICE OF PROPRIETARY PROPERTY:

THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING:

I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE

II NOT TO REPRODUCE OR COPY IT

III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART

IV ALL RIGHTS RESERVED

D

C

B

A

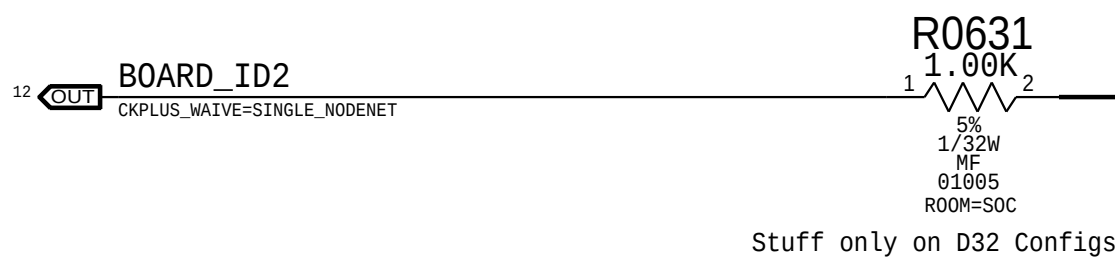
D

C

B

A

BOOTSTRAPPING:BOARD ID[3]



PP1V8_S2 6 11 14 15

SELECTED-->

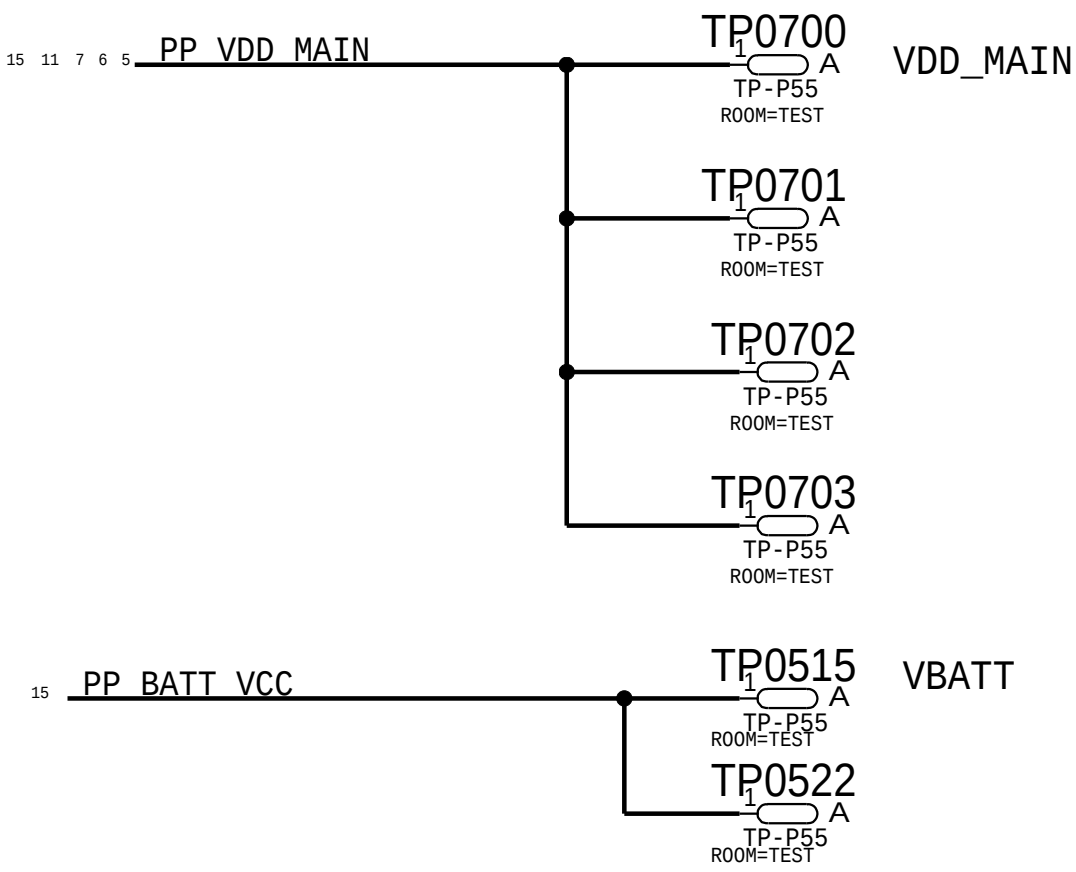
Board ID[4:0]					
Float = Low PU = High					
	4	3	2	1	0
	Denali = 0, Imola = 1	Mav = 0, Ice = 1	00=Open 01=D33 10=N84 11=D32	MLB = 0, Dev = 1	
D33 MLB Mav	0	0	0	1	0
D33p MLB Mav	1	0	0	1	0
D33 Dev Mav	0	0	0	1	1
D33p Dev Mav	1	0	0	1	1
D33 MLB Ice	0	1	0	1	0
D33p MLB Ice	1	1	0	1	0
D33 Dev Ice	0	1	0	1	1
D33p Dev Ice	1	1	0	1	1
D32 MLB Mav	0	0	1	1	0
D32 Dev Mav	0	0	1	1	1
D32 MLB Ice	0	1	1	1	0
D32 Dev Ice	0	1	1	1	1

			SYNC_DATE=11/03/2016				
PAGE TITLE			BOOTSTRAPPING				
			DRAWING NUMBER	051-02613	SIZE	D	
			REVISION	4.0.0			
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			BRANCH				
			PAGE			6 OF 85	
			SHEET			4 OF 49	

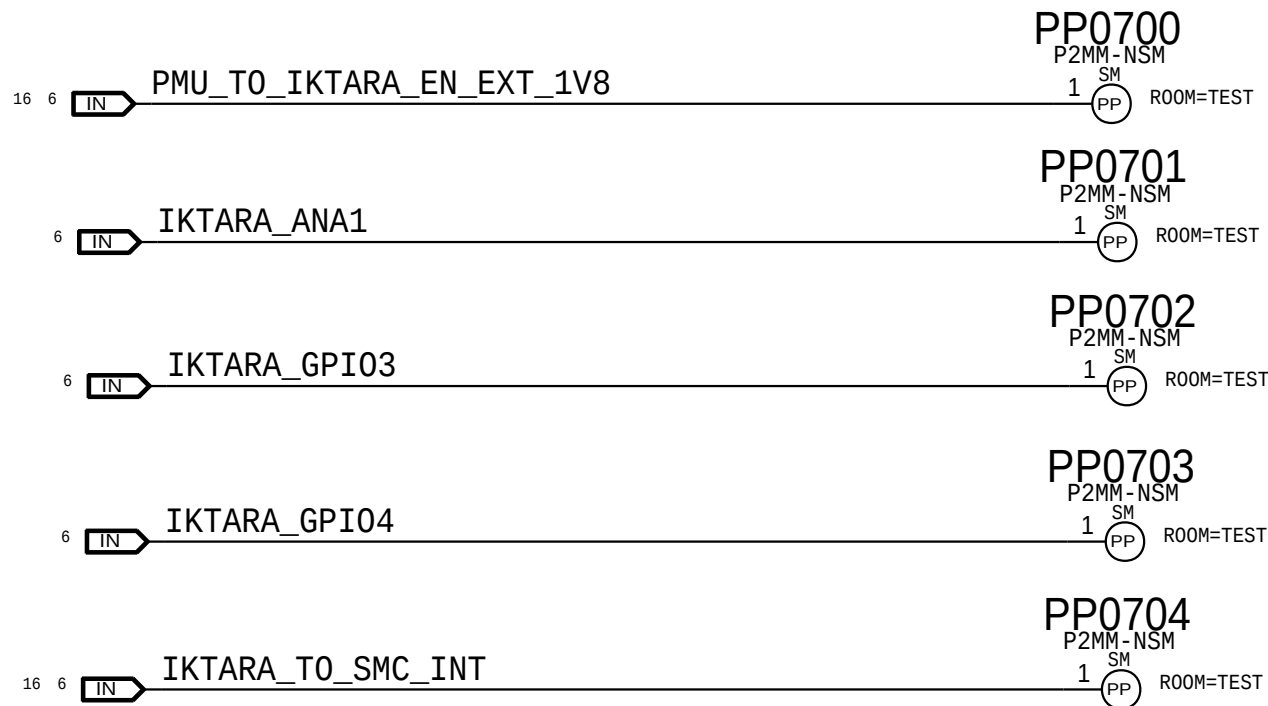


Test Points

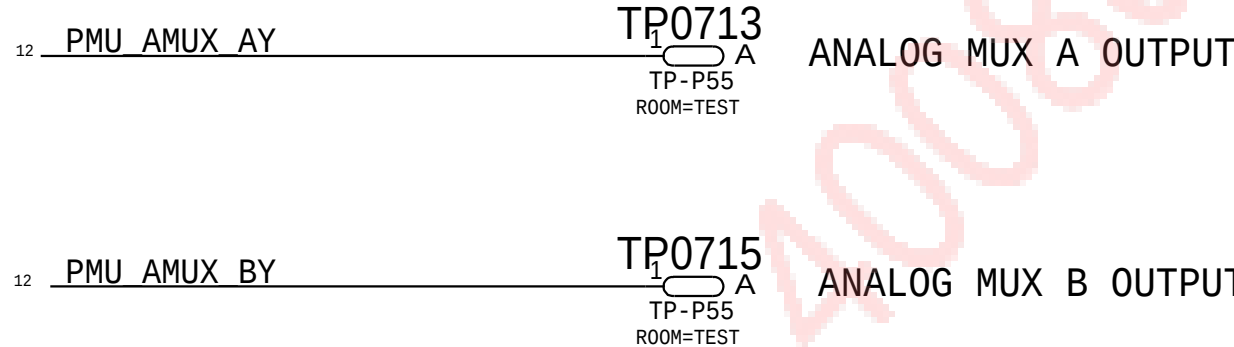
Probe Points



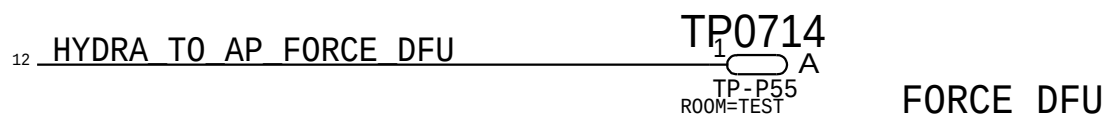
Iktara Debug



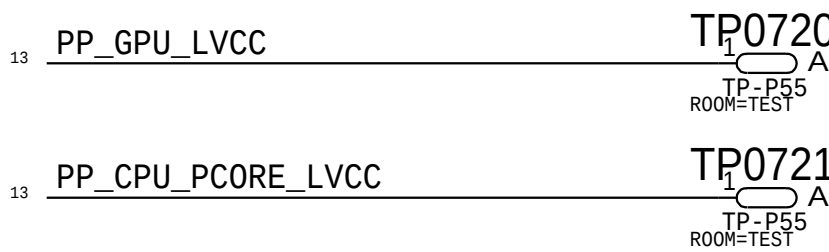
AMUX



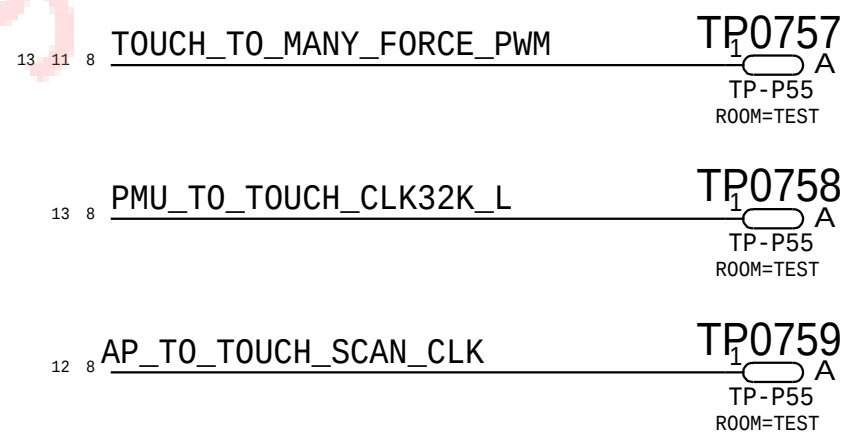
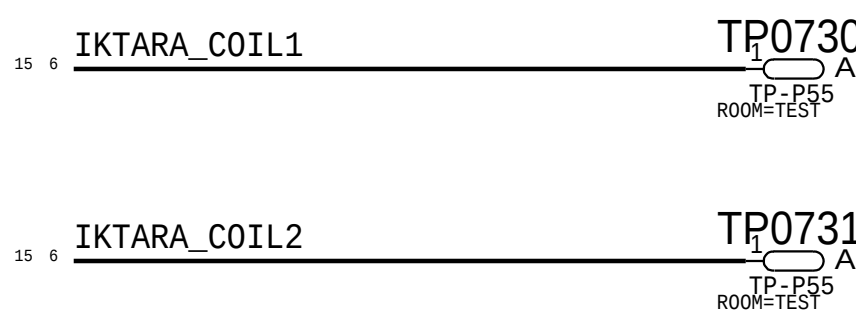
DFU



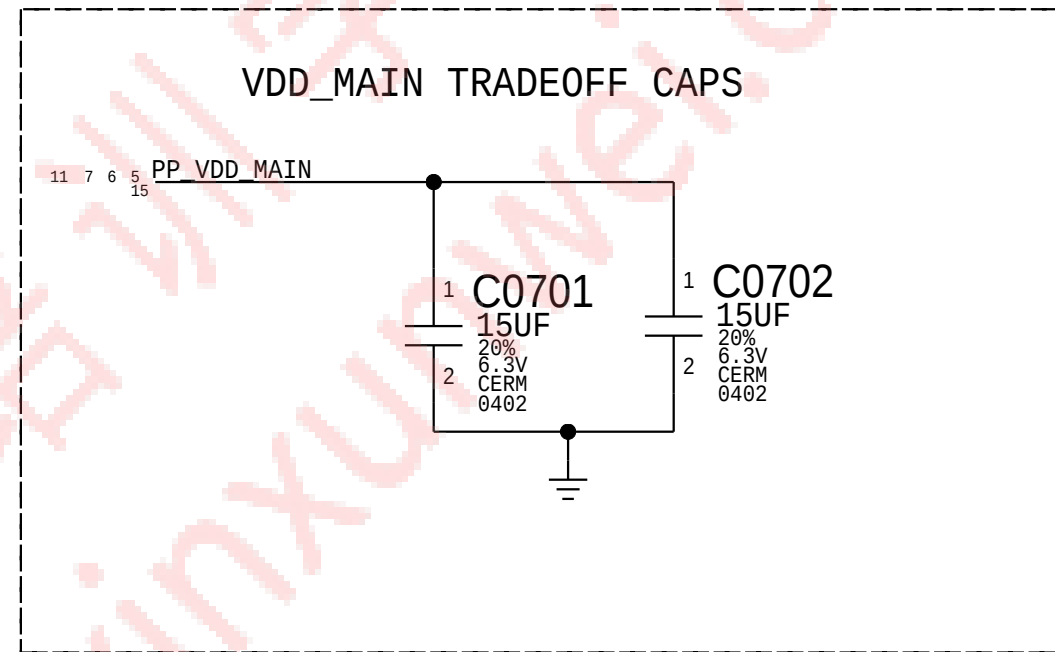
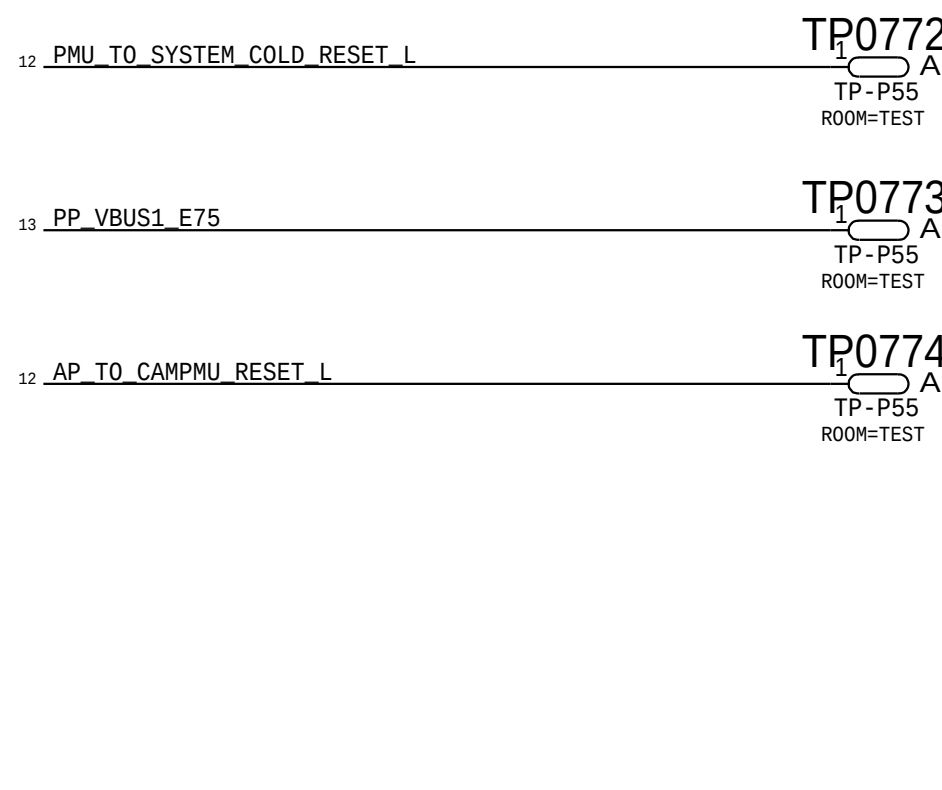
LVCC



COIL



REL BIAS

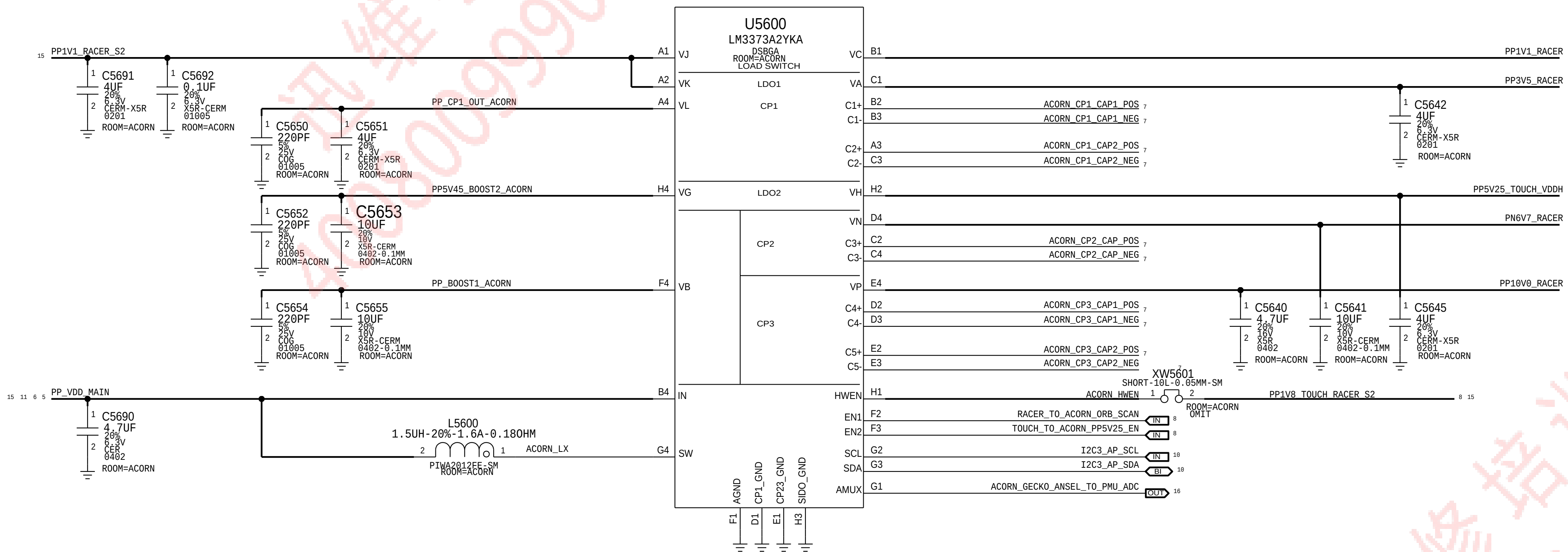
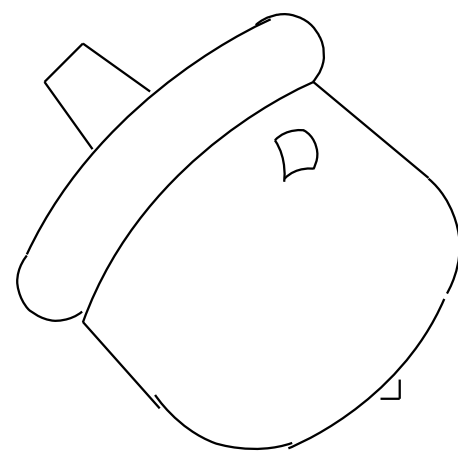


SYNC_MASTER=test_#1b			SYNC_DATE=06/28/2017		
PAGE TITLE			SYSTEM: Testpoints (Bottom)		
			DRAWING NUMBER	051-02613	SIZE
			REVISION	4.0.0	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			BRANCH		
			PAGE		
			7 OF 85		
			SHEET		
			5 OF 49		

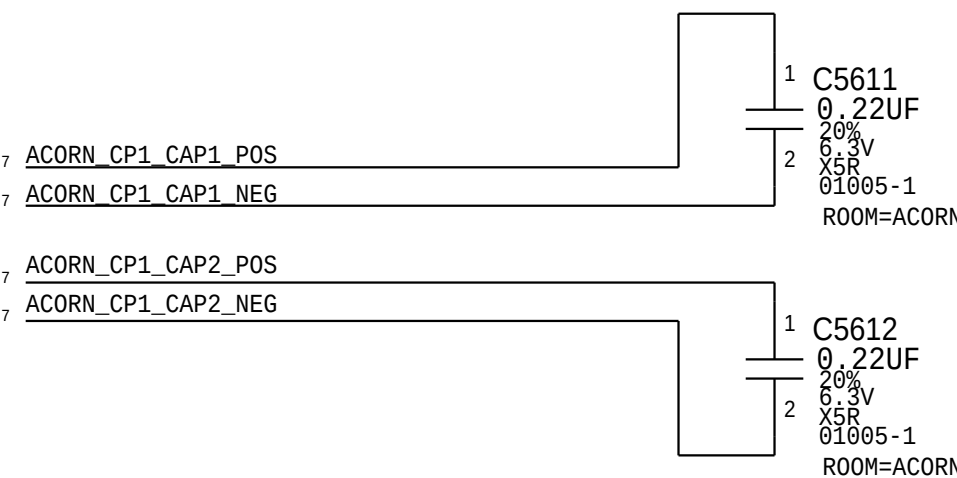


2	1
---	---

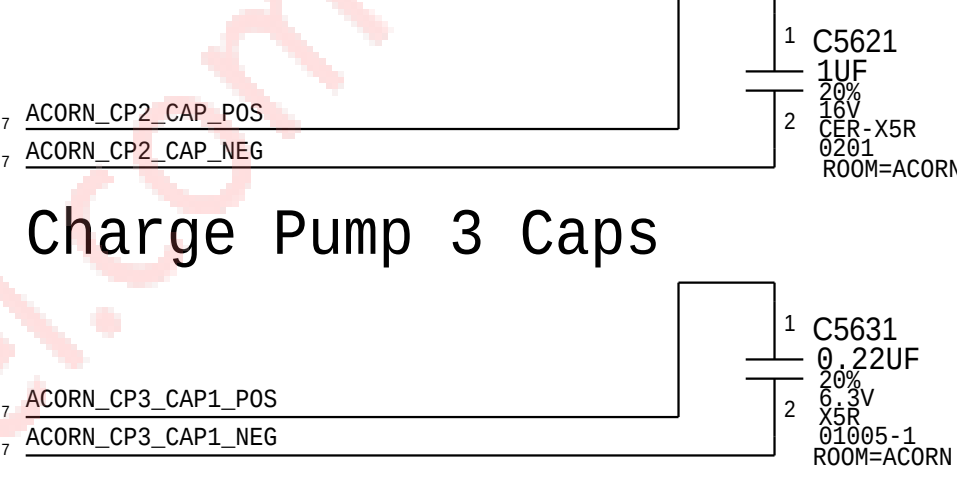
Acorn PMU



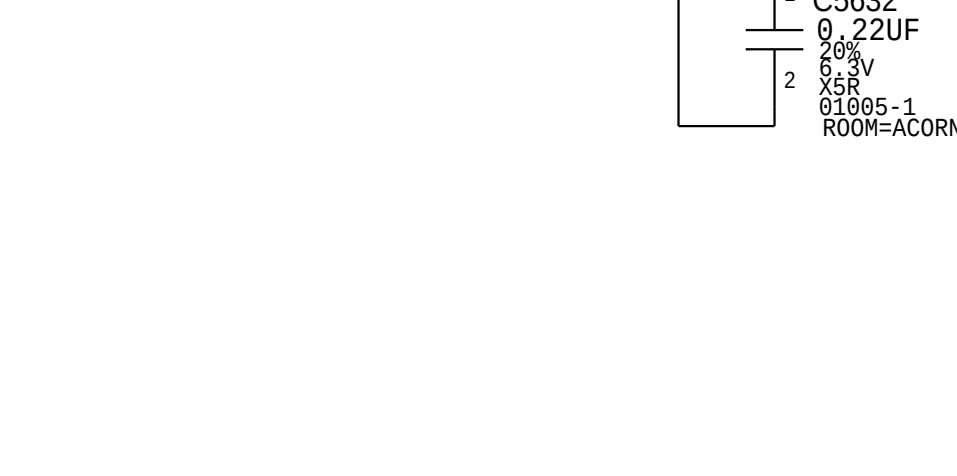
Charge Pump 1 Caps



Charge Pump 2 Caps



Charge Pump 3 Caps



PAGE TITLE		
CG: Power Supplies - Touch & Display		
DRAWING NUMBER	051-02613	SIZE
REVISION	4.0.0	D
NOTICE OF PROPRIETARY PROPERTY:		
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING:		
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		
II NOT TO REPRODUCE OR COPY IT		
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		
IV ALL RIGHTS RESERVED		
BRANCH		
PAGE	56 OF 85	
SHEET	7 OF 49	



Top Board Interposer APN:998-12513
Bot Board Interposer APN:998-12514 <--- Stuffed

D

C

B

A

D

C

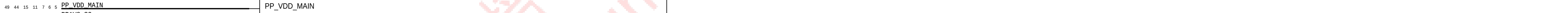
B

A

J_INT_TOP			J_INT_TOP		
SMT-PAD			SMT-PAD		
SYM 1 OF 2			SYM 2 OF 2		
INTERPOSER-MLB-TOP-V3-D32			INTERPOSER-MLB-TOP-V3-D32		
1	IO1	IO95	95	AP TO NFC_DEV_WAKE	12
2	IO2	IO96	96	AP TO NFC_FW_DWLD_REQ	12
3	IO3	IO97	97	GND	12
4	IO4	IO98	98	PMU TO NFC_VDD_MAIN_EN	12
5	IO5	IO99	99	UART_AOP_TO_BB_TXD	12
6	IO6	IO100	100	UART_AP_TO_GNSS_TXD	12
7	IO7	IO101	101	GND	12
8	IO8	IO102	102	AP TO BB_COREDUMP_TRIG	12
9	IO9	IO103	103	UART_AP_TO_NFC_TXD	12
10	IO10	IO104	104	UART_NFC_TO_AP_RXD	12
11	IO11	IO105	105	BB TO AP_RESET_DETECT_L	12
12	IO12	IO106	106	GND	12
13	IO13	IO107	107	BOARD_ID2	12
14	IO14	IO108	108	AP TO_GNSS_TIME_MARK	12
15	IO15	IO109	109	PMU_PMGR_TO_PMU_IKTARA_SCLK	12
16	IO16	IO110	110	BB TO_BB_PEAK_POWER_INDICATOR	12
17	IO17	IO111	111	AP TO_BBPMU_RADIO_ON_L	12
18	IO18	IO112	112	PP_VDD_MAIN	12
19	IO19	IO113	113	PP_VDD_MAIN	12
20	IO20	IO114	114	PP_VDD_MAIN	12
21	IO21	IO115	115	GND	12
22	IO22	IO116	116	90_PCIE_BB_TO_AP_RXD_N	12
23	IO23	IO117	117	90_PCIE_BB_TO_AP_RXD_P	12
24	IO24	IO118	118	GND	12
25	IO25	IO119	119	90_PCIE_AP_TO_BB_TXD_N	12
26	IO26	IO120	120	90_PCIE_AP_TO_BB_TXD_P	12
27	IO27	IO121	121	GND	12
28	IO28	IO122	122	90_PCIE_AP_TO_BB_REFCLK_P	12
29	IO29	IO123	123	90_PCIE_AP_TO_BB_REFCLK_N	12
30	IO30	IO124	124	GND	12
31	IO31	IO125	125	UART_BB_TO_AOP_RXD	12
32	IO32	IO126	126	UART_GNSS_TO_AP_RXD	12
33	IO33	IO127	127	PCIE_AP_TO_BB_PERST_L	12
34	IO34	IO128	128	UART_NFC_TO_AP_CTS_L	12
35	IO35	IO129	129	GND	12
36	IO36	IO130	130	UART_AP_TO_NFC_RTS_L	12
37	IO37	IO131	131	PMU_AMUX_BY	12
38	IO38	IO132	132	PMU_AMUX_AY	12
39	IO39	IO133	133	GND	12
40	IO40	IO134	134	PCIE_BB_BI_AP_CLKREQ_L	12
41	IO41	IO135	135	PMU_IKTARA_BI_PMGR_SDATA	12
42	IO42	IO136	136	BB TO AP PEAK_POWER_INDICATOR	12
43	IO43	IO137	137	GND	12
44	IO44	IO138	138	PP_VDD_MAIN	12
45	IO45	IO139	139	PP_VDD_MAIN	12
46	IO46	IO140	140	PP_VDD_MAIN	12
47	IO47	IO141	141	GND	12
48	IO48	IO142	142	GND	12
49	IO49	IO143	143	GND	12
50	IO50	IO144	144	GND	12
51	IO51	IO145	145	GND	13
52	IO52	IO146	146	GND	13
53	IO53	IO147	147	GND	13
54	IO54	IO148	148	GND	13
55	IO55	IO149	149	GND	13
56	IO56	IO150	150	GND	13
57	IO57	IO151	151	GND	13
58	IO58	IO152	152	GND	13
59	IO59	IO153	153	GND	13
60	IO60	IO154	154	PP_VDD_MAIN	13
61	IO61	IO155	155	PP_VDD_MAIN	13
62	IO62	IO156	156	GND	13
63	IO63	IO157	157	PP_VDD_MAIN	13
64	IO64	IO158	158	PP_VDD_MAIN	13
65	IO65	IO159	159	GND	13
66	IO66	IO160	160	PMU TO_NFC_EN	13
67	IO67	IO161	161	GND	13
68	IO68	IO162	162	PMU TO_BBPMU_RESET_L	13
69	IO69	IO163	163	GND	13
70	IO70	IO164	164	PMU TO_TOUCH_CLK32K_L	13
71	IO71	IO165	165	GND	13
72	IO72	IO166	166	PCIE_WLAN_BI_AP_CLKREQ_L	13
73	IO73	IO167	167	GND	13
74	IO74	IO168	168	GND	13
75	IO75	IO169	169	BB TO_PMU_PCIE_HOST_WAKE_L	13
76	IO76	IO170	170	GND	13
77	IO77	IO171	171	GND	13
78	IO78	IO172	172	WLAN TO_PMU_HOST_WAKE	13
79	IO79	IO173	173	GND	13
80	IO80	IO174	174	PMU TO_WLAN_CLK32K	13
81	IO81	IO175	175	GND	13
82	IO82	IO176	176	NFC TO_AOP_HOST_WAKE	13
83	IO83	IO177	177	GND	13
84	IO84	IO178	178	TOUCH_TO_MANY_FORCE_PWM	13
85	IO85	IO179	179	GND	13
86	IO86	IO180	180	UART_AP_TO_BT_TXD	13
87	IO87	IO181	181	GND	13
88	IO88	IO182	182	UART_AP_TO_BT_RTS_L	13
89	IO89	IO183	183	GND	13
90	IO90	IO184	184	GND	13
91	IO91	IO185	185	GND	13
92	IO92	IO186	186	GND	13
93	IO93	IO187	187	GND	13
94	IO94	IO188	188	GND	13

J_INT_TOP			J_INT_TOP		
SMT-PAD			SMT-PAD		
SYM 2 OF 2			SYM 2 OF 2		
INTERPOSER-MLB-TOP-V3-D32			INTERPOSER-MLB-TOP-V3-D32		
189	IO189	IO282	282	GND	13
190	IO190	IO283	283	GND	13
191	IO191	IO284	284	GND	13
192	IO192	IO285	285	GND	13
193	IO193	IO286	286	GND	16
194	IO194	IO287	287	APORN_GECKO_ANSEL_TO_PMU_ADC	16
195	IO195	IO288	288	GND	16
196	IO196	IO289	289	RACER_TO_AOP_INT_L	16
197	IO197	IO290	290	GND	16
198	IO198	IO291	291	HALL_CASE_TO_AOP_SOUTH_L	16
199	IO199	IO292	292	GND	16
200	IO200	IO293	293	PMU_TO_IKTARA_EN_EXT_1V8	16
201	IO201	IO294	294	GND	16
202	IO202	IO295	295	IKTARA_TO_SMC_INT	16
203	IO203	IO296	296	GND	16
204	IO204	IO297	297	I2C0_SMC_SCL	16
205	IO205	IO298	298	I2C0_SMC_SDA	16
206	IO206	IO299	299	GND	16
207	IO207	IO300	300	IKTARA_COIL2	16
208	IO208	IO301	301	IKTARA_COIL2	16
209	IO209	IO302	302	IKTARA_COIL2	16
210	IO210	IO303	303	IKTARA_COIL2	16
211	IO211	IO304	304	IKTARA_COIL1	16
212	IO212	IO305	305	IKTARA_COIL1	16
213	IO213	IO306	306	IKTARA_COIL1	16
214	IO214	IO307	307	IKTARA_COIL1	16
215	IO215	IO308	308	GND	16
216	IO216	IO309	309	NC_PAPER_TIGER1	16
217	IO217	IO310	310	GND	16
218	IO218	IO311	311	NC_PAPER_TIGER2	16
219	IO219	IO312	312	GND	16
220	IO220	IO313	313	GND	16
221	IO221	IO314	314	GND	16
222	IO222	IO315	315	GND	16
223	IO223	IO316	316	GND	16
224	IO224	IO317	317	GND	16
225	IO225	IO318	318	GND	16
226	IO226	IO319	319	GND	16
227	IO227	IO320	320	GND	16
228	IO228	IO321	321	GND	16
229	IO229	IO322	322	GND	16
230	IO230	IO323	323	GND	16
231	IO231	IO324	324	GND	16
232	IO232	IO325	325	GND	16
233	IO233	IO326	326	GND	16
234	IO234	IO327	327	GND	16
235	IO235	IO328	328	GND	16
236	IO236	IO329	329	GND	16
237	IO237	IO330	330	GND	16
238	IO238	IO331	331	GND	16
239	IO239	IO332	332	GND	16
240	IO240	IO333	333	GND	16
241	IO241	IO334	334	GND	16
242	IO242	IO335	335	GND	16
243	IO243	IO336	336	GND	16
244	IO244	IO337	337	GND	16
245	IO245	IO338	338	GND	16
246	IO246	IO339	339	GND	16
247	IO247	IO340	340	GND	16
248	IO248	IO341	341	GND	16
249	IO249	IO342	342	GND	16
250	IO250	IO343	343	GND	16
251	IO251	IO344	344	GND	16
252	IO252	IO345	345	GND	16
253	IO253	IO346	346	GND	16
254	IO254	IO347	347	GND	16
255	IO255	IO348	348	GND	16
256	IO256	IO349	349	GND	16
257	IO257	IO350	350	GND	16
258	IO258	IO351	351	GND	16
259	IO259	IO352	352	GND	16
260	IO260	IO353	353	GND	16
261	IO261	IO354	354	GND	16
262	IO262	IO355	355	GND	16
263	IO263	IO356	356	GND	16
264	IO264	IO357	357	GND	16
265	IO265	IO358	358	GND	16
266	IO266				
267	IO267				
268	IO268				
269	IO269				
270	IO270				
271	IO271				
272	IO272				
273	IO273				
274	IO274				
275	IO275				
276	IO276				
277	IO277				
278	IO278				
279	IO279				
280	IO280				
281	IO281				

PAGE TITLE		
?		
DRAWING NUMBER		SIZE
051-02613		D
REVISION		4.0.0
NOTICE OF PROPRIETARY PROPERTY:		BRANCH
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		66 OF 85
II NOT TO REPRODUCE OR COPY IT		SHEET
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		9 OF 49
IV ALL RIGHTS RESERVED		



		DRAWING NUMBER 051-02613		SIZE D	
		REVISION		4.0.0	
NOTICE OF PROPRIETARY PROPERTY:			BRANCH		
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING:			PAGE		
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			80 OF 85		
II NOT TO REPRODUCE OR COPY IT			SHEET		
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART			11 OF 49		
IV ALL RIGHTS RESERVED					

[illegible]

THIS SIDE HAS ATTRIBUTE
MAKE **BASE=TRUE**

9	PCIE AP TO WLAN PERST L	NAME: BASE/TRUE		PCIE AP TO WLAN PERST L	9
9	GND	NAME: BASE/TRUE	GND	12	15
9	AP TO RACER RESET L	NAME: BASE/TRUE		AP TO RACER RESET L	9
9	GND	NAME: BASE/TRUE	GND	12	15
9	AP TO WLAN TIME SYNC	NAME: BASE/TRUE		AP TO WLAN TIME SYNC	11
9	GND	NAME: BASE/TRUE	GND	12	15
9	GNSS TO AP LOW PWR TND	NAME: BASE/TRUE		GNSS TO AP LOW PWR TND	11
9	GND	NAME: BASE/TRUE	GND	12	15
9	HYDRA TO AP FORCE DFU	NAME: BASE/TRUE		HYDRA TO AP FORCE DFU	9
9	GND	NAME: BASE/TRUE	GND	12	15
9	PP1V8_S2	NAME: BASE/TRUE		PP1V8_S2	15
9	PP1V8_S2	NAME: BASE/TRUE		PP1V8_S2	15
9	GND	NAME: BASE/TRUE	GND	12	15
9	GND	NAME: BASE/TRUE	GND	15	
9	GND	NAME: BASE/TRUE	GND	15	
9	GND	NAME: BASE/TRUE	GND	15	
9	GND	NAME: BASE/TRUE	GND	12	13
12	INTERPOSER PIN_90	NAME: BASE/TRUE		INTERPOSER PIN_90	9
9	GND	NAME: BASE/TRUE	GND	15	
9	AP TO BB COEX	NAME: BASE/TRUE		AP TO BB COEX	11
9	BB TO AP COEX	NAME: BASE/TRUE		BB TO AP COEX	11
9	GND	NAME: BASE/TRUE	GND	15	
9	AP TO NFC DEV WAKE	NAME: BASE/TRUE		AP TO NFC DEV WAKE	11
9	AP TO NFC FW DWLD REQ	NAME: BASE/TRUE		AP TO NFC FW DWLD REQ	11
9	GND	NAME: BASE/TRUE	GND	15	
9	PMU TO NFC VDD MAIN EN	NAME: BASE/TRUE		PMU TO NFC VDD MAIN EN	11
9	UART AOP TO BB TXD	NAME: BASE/TRUE		UART AOP TO BB TXD	13
9	UART AP TO GNSS TXD	NAME: BASE/TRUE		UART AP TO GNSS TXD	11
9	GND	NAME: BASE/TRUE	GND	15	
9	AP TO BB COREDUMP TRIG	NAME: BASE/TRUE		AP TO BB COREDUMP TRIG	11
9	UART AP TO NFC TXD	NAME: BASE/TRUE		UART AP TO NFC TXD	11
9	UART NFC TO AP RXD	NAME: BASE/TRUE		UART NFC TO AP RXD	11
9	BB TO AP RESET DETECT L	NAME: BASE/TRUE		BB TO AP RESET DETECT L	11
9	GND	NAME: BASE/TRUE	GND	15	
9	BOARD ID2	NAME: BASE/TRUE		BOARD ID2	4
9	AP TO GNSS TIME MARK	NAME: BASE/TRUE		AP TO GNSS TIME MARK	11
AP	SPMI PMGR TO PMU IKTARA SCLK	NAME: BASE/TRUE		SPMI PMGR TO PMU IKTARA SCLK	11
AP	TO_BB_PEAK_POWER_INDICATOR	NAME: BASE/TRUE		TO_BB_PEAK_POWER_INDICATOR	11
9	AP TO BBPMU RADIO ON L	NAME: BASE/TRUE		AP TO BBPMU RADIO ON L	11
9	PP_VDD_MAIN	NAME: BASE/TRUE		PP_VDD_MAIN	15
9	PP_VDD_MAIN	NAME: BASE/TRUE		PP_VDD_MAIN	15
9	PP_VDD_MAIN	NAME: BASE/TRUE		PP_VDD_MAIN	15
9	GND	NAME: BASE/TRUE	GND	12	13
9	90_PCIE_BB_TO_AP_RXD_N	NAME: BASE/TRUE		90_PCIE_BB_TO_AP_RXD_N	11
9	90_PCIE_BB_TO_AP_RXD_P	NAME: BASE/TRUE		90_PCIE_BB_TO_AP_RXD_P	11
9	GND	NAME: BASE/TRUE	GND	12	15
9	90_PCIE_AP_TO_BB_TXD_N	NAME: BASE/TRUE		90_PCIE_AP_TO_BB_TXD_N	11
9	90_PCIE_AP_TO_BB_TXD_P	NAME: BASE/TRUE		90_PCIE_AP_TO_BB_TXD_P	11
9	GND	NAME: BASE/TRUE	GND	12	15
9	90_PCIE_AP_TO_BB_REFCLK_P	NAME: BASE/TRUE		90_PCIE_AP_TO_BB_REFCLK_P	11
9	90_PCIE_AP_TO_BB_REFCLK_N	NAME: BASE/TRUE		90_PCIE_AP_TO_BB_REFCLK_N	11
9	GND	NAME: BASE/TRUE	GND	12	15
9	UART_BB_TO_AOP_RXD	NAME: BASE/TRUE		UART_BB_TO_AOP_RXD	11
9	UART_GNSS_TO_AP_RXD	NAME: BASE/TRUE		UART_GNSS_TO_AP_RXD	11
9	PCIE_AP_TO_BB_PERST_L	NAME: BASE/TRUE		PCIE_AP_TO_BB_PERST_L	11
9	UART_NFC_TO_AP_CTS_L	NAME: BASE/TRUE		UART_NFC_TO_AP_CTS_L	11
9	GND	NAME: BASE/TRUE	GND	12	15
9	UART_AP_TO_NFC_RTS_L	NAME: BASE/TRUE		UART_AP_TO_NFC_RTS_L	11
9	PMU_AMUX_BY	NAME: BASE/TRUE		PMU_AMUX_BY	5
9	PMU_AMUX_AY	NAME: BASE/TRUE		PMU_AMUX_AY	5
9	GND	NAME: BASE/TRUE	GND	12	15
9	PCIE_BB_BI_AP_CLKREQ_L	NAME: BASE/TRUE		PCIE_BB_BI_AP_CLKREQ_L	11
SPMI	PMU_IKTARA_BI_PMGR_SDATA	NAME: BASE/TRUE		PMU_IKTARA_BI_PMGR_SDATA	11
BB	TO_AP_PEAK_POWER_INDICATOR	NAME: BASE/TRUE		TO_AP_PEAK_POWER_INDICATOR	11
9	GND	NAME: BASE/TRUE	GND	12	15
9	PP_VDD_MAIN	NAME: BASE/TRUE		PP_VDD_MAIN	15
9	PP_VDD_MAIN	NAME: BASE/TRUE		PP_VDD_MAIN	15
9	PP_VDD_MAIN	NAME: BASE/TRUE		PP_VDD_MAIN	15
9	GND	NAME: BASE/TRUE	GND	12	13
9	GND	NAME: BASE/TRUE	GND	12	15
9	GND	NAME: BASE/TRUE	GND	12	15
9	GND	NAME: BASE/TRUE	GND	12	15
9	GND	NAME: BASE/TRUE	GND	12	15

PAGE TITLE		SYN. DATE: 10/26/20	
Interposer: Pins 1-144			
	DRAWING NUMBER	SIZE	
	051-02613	D	
	REVISION	4.0.0	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH		
	PAGE		
	81 OF 85		
SHEET			
			12 OF 49

[illegible]

THIS SIDE HAS ATTRIBUTE
MAKE BASE=TRUE

[illegible]

PAGE TITLE		DRAWING NUMBER 051-02613		SIZE D	
Interposer: Pins 145-285		REVISION 4.0.0			
NOTICE OF PROPRIETARY PROPERTY:		BRANCH			
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING:		PAGE		82 OF 85	
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		SHEET		13 OF 49	
II NOT TO REPRODUCE OR COPY IT					
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART					
IV ALL RIGHTS RESERVED					

D

C

B

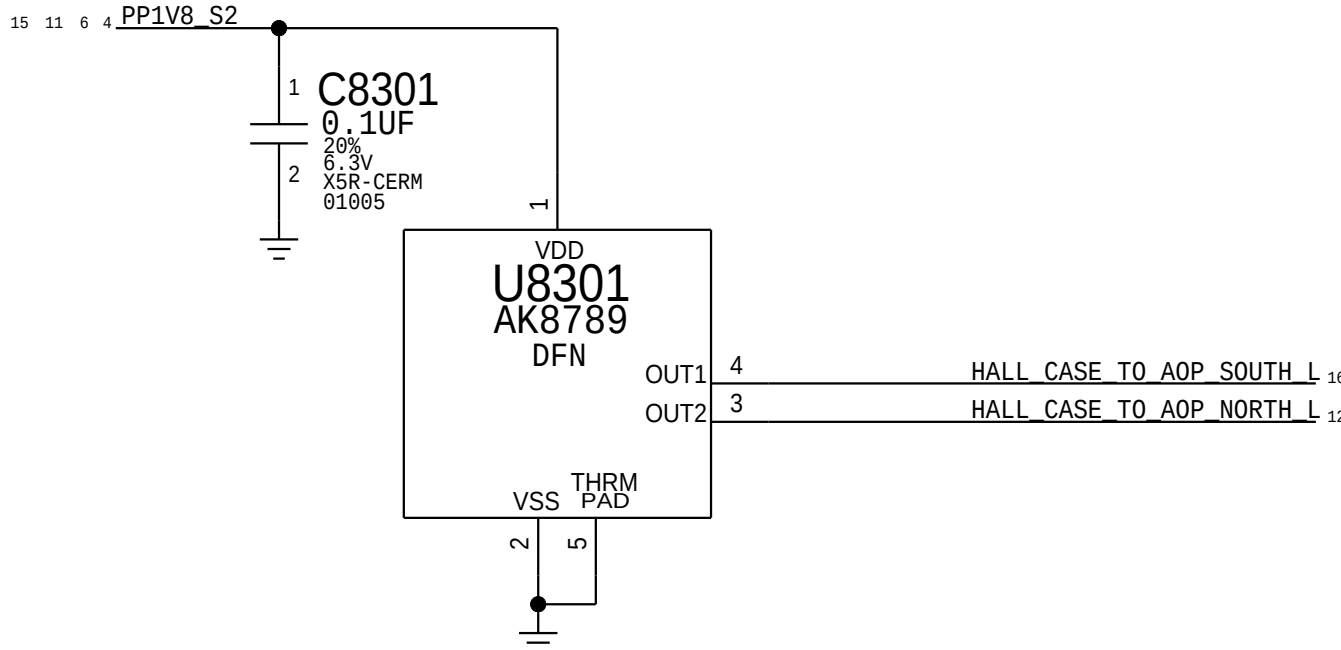
A

D

C

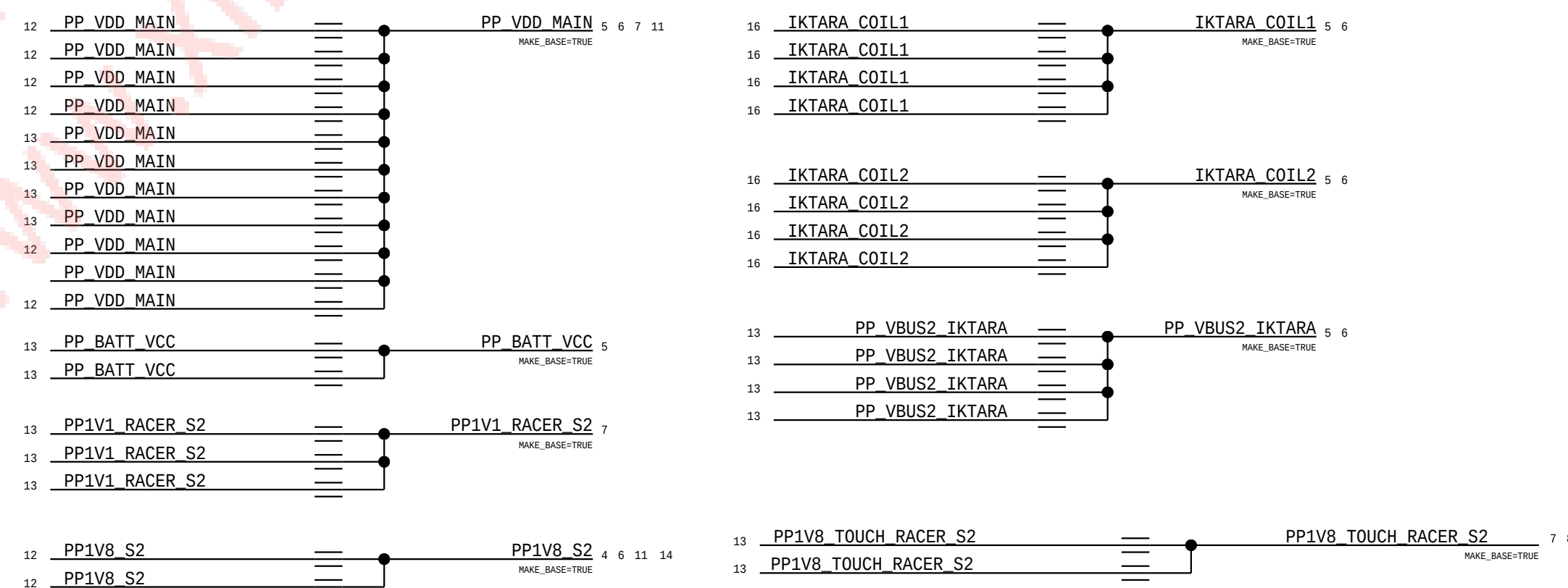
B

A

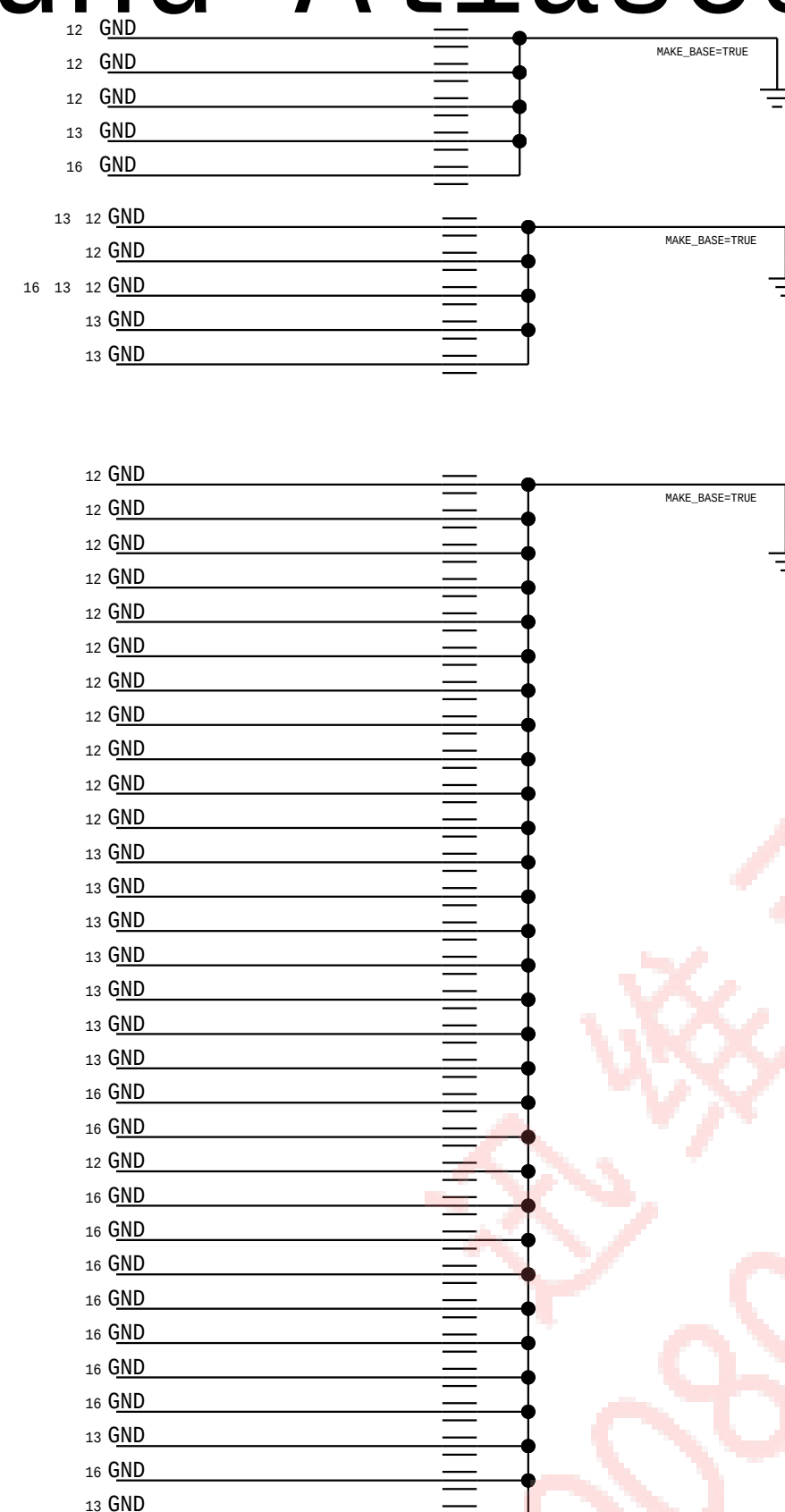


PAGE TITLE			Hall			SYNC_DATE=04/20/2017			
0			DRAWING NUMBER		051-02613	SIZE			
			REVISION		4.0.0	D			
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			BRANCH						
			PAGE		83	OF	85		
			SHEET		14	OF	49		

Power Aliases



Ground Aliases



PAGE TITLE		DRAWING NUMBER 051-02613		SIZE D
Interposer: Top Aliases		REVISION 4.0.0		
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH		
		PAGE 84 OF 85		
		SHEET 15 OF 49		

THIS SIDE HAS ATTRIBUTE
MAKE_BASE=TRUE

9 GND MAKE_BIAS=TRUE GND 12 13 15 1

9 GND MAKE_BIAS=TRUE GND 12 13 15 1

		SNC_DATE=18/02/2017	
PAGE TITLE			
Interposer: Pins 286-359			
		DRAWING NUMBER	SIZE
		051-02613	D
		REVISION	
		4.0.0	
NOTICE OF PROPRIETARY PROPERTY:		BRANCH	
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING:			
I. TO MAINTAIN THIS DOCUMENT IN CONFIDENCE			
II. NOT TO REPRODUCE OR COPY IT			
III. NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		PAGE	
IV. ALL RIGHTS RESERVED		85	OF 85
		SHEET	
		16	OF 49

8		7		6		5		4		3		2		1	
1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%. 2. ALL CAPACITANCE VALUES ARE IN MICROFARADS. 3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.												REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
												4	0011179651	ENGINEERING RELEASED	2018-02-06

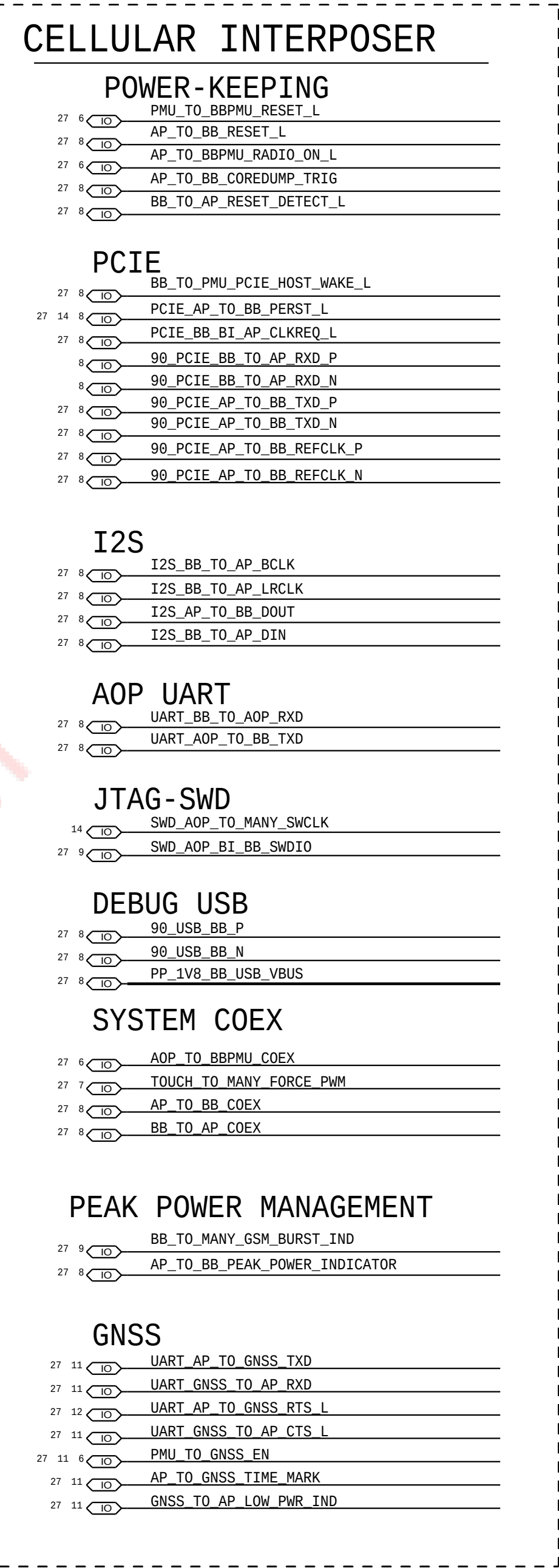
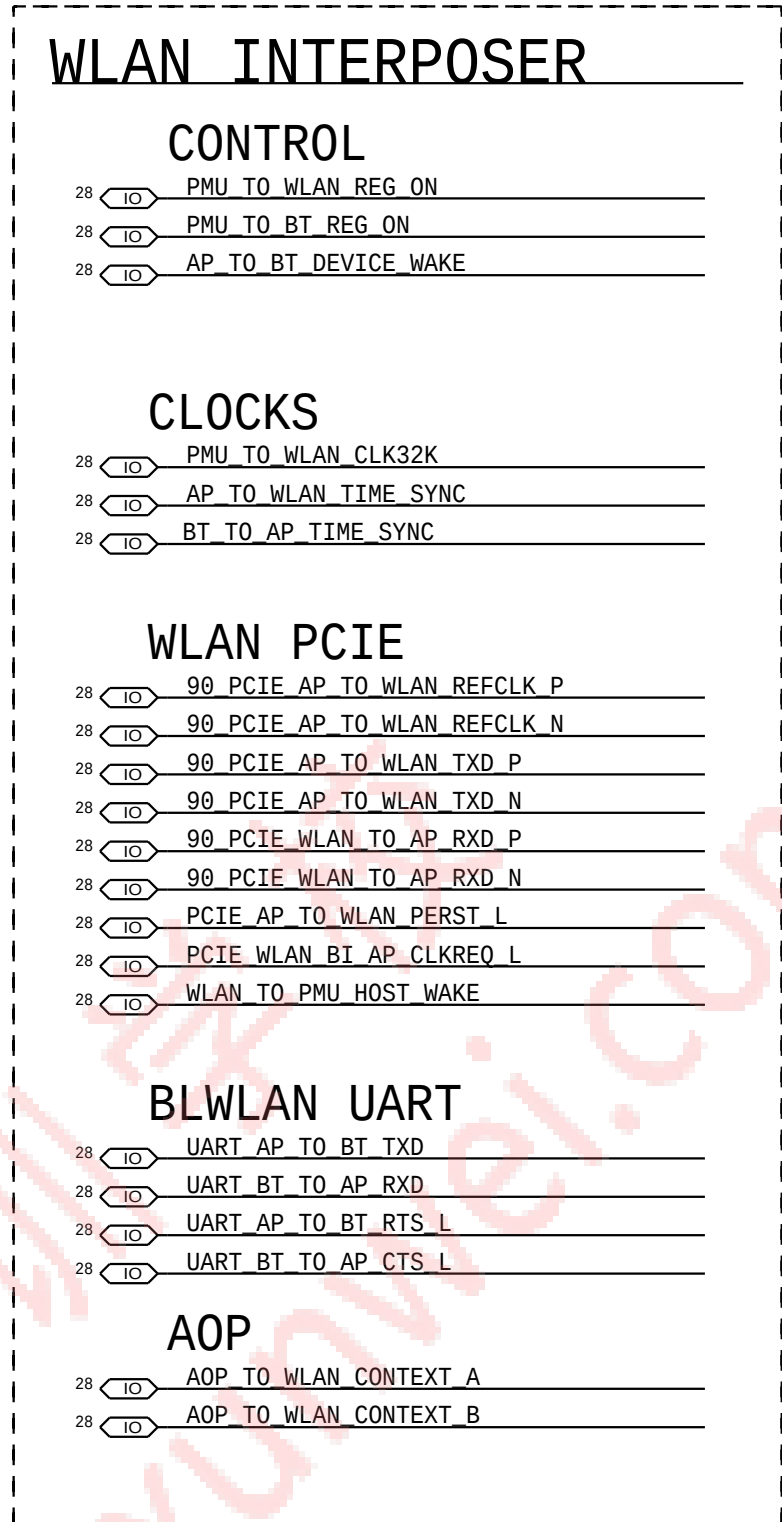
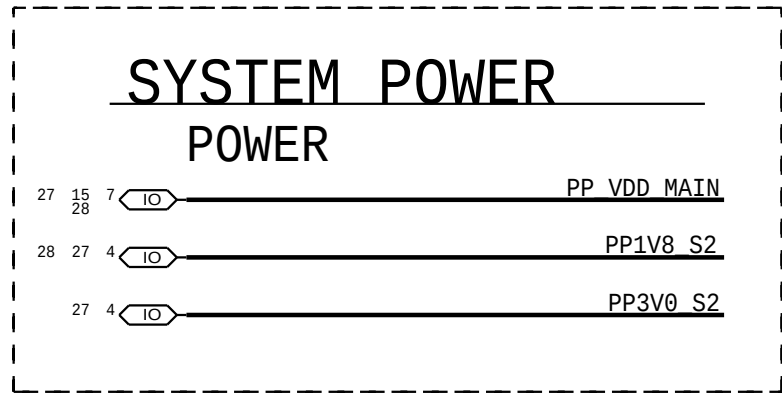
</

D

C

B

A



?

DRAWING TITLE			
SCH, MLB, BOT, ICE, D32			
		DRAWING NUMBER	051-02613
		REVISION	4.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AND AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	1 OF 28
		SHEET	17 OF 49

BOM TABLES

ROW SKU

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
118S0724	1	RES,MF,0.0 OHM,1/20W,0201,HIGH FREQ	R2016_K	ROW
118S0724	1	RES,MF,0.0 OHM,1/20W,0201,HIGH FREQ	R2003_K	ROW
152S2036	1	IND,FILM,2.5NH, +/-0.1NH,600MA,UH-Q,0201	R2015_K	ROW
118S0636	1	RES,MF,4.7K OHM,1/32W,01005	R312_K	ROW

JP SKU

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
152S2054	1	IND,FILM,9.1NH, +/-3%,300MA,UH-Q,0201	R2016_K	JP
118S0724	1	RES,MF,0.0 OHM,1/20W,0201,HIGH FREQ	R2002_K	JP
152S2042	1	IND,FILM,1.0NH, +/-0.1NH,600MA,UH-Q,0201	R1403_K	JP
117S0167	1	RES,MF,3.3K OHM,1/32W,01005	R312_K	JP

NA SKU

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
118S0724	1	RES,MF,0.0 OHM,1/20W,0201,HIGH FREQ	R2016_K	NA
118S0724	1	RES,MF,0.0 OHM,1/20W,0201,HIGH FREQ	R2003_K	NA
152S2036	1	IND,FILM,2.5NH, +/-0.1NH,600MA,UH-Q,0201	R2015_K	NA
118S00124	1	RES,MF,47K OHM,1/32W,01005	R312_K	NA

4FF + VINYL

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
512S00030	1	SIM,TALL,PTH,4X3,D33	J_SIM	VINYL
990-12938	1	VINYL,NIB,BIRCH 4.0	U_SIM	VINYL
117S0158	1	RES,MF,100K OHM,1/32W,01005	R550_K	VINYL
118S0636	1	RES,MF,4.7K OHM,1/32W,01005	R30_SIM	VINYL

SINGLE 4FF

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	BOM OPTION
512S00030	1	SIM,TALL,PTH,4X3,D33	J_SIM	SINGLE
117S0158	1	RES,MF,100K OHM,1/32W,01005	R551_K	SINGLE

ETIC CAP ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S00167	138S00041	BOM_TABLE_ALTS	C1204_K	CAP,2.2UF
138S00167	138S00041	BOM_TABLE_ALTS	C1205_K	CAP,2.2UF
138S00167	138S00041	BOM_TABLE_ALTS	C1206_K	CAP,2.2UF

CAP ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
138S0031	138S00049	BOM_TABLE_ALTS	ALL	CAP,2.2UF
138S00006	138S00084	BOM_TABLE_ALTS	ALL	CAP,20UF
138S0719	138S1103	BOM_TABLE_ALTS	ALL	CAP,4.7UF
138S00128	138S00133	BOM_TABLE_ALTS	ALL	CAP,0.47UF

EEPROM ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
335S00013	335S00094	BOM_TABLE_ALTS	EEPROM_K	EEPROM,8KBIT,I2C

RULER_RULE_SET=MSAP_2017		MANUFACTURING CONFIGURATION						
SPACING MULTIPLES	DIELECTRIC MULTIPLES	PAD TO X RATIO	VIA TO X RATIO	VOID SPACE RATIO	MECH HOLE ASPECT RATIO	LAZER HOLE ASPECT RATIO	CURRENT DENSITY (AMPS/SQR MM)	BOARD SIZE (LONGEST SIDE LENGTH MM)
1	1.65, 2, 2D, 3.2D, 4D	?	?		?	?	2	? 50
LAYERS				MINIMUM CU WIDTH RATIO		MINIMUM CU SPACING RATIO		MINIMUM TO DEFAULT RATIO
TOP,BOTTOM,ISL4,ISL5				1.0		1.0		1.0
ISL2,ISL3,ISL6,ISL7				1.0		1.0		1.0

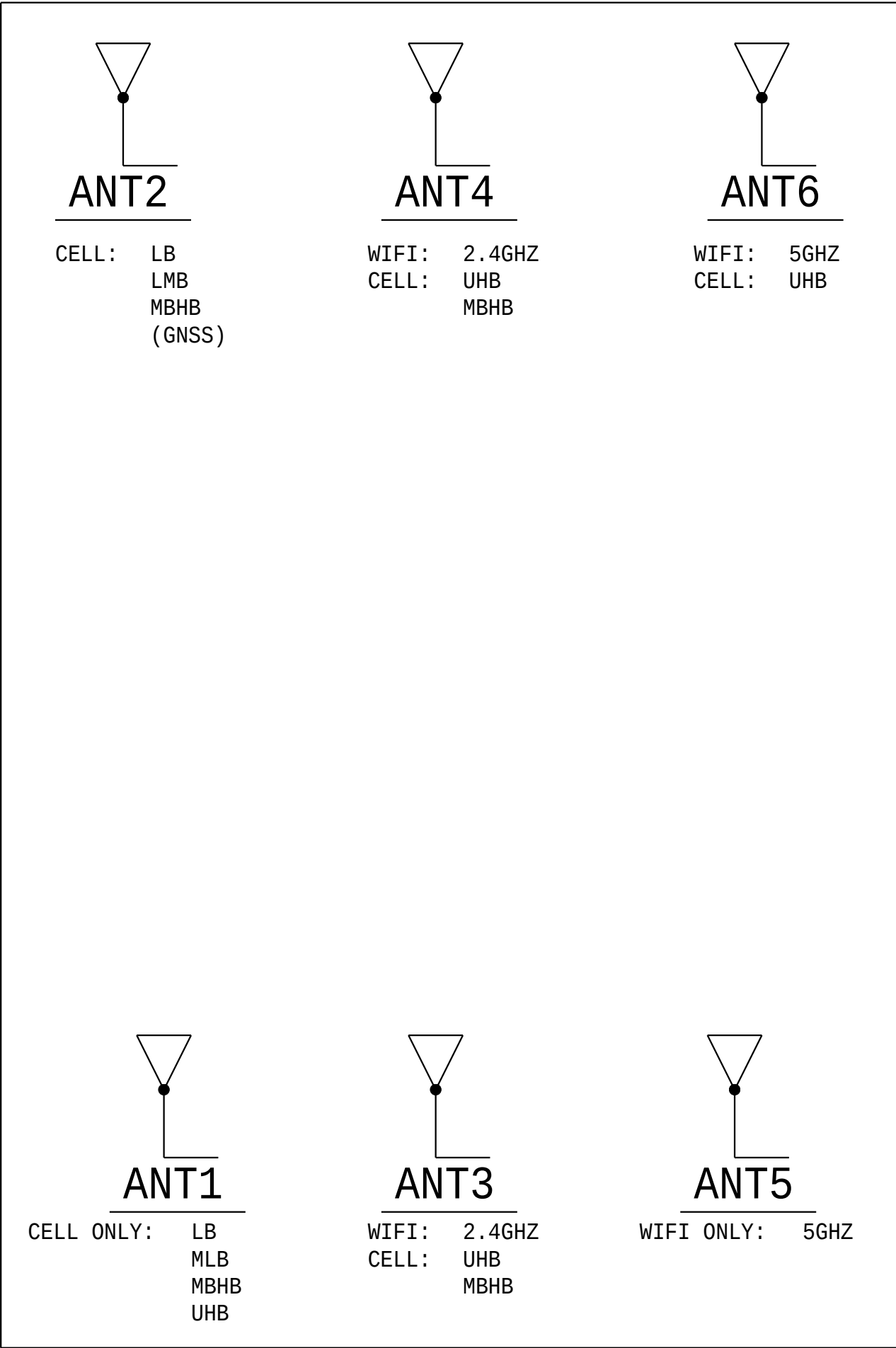
LOAD_CURRENT=50
LOAD_CURRENT=150
LOAD_CURRENT=500
LOAD_CURRENT=1000
LOAD_CURRENT=3000

TABLE_IMPEDANCE_HYBRID=50_HYBRID	
LAYERS	IMPEDANCE RULE
TOP	50_OHM_SE
ISL3	50_OHM_SE
ISL6	50_OHM_SE_6R4R8
ISL7	50_OHM_SE
ISL5	50_OHM_SE

TABLE_IMPEDANCE_HYBRID=50_HYBRID_WIDE	
LAYERS	IMPEDANCE RULE
TOP	50_OHM_SE_1R3
ISL3	50_OHM_SE
ISL6	50_OHM_SE_6R4R8
ISL7	50_OHM_SE
ISL5	50_OHM_SE

PAGE TITLE			
BOM TABLES			
		DRAWING NUMBER	051-02613
		REVISION	4.0.0
		BRANCH	
		PAGE	2 OF 28
		SHEET	18 OF 49
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			

RADIO ANTENNAS



UAT

LAT

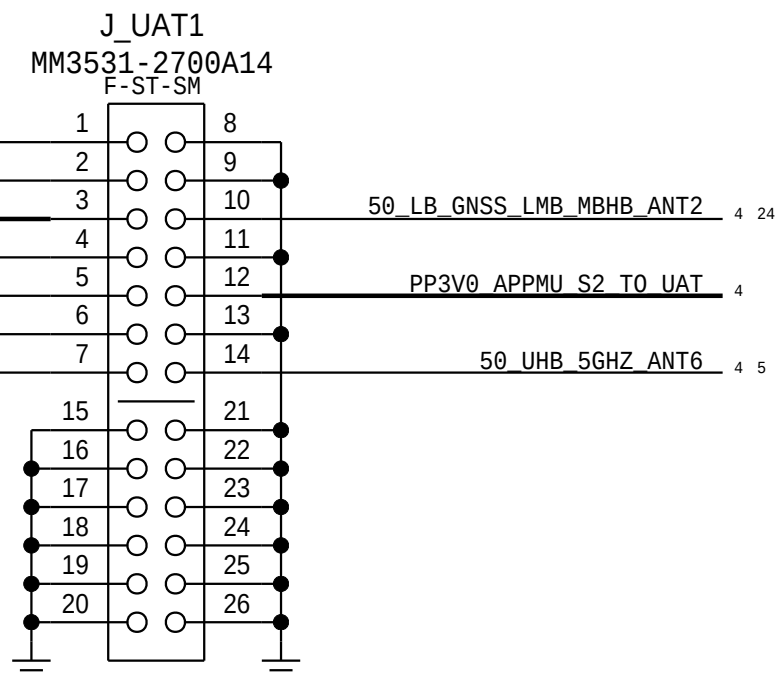
PAGE TITLE			ANTENNA DIAGRAM		
		DRAWING NUMBER	051-02613	SIZE	D
		REVISION	4.0.0		
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH			
		PAGE			
		3 OF 28			
		SHEET			
		19 OF 49			

ANTENNA B2BS

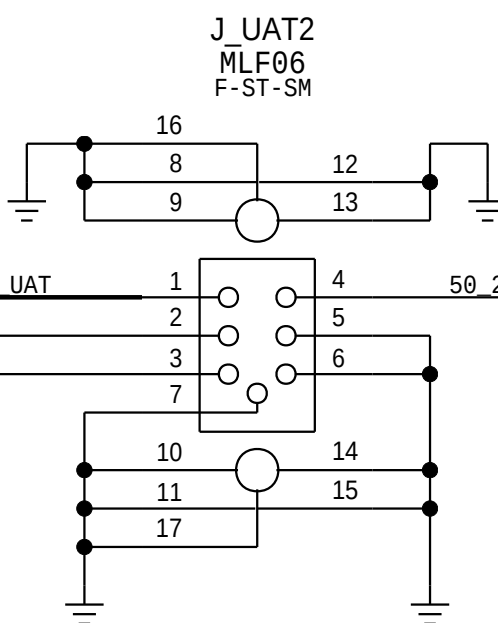
UAT

D3X UAT B2BS

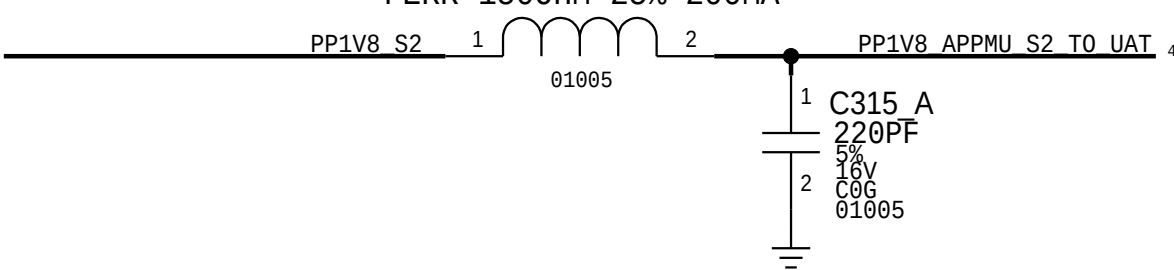
RCPT ON MLB_B0T (518S00167)
PLUG ON FLEX (518S00166)



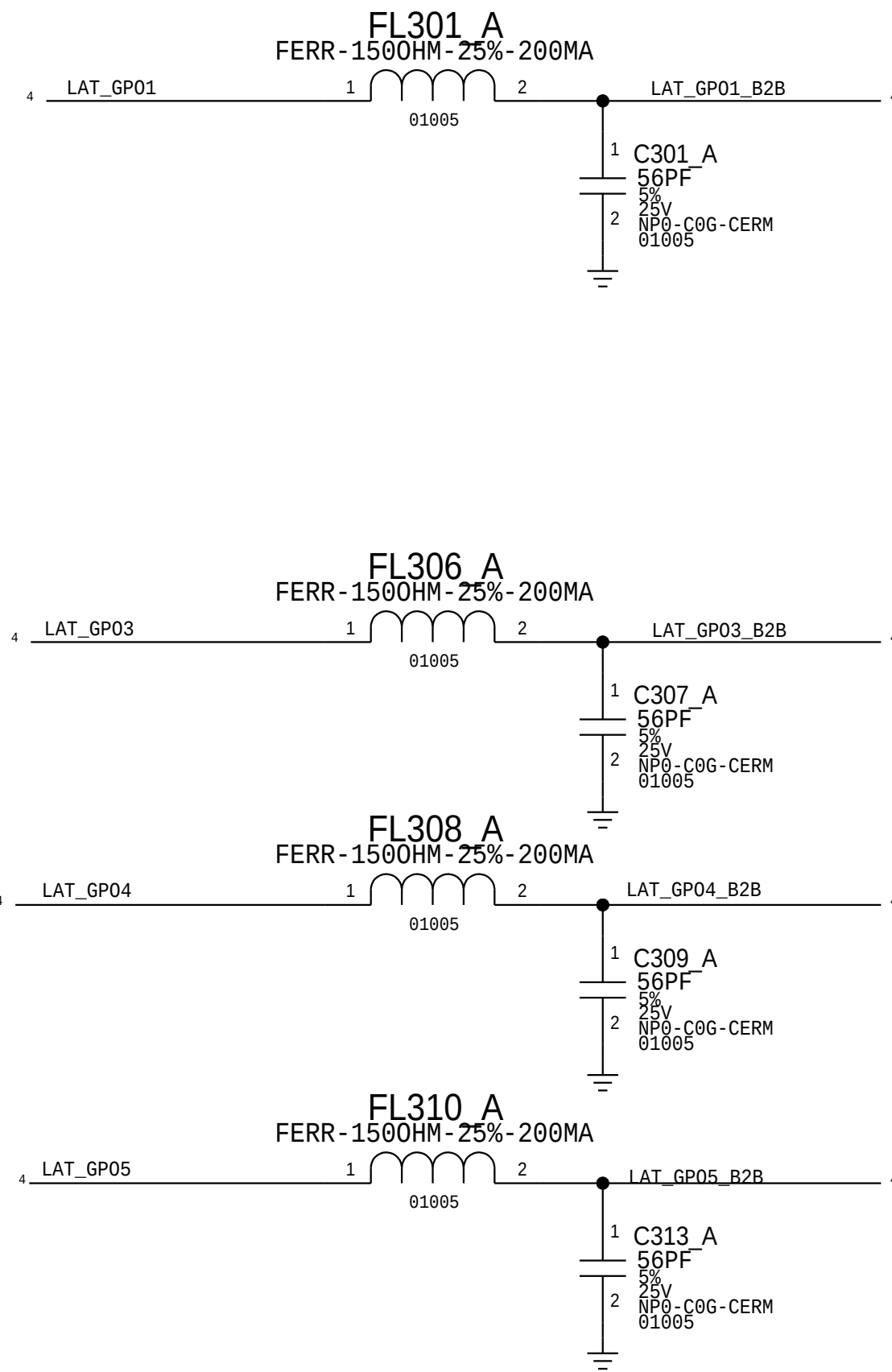
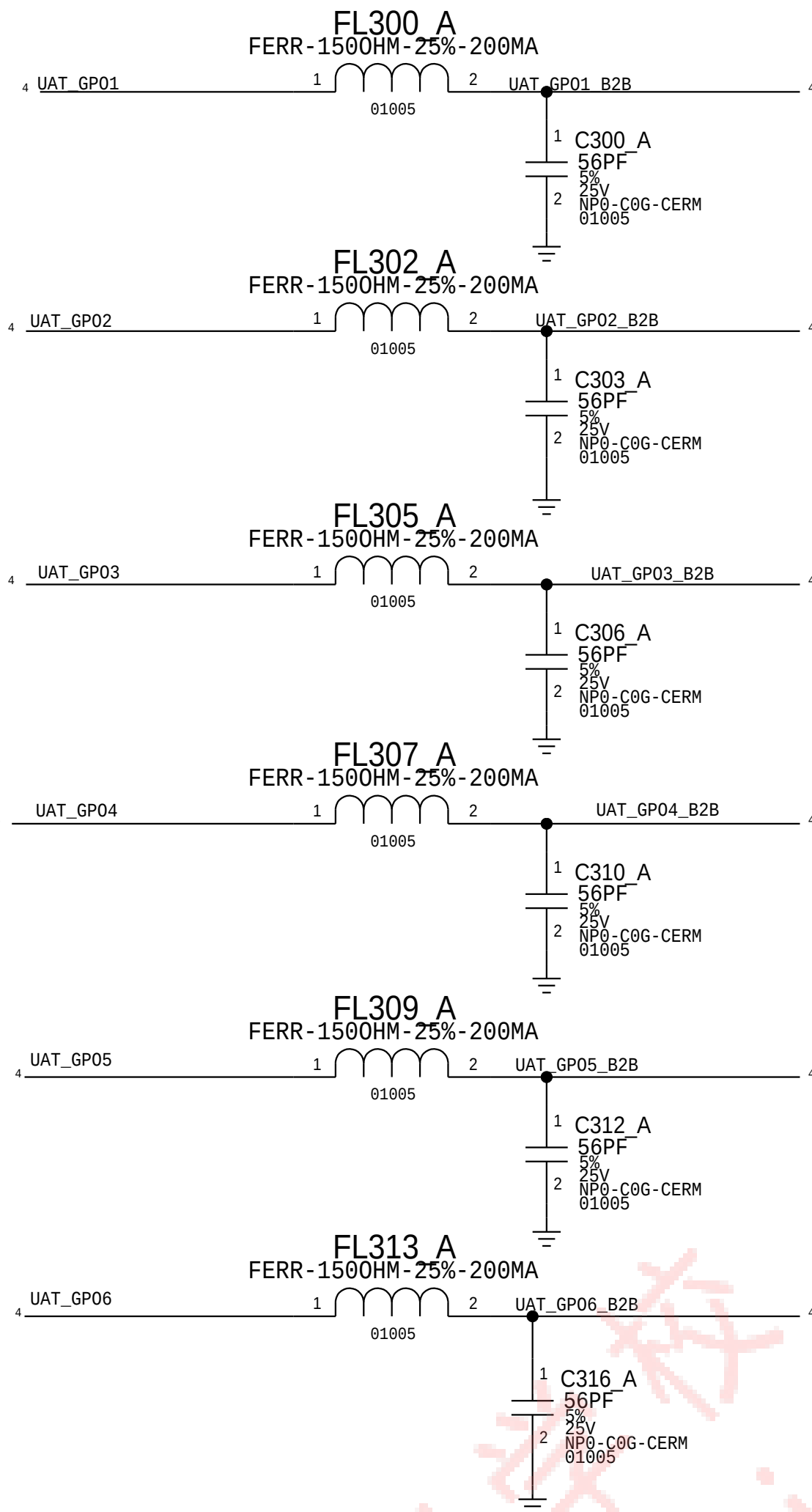
RCPT ON MLB_B0T (516S00434)
PLUG ON FLEX (516S00435)



FL312 A
FERR-1500HM-25%-200MA



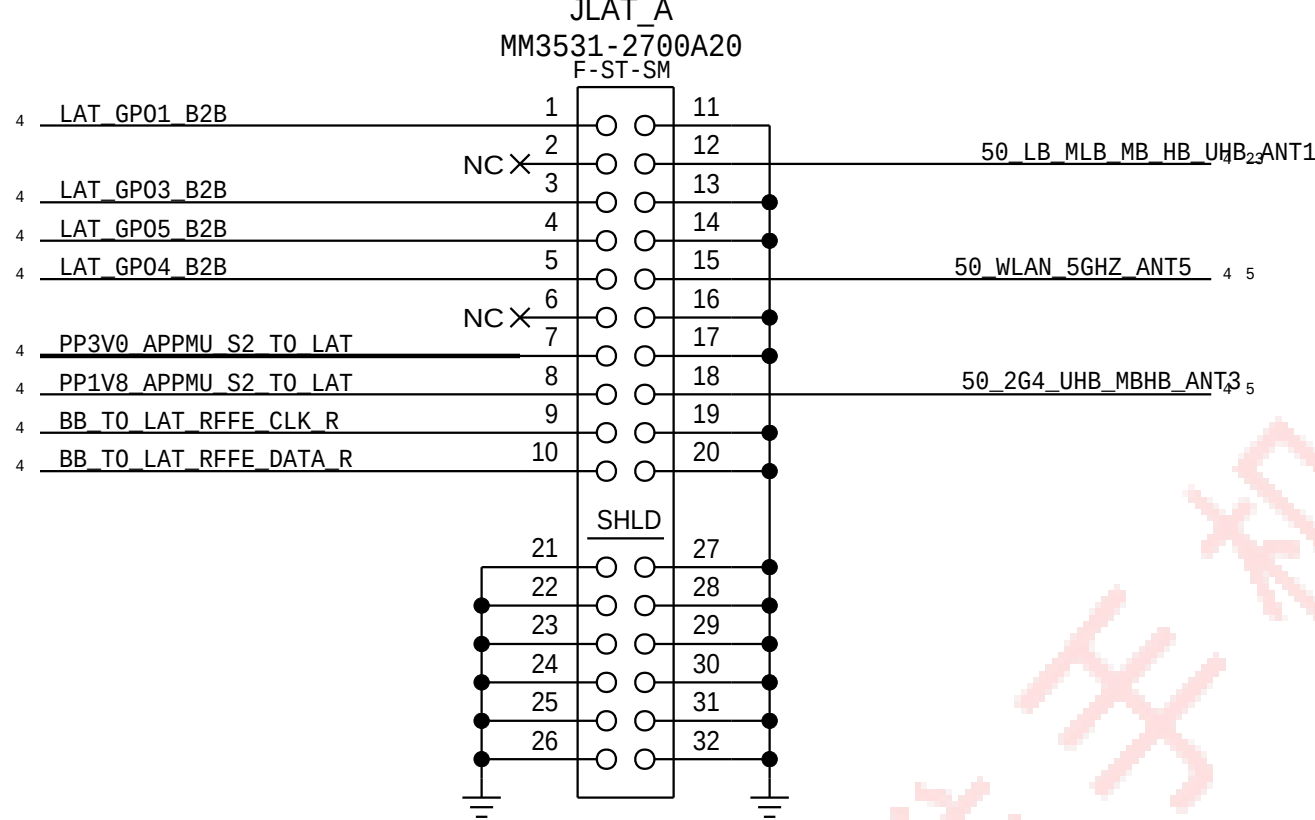
GPO FILTERS



LAT

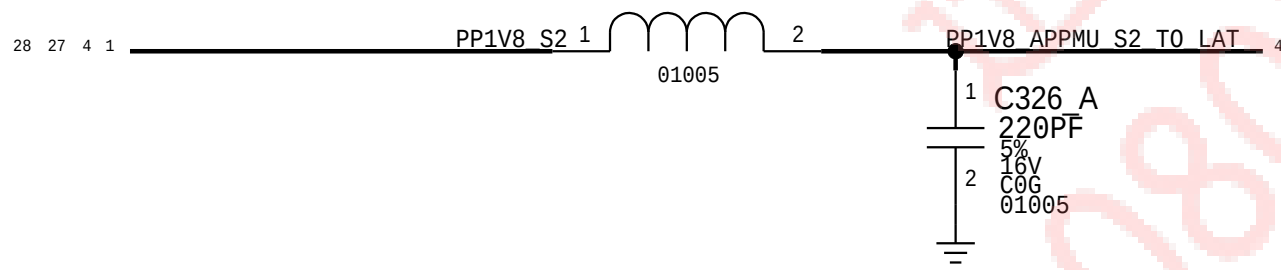
D32 LAT B2B

RCPT ON MLB_B0T (518S00149)
PLUG ON FLEX (518S00150)

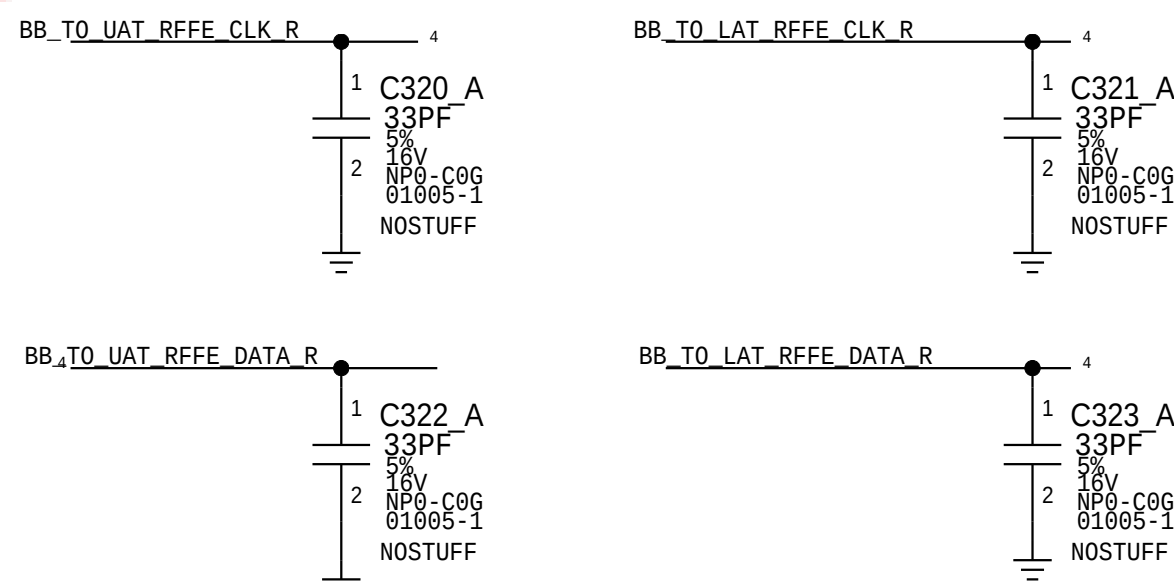


D32 ONLY

FL316 A
FERR-1500HM-25%-200MA



RFFE FILTERS



ANTENNA CONNECTIONS

50_LB_MLB_MB_HB_UHB_ANT1	50_LB_MLB_MB_HB_UHB_ANT1	23
MAKE_BASE=TRUE		
50_LB_GNSS_LMB_MBHB_ANT2	50_LB_GNSS_LMB_MBHB_ANT2	24
MAKE_BASE=TRUE		
50_2G4_UHB_MBHB_ANT3	50_2G4_UHB_MBHB_ANT3	5
MAKE_BASE=TRUE		
50_2G4_UHB_MBHB_ANT4	50_2G4_UHB_MBHB_ANT4	5
MAKE_BASE=TRUE		
50_WLAN_5GHZ_ANT5	50_WLAN_5GHZ_ANT5	5
MAKE_BASE=TRUE		
50_UHB_5GHZ_ANT6	50_UHB_5GHZ_ANT6	5
MAKE_BASE=TRUE		

PAGE TITLE		
ANTENNA: B2BS		
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	DRAWING NUMBER	051-02613
	REVISION	4.0.0
PAGE	BRANCH	
	SHEET	20 OF 49

SHARED N-PLEXERS

UAT

ANT4

ANT6

MHB-UHB/2.4G WLAN TRIPLEXER2

UHB-5G WLAN DIPLEXER3

ANT3

LAT

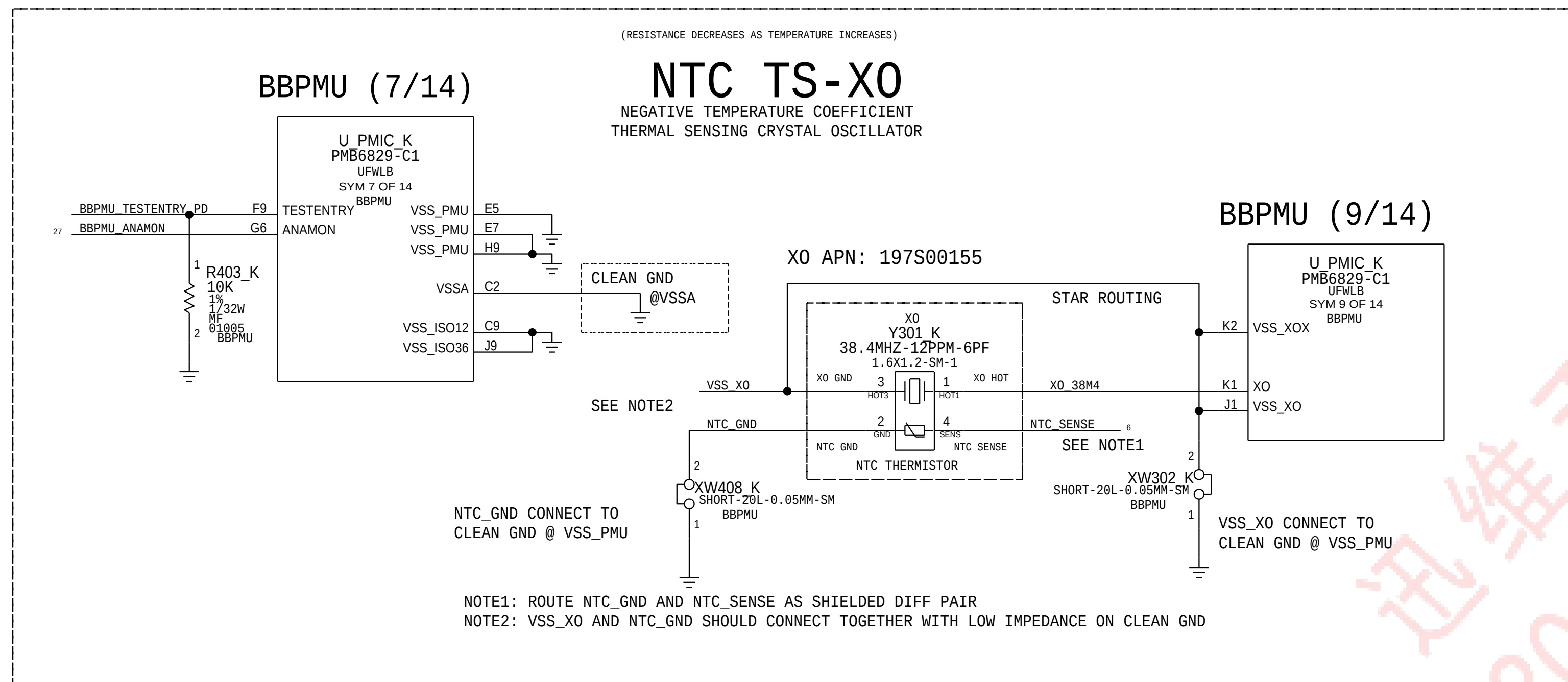
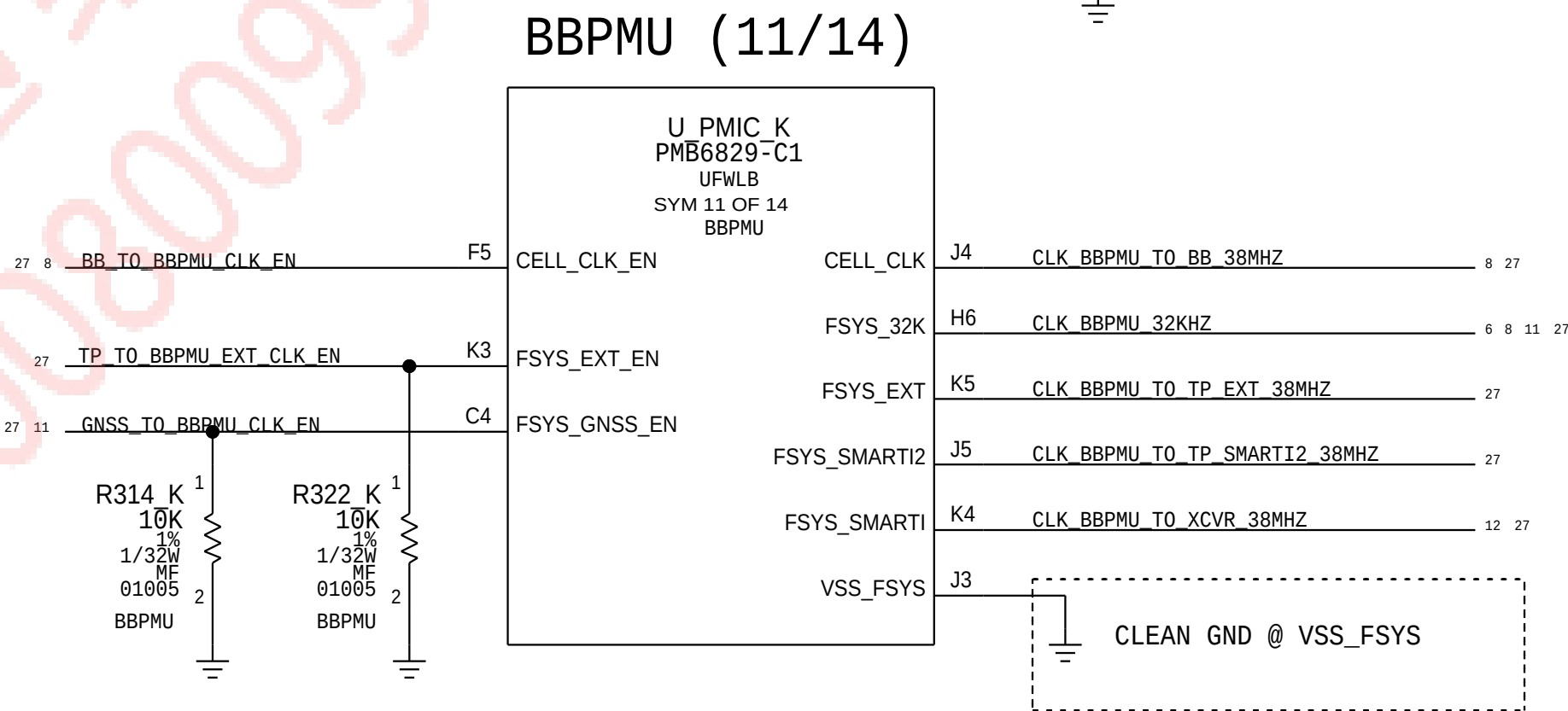
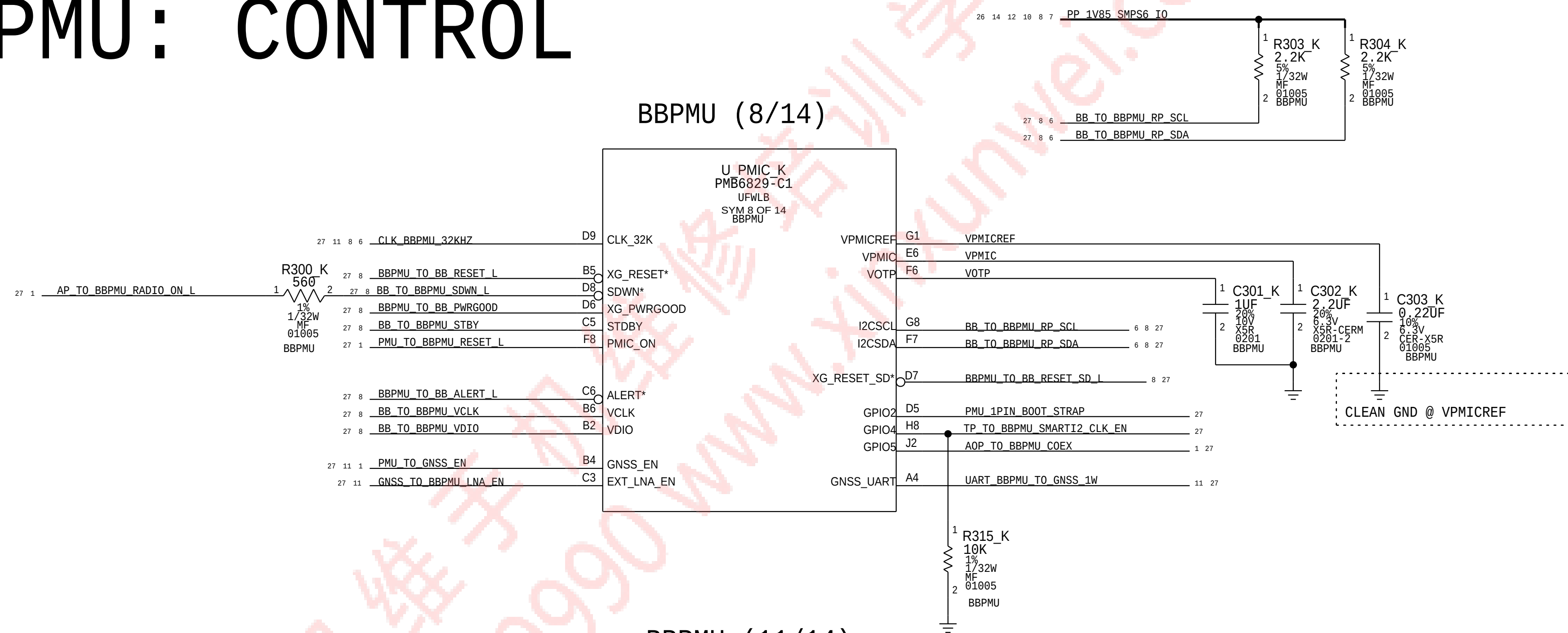
ANT5

MHB-UHB/2.4G WLAN TRIPLEXER2

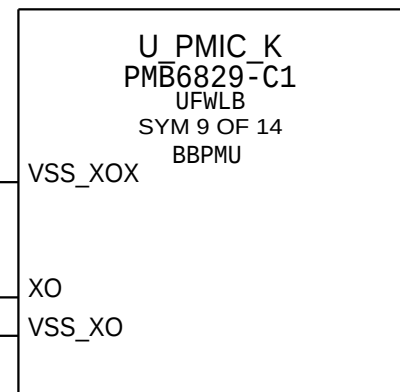
5GHZ WLAN BPF

PAGE TITLE		
ANTENNA: N-PLEX SHARED		
DRAWING NUMBER		051-02613
REVISION		4.0.0
BRANCH		
PAGE		5 OF 28
SHEET		21 OF 49
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR PUBLISH IT I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART I ALL RIGHTS RESERVED		

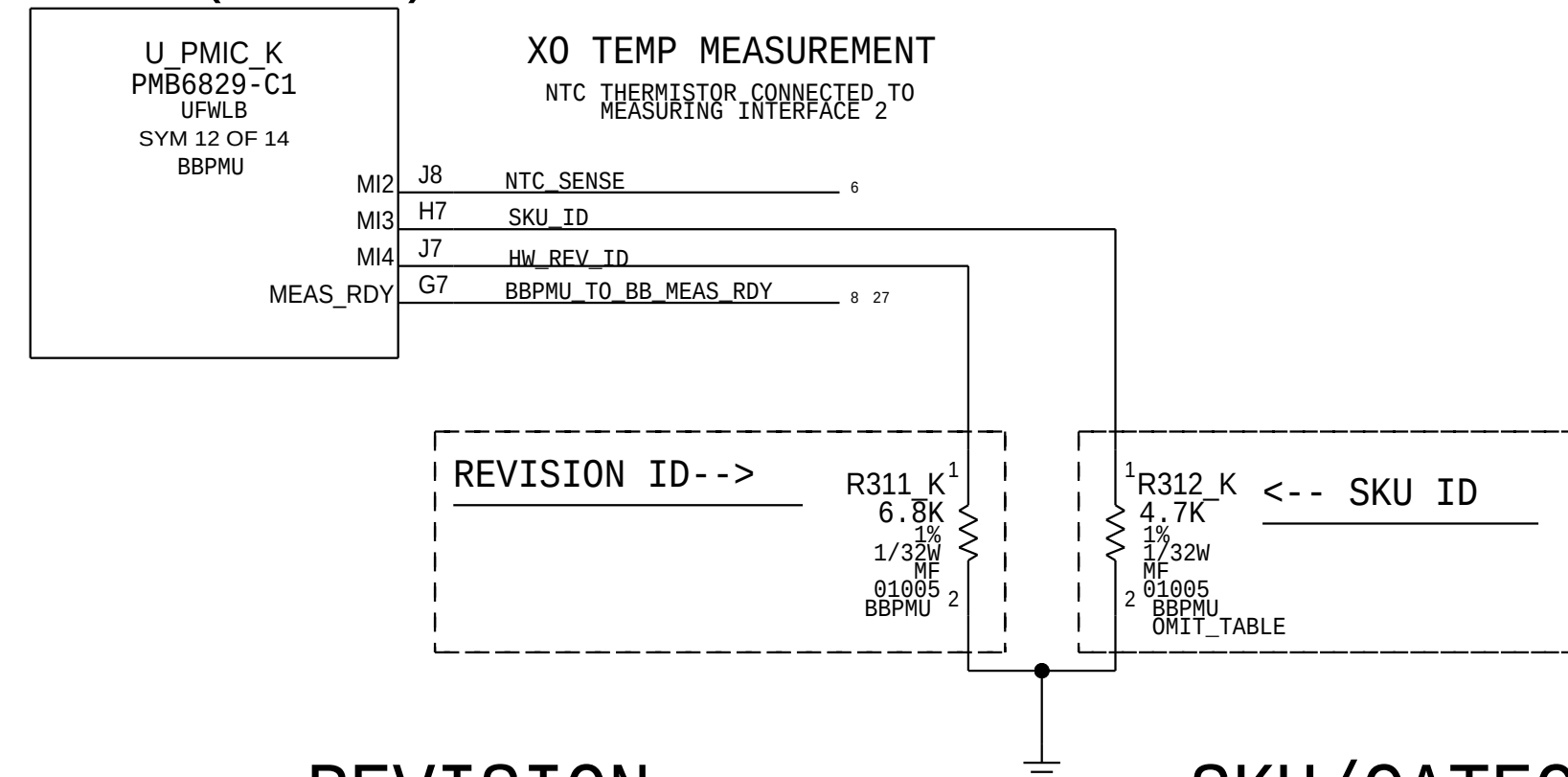
BBPMU: CONTROL



BBPMU (9/14)



BBPMU (12/14)



REVISION

R311_K	MLB	RF DEV
0.0	T/POC/PRE-PROT00	1.0
1.2K	P0	1.1
2.2K	P1	2.0
3.3K	P2	2.1
4.7K	P2 WITH S7.1	3.0
6.8K	EVT	
8.2K	EVT 1.5	4.0
10K	CARRIER	
12K	DVT	5.0
15K	PVT	5.1
18K		
22K		6.0
27K		6.0
33K		
39K		
47K		
56K	DARWIN	
68K	JP_SKWS	
82K		
100K	P1 DOE2, TXC	
120K	P1 DOE1, S7 B1	
150K		

SKU/CATEGORY

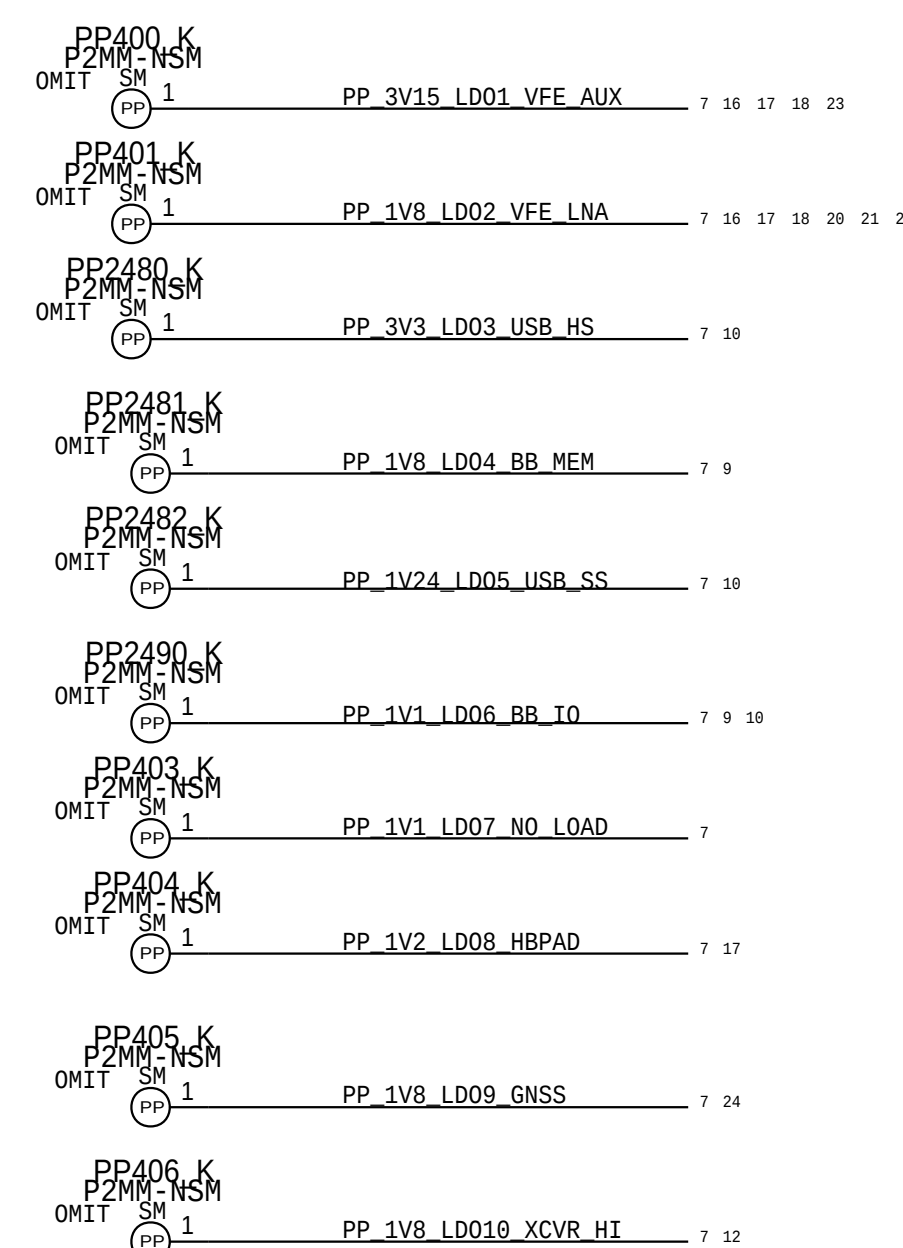
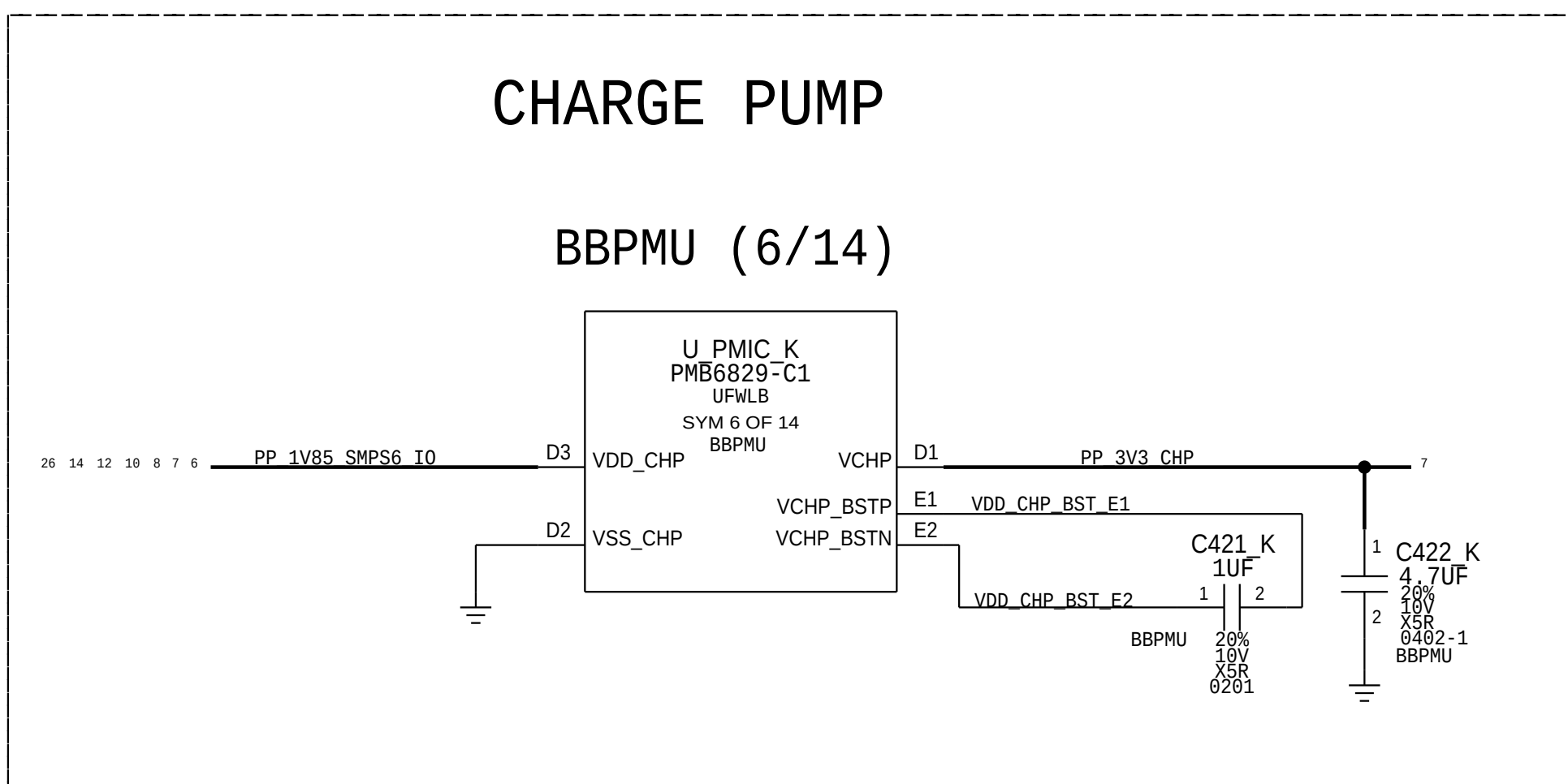
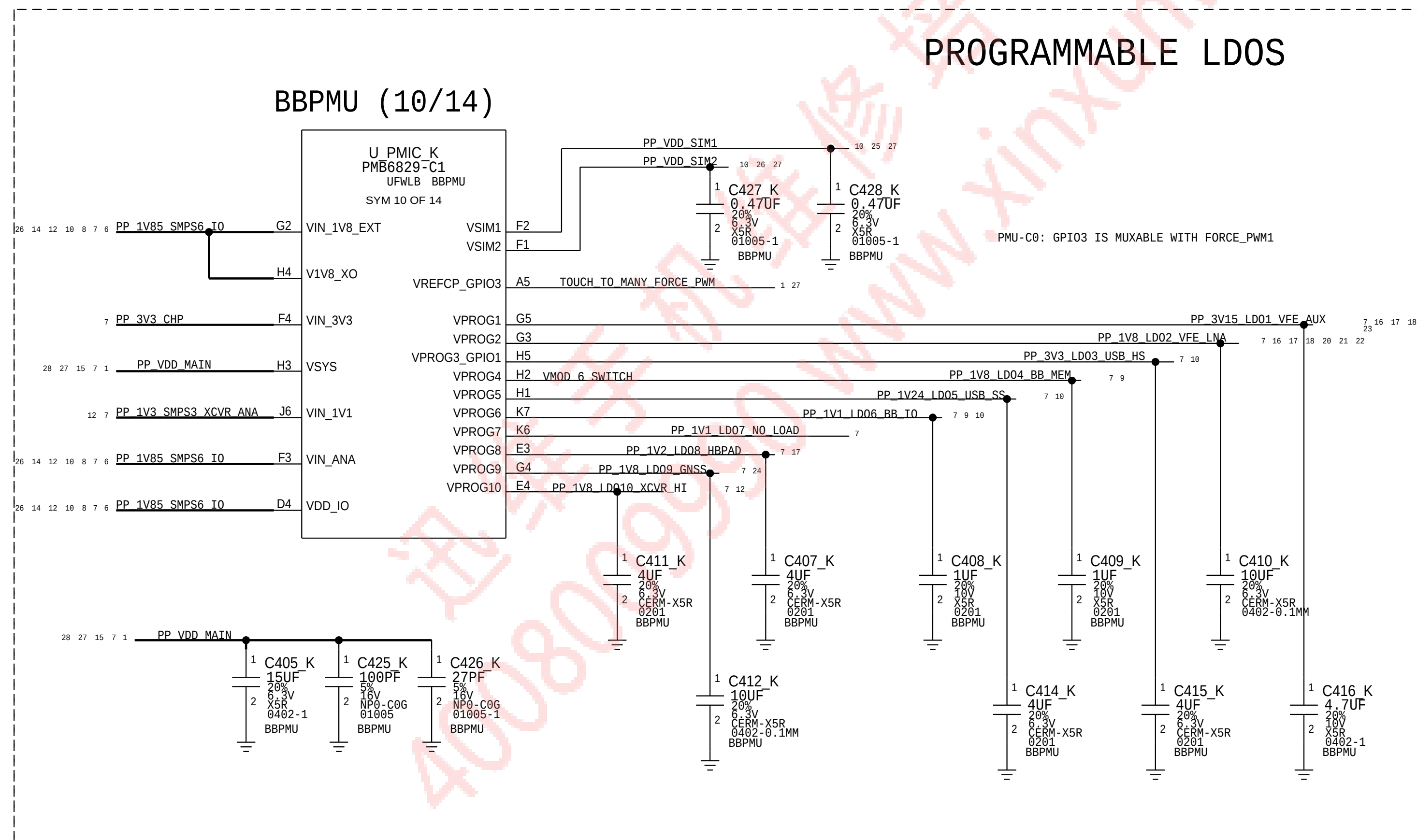
R312_K	HW REVISION	X-CODE
0.0	RFDEV JP	
1.2K	ICE 18.0 JP	
2.2K	ICE 18.0 ROW	X1344 (ICE 18.0)
3.3K	ICE 18.1 JP	
4.7K	ICE 18.1 ROW	X1049 (ICE 18.1)
6.8K	ICE 18.2 JP	
8.2K	ICE 18.2 ROW	X1210 (ICE 18.2)
10K	RFDEV ROW	
12K	ICE 18.5 JP	X1170
15K	ICE 18.6 JP	X1176
18K		
22K	ICE 18.5 US	X1170
27K	ICE 18.6 US	X1176
33K	ICE 18.0 NA	X1344
39K	RFDEV NA	
47K	ICE 18.1 NA	X1049
56K	ICE 18.2 NA	X1210
82K	KAR00 INTERNAL	
100K	KAR00 INTERNAL	
120K	KAR00 INTERNAL	
150K		

HWID TABLE RDAR://32880011
ADC TABLE RDAR://27081897

BBPMU ADC TABLES

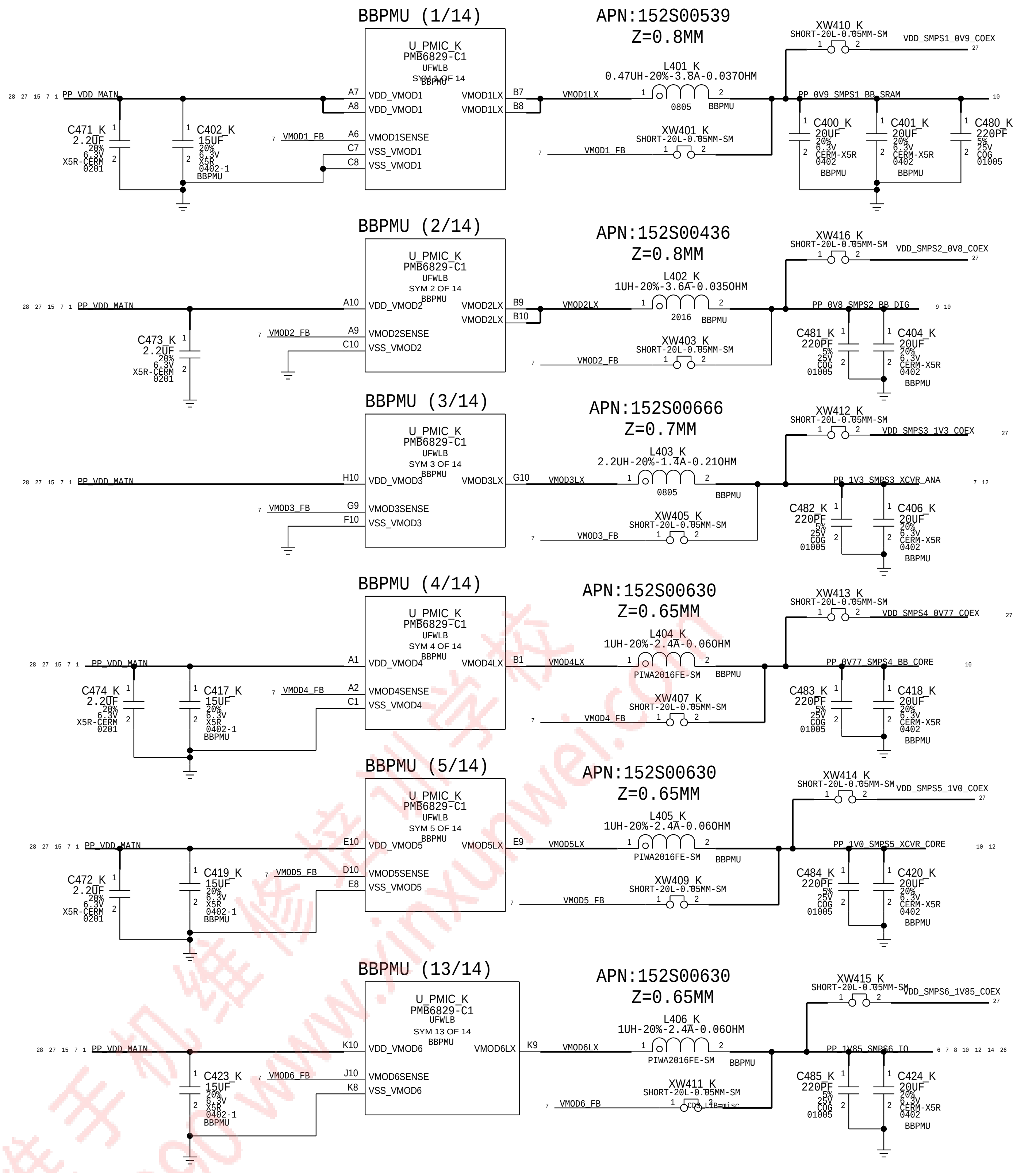
PAGE TITLE					
BBPMU: CONTROL					
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	DRAWING NUMBER	051-02613	SIZE	D	
	REVISION			4.0.0	
	BRANCH				
	PAGE			6 OF 28	
	SHEET			22 OF 49	

BBPMU: RAILS



SWITCHERS

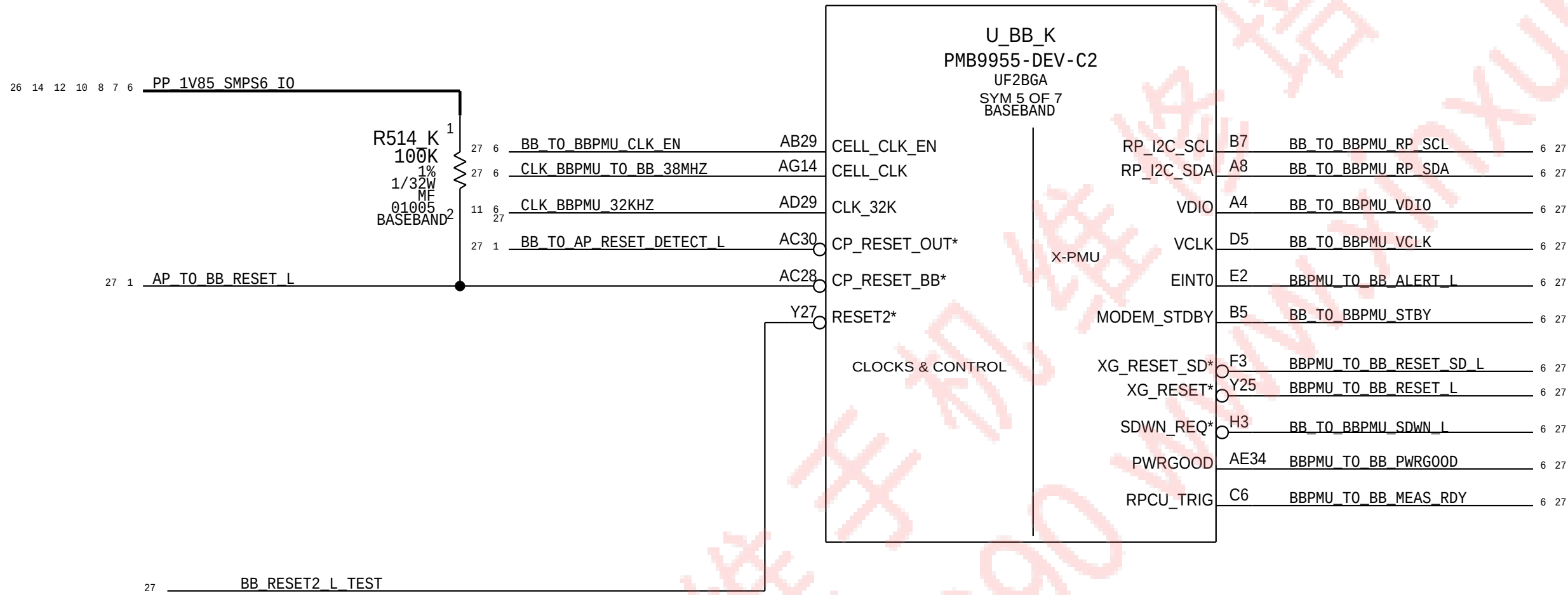
SMPS1/2/4	SHARED	VDD_MAIN	INTERNALLY
SMPS3/5/6	SHARED	VDD_MAIN	INTERNALLY



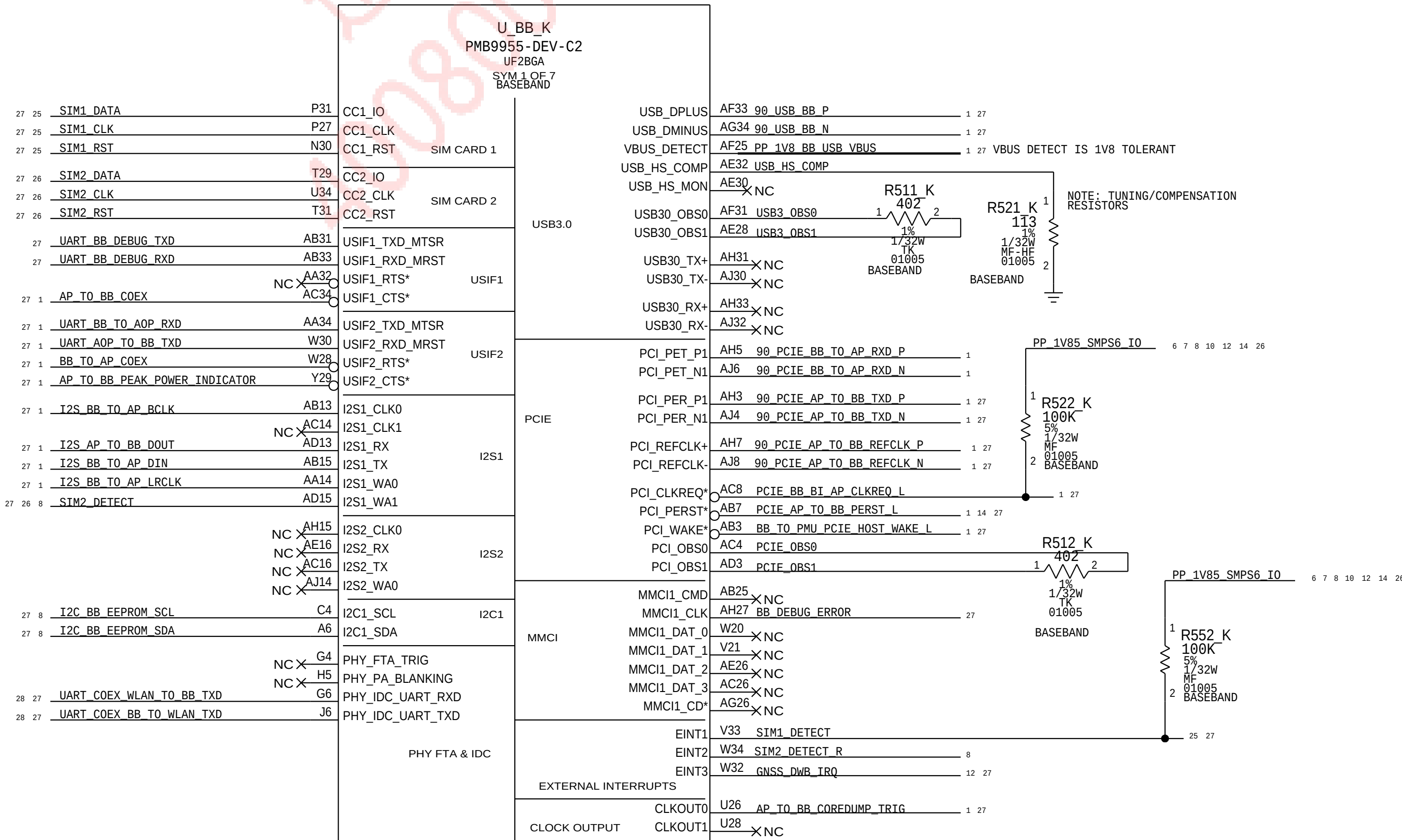
PAGE TITLE		BBPMU: RAILS	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I. TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II. NOT TO REPRODUCE OR COPY IT III. NOT TO BE REVEALED OR PUBLISHED IN WHOLE OR PART IV. ALL RIGHTS RESERVED		DRAWING NUMBER	051-02613
		SIZE	D
		REVISION	4.0.0
		BRANCH	
		PAGE	7 OF 28
		SHEET	23 OF 49

BB: INTERFACE

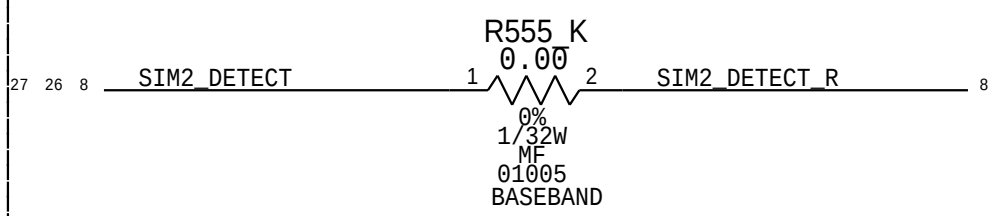
XG756 (5/7)



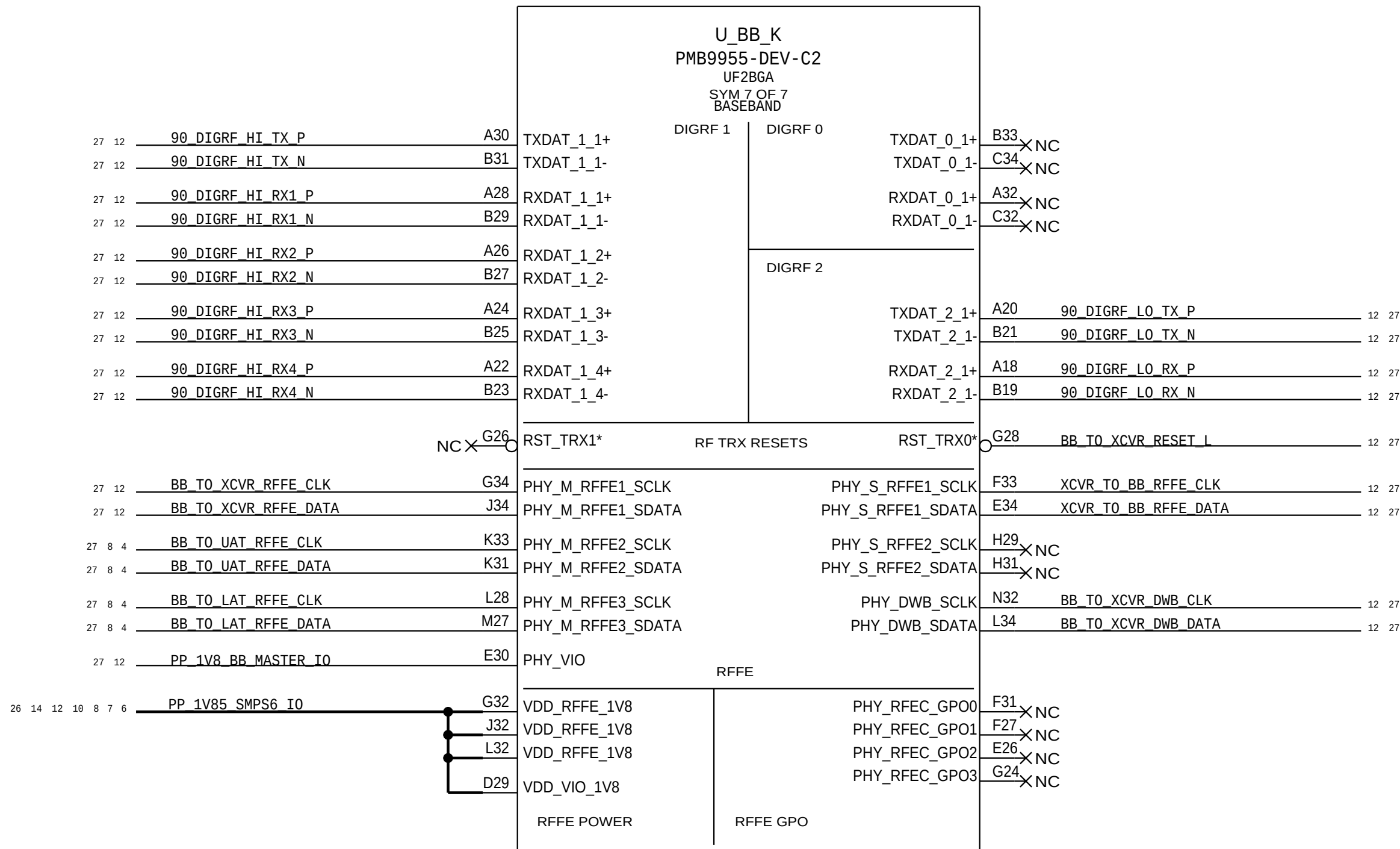
XG756 (1/7)



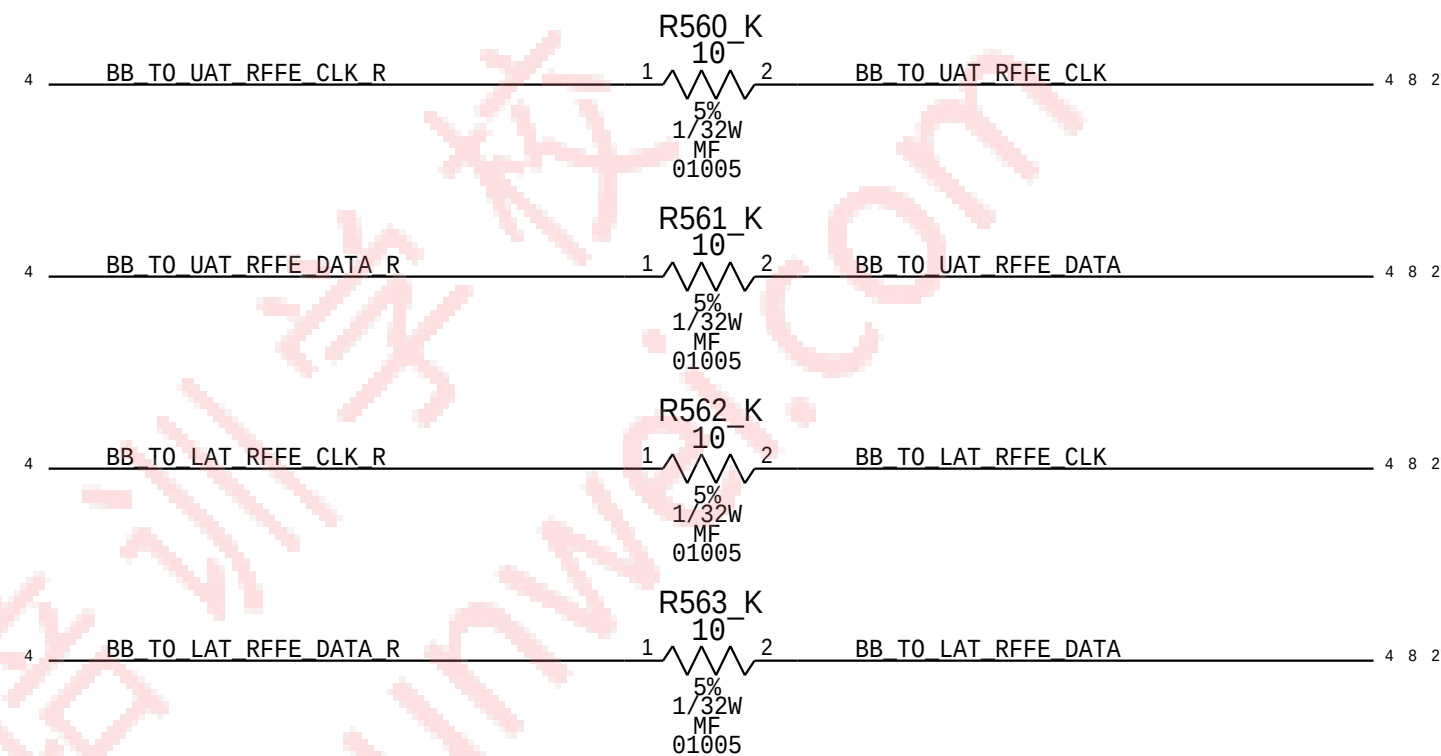
SIM2_DETECT TO EINT2



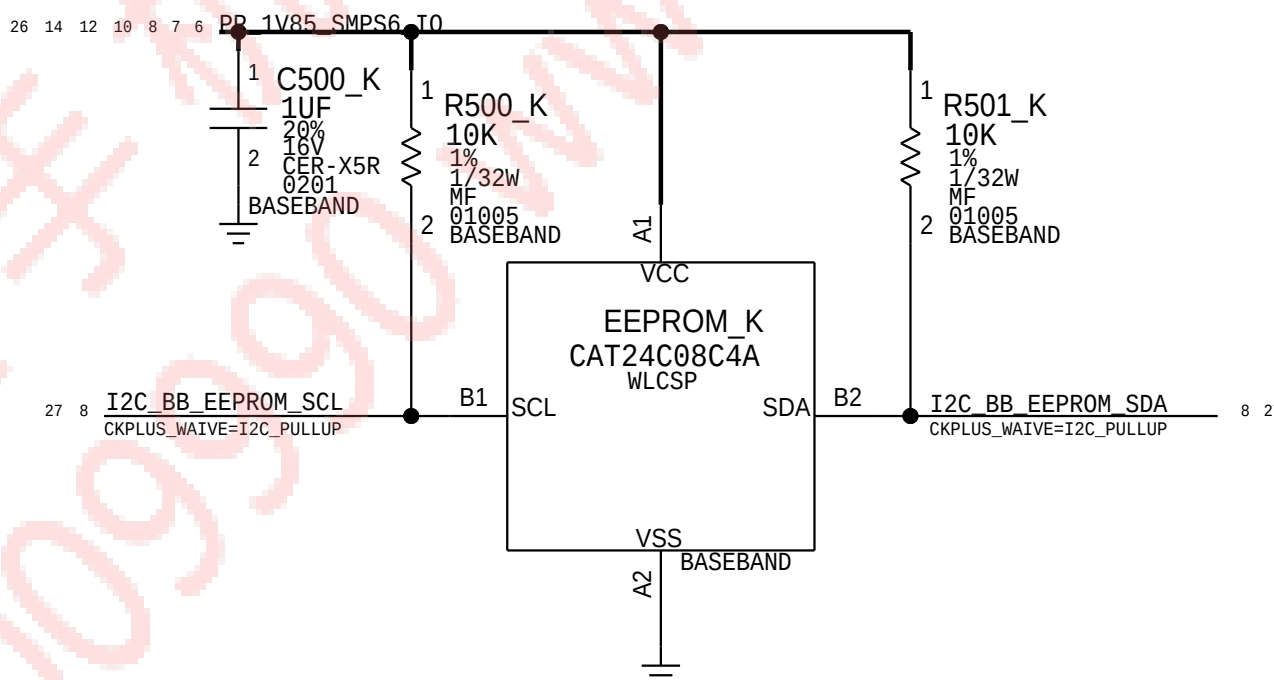
XG756 (7/7)



BB RFFE DAMPING RESISTORS



BB EEPROM



PAGE TITLE		
BB: INTERFACE		
DRAWING NUMBER	051-02613	SIZE
REVISION	4.0.0	D
NOTICE OF PROPRIETARY PROPERTY:		
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING:		
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		
I NOT TO REPRODUCE OR COPY IT		
I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		
I ALL RIGHTS RESERVED		
BRANCH		
PAGE	8 OF 28	
SHEET	24 OF 49	

BB: DDR PWR & JTAG

HOOKS FOR DDR4 TESTING/CAL

XG756 (2/7)

XG756 (6/7)

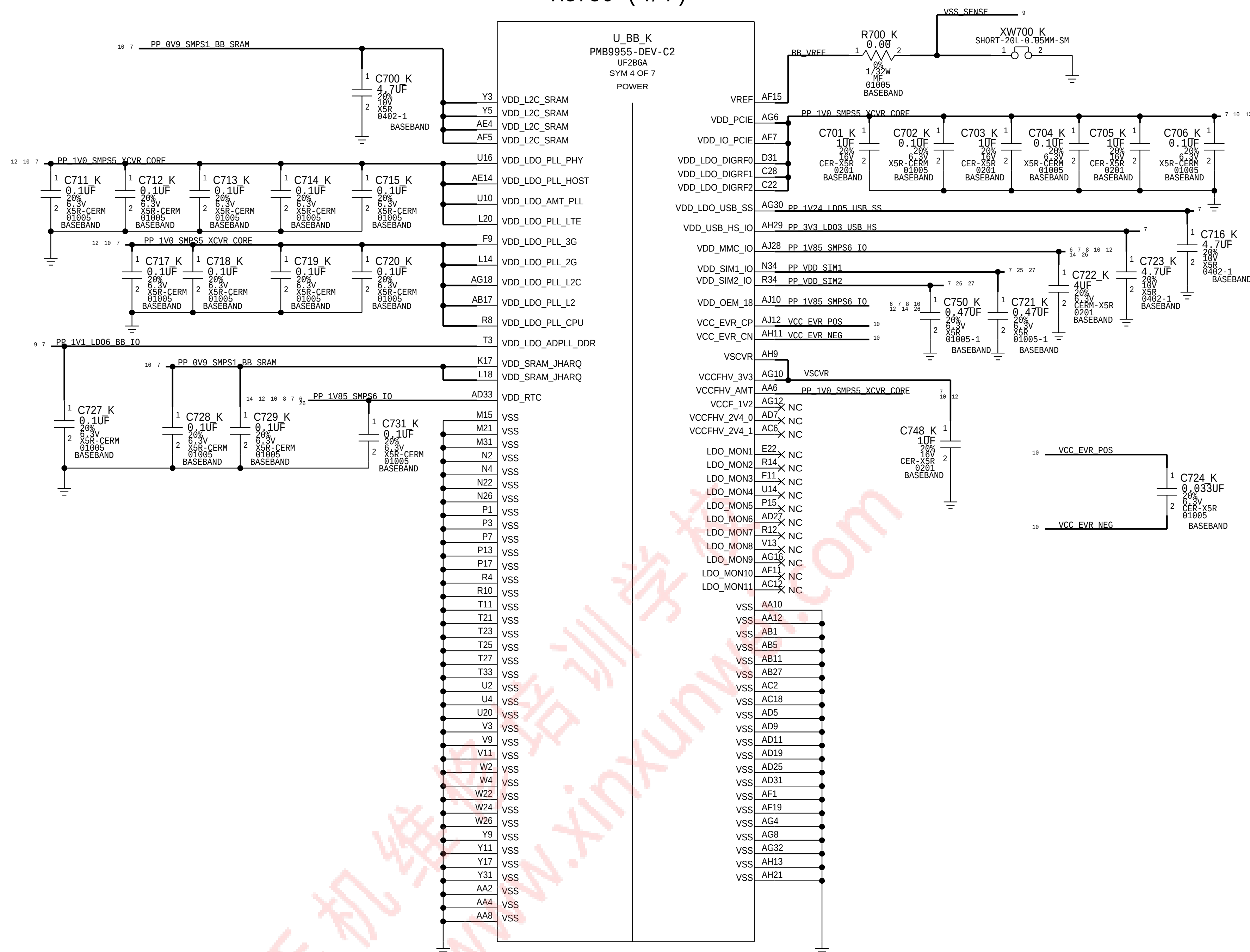
SWD, SEE PAGE 14

PAGE TITLE		
BB: DDR PWR & JTAG		
	DRAWING NUMBER	051-02613
	REVISION	4.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH	
	PAGE	9 OF 28
	SHEET	25 OF 49

XG756 (3/7)



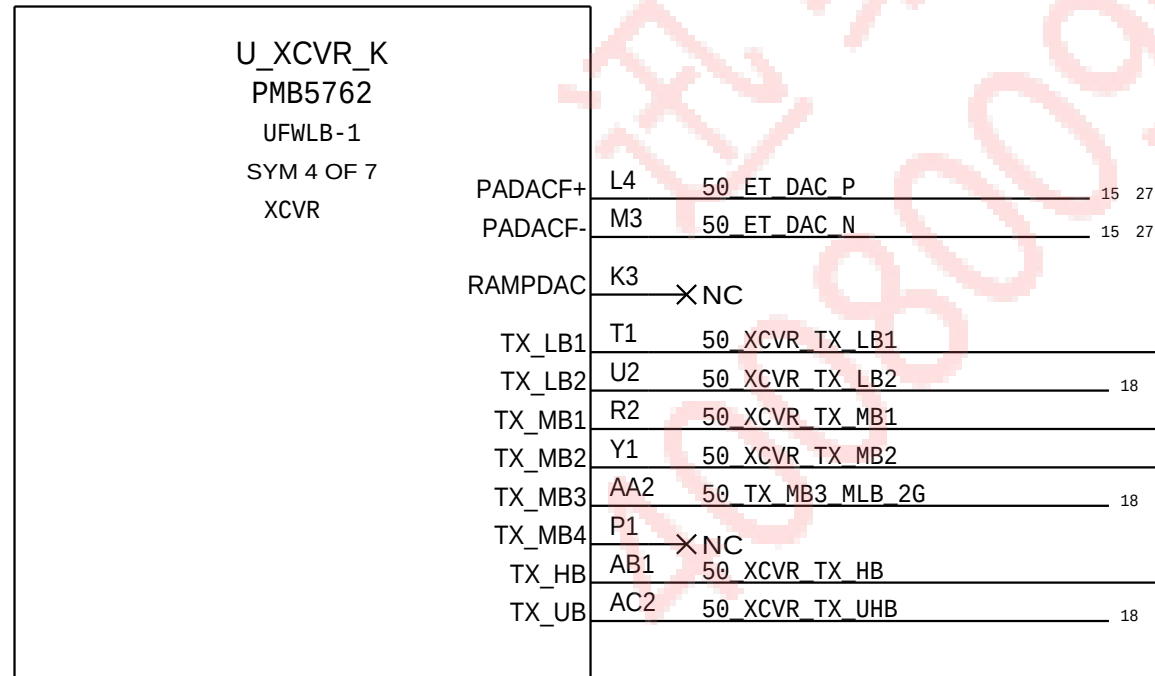
XG756 (4/7)



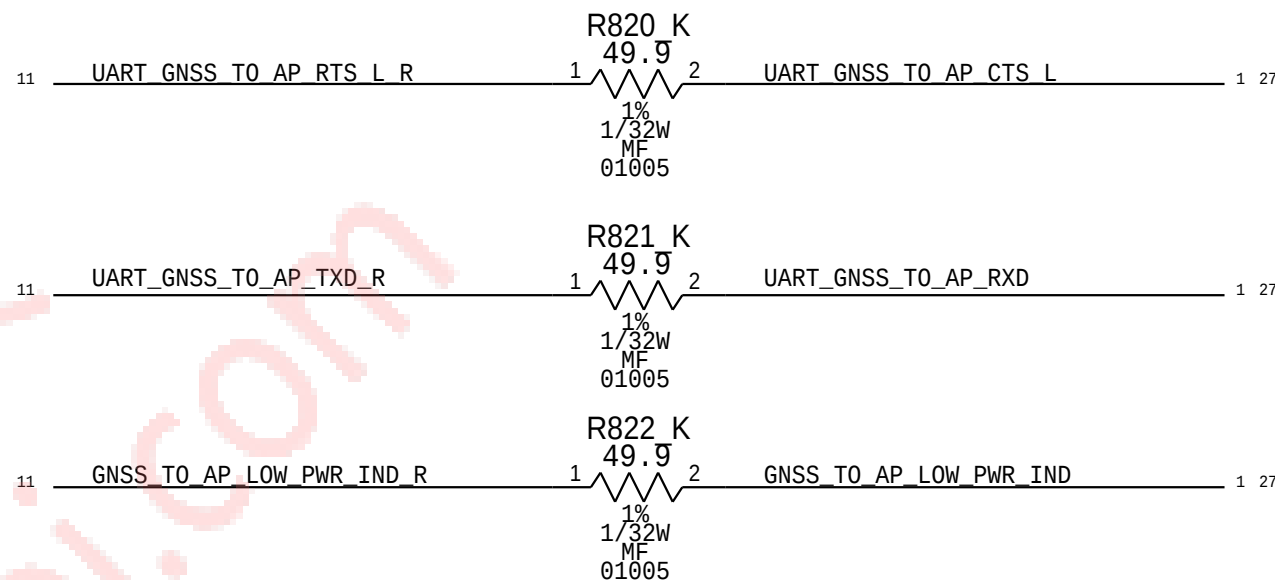
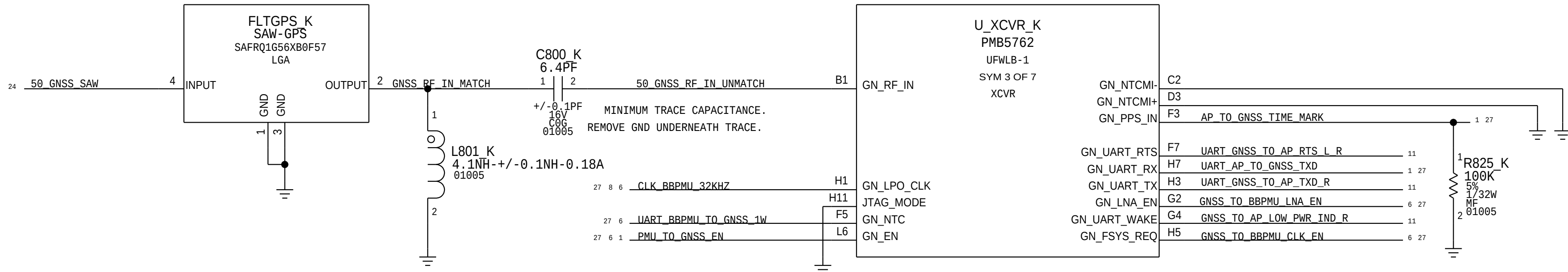
PAGE TITLE	BB: DIGITAL PWR	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE I NOT TO REPRODUCE OR COPY IT I NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	DRAWING NUMBER	051-02613
	REVISION	4.0.0
	BRANCH	
	PAGE	10 OF 28
	SHEET	26 OF 49

XCVR: TRANSMIT & GNSS

SMARTI7.1 (4/7)



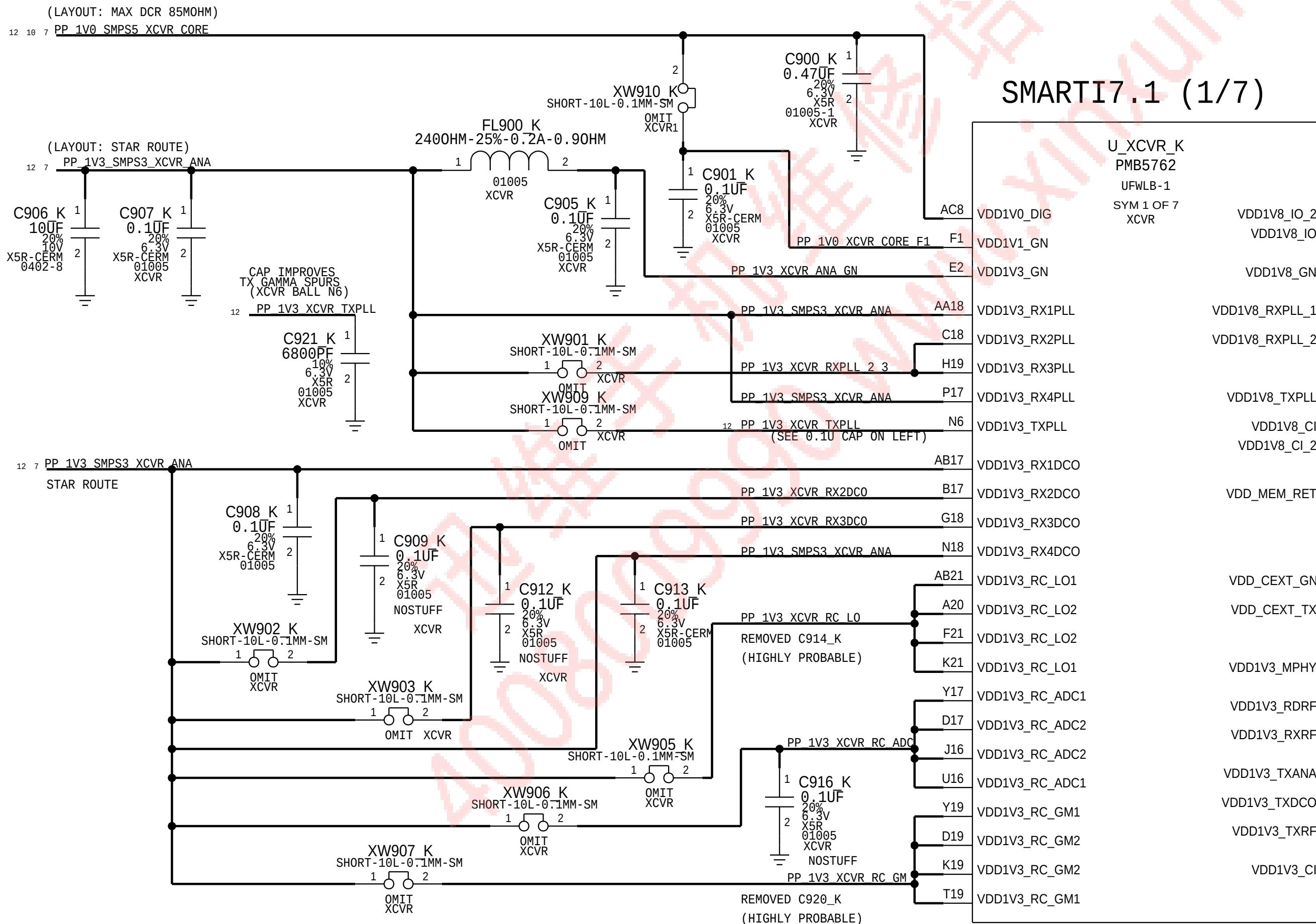
SMARTI7.1 (3/7)



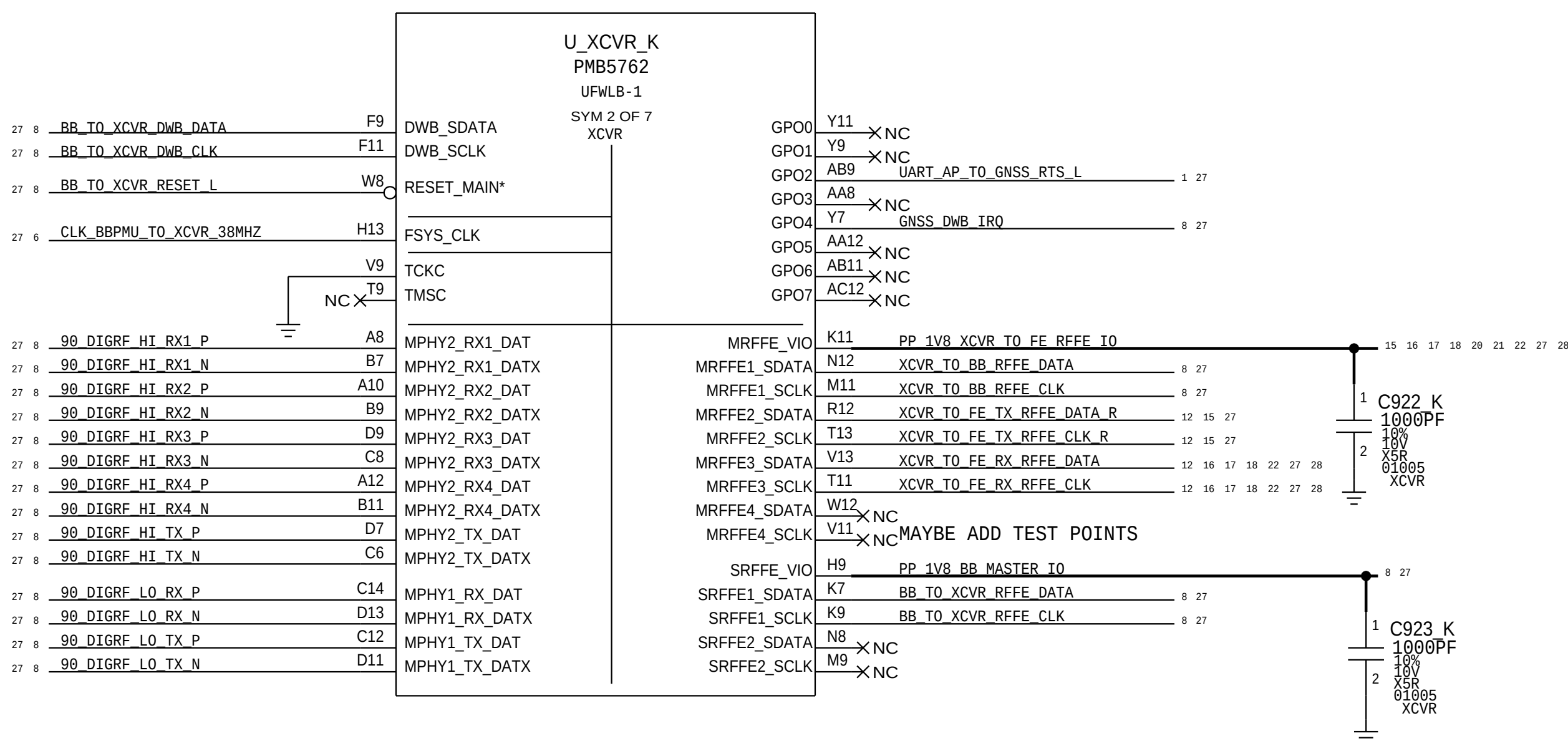
PAGE TITLE		
XCVR: TX & GNSS		
	DRAWING NUMBER	051-02613
	REVISION	4.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH	
	PAGE	11 OF 28
	SHEET	27 OF 49

XCVR: INTERFACE & PWR

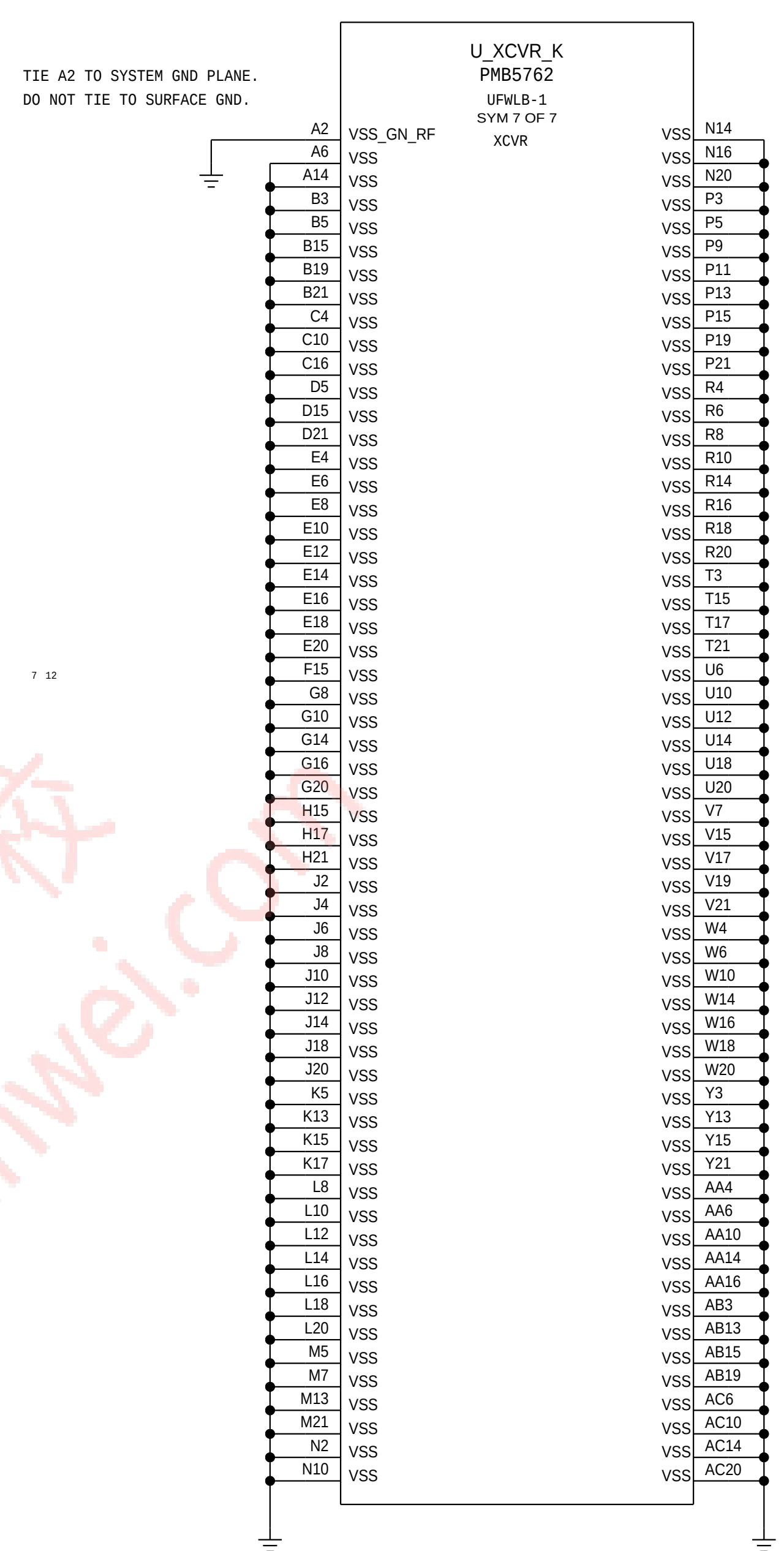
SMARTI7.1 (1/7)



SMARTI7.1 (2/7)



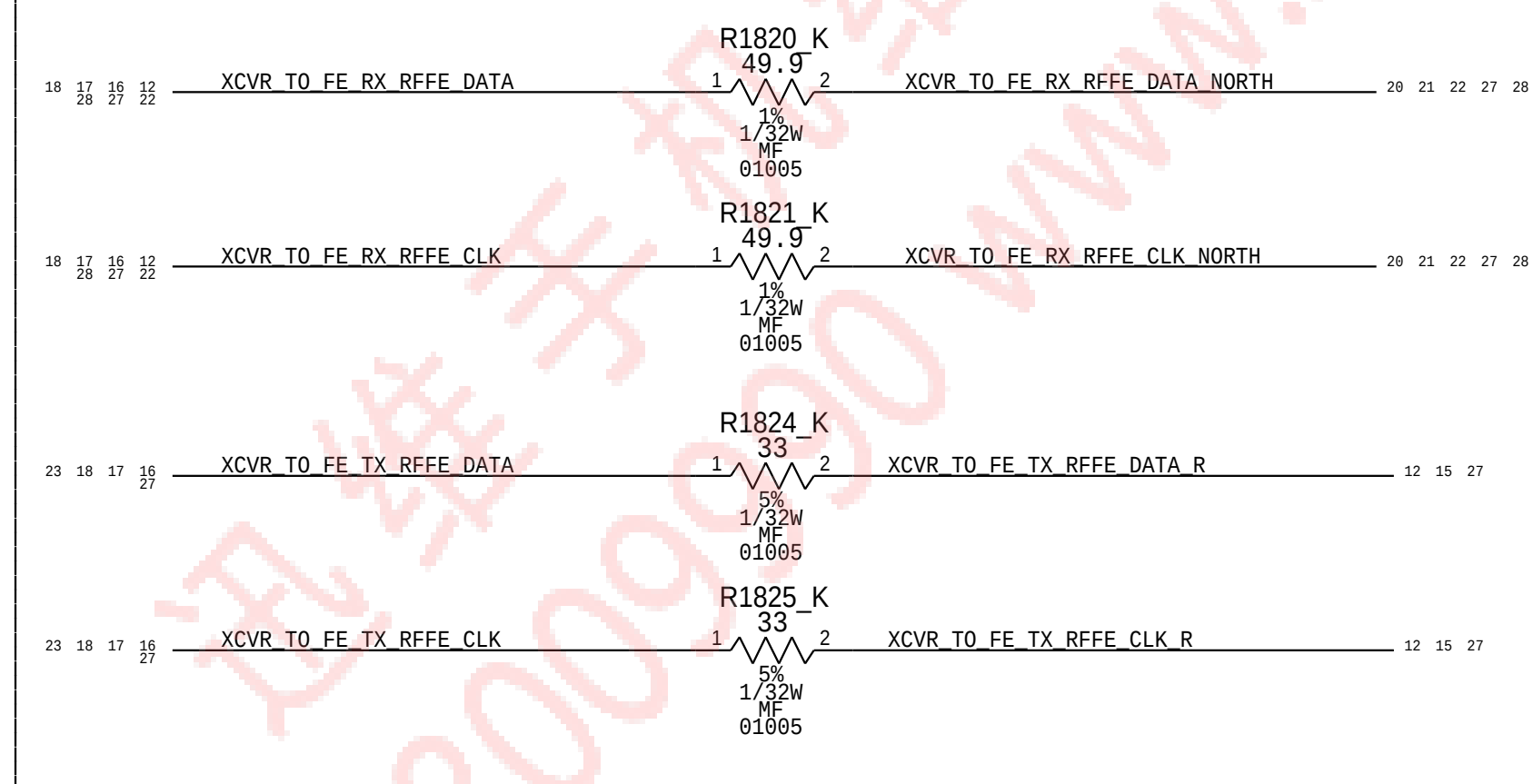
SMARTI7.1 (7/7)



XCVR + FRONT END POWER DISTRIBUTION

CONFIGURATION 4 (RDAR://28679883)

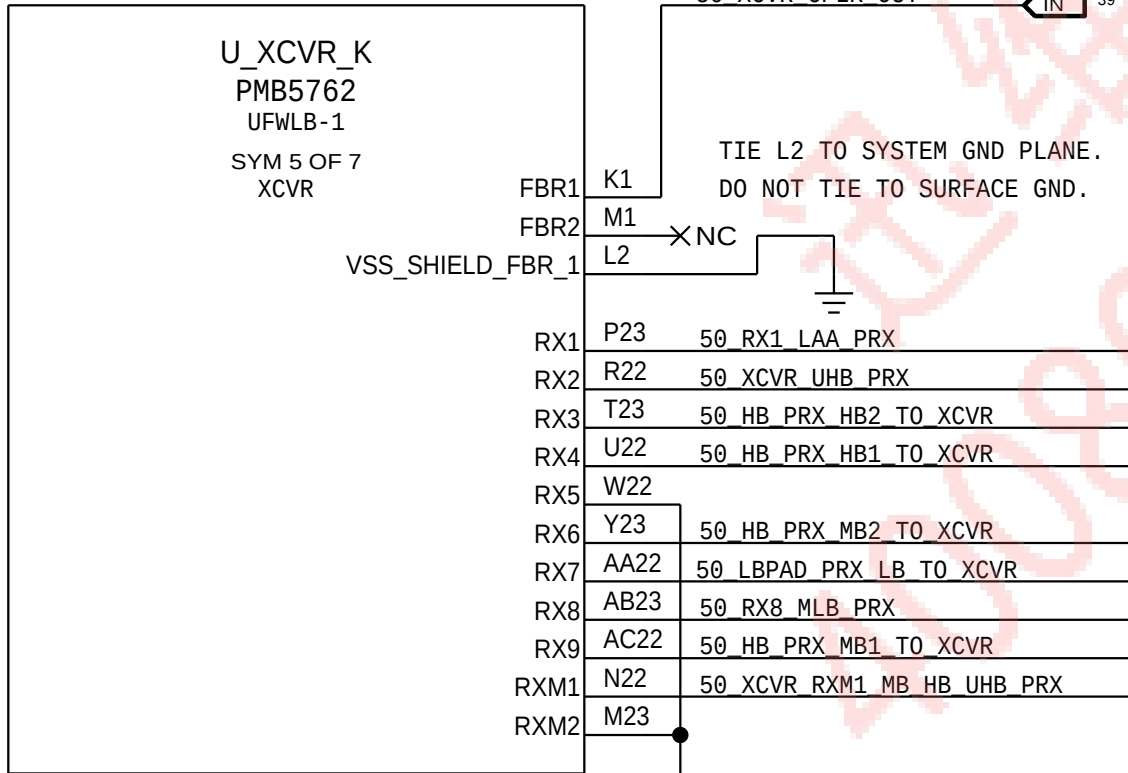
XCVR RFFE DAMPING RESISTORS



PAGE TITLE		
XCVR: INTERFACE & PWR		
DRAWING NUMBER	051-02613	SIZE
REVISION	4.0.0	D
NOTICE OF PROPRIETARY PROPERTY:		
THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING:		
I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE		
II NOT TO REPRODUCE OR COPY IT		
III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART		
IV ALL RIGHTS RESERVED		
BRANCH		
PAGE	12 OF 28	
SHEET	28 OF 49	

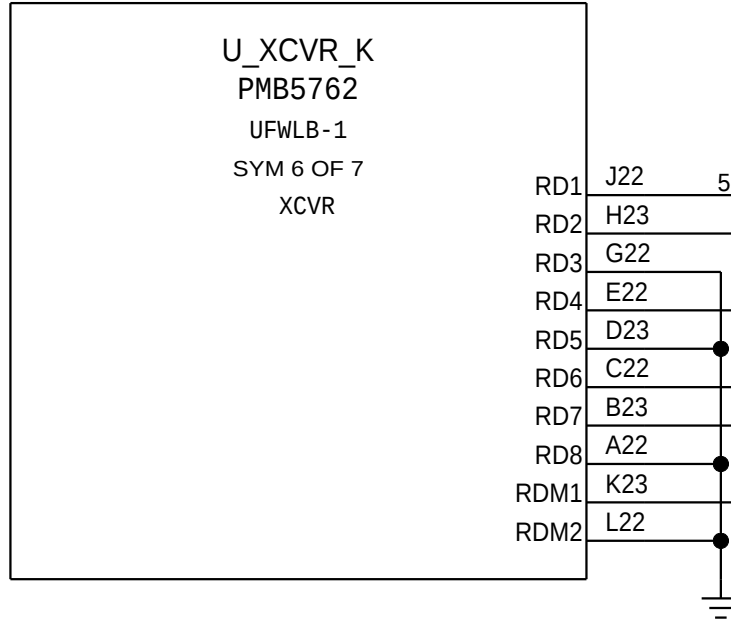
XCVR: PRIMARY/DIVERSITY RX

SMARTI7.1 PRX (5/7)



CONNECT ALL UNUSED RX PORTS TO GND.

SMARTI7.1 DRX (6/7)

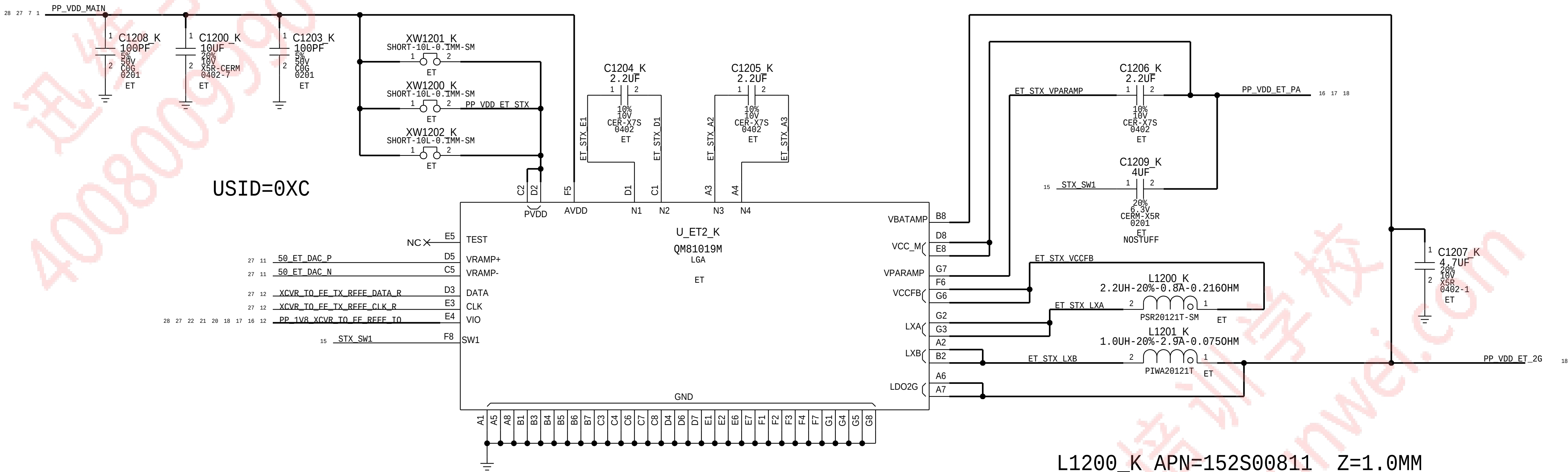


S7 PORT MAP CHANGE 10/11/2016:
USE RD6 INSTEAD OF RD3.
CONNECT ALL UNUSED RD PORTS TO GND.

PAGE TITLE		
XCVR: PRX DRX		
	DRAWING NUMBER	051-02613
	REVISION	4.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH	
	PAGE	13 OF 28
	SHEET	29 OF 49

ET MODULATOR

ALPES STX E7.2.1 MODULE



USID=0XC

L1200_K APN=152S00811 Z=1.0MM
L1201_K APN=152S00814 Z=1.0MM

C1204_K, C1205_K, C1206_K: KYOCERA APN=138S00167
C1204_K, C1205_K, C1206_K ALTERNATE: MURATA APN=138S00041

PAGE TITLE		
ET		
	DRAWING NUMBER	051-02613
	REVISION	4.0.0
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR PUBLISH IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH	
	PAGE	15 OF 28
	SHEET	31 OF 49

INTENTIONALLY
NOT CONNECTED



PAGE TITLE		
LB SPAD		
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	DRAWING NUMBER	SIZE
	051-02613	D
	REVISION	
	4.0.0	
	BRANCH	
	PAGE	16 OF 28
	SHEET	32 OF 49

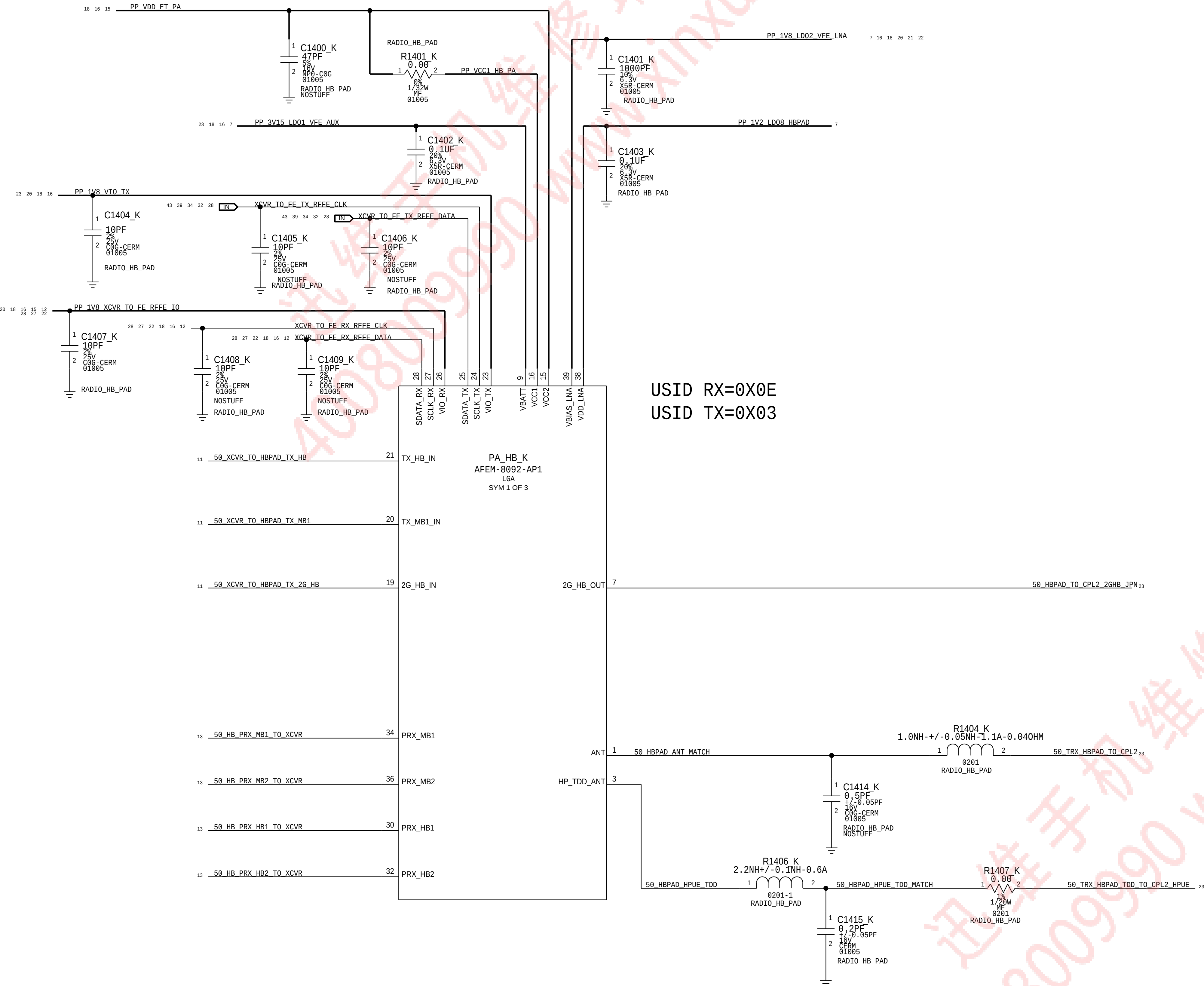
HB SPAD

D

C

B

A



D

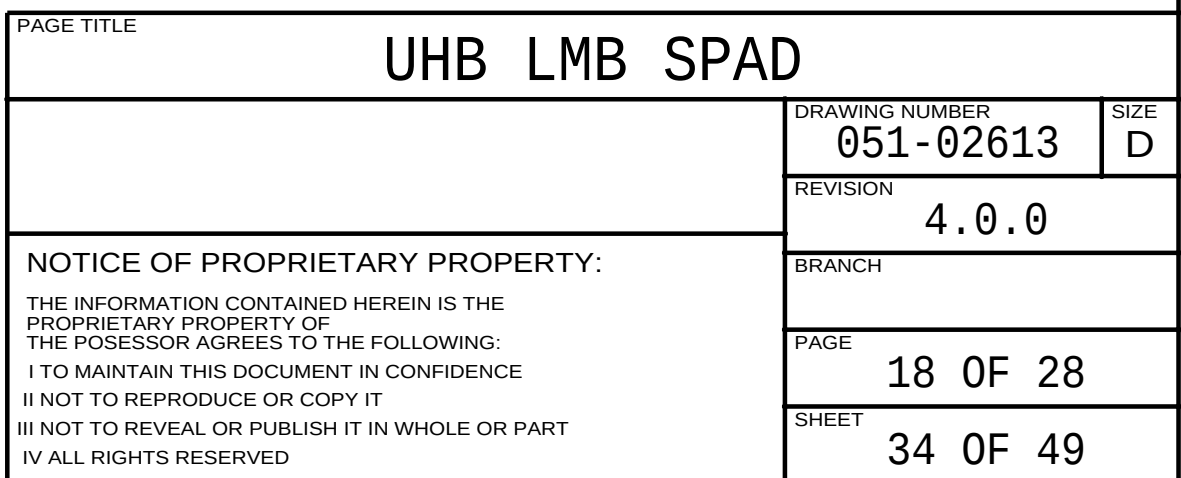
C

B

A

PAGE TITLE		
HB SPAD		
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	DRAWING NUMBER	051-02613
	REVISION	4.0.0
	BRANCH	
	PAGE	17 OF 28
	SHEET	33 OF 49

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSE ONLY - NOT A CHANGE REQUEST





BLANK PAGE

迅维手机维修培训学校
4008009990 www.xinxunwei.com

PAGE TITLE			BLANK PAGE		
			DRAWING NUMBER	051-02613	SIZE
			REVISION	4.0.0	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED			BRANCH		
			PAGE		
			19 OF 28		
			SHEET		
			35 OF 49		

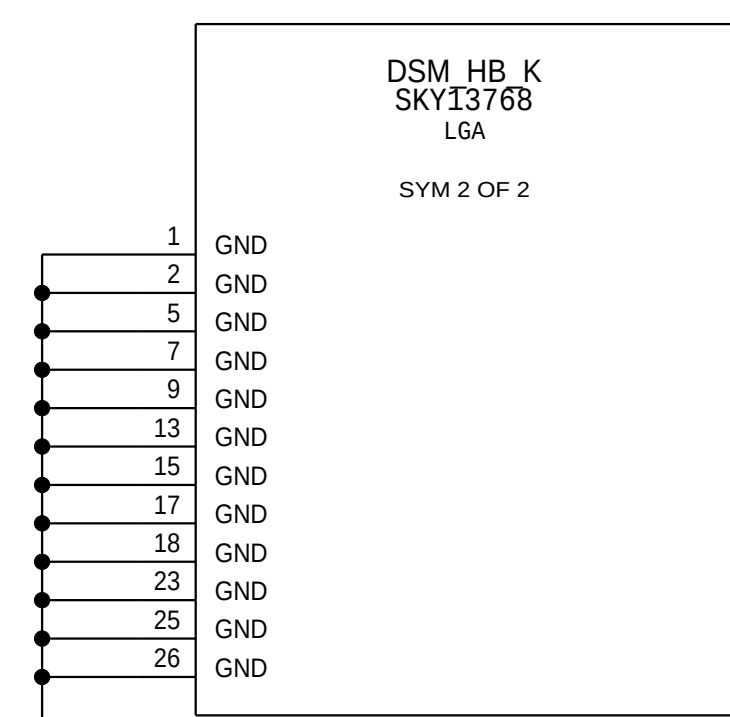
LB DIVERSITY RECEIVE LNA

USID RX=0X09

RFFE FILTERING

PAGE TITLE		
LB DIVERSITY RECEIVE LNA		
DRAWING NUMBER	051-02613	SIZE
REVISION	4.0.0	D
BRANCH		
PAGE	20 OF 28	
SHEET	36 OF 49	

CONFIDENTIAL AND PROPRIETARY APPLE SYSTEM DESIGN. FOR REFERENCE PURPOSE ONLY - NOT A CHANGE REQUEST

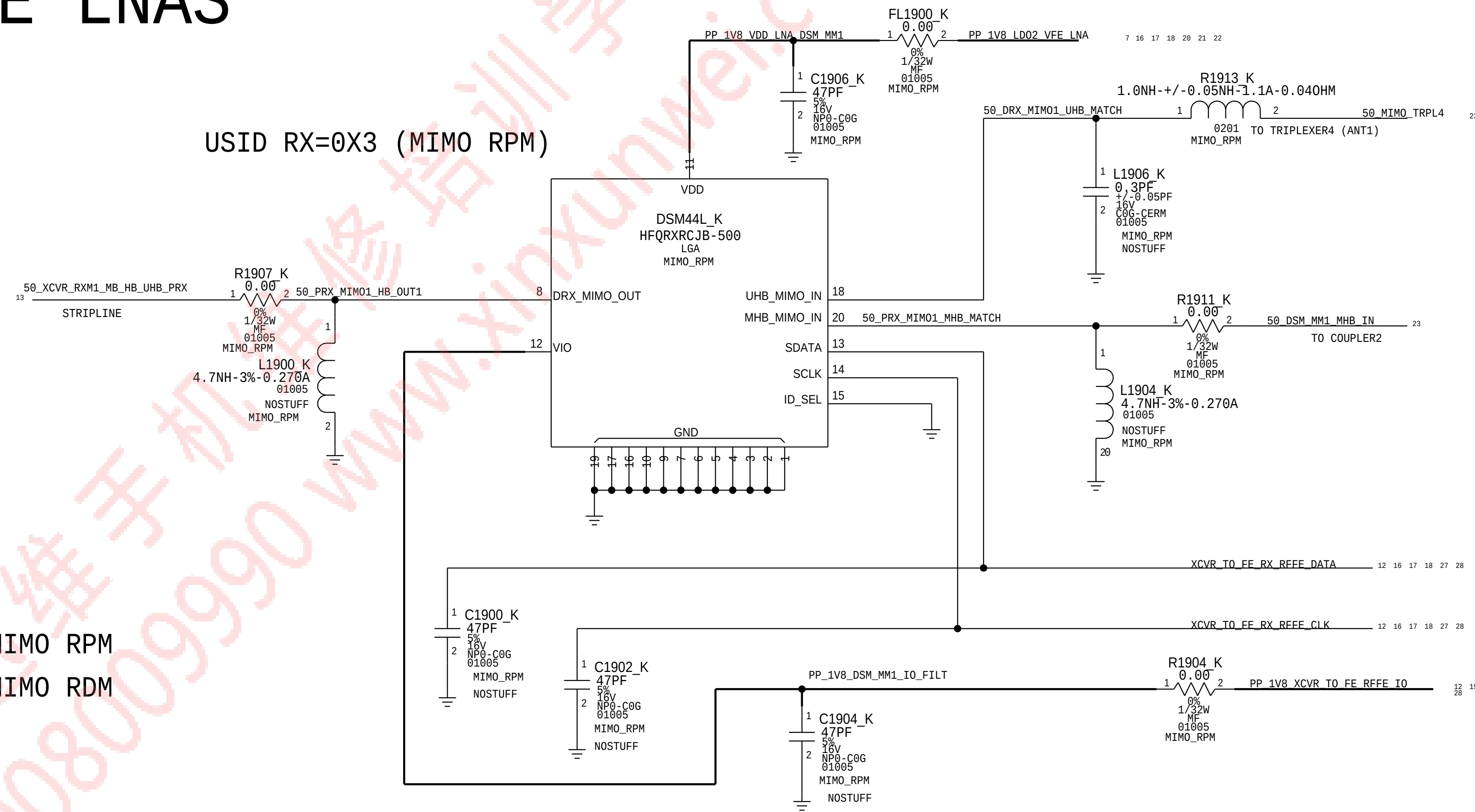


PAGE TITLE		
HB DIVERSITY RECEIVING LNA		
0	DRAWING NUMBER	SIZE
	051-02613	D
	REVISION	
	4.0.0	
IV ALL RIGHTS RESERVED	BRANCH	
	PAGE	
	21 OF 28	
	SHEET	
	37 OF 49	

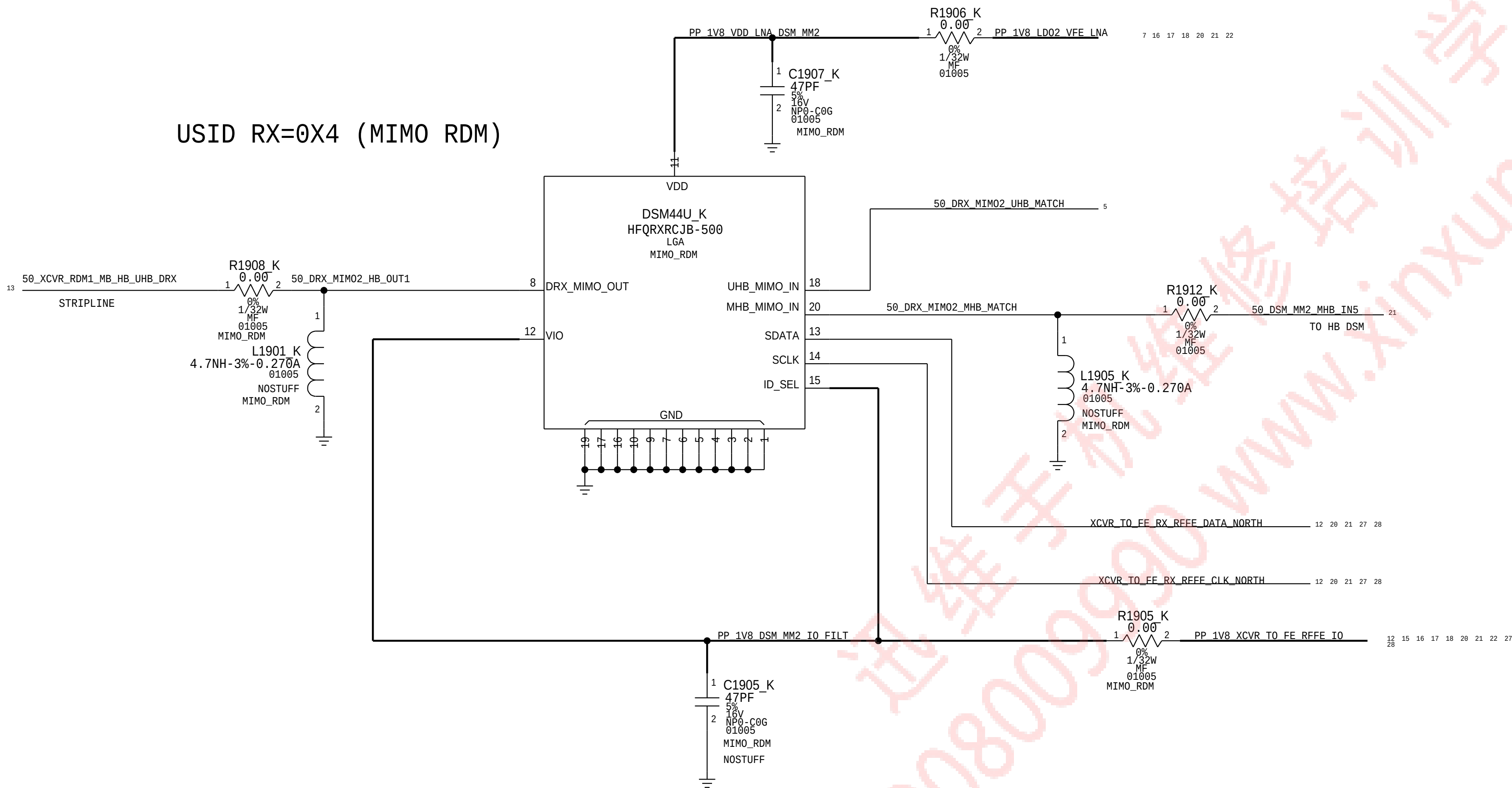
MIMO RECEIVE LNAS

ID_SEL = 0, MIMO RPM
ID_SEL = 1, MIMO RDM

USID RX=0X3 (MIMO RPM)



USID RX=0X4 (MIMO RDM)



PAGE TITLE		
MIMO RECEIVE LNAS		
DRAWING NUMBER	051-02613	SIZE
	4.0.0	D
REVISION		
BRANCH		
PAGE		
22 OF 28		
SHEET		
38 OF 49		

USID TX=0X06



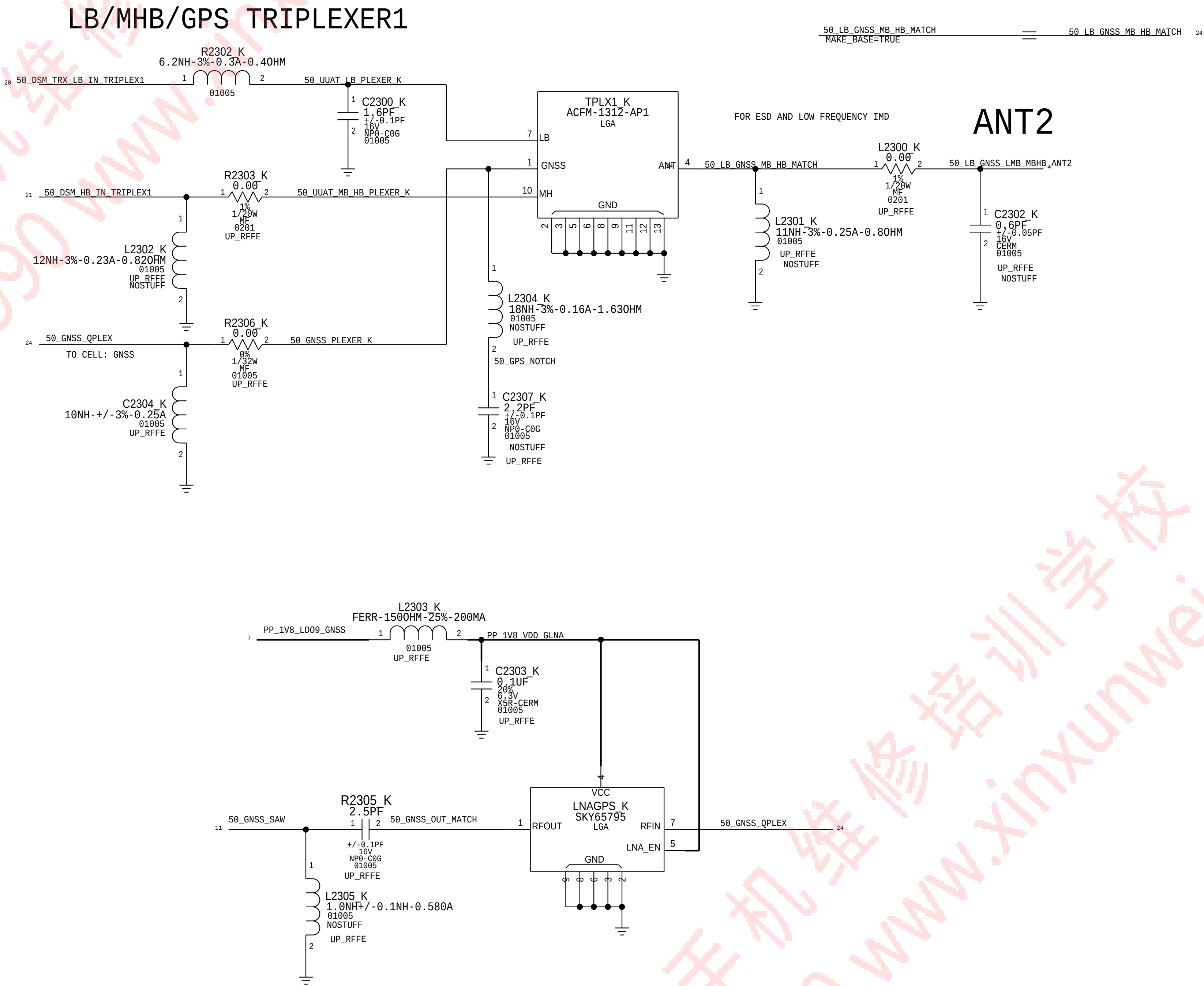
LB-MHB-UHB TRIPLEXER4 (ANT1)

ANT1

ALIAS FOR LAYOUT PURPOSES

PAGE TITLE		LOWER ANTENNA & COUPLER	
	DRAWING NUMBER 051-02613		SIZE D
	REVISION 4.0.0		
	BRANCH		
	PAGE 23 OF 28		
	SHEET 39 OF 49		

UPPER ANTENNA FEEDS



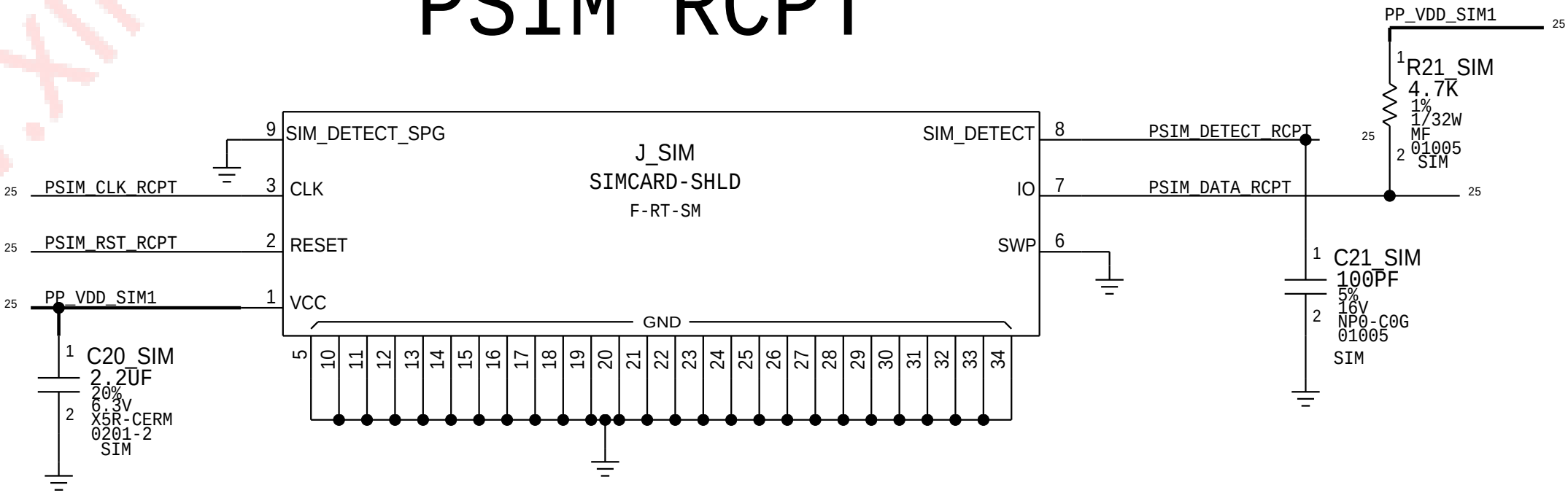
UPPER ANTENNA FEEDS		
	DRAWING NUMBER	051-02613
	REVISION	4.0.0
	BRANCH	
	PAGE	24 OF 28
	SHEET	40 OF 49
IV ALL RIGHTS RESERVED		

PSIM

ICE18.1 PHYSICAL SIM

ICE17 PART OFFERS MORE SHIELDING AND PLACEMENT AREA

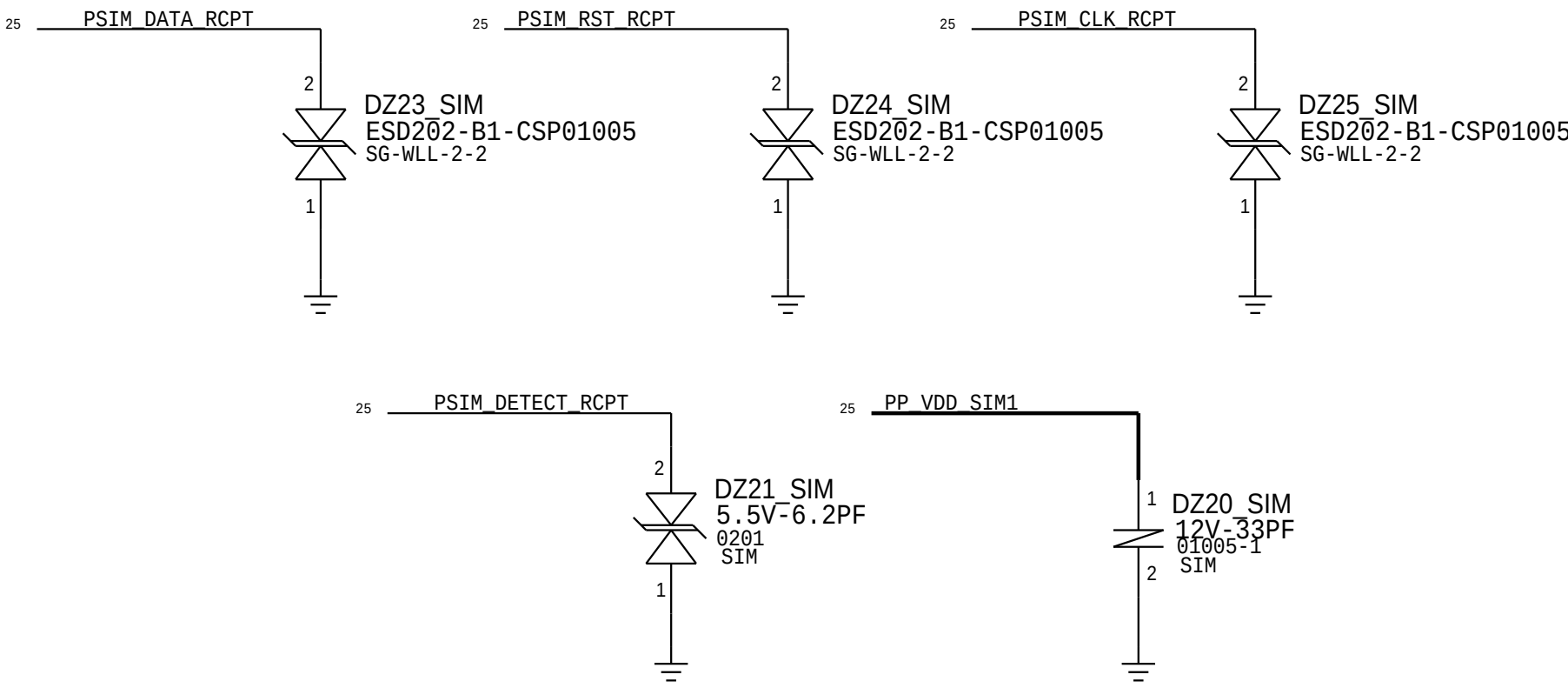
PSIM RCPT



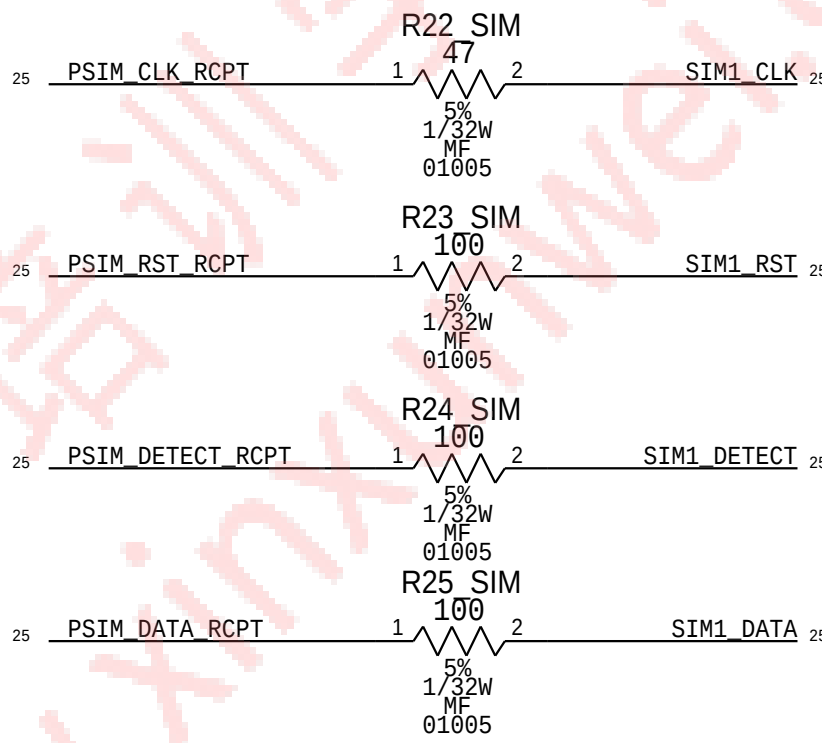
ICE18.1 ONLY

APN:512S00030

ESD DIODES



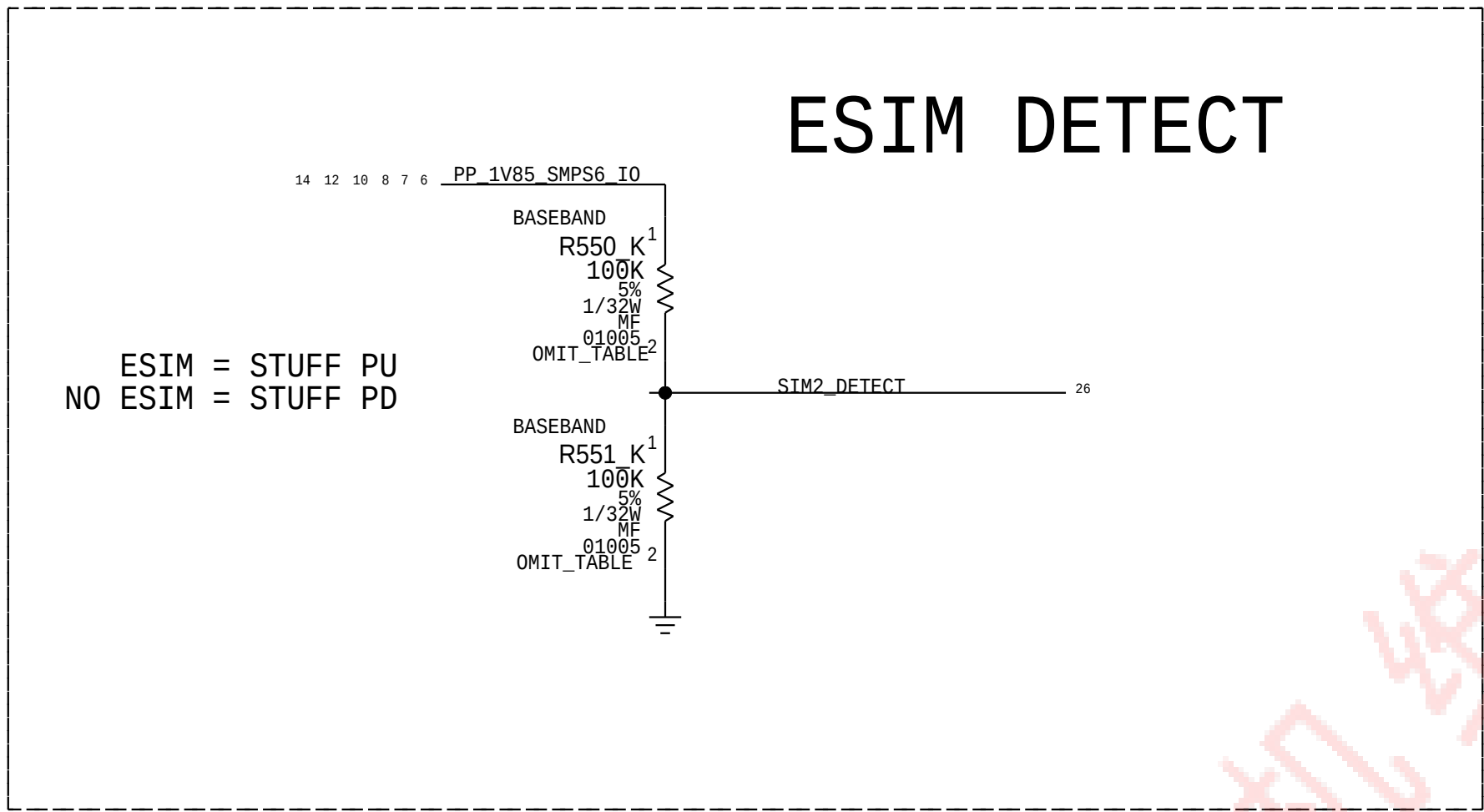
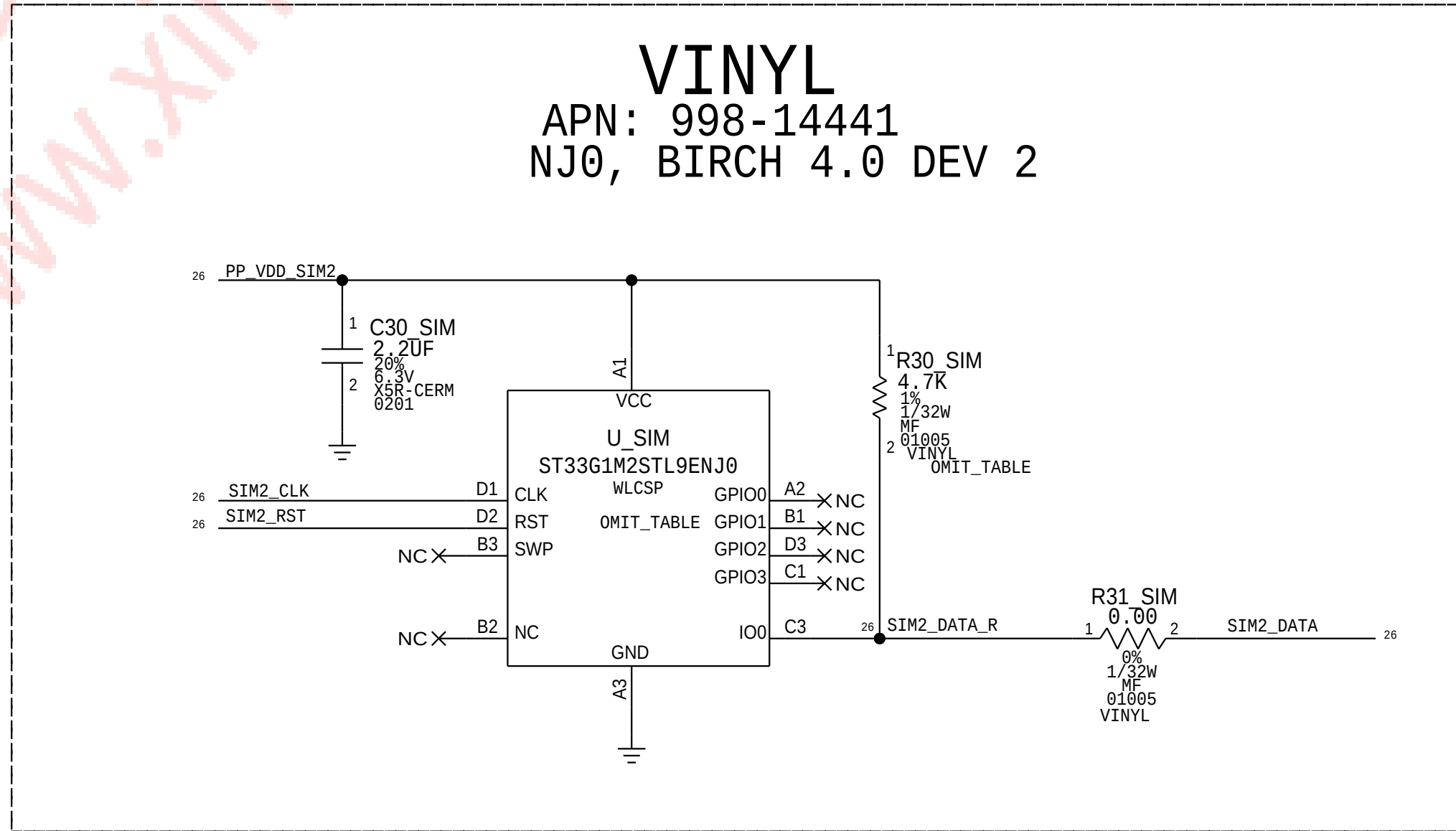
EOS RESISTORS



27	8	SIM1_CLK	==	SIM1_CLK	25
		MAKE_BASE=TRUE			
27	8	SIM1_DATA	==	SIM1_DATA	25
		MAKE_BASE=TRUE			
27	8	SIM1_RST	==	SIM1_RST	25
		MAKE_BASE=TRUE			
27	8	SIM1_DETECT	==	SIM1_DETECT	25
		MAKE_BASE=TRUE			
27	10	PP_VDD_SIM1	==	PP_VDD_SIM1	25
		MAKE_BASE=TRUE			

PAGE TITLE			SIM: PSIM	
		DRAWING NUMBER	051-02613	SIZE
		REVISION	4.0.0	D
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH		
		PAGE	25 OF 28	
		SHEET	41 OF 49	

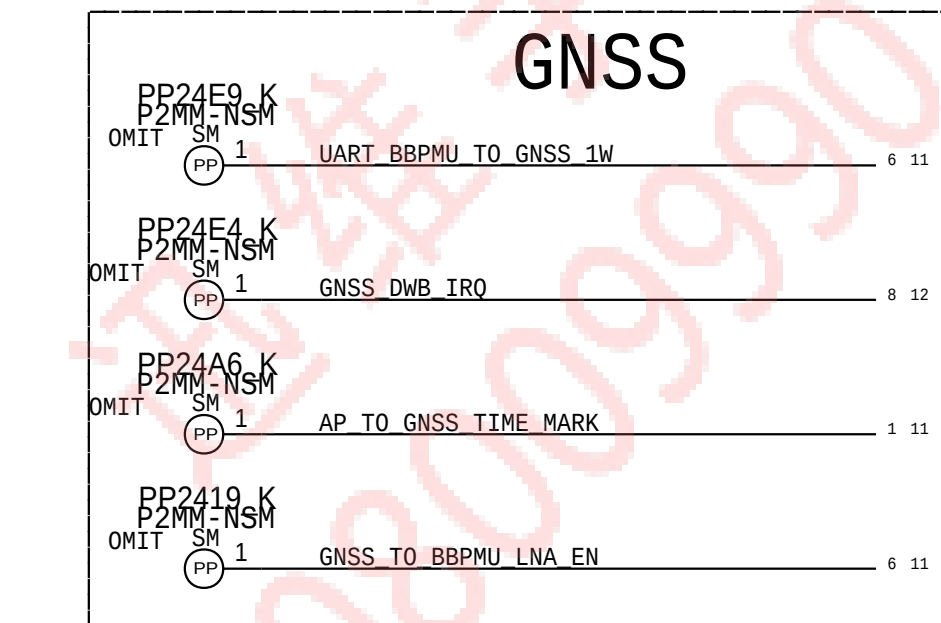
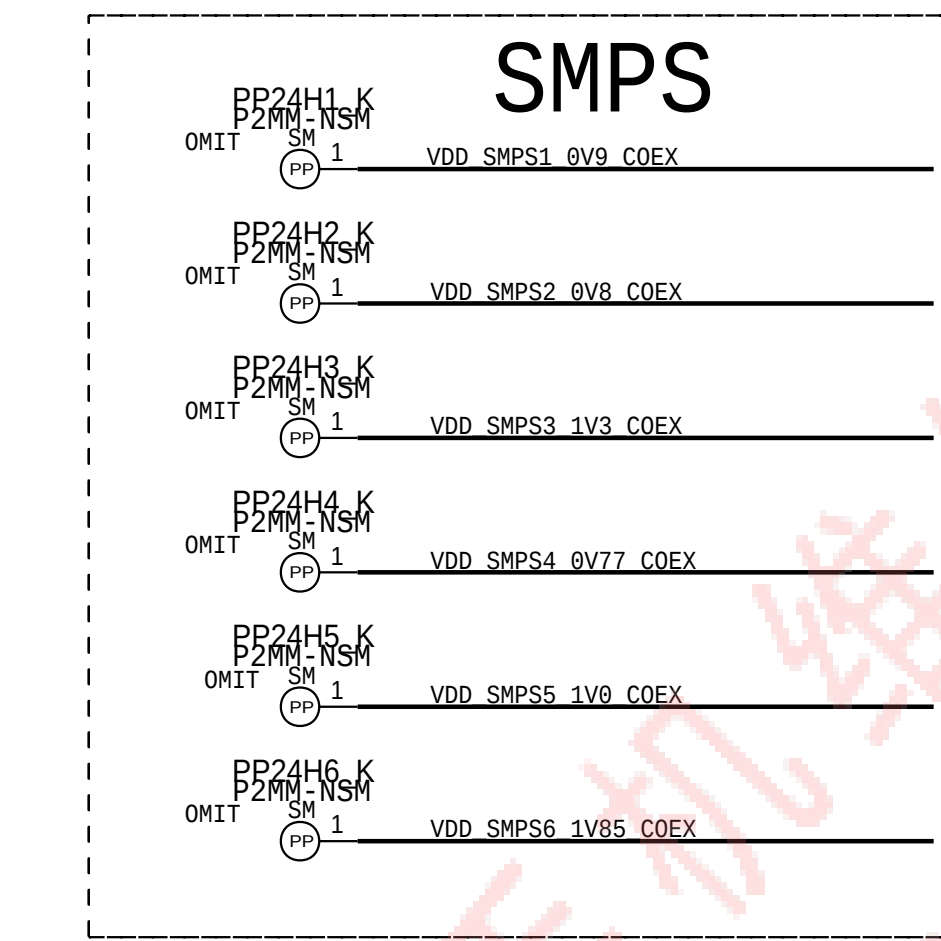
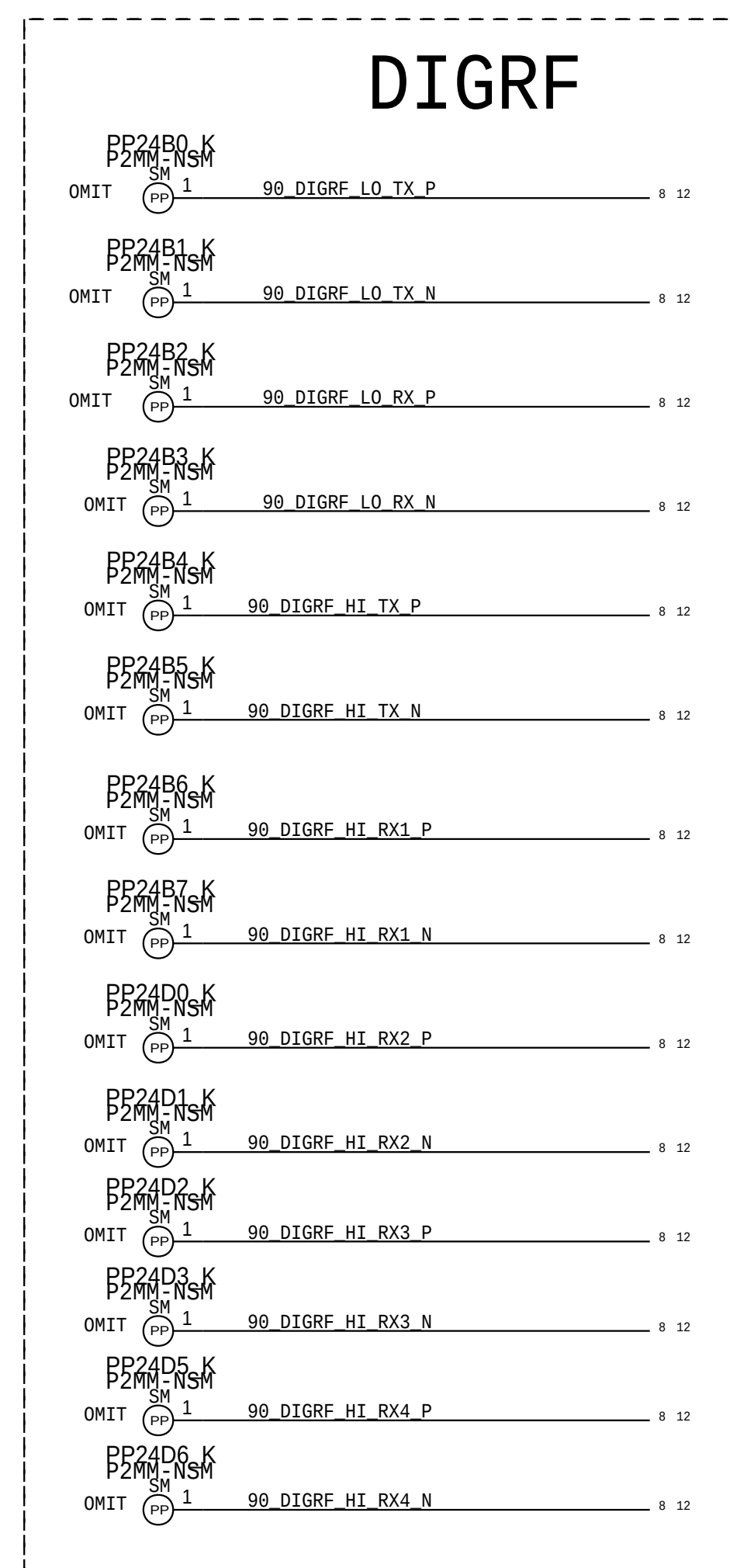
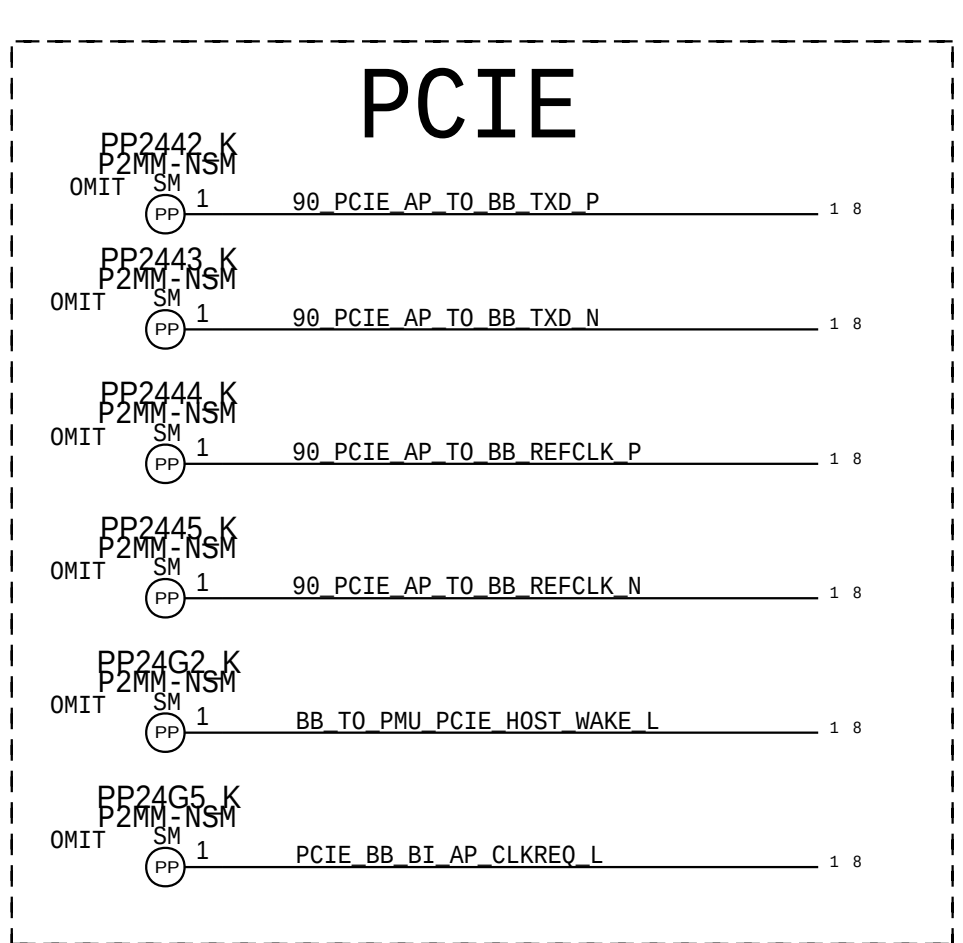
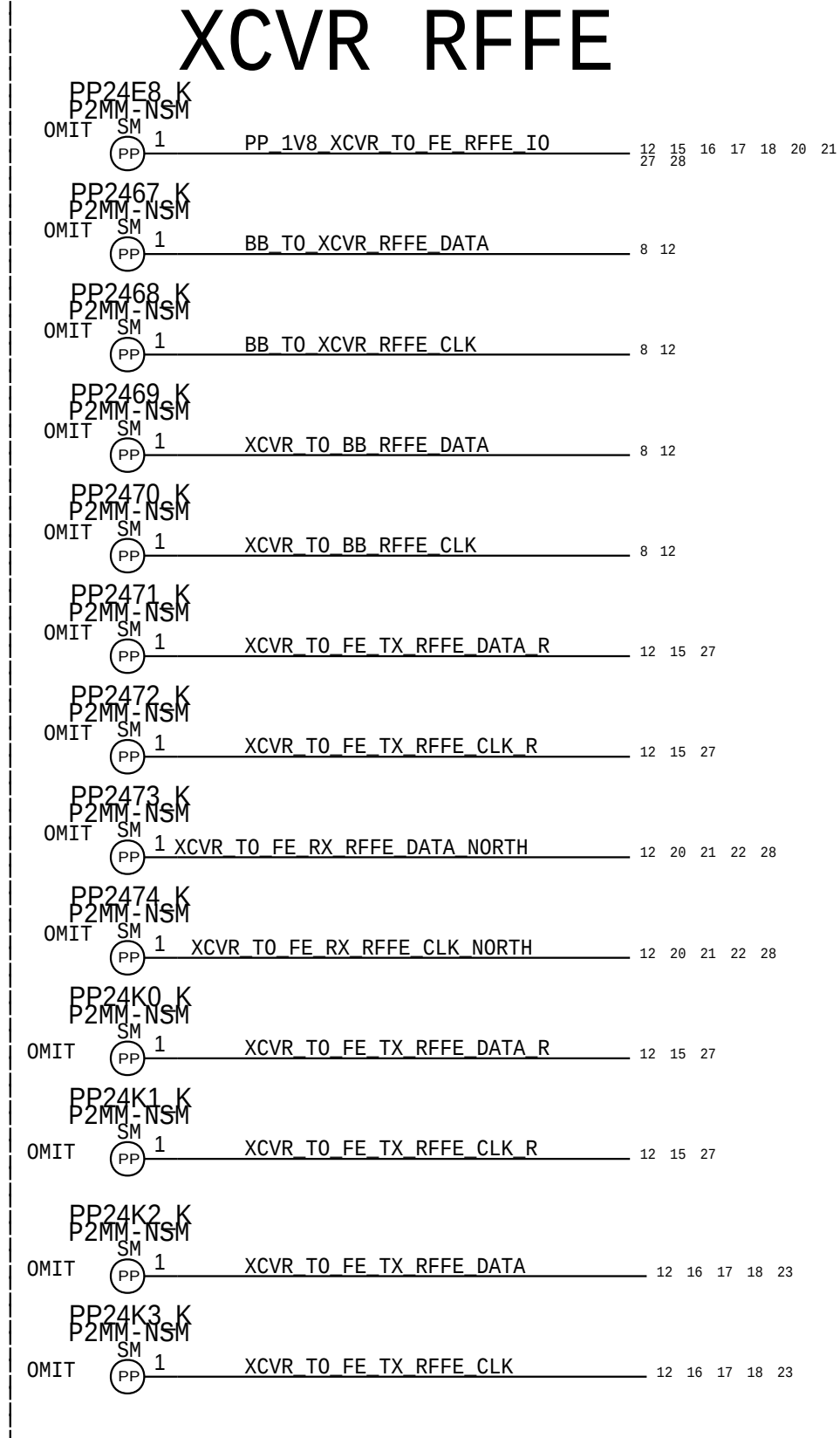
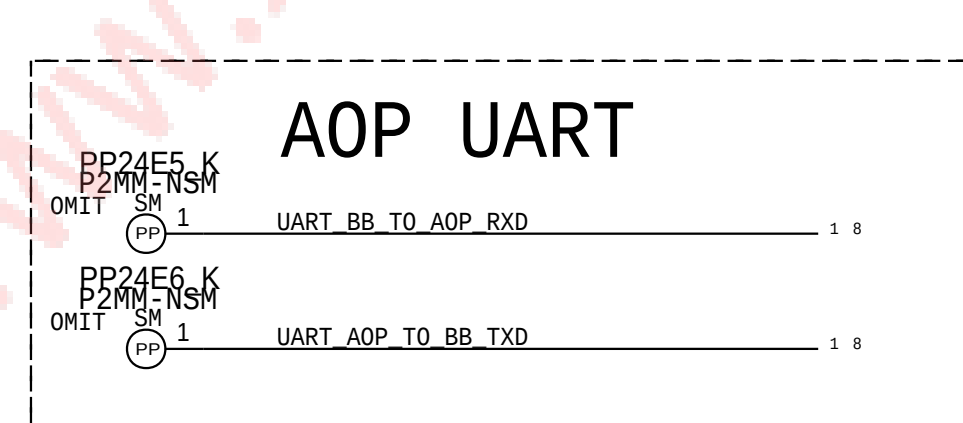
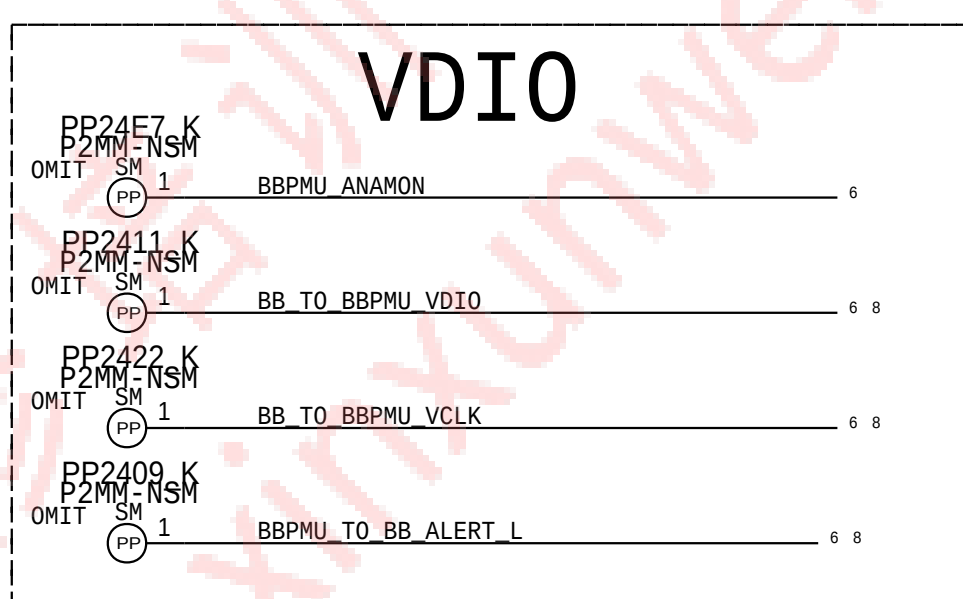
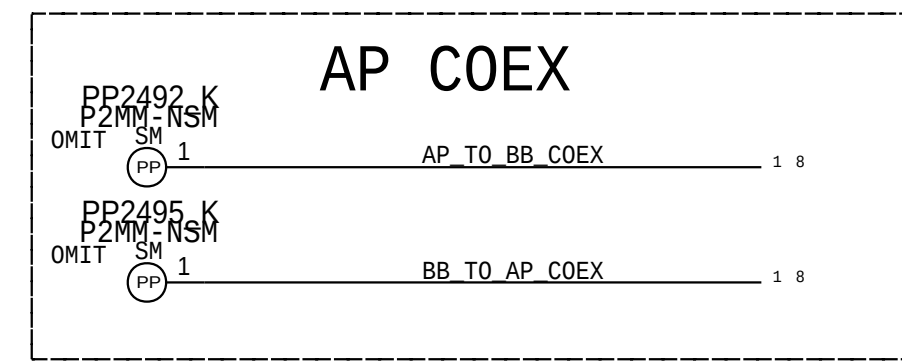
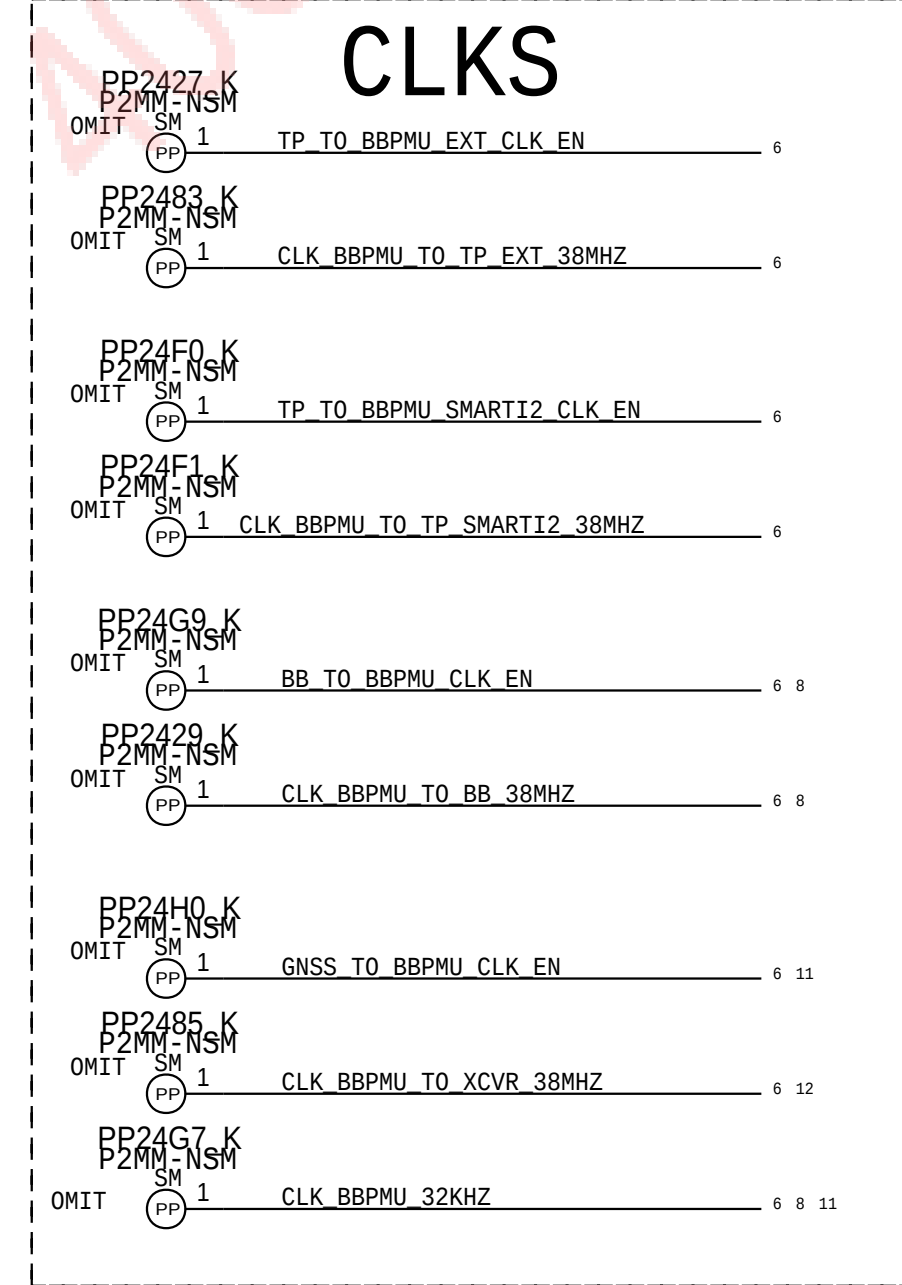
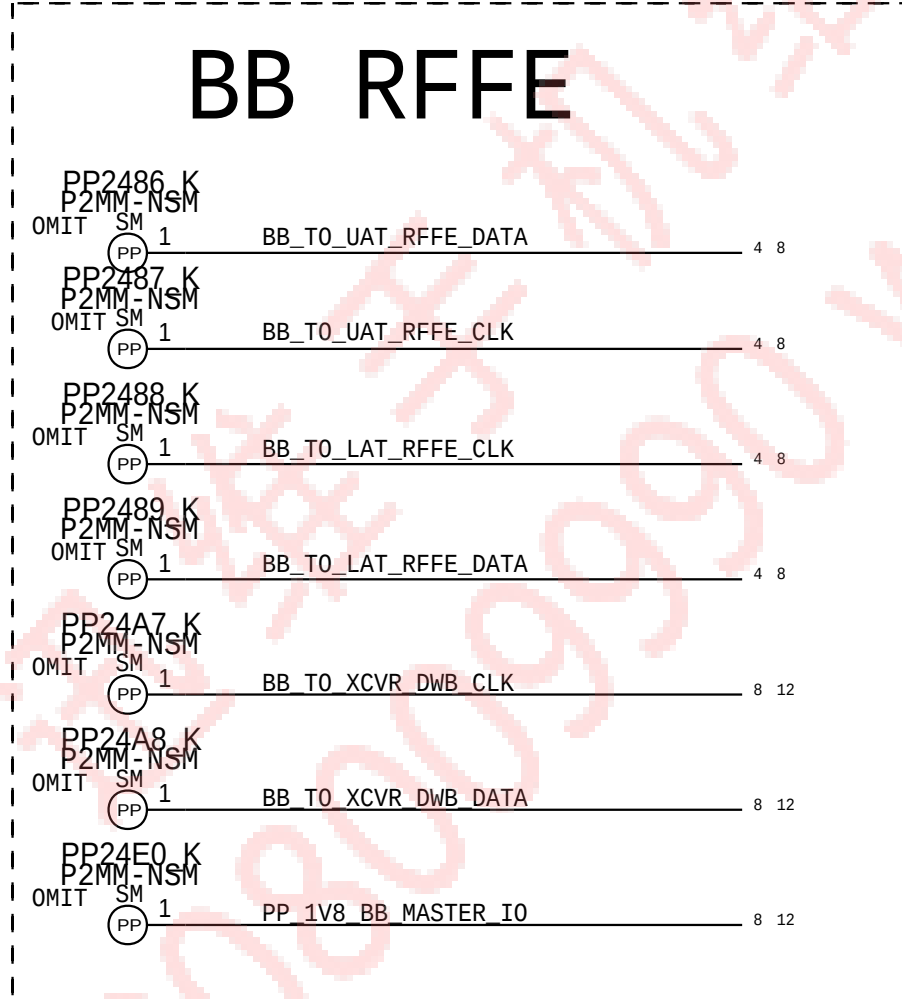
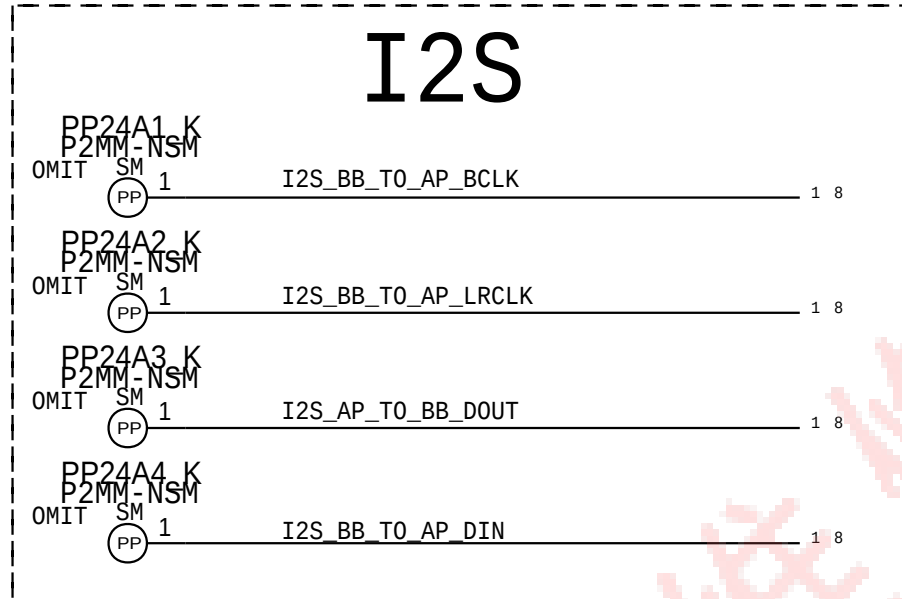
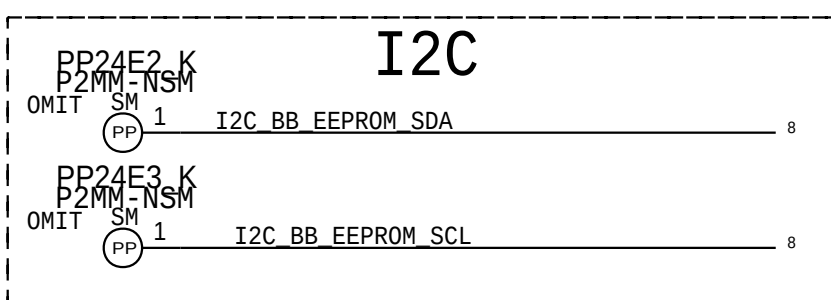
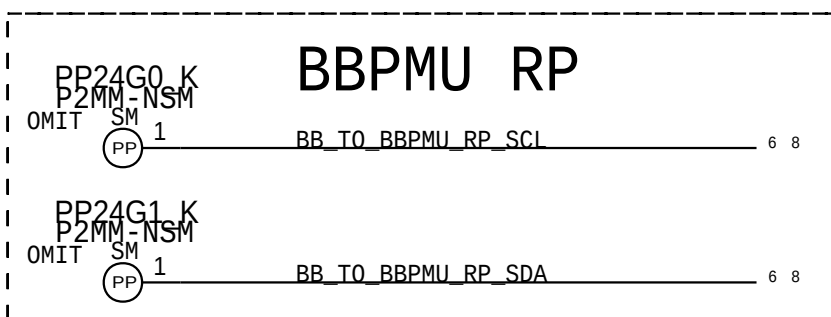
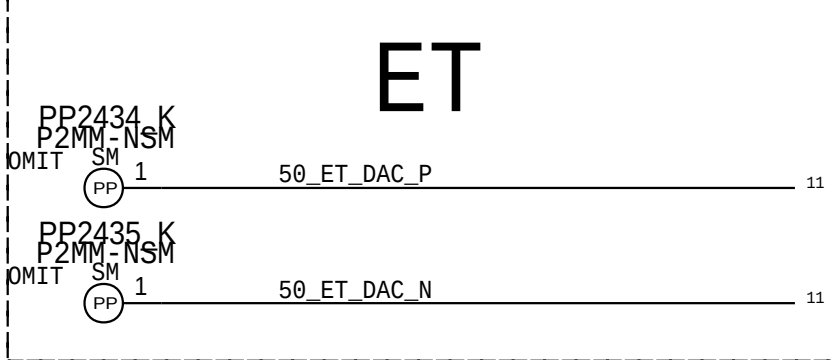
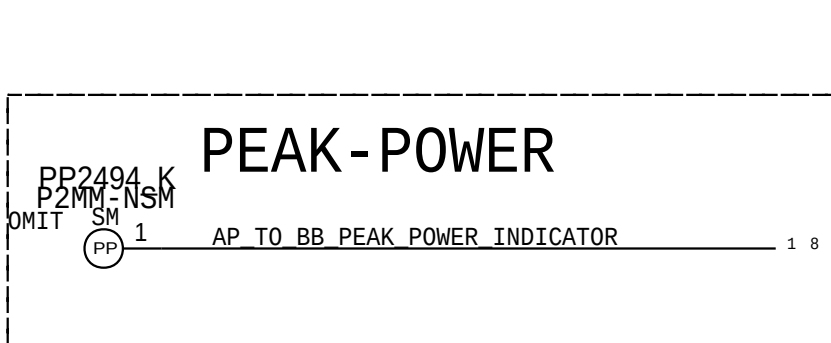
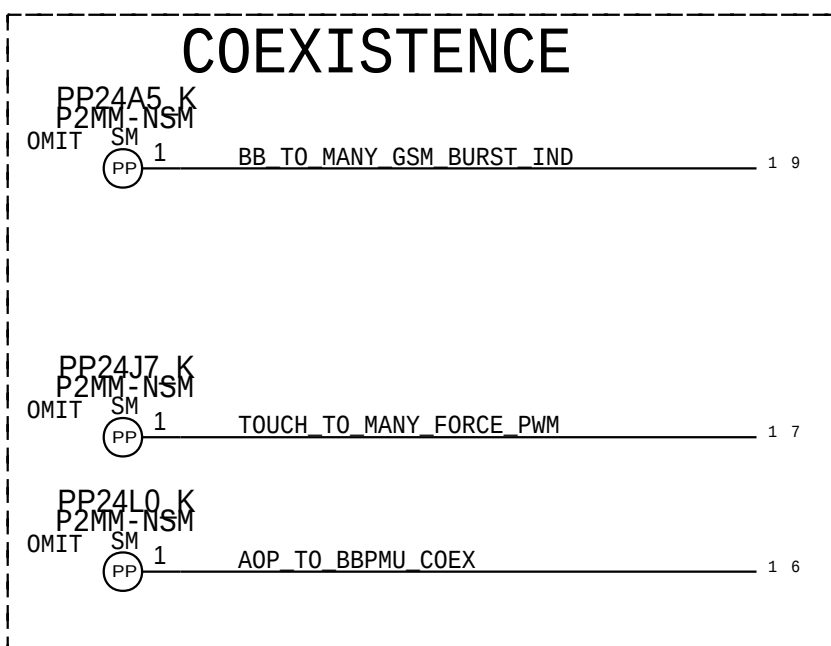
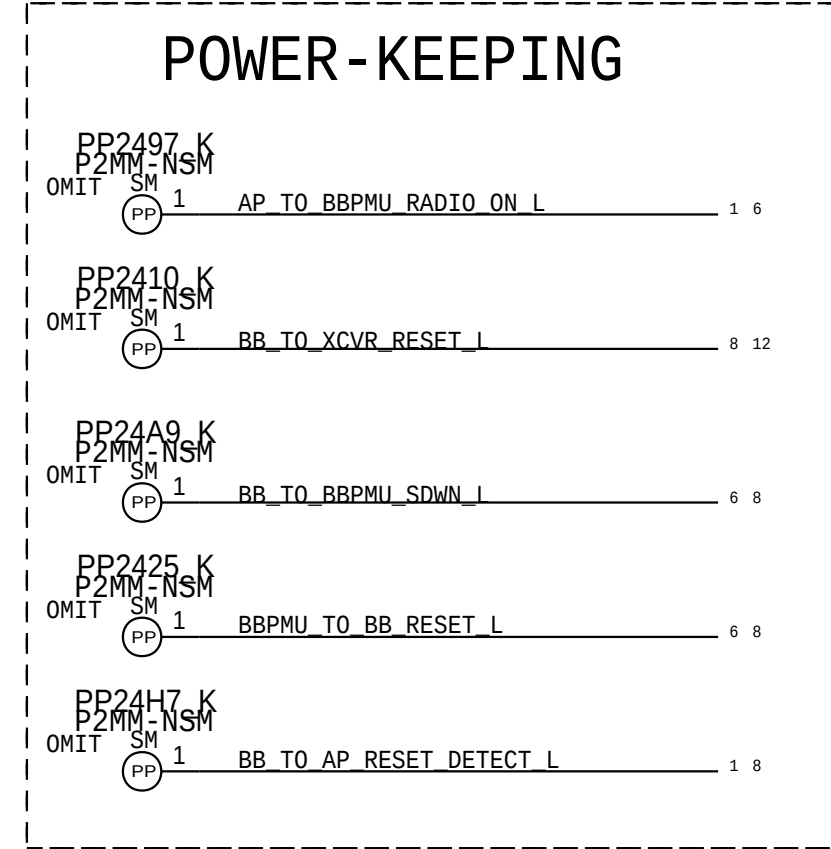
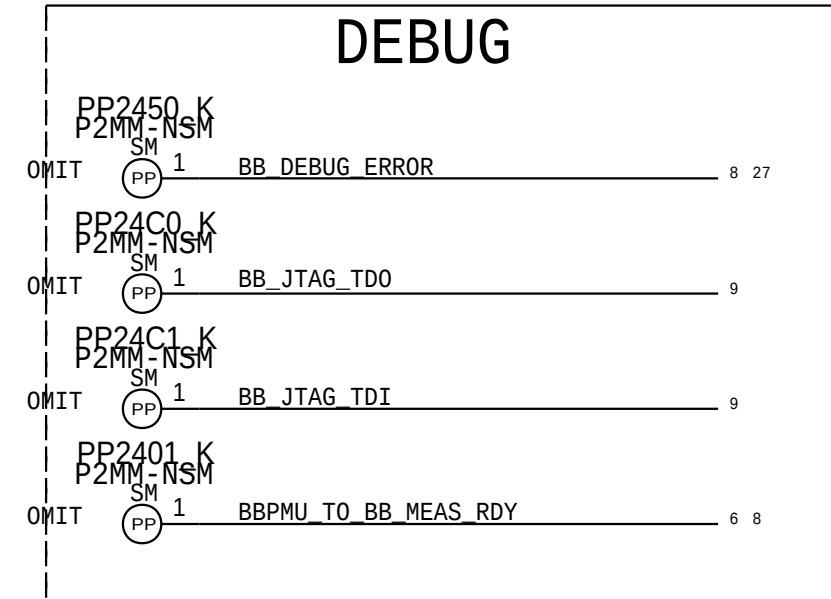
ESIM



27	8	SIM2_CLK	==	SIM2_CLK	26
27	8	SIM2_DATA	==	SIM2_DATA	26
27	8	SIM2_RST	==	SIM2_RST	26
27	8	SIM2_DETECT	==	SIM2_DETECT	26
27	10	PP_VDD_SIM2	==	PP_VDD_SIM2	26
27	27	SIM2_DATA_R	==	SIM2_DATA_R	26

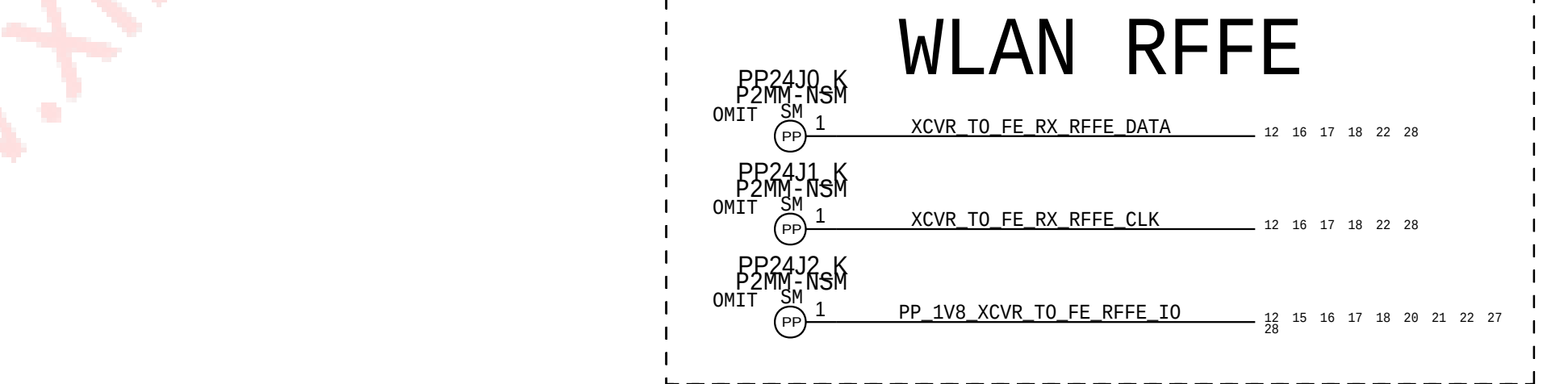
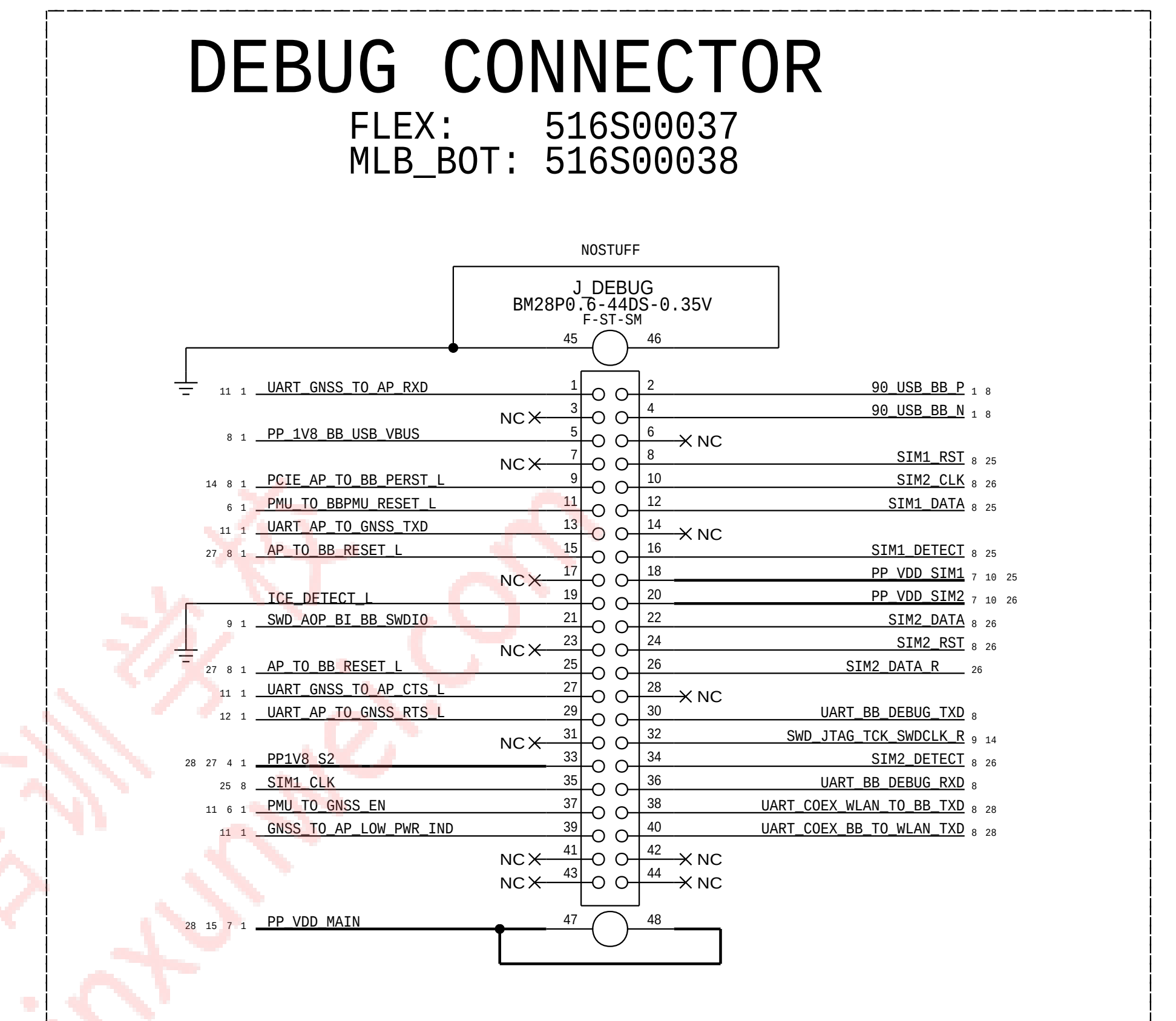
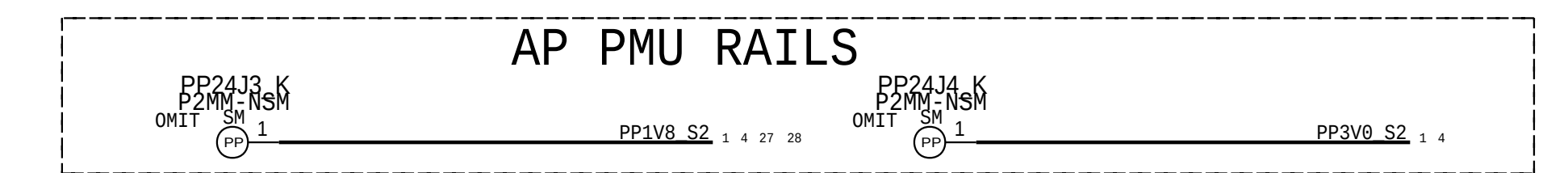
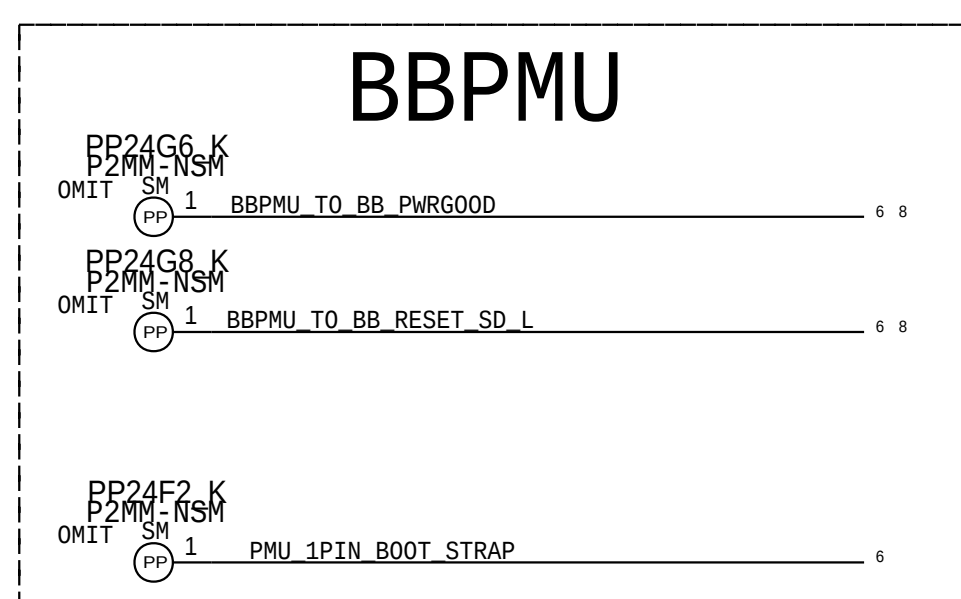
PAGE TITLE			SIM: ESIM		
		DRAWING NUMBER	051-02613		SIZE
		REVISION	4.0.0		
		BRANCH			
		PAGE	26 OF 28		
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		SHEET	42 OF 49		

DEBUG: TEST POINTS



LO

HI



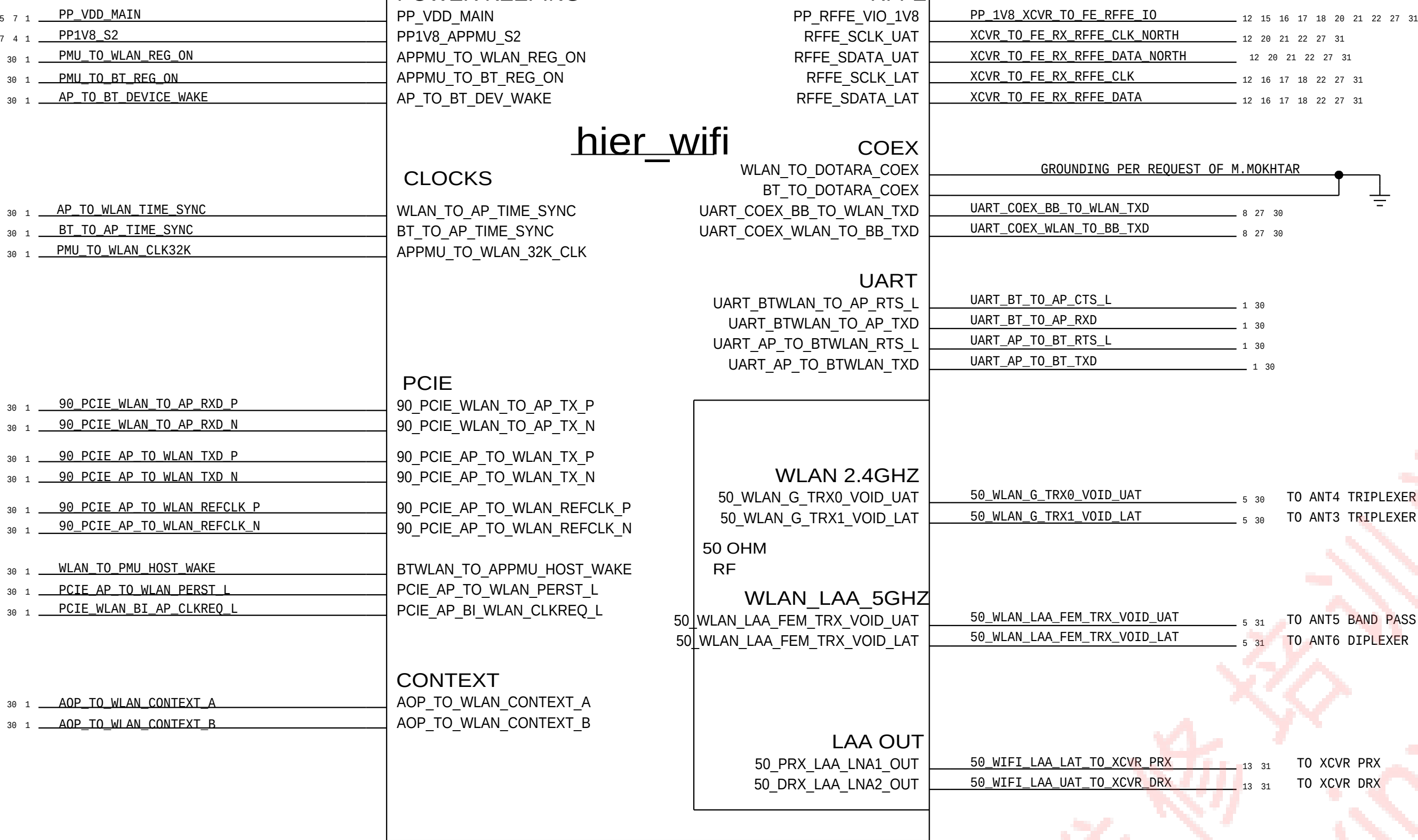
PAGE TITLE			
TEST POINTS			
DRAWING NUMBER		051-02613	SIZE D
REVISION		4.0.0	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		BRANCH	
		PAGE	27 OF 28
		SHEET	43 OF 49



SYMBOL: WIFI

HIERARCHICAL SYMBOL FOR HIER_WIFI

DIETCOKE HIER WIFI



PAGE TITLE		
SYMBOL: WIFI		
	DRAWING NUMBER	SIZE
	051-02613	D
	REVISION	
	4.0.0	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	BRANCH	
	PAGE	28 OF 28
	SHEET	44 OF 49

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.
2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.
3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV	ECN	DESCRIPTION OF REVISION	CK APPD DATE
4	0011179651	ENGINEERING RELEASED	2018-02-06

X1049 HIER_WIFI (DIETCOKE)

LAST_MODIFICATION=Tue Feb 6 10:49:37 2018

PAGE	CSA	CONTENTS	SYNC	DATE
45	1	WIFI: TABLE OF CONTENTS		
46	2	DIETCOKE		
47	3	FEM MODULES		

WLAN SYMBOL IO PORTS

POWER

```

31 30 28 IO PP VDD MAIN
VOLTAGE=5V
31 30 28 IO PP1V8 S2
VOLTAGE=1.8V

```

CONTROL

```

30 28 IO PMU TO WLAN REG ON
30 28 IO PMU TO BT REG ON
30 28 IO AP TO BT DEVICE WAKE

```

CLOCKS

```

30 28 IO PMU TO WLAN CLK32K
30 28 IO AP TO WLAN TIME SYNC
30 28 IO BT TO AP TIME SYNC

```

WLAN PCIE

36	IO	90	PCIE AP TO WLAN REFCLK P
36	IO	90	PCIE AP TO WLAN REFCLK N
36	IO	90	PCIE AP TO WLAN TXD P
36	IO	90	PCIE AP TO WLAN TXD N
36	IO	90	PCIE WLAN TO AP RXD P
36	IO	90	PCIE WLAN TO AP RXD N
36	IO		PCIE AP TO WLAN PERST L
36	IO		PCIE WLAN BI AP CLKREQ L
36	IO		WLAN TO PMU HOST WAKE

BLWLAN UART

```

30 28  UART AP TO BT TXD
30 28  UART BT TO AP RXD
30 28  UART AP TO BT RTS L
30 28  UART BT TO AP CTS L

```

AOP

```

30 28 [IO] AOP TO WLAN CONTEXT A
30 28 [IO] AOP TO WLAN CONTEXT B

```

COEX

30 28 UART COEX BB TO WLAN TXD

30 28 UART COEX WLAN TO BB TXD

30 30 GND

30 30 GND

ANTENNA

```

30 28 IO 50 WLAN G TRX0 VOID UAT
30 28 IO 50 WLAN G TRX1 VOID LAT
31 28 IO 50 WLAN LAA FEM TRX VOID UAT
32 28 IO 50 WLAN LAA FEM TRX VOID LAT

```

RF

```

31 28 IO 50 WIFI LAA LAT TO XCVR PRX
31 28 IO 50 WIFI LAA UAT TO XCVR DRX

```

RFFE

```

31 28 10 PP 1V8 XCVR TO FE RFFE IO

```

```

31 28  (IO)  XCVR_TO_FE_RX_RFFE_CLK_NORTH
28 28  (IO)  XCVR_TO_FE_RX_RFFE_DATA_NORTH
31 28  (IO)  XCVR_TO_FE_RX_RFFE_CLK
31 28  (IO)  XCVR_TO_FE_RX_RFFE_DATA

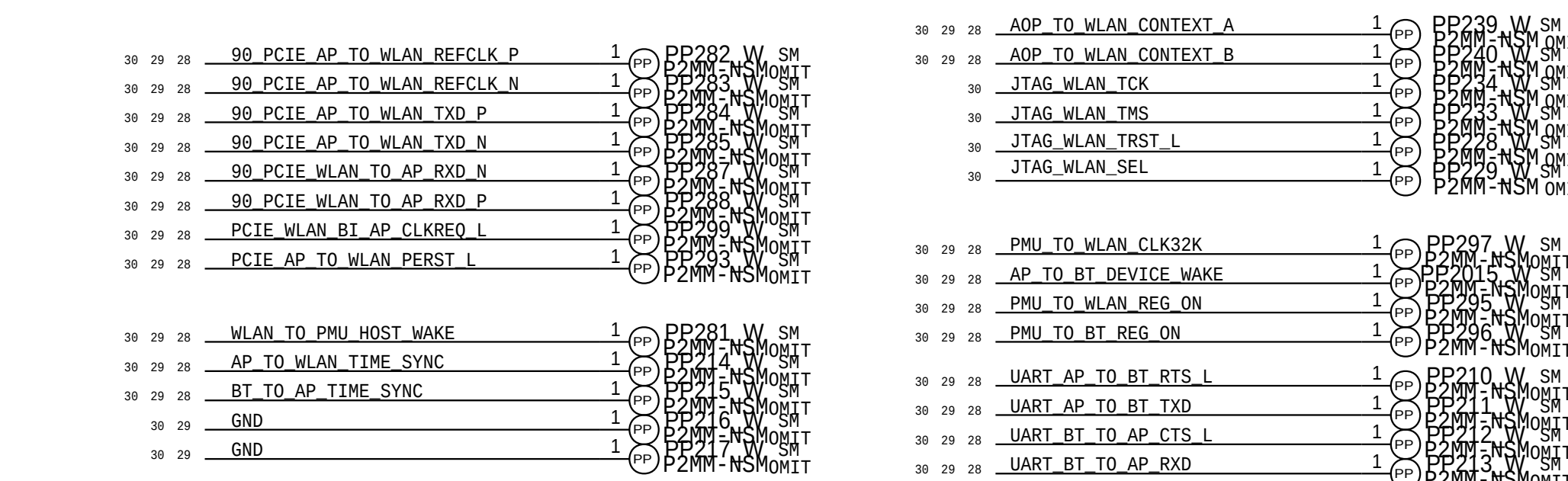
```

U_WLAN_W ALTERNATES

PART NUMBER	ALTERNATE FOR PART NUMBER	BOM OPTION	REF DES	COMMENTS:
998-14250	998-14229	BOM_TABLE_ALTS	U_WLAN_W	WIFI/BT MODULE

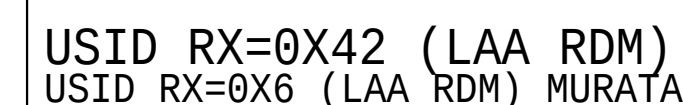
SCHEMATIC APN: 051-02623

DRAWING TITLE			
SCH, MLB, BOT, ICE, D32			
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	DRAWING NUMBER	051-02613	SIZE
	REVISION	4.0.0	
	BRANCH		
	PAGE	1 OF 3	
	SHEET	45 OF 49	

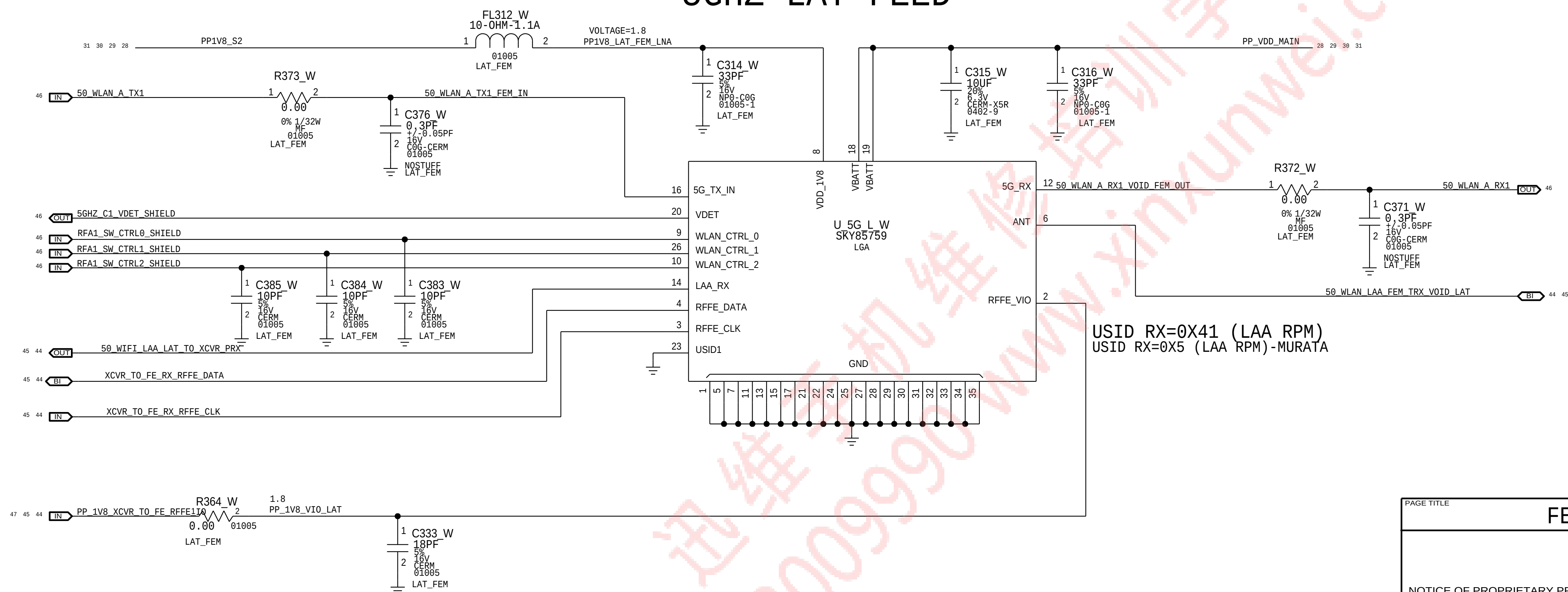
[illegible]

PAGE TITLE	DIETCOKE	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED	DRAWING NUMBER	SIZE
	051-02613	D
	REVISION	
	4.0.0	
	BRANCH	
	PAGE	2 OF 3
	SHEET	46 OF 49

5GHZ UAT FEED



5GHZ LAT FEED



PAGE TITLE		FEM MODULES	
NOTICE OF PROPRIETARY PROPERTY: THE INFORMATION CONTAINED HEREIN IS THE PROPRIETARY PROPERTY OF THE POSSESSOR AGREES TO THE FOLLOWING: I TO MAINTAIN THIS DOCUMENT IN CONFIDENCE II NOT TO REPRODUCE OR COPY IT III NOT TO REVEAL OR PUBLISH IT IN WHOLE OR PART IV ALL RIGHTS RESERVED		DRAWING NUMBER	SIZE
		051-02613	D
		REVISION	
		4.0.0	
		BRANCH	
		PAGE	
		3	OF 3
		SHEET	
		47	OF 49

8

7

6

5

4

3

2

1

NFC CONTROLLER

